



Honda CM 185 & 200 Twins

184 cc. 1977 to 1980

OWNERS WORKSHOP MANUAL

THE
BOOK

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Honda CD185/200 and CM185/200 Owners Workshop Manual

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Martyn Meek

Models covered

CD185T. 180 cc. UK only, May 1978 – July 1979
CM185T. 180 cc. USA only, August 1977 – September 1978
CD200T. 180 cc. Introduced UK only, August 1979
CM200T. 180 cc. Introduced UK February 1980,
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who kindly supplied information and technical assistance on tyre fitting; NGK Spark Plugs (UK) Ltd for information on sparking plug maintenance and electrode conditions; and Re-nold Ltd for advice on chain care and renewal.

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About this manual

The purpose of this manual is to present the owner with a concise and graphic guide which will enable him to tackle any operation from basic routine maintenance to a major overhaul. It has been assumed that any work will be undertaken without the luxury of a well-equipped workshop and a range of manufacturer's service tools.

To this end, the machine featured in the manual was stripped and rebuilt in our own workshop, by a team comprising a mechanic, a photographer and the author. The resulting photographic sequence depicts events as they took place, the hands shown being those of the author and the mechanic.

The use of specialised, and expensive, service tools was avoided unless their use was considered to be essential due to risk of breakage or injury. There is usually some way of improvising a method of removing a stubborn component, provided that a suitable degree of care is exercised.

The author learnt his motorcycle mechanics over a number of years, faced with the same difficulties and using similar facilities to those encountered by most owners. It is hoped that this practical experience can be passed on through the pages of this manual.

Where possible, a well-used example of the machine is chosen for a workshop project, as this highlights any areas which might be particularly prone to giving rise to problems. In this way, any such difficulties are encountered and resolved

before the text is written, and the techniques used to deal with them can be incorporated in the relevant Section. Armed with a working knowledge of the machine, the author undertakes a considerable amount of research in order that the maximum amount of data can be included in this manual.

Each Chapter is divided into numbered Sections. Within these Sections are numbered paragraphs. Cross reference throughout the manual is quite straightforward and logical. When reference is made 'See Section 6.10' it means Section 6, paragraph 10 in the same Chapter. If another Chapter were intended the reference would read, for example, 'See Chapter 2, Section 6.10'. All the photographs are captioned with a section/paragraph number to which they refer and are relevant to the Chapter text adjacent.

Figures (usually line illustrations) appear in a logical but numerical order, within a given Chapter. Fig. 1.1 therefore refers to the first figure in Chapter 1.

Left-hand and right-hand descriptions of the machine and their components refer to the left and right of a given machine when the rider is seated normally.

Whilst every care is taken to ensure that the information in this manual is correct no liability can be accepted by the author or publishers for loss, damage or injury caused by any errors in or omissions from the information given.

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Note: General descriptions and specifications are given in each Chapter immediately after the list of contents. Fault diagnosis is given at the end of the Chapter.

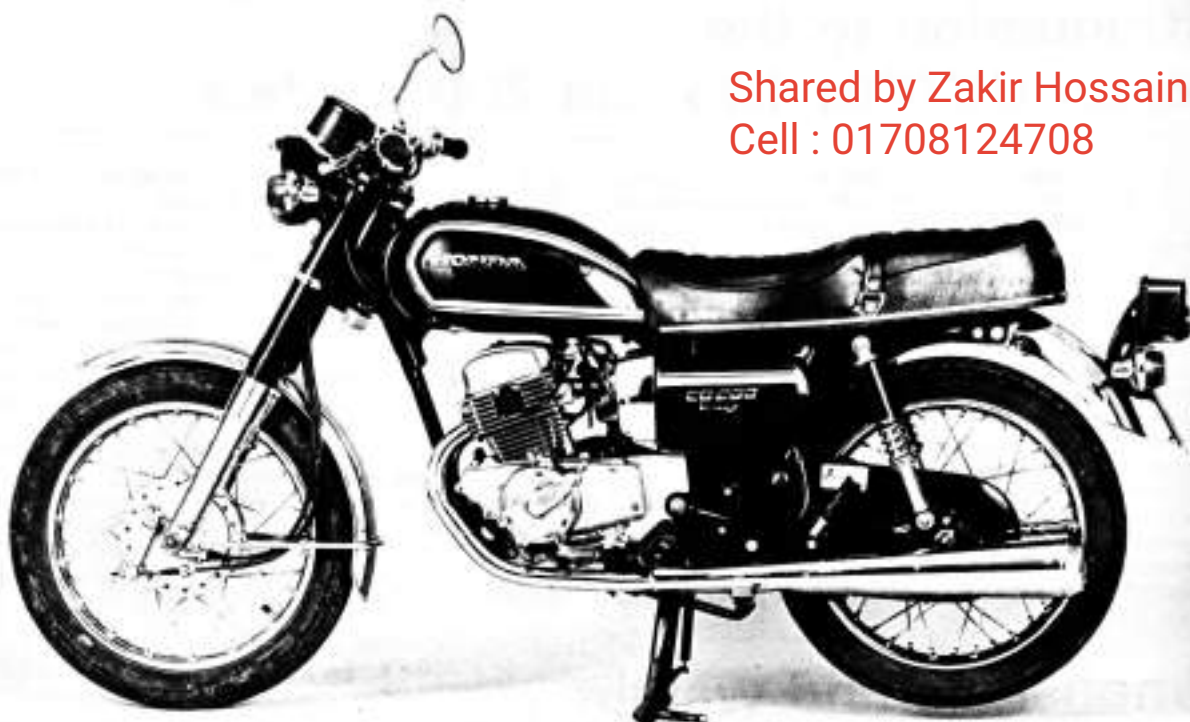
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Right-hand view of 1980 Honda CM200 Twinstar

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Left-hand view of 1980 Honda CD200



Close-up of engine/gearbox unit

Introduction to the Honda CD/CM 185 and 200 models

The present Honda empire, which started in a wooden shack in 1947, now occupies vast factory space, covering all aspects of modern motorcycle design, research, testing and production. The facilities available to the large work force are second to none.

The first motorcycle to be imported into the UK in the early 1960s, the 250 cc twin cylinder 'Dream', was the thin edge of the wedge which has been the Japanese domination of the motorcycle industry. Strange it looked too, to Western eyes, with pressed steel frame, and 'square' styling.

There was, however, nothing strange about the way these modern machines were accepted, and sales were impressive. The trend set by these early machines, that of almost unheard of sophistication, especially on machines of small engine capacity, allied to a remarkably reasonable selling price and a lively performance, ensured their success and started the name Honda on its way to becoming known in virtually every home in the country. Honda now offer machines to the public to cover

every conceivable aspect of motorcycling from tiny 50 cc fun bikes to 1100 cc six-cylinder machines.

The first model of the CD/CM range to be introduced was the CM185T Twinstar which became available in the USA only in August 1977. Using a conventional twin-cylinder, overhead camshaft engine but with 'custom' styling with a stepped seat and high handlebars this machine combines economy with a fashionable appearance more usually seen on machines of much larger engine capacity. The CM185T was superseded in October 1979 by the CM200T Twinstar which remains the only model available in the USA.

The CD models, which have been available only in the UK, share the engine and many cycle parts of the CM variants but are styled more conservatively in keeping with the theme of economy and simplicity. The CD185T, which was introduced in May 1978, was discontinued on introduction of the CD200T in August 1979. The US styled CM200T was introduced into the UK in February 1980.

Dimensions and weight

Overall length:

CD185T	1975 mm (77.8 in)
CD200T	1990 mm (78.3 in)
CM185T	1960 mm (77.2 in)
CM200T	1990 mm (78.3 in)

Overall width:

CD185T	780 mm (30.7 in)
CD200T	845 mm (33.3 in)
CM185T	830 mm (32.7 in)
CM200T	845 mm (33.3 in)

Overall height:

CD185T	1055 mm (41.5 in)
CD200T	1105 mm (43.5 in)
CM185T	1130 mm (44.5 in)
CM200T	1105 mm (43.5 in)

Wheelbase:

CD185T	1275 mm (50.0 in)
CD200T	1280 mm (50.4 in)
CM185T	1280 mm (50.4 in)
CM200T	1280 mm (50.4 in)

Seat height:

CD185T	745 mm (29.3 in)
CD200T	744 mm (29.3 in)
CM185T	730 mm (28.7 in)
CM200T	730 mm (28.7 in)

Ground clearance:

CD185T	145 mm (5.7 in)
CD200T	150 mm (5.9 in)
CM185T	180 mm (7.1 in)
CM200T	175 mm (6.9 in)

Dry weight:

CD185T	131 kg (289 lb)
CD200T	122 kg (269 lb)
CM185T	124 kg (273 lb)
CM200T	122 kg (269 lb)

Ordering spare parts

When ordering spare parts for the CD/CM 185 or 200 models, it is advisable to deal direct with an official Honda agent, who will be able to supply many of the items required ex-stock. It is advisable to get acquainted with the local Honda agent, and to rely on his advice when purchasing spares. He is in a better position to specify exactly the parts required and to identify the relevant spart part numbers so that there is less chance of the wrong part being supplied by the manufacturer due to a vague or incomplete description.

When ordering spares, always quote the frame and engine numbers in full, together with any prefixes or suffixes in the form of letters. The frame number is found stamped on the right-hand side of the steering head, in line with the forks. The engine number is stamped on the left-hand side of the crankcase, immediately in front of the gearchange lever shaft.

Use only parts of genuine Honda manufacture. A few pattern parts are available, sometimes at cheaper prices, but there is no guarantee that they will give such good service as the originals they replace. Retain any worn or broken parts until the replacements have been obtained; they are sometimes needed as a pattern to help identify the correct replacement when design changes have been made during a production run.

Some of the more expendable parts such as sparking plugs, bulbs, tyres, oils and greases etc., can be obtained from accessory shops and motor factors, who have convenient opening hours, and can often be found not far from home. It is also possible to obtain parts on a Mail Order basis from a number of specialists who advertise regularly in the motorcycle magazines.



Frame number location



Engine number location

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Safety First!

Professional motor mechanics are trained in safe working procedures. However enthusiastic you may be about getting on with the job in hand, do take the time to ensure that your safety is not put at risk. A moment's lack of attention can result in an accident, as can failure to observe certain elementary precautions.

There will always be new ways of having accidents, and the following points do not pretend to be a comprehensive list of all dangers; they are intended rather to make you aware of the risks and to encourage a safety-conscious approach to all work you carry out on your vehicle.

Essential DOs and DON'Ts

DON'T start the engine without first ascertaining that the transmission is in neutral.

DON'T suddenly remove the filler cap from a hot cooling system – cover it with a cloth and release the pressure gradually first, or you may get scalded by escaping coolant.

DON'T attempt to drain oil until you are sure it has cooled sufficiently to avoid scalding you.

DON'T grasp any part of the engine, exhaust or silencer without first ascertaining that it is sufficiently cool to avoid burning you.

DON'T syphon toxic liquids such as fuel, brake fluid or antifreeze by mouth, or allow them to remain on your skin.

DON'T inhale brake lining dust – it is injurious to health.

DON'T allow any spilt oil or grease to remain on the floor – wipe it up straight away, before someone slips on it.

DON'T use ill-fitting spanners or other tools which may slip and cause injury.

DON'T attempt to lift a heavy component which may be beyond your capability – get assistance.

DON'T rush to finish a job, or take unverified short cuts.

DON'T allow children or animals in or around an unattended vehicle.

DON'T inflate a tyre to a pressure above the recommended maximum. Apart from overstressing the carcase and wheel rim, in extreme cases the tyre may blow off forcibly.

DO ensure that the machine is supported securely at all times. This is especially important when the machine is blocked up to aid wheel or fork removal.

DO take care when attempting to slacken a stubborn nut or bolt. It is generally better to pull on a spanner, rather than push, so that if slippage occurs you fall away from the machine rather than on to it.

DO wear eye protection when using power tools such as drill, sander, bench grinder etc.

DO use a barrier cream on your hands prior to undertaking dirty jobs – it will protect your skin from infection as well as making the dirt easier to remove afterwards; but make sure your hands aren't left slippery.

DO keep loose clothing (cuffs, tie etc) and long hair well out of the way of moving mechanical parts.

DO remove rings, wristwatch etc, before working on the vehicle – especially the electrical system.

DO keep your work area tidy – it is only too easy to fall over articles left lying around.

DO exercise caution when compressing springs for removal or installation. Ensure that the tension is applied and released in a controlled manner, using suitable tools which preclude the possibility of the spring escaping violently.

DO ensure that any lifting tackle used has a safe working load rating adequate for the job.

DO get someone to check periodically that all is well, when working alone on the vehicle.

DO carry out work in a logical sequence and check that everything is correctly assembled and tightened afterwards.

DO remember that your vehicle's safety affects that of yourself and others. If in doubt on any point, get specialist advice.

IF, in spite of following these precautions, you are unfortunate enough to injure yourself, seek medical attention as soon as possible.

Fire

Remember at all times that petrol (gasoline) is highly flammable. Never smoke, or have any kind of naked flame around, when working on the vehicle. But the risk does not end there – a spark caused by an electrical short-circuit, by two metal surfaces contacting each other, or even by static electricity built up in your body under certain conditions, can ignite petrol vapour, which in a confined space is highly explosive.

Always disconnect the battery earth (ground) terminal before working on any part of the fuel system, and never risk spilling fuel on to a hot engine or exhaust.

It is recommended that a fire extinguisher of a type suitable for fuel and electrical fires is kept handy in the garage or workplace at all times. Never try to extinguish a fuel or electrical fire with water.

Fumes

Certain fumes are highly toxic and can quickly cause unconsciousness and even death if inhaled to any extent. Petrol (gasoline) vapour comes into this category, as do the vapours from certain solvents such as trichloroethylene. Any draining or pouring of such volatile fluids should be done in a well ventilated area.

When using cleaning fluids and solvents, read the instructions carefully. Never use materials from unmarked containers – they may give off poisonous vapours.

Never run the engine of a motor vehicle in an enclosed space such as a garage. Exhaust fumes contain carbon monoxide which is extremely poisonous; if you need to run the engine, always do so in the open air or at least have the rear of the vehicle outside the workplace.

If you are fortunate enough to have the use of an inspection pit, never drain or pour petrol, and never run the engine, while the vehicle is standing over it; the fumes, being heavier than air, will concentrate in the pit with possibly lethal results.

The battery

Never cause a spark, or allow a naked light, near the vehicle's battery. It will normally be giving off a certain amount of hydrogen gas, which is highly explosive.

Always disconnect the battery earth (ground) terminal before working on the fuel or electrical systems.

If possible, loosen the filler plugs or cover when charging the battery from an external source. Do not charge at an excessive rate or the battery may burst.

Take care when topping up and when carrying the battery. The acid electrolyte, even when diluted, is very corrosive and should not be allowed to contact the eyes or skin.

If you ever need to prepare electrolyte yourself, always add the acid slowly to the water, and never the other way round. Protect against splashes by wearing rubber gloves and goggles.

Mains electricity

When using an electric power tool, inspection light etc which works from the mains, always ensure that the appliance is correctly connected to its plug and that, where necessary, it is properly earthed (grounded). Do not use such appliances in damp conditions and, again, beware of creating a spark or applying excessive heat in the vicinity of fuel or fuel vapour.

Ignition HT voltage

A severe electric shock can result from touching certain parts of the ignition system, such as the HT leads, when the engine is running or being cranked, particularly if components are damp or the insulation is defective. Where an electronic ignition system is fitted, the HT voltage is much higher and could prove fatal.

Routine Maintenance

Periodic routine maintenance is a continuous process which should commence immediately the machine is used. The object is to maintain all adjustments and to diagnose and rectify minor defects before they develop into more extensive, and often more expensive, problems.

It follows that if the machine is maintained properly, it will both run and perform with optimum efficiency, and be less prone to unexpected breakdowns. Regular inspection of the machine will show up any parts which are wearing, and with a little experience, it is possible to obtain the maximum life from any one component, renewing it when it becomes so worn that it is liable to fail.

Maintenance should, therefore, be regarded as an insurance policy to help keep the machine in peak condition and to ensure long, trouble-free service. It has the additional benefit of giving early warning of any faults that may develop and will act as a regular safety check, to the obvious advantage of both rider and machine alike.

Regular cleaning can be considered as important as mechanical maintenance. This will ensure that all the cycle parts are inspected regularly and are kept free from accumulations of road dirt and grime.

Cleaning is especially important during the winter months, despite its appearance of being a thankless task which very soon appears pointless. On the contrary, it is during these months that the paintwork, chromium plating, and the alloy

casings suffer the ravages of abrasive grit, rain and road salt. A couple of hours spent weekly on cleaning the machine will maintain its appearance and value, and highlight small points, like chipped paint, before they become a serious problem.

The various maintenance tasks are described under their respective mileage and calendar headings. Accompanying diagrams are provided where necessary. It should be remembered that the interval between the various maintenance tasks serves only as a guide. As the machine gets older or is used under particularly adverse conditions, it would be advisable to reduce the period between each check.

For ease of reference each service operation is described in detail under the relevant heading. However, if further general information is required, it can be found within the manual under the pertinent section heading in the relevant Chapter.

In order that the routine maintenance tasks are carried out with as much ease as possible, it is essential that a good selection of general workshop tools is available.

Included in the kit must be a range of metric ring or combination spanners and a selection of crosshead screwdrivers.

Additionally, owing to the extreme tightness of most casing screws on Japanese machines, an impact screwdriver, together with a choice of large and small crosshead screw bits, is absolutely indispensable. This is particularly so if the engine has not been dismantled since leaving the factory.

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Check engine oil level with dipstick



Replenish engine with the correct amount of oil

Weekly or every 200 miles (320 km)

1 Checking the engine/transmission oil level

Unscrew the combined filler plug and dipstick, which is situated to the front of the right-hand crankcase cover. Wipe off the dipstick using a clean, lint-free rag. Place the plug back in position, but do not screw it home; allow it to rest in position on the edge of the orifice. Remove the plug and note the level of the oil on the dipstick, which should be between the two level marks. If necessary, top up using SAE 10W/40 engine oil to bring the oil level up to the upper mark on the dipstick. Note that if the machine has been ridden recently, it should be allowed to stand for a few minutes to allow the oil clinging to the internal surfaces to drain down into the sump. Note also that the machine should be on its centre stand and on level ground when the oil level is being checked.

2 Tyre pressures

Check the tyre pressures with a gauge that is known to be accurate. It is worthwhile purchasing a pocket gauge for this purpose, because the gauges on garage forecourts cannot always be relied upon to be accurate. Check the pressures when the tyres are cold. If the machine has travelled a number of miles, the tyres will have become hot, and the pressure will have increased. A false reading will therefore result.

Tyre pressures – CD185T/CD200T models

Up to 90 kg (200 lb) load:

Front 25 psi (1.75 kg/cm²)

Rear 28 psi (2.00 kg/cm²)

Over 90 kg (200 lb) load (up to vehicle capacity load):

Front 25 psi (1.75 kg/cm²)

Rear 40 psi (2.80 kg/cm²)

Tyre pressures – CM185T/CM200T models

Up to 90 kg (200 lb) load:

Front 24 psi (1.75 kg/cm²)

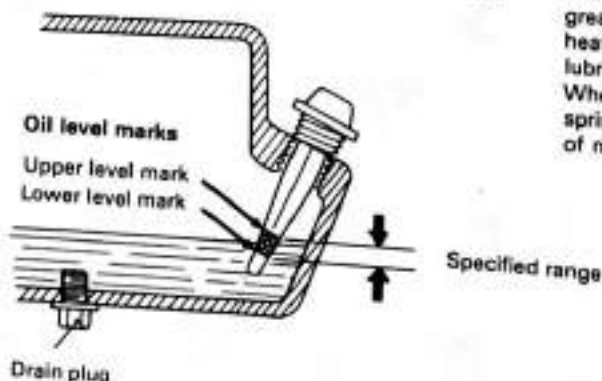
Rear 28 psi (2.0 kg/cm²)

Over 90 kg (200 lb) load (up to vehicle capacity load):

Front 24 psi (1.75 kg/cm²)

Rear 32 psi (2.25 kg/cm²)

At the same time as the tyre pressures are checked, examine the tyres themselves. Check them for damage, especially splitting of the sidewalls. Remove any small stones or other road debris caught between the treads. This is particularly important on the rear tyre, where rapid deflation due to penetration of the inner tube will almost certainly cause total loss of control. When checking the tyres for damage, they should be examined for tread depth in view of both the legal and safety aspects. It is vital to keep the tread depth within the UK legal limits of 1 mm of depth over three-quarters of the tread breadth and around the entire circumference. Many riders, however, consider nearer 2 mm to be the limit for secure road-



Use pocket pressure gauge for tyre pressure checks

holding, traction and braking, especially when riding in adverse weather and road conditions. Honda themselves recommend replacement of the front tyre when there is 1.5 mm (0.05 in) of tread remaining, and replacement of the rear tyre when wear has reached the 2.0 mm (0.08 in) level.

3 Control cable lubrication

Apply a few drops of motor oil to the exposed inner portion of each control cable. This will prevent drying-up of the cables between the more thorough lubrication that should be carried out during the 6 monthly/4500 mile service. The choke control cable should not be neglected when lubricating the cables.

4 Final drive chain lubrication and adjustment – CM models

To prolong the life of the final drive chain and sprockets regular lubrication and adjustment is required. This is particularly the case with the CM models where a full enclosure is not fitted.

Although ordinary engine oil can be used for chain lubrication, its effective life is limited by the speed at which it is flung off the rotating chain. It is recommended that the chain is lubricated whilst in place on the machine, by the application of one of the proprietary chain greases contained in an aerosol can.

A more thorough lubrication should be made at greater intervals or when the machine is used in bad conditions. After removing the chain from the machine it should be cleaned thoroughly in a petrol/paraffin mixture and then allowed to dry. The chain can now be immersed in one of the special chain greases, such as Linklyte or Chainguard. The grease must be heated following the instructions on the can, so that the lubricant penetrates the area between the link pins and rollers. When refitting the chain ensure that the horseshoe-shaped spring link is fitted so that the closed end points in the direction of normal chain travel.

On completion of lubrication check the slack in the chain, measured at the mid-point in the chain lower run. Always check the chain at the tightest point; a chain rarely wears evenly in service. The correct slack is 15–25 mm ($\frac{1}{2}$ – 1 in).

If adjustment is required remove the split pin from the castellated wheel spindle nut and slacken off the nut. Loosen the locknuts on the chain adjuster bolts and, turning the bolts an equal number of times to maintain wheel alignment, adjust the chain slack. Turn the bolts clockwise to tighten the chain. Wheel alignment can be verified by checking that the mark on the outer face of each adjuster is in alignment with the same relative index mark on the fork ends. When adjustment is complete tighten the locknuts and wheel spindle nut and check the adjustment. Finally, tighten the wheel spindle nut to a torque value of 4.0 – 5.0 kgf m (29 – 36 lbf ft) on 185 models, or 5.0 – 7.0 kgf m (36 – 50 lbf ft) on 200 models. Do not forget to secure the nut with a new split pin.

5 Safety check

Besides checking the tyres as already described, give the machine a close visual inspection, checking for loose nuts and fittings, frayed control cables etc.

6 Legal check

Ensure that the lights, horn, and flashing indicators all function correctly, and check that the speedometer is working and giving an accurate reading of road speed.

Monthly or every 800 miles (1200 km)

Carry out the operations listed under the weekly/200 mile heading, then carry out the following:

1 Battery electrolyte level

A conventional lead/acid battery is employed on these models. It is rated at 6 volts 12 Ah.

The transparent plastic case of the battery permits the upper and lower levels of the electrolyte to be observed when the right-hand side panel is removed. Maintenance is normally limited to keeping the electrolyte level between the prescribed

upper and lower limits and by making sure that the vent pipe is not blocked. The lead plates and their separators can be seen through the transparent case, a further guide to the general condition of the battery.

Unless acid is split, as may occur if the machine falls over, the electrolyte should always be topped up with distilled water, to restore the correct level. If acid is spilt on any part of the machine, it should be neutralised with an alkali such as washing soda and washed away with plenty of water, otherwise serious corrosion will occur. Top up with sulphuric acid of the correct specific gravity (1.260 – 1.280) only when spillage has occurred. Check that the vent pipe is well clear of the frame tubes or any other of the cycle parts, for obvious reasons.

Two monthly or every 1500 miles (2400 km)

Carry out the operations listed under the weekly/200 mile and monthly/800 mile headings, then perform the following:

1 Changing the engine/transmission oil

It is recommended that the oil be changed after a run when the engine has been warmed to its normal operating temperature. This assists the oil to drain thoroughly. Place the machine on its centre stand or side stand (CM models), and obtain a suitable container into which the oil may be drained. The sump's oil capacity is approximately 1.5 litre (2.6 Imp pint/1.6 US quart), so ensure that the container is of adequate capacity.

Slacken and remove the hexagon-headed drain plug. It is located on the lower left-hand side wall of the crankcase, below the alternator cover. Also unscrew and remove the combined filler cap/dipstick from the right-hand crankcase cover. Allow the oil to drain completely.

When the oil has drained completely, refit the drain plug. Check that the sealing ring is in good condition, renewing it if necessary. Refill the engine with the correct quantity (1.5 litre/2.6 Imp pint/1.6 US quart) of good quality SAE 10W/40 engine oil. Refit the filler cap/dipstick. Run the engine for a few minutes, then allow the level to settle and recheck the level on the dipstick. If necessary, top up to bring the oil level up to the maximum level mark on the dipstick.

2 Final drive chain lubrication and adjustment – CD models

Because the CD models are fitted with a final drive chain full enclosure, and thus the chain is protected from the worst ravages of rain and road grime, the intervals at which lubrication and adjustment are required can be increased.

Access to the chain for both adjustment and lubrication is made through an inspection hole in the chaincase upper half, which is closed by a rubber bung. Place the machine on the centre stand and remove the bung. Rotate the wheel so that the tightest point in the chain is found and then push the machine off the centre stand so that the weight of the machine is taken by the rear suspension. The correct play in the chain is 20 mm (0.8 in). If the play is incorrect, remove the split pin from the castellated wheel nut and loosen the wheel and also the larger sleeve nut. Slacken off the locknuts on the chain adjuster bolts. Adjustment is achieved by screwing the bolts in or out. Rotate the bolts clockwise to tighten the chain. Always turn the two bolts an equal amount to preserve wheel alignment. This can be checked by ensuring that the index mark on each adjuster is aligned with the same relative vertical alignment mark on both fork ends. On completion of adjustment tighten the sleeve nut to 8.0 – 7.0 kgf m (43 – 50 lbf ft) and tighten the wheel nut. Recheck the adjustment. If all is well, tighten the wheel nut to 4.0 – 5.0 kgf m (29 – 36 lbf ft) and secure the nut with a split pin. Tighten the adjuster bolt locknuts.

Place the machine on the centre stand again and attend to the lubrication. Because of the full enclosure removal of the chain from the machine is unlikely to be necessary for cleaning



Oiling a control cable

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and lubrication, except perhaps at yearly intervals during a full major service. The most suitable lubrication for application whilst the chain is on the machine is one of the proprietary chain lubricants contained in an aerosol can. Engine oil can be used, but its effective life is limited because of the speed with which it is thrown off the rotating chain.

Always keep the drain holes in the chain case free of obstruction to prevent the accumulation of rain water. Apart from being an unsatisfactory lubricant water will promote corrosion and wear.

Four monthly or every 3000 miles (4800 km)

Complete the tasks listed under the previous headings and then perform the following tasks:

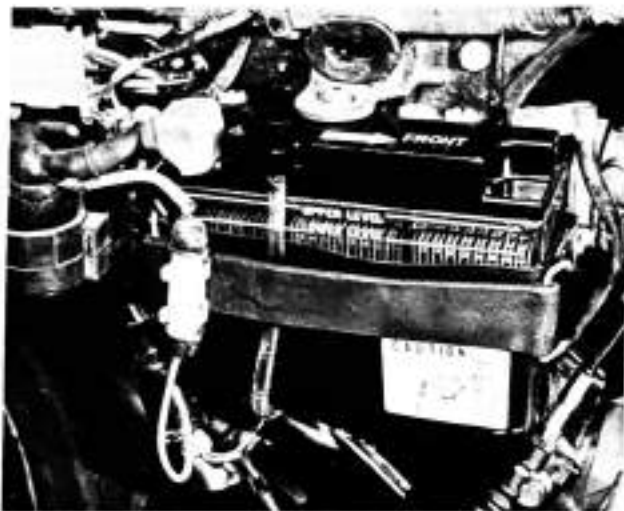
1 Cleaning the air filter element

The air filter element must be removed at regular intervals for cleaning to prevent excessive clogging and a resultant increase in fuel consumption and loss of engine performance. If

the machine is used in particularly dusty conditions, the maintenance periods should be shortened accordingly.

Remove the left-hand frame side panel to gain access to the air filter case. The element can be removed after the two case cover securing nuts and the cover have been detached and the spring blade which retains the element has been withdrawn. Pull the foam sleeve off its supporting case. The foam sleeve and the inner core of the filter should be washed out with a solvent such as methylated spirit or de-natured alcohol. Alternatively warm water and detergent may be used. Dry all the components carefully, then impregnate the foam sleeve with clean gear oil (SAE 89 or 90), squeezing the foam to remove the excess oil. Do not wring out the element because this may damage the fabric, necessitating renewal. If the air filter element is perforated or perished it should be renewed forthwith. Reassemble the filter components ensuring the cover makes a good seal with the case.

On no account run without the air filter attached, or with the element missing. The jetting of the carburettor takes into account the presence of the air filter and engine performance will be seriously affected if this balance is upset.



Maintain battery electrolyte level between upper and lower level lines



Engine oil drain plug

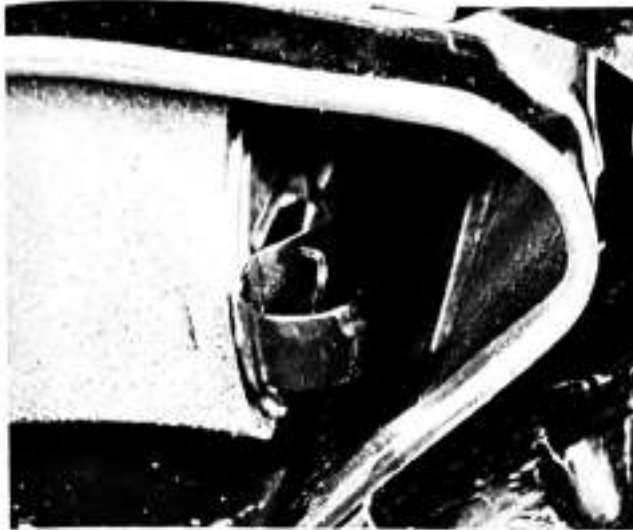


Check chain tension through aperture in chain case (CD models)



Lubricate chain at regular intervals with a recommended lubricant

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Air filter element is held in by spring blade



Cam chain tensioner adjuster stud and locknut

2 Adjusting the cam chain tension

Adjustment of the cam chain tension must be performed with the engine stopped. From the left-hand side of the engine, remove two screws (185 models) or three screws (200 models) and detach the circular inspection cover. Using a spanner on the alternator rotor centre bolt (185 models) or on the large engine turning hexagon (200 models), turn the engine anti-clockwise until the 'T1' mark (185 models) or 'T' mark (200 models) on the automatic timing unit (ATU) is aligned with the fixed index mark on the left hand crankcase cover. These marks may be viewed through the aperture in the contact breaker baseplate.

Slacken the cam chain tensioner locknut, located at the rear of the cylinder block, and then retighten the locknut. The cam chain tensioner will automatically position itself to provide the correct tension when the locknut is slackened. Note that when retightening the locknut, the adjuster stud must be prevented from turning.



Cam chain tensioning alignment mark on ATU

3 Checking and adjusting the valve clearances (tappets)

It is important that the correct valve clearances are maintained to ensure proper operation of the valve assemblies. A small amount of free play is designed into each valve train to allow for expansion of the various engine components. If the setting deviates greatly from that specified, a marked drop in performance will be evident. In the case of the clearance becoming too great, it will be found that valve operation will be noisy, and performance will drop off as a result of the valves not opening fully. If on the other hand, the clearance is too small, the valves may not close completely. This will not only cause loss of compression, but will also cause the exhaust valve to burn out very quickly. In extreme cases, the valve head may strike the piston crown, causing extensive damage to the engine. The clearances should be checked and adjusted with a cold engine.

To gain access to the rocker assemblies so that inspection and adjustment can take place, the cylinder head cover must first be detached. If considered necessary, to facilitate easier access to the rocker assemblies, the petrol tank may be removed. Removal of the petrol tank will necessitate removal of the dualseat on all models and also the side panels on the CM models.

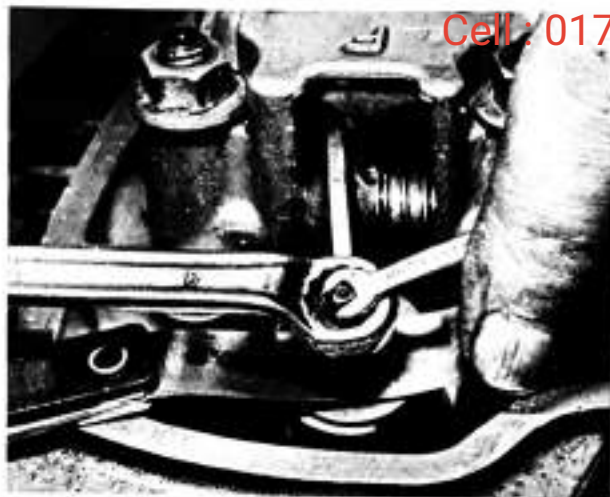
The cylinder head cover is held by two special bolts under which fit rubber sealing plugs. Remove the bolts and lift the cover and sealing ring away. Remove the circular inspection cover from the left-hand crankcase cover, if it was refitted after adjusting the cam chain tension. Remove both sparking plugs so that the engine can be turned over with ease. Using a spanner on the alternator rotor centre bolt (185 models) or the large engine turning hexagon (200 models), rotate the engine anti-clockwise until the 'T1' mark (185 models) or 'T' mark (200 models) on the automatic timing unit (ATU) is aligned with the fixed index mark on the left-hand crankcase cover. These marks may be viewed through the aperture in the contact breaker baseplate. With the marks aligned in this way, the piston of one cylinder will be at top dead centre (TDC) on the compression stroke. In this position, there will be free play in the rockers. If, when the rockers are checked for free play, this is not evident, it is likely that the camshaft is positioned 180° out. This can be corrected by turning the engine through one revolution (360°) and re-aligning the timing marks. There is a further method for ensuring that the engine is in the correct position for valve adjustment. The milled slot in the left-hand end of the camshaft

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must be in the 3 o'clock position with the timing marks aligned.

Using a 0.05 mm (0.002 in) feeler gauge, check the clearance between the top of each valve stem and its corresponding rocker. The feeler gauge must be a light sliding fit with the rocker and valve stem just nipping it. If adjustment is necessary slacken the locknut, and turn the small square-headed adjuster to obtain the correct setting. Tighten the locknut, holding the adjuster at the same time to prevent it from moving. Recheck the gap after tightening the locknut. When both inlet and exhaust valves are correctly set, rotate the engine one full revolution (360°) and realign the marks. The piston of the second cylinder should now be at TDC on the compression stroke. Check, and if necessary, adjust the valve clearances of the second cylinder using the same procedure as for the first pair of valves.

Refit the cylinder head cover checking that the sealing ring and sealing plugs are in good condition, and fit and tighten the bolts. If a torque wrench is available, tighten the cover retaining bolts to a torque setting of 0.8 – 1.2 kgf m (5.8 – 7.8 lbf ft). Refit the petrol tank, dualseat and side panels (CM models), if these items were removed. Do not refit the sparking plugs until the contact breaker adjustment has been checked and ignition timing is completed.



Valve clearance adjustment

4 Checking and adjusting the contact breaker points

It is advisable to check, and if necessary reset, the contact breaker gap at the same time as the valve clearances are dealt with because the left-hand inspection cover needs to be removed for both operations. Rotate the engine slowly by means of the alternator rotor centre bolt (185 models) or the engine turning hexagon (200 models), until the points are in the fully open position. Examine the contact faces; if they are dirty, burnt or pitted, they should be removed for renewal as described in Section 5 of Chapter 3. If the points are in sound condition, proceed with adjustment as follows.

Adjustment is carried out by slackening the two screws which retain the fixed contact breaker point and inserting a screwdriver on the adjusting lugs provided. Turn in the appropriate direction until the gap is within the range 0.3 – 0.4 mm (0.012 – 0.016 in) and then retighten the two retaining screws. It is imperative that the points are open FULLY whilst this adjustment is made, or a false reading will result. Recheck the setting after the screws have been tightened; it is not unknown for the setting to move slightly during the retightening operation.

If adjustment has been made, the ignition timing must now be checked and adjusted as described below.



Correct camshaft notch position for valve adjustment

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5 Checking and adjusting the ignition timing

Whenever the contact breaker points receive attention, the ignition timing should be checked as a matter of course and adjusted, if necessary.

Provided that the contact breaker points are in good condition and care is taken, static adjustment of the ignition timing should be acceptably accurate. If at all possible, however, the timing should be checked with the engine running using a stroboscopic lamp.

Place a spanner on the rotor centre bolt (185 models) or on the engine turning hexagon (200 models), and turn the engine anti-clockwise whilst viewing the ATU through the aperture in the contact breaker points baseplate. If the ignition timing is correct, the 'F1' (185 models) or 'F' (200 models) line on the ATU will coincide exactly with the fixed index line, when the points are just beginning to separate.

The moment at which the contact breaker points separate can be determined using a bulb with two leads soldered to its contact. Connect one lead to the moving contact on the contact breaker assembly and the other lead to a good earth point on the engine. Note that if the lead ends are fitted with crocodile clips, connection and disconnection will be made more convenient. If a multimeter is available, this may be used in preference to the battery and bulb arrangement. Set the multimeter to the 0 – 10 volts dc scale, connect the negative probe to a good earthing point on the engine, and the positive probe to the moving contact of the contact breaker assembly. With the ignition switched on, it will be found that the bulb will light at the exact instant the points separate. In the case of the multimeter, a corresponding needle deflection will occur.

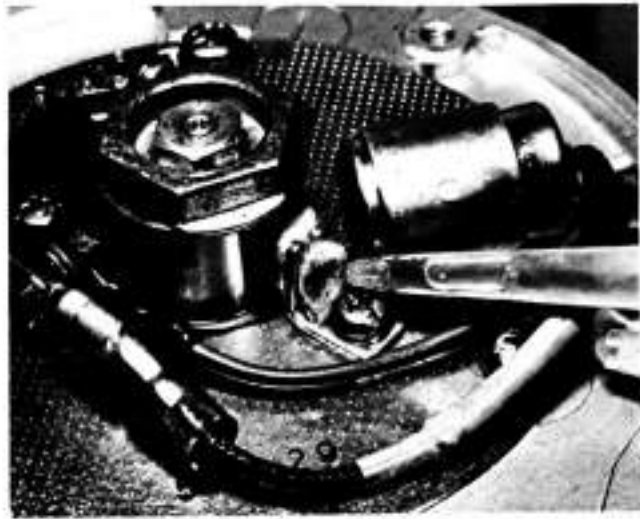
If the ignition timing is incorrect, slacken the three crosshead screws at the periphery of the contact breaker baseplate. The whole baseplate can then be rotated until the points commence to separate as the timing marks on the ATU coincide with the fixed index mark. If the timing is too early (advanced), rotate the baseplate anti-clockwise and if the timing is too late (retarded), rotate the baseplate clockwise, to remedy the situation.

If a stroboscope and a test tachometer are possessed a further test can be performed to check the accuracy of the ignition timing. Full details of this test will be found in Section 6 of Chapter 3.

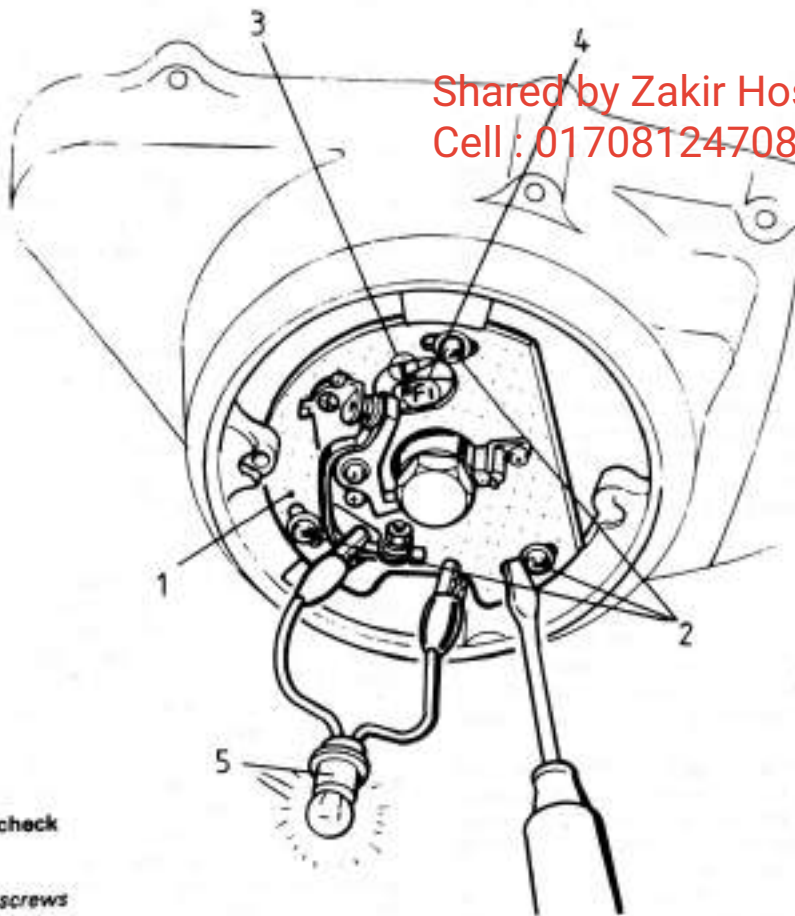
Before refitting the inspection cover, apply a small quantity of light oil or grease to the cam lubricating felt wick. Do not overlubricate, because excess oil may find its way onto the points, causing ignition failure.



Check and adjust if necessary the contact breaker gap



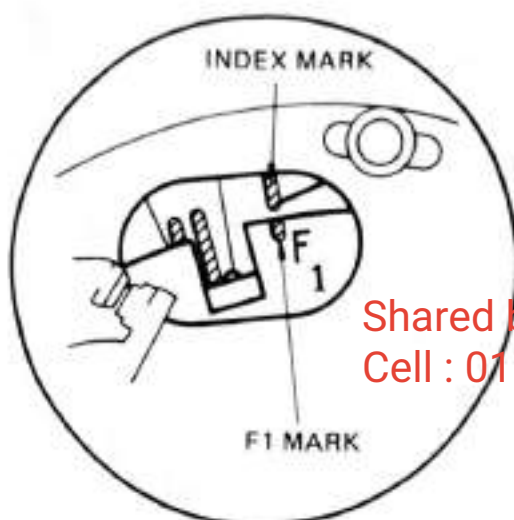
Apply a few drops of thin lubricating oil to the cam felt wick



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Ignition timing check

- 1 Baseplate
- 2 Adjustment screws
- 3 Fixed index mark
- 4 'F' or 'F1' mark
- 5 Test bulb



Ignition timing alignment marks



Clutch operating lever cable adjuster

6 Cleaning and adjusting the sparking plugs

Pull off the sparking plug caps, and using the correct spanner, remove the plugs.

Before cleaning, compare the plugs with those shown in the sparking plug condition chart in Chapter 3 to obtain an indication of the running condition of each cylinder. Clean both plugs, using a wire brush – the type used for cleaning suede shoes is ideal – and carefully clean the electrode faces with a strip of fine emery cloth or paper. Check the plug gap with a feeler gauge, adjusting it if necessary to within the range 0.6 – 0.7 mm (0.024 – 0.028 in). Make adjustments by bending the outer electrode, ensuring that the two electrode faces are kept square. On no account should the centre electrode be bent as this will only succeed in cracking the ceramic insulator, rendering the plug useless. The sparking plug must be of the correct length (reach) and grade.

When refitting the sparking plugs, smear the threads with a little graphited grease; this will aid subsequent removal.

7 Carburettor adjustment

If rough running of the engine has developed some adjustment of the carburettor pilot setting and tick-over speed may be required. If this is the case refer to Chapter 2, Section 7 for details. Do not make these adjustments unless they are obviously required, there is little to be gained by unwarranted attention to the carburettor.

8 Clutch adjustment

Accurate adjustment of the clutch is necessary to ensure efficient operation of the unit. There is no provision for adjustment of the clutch itself. Adjustment is possible at two points on the clutch cable, one at the operating lever on the handlebar and the other at the operating arm on the right-hand crankcase cover.

Normal adjustment is made using the cable lower end adjuster, at the operating arm on the crankcase cover. Slacken the locknut and turn the adjuster nut. This tensions the inner cable and reduces the free play at the handlebar lever. When the free play at the handlebar lever is between 10 – 20 mm (0.4 – 0.8 in), the adjustment is correct. Retighten the lower end locknut. Fine adjustment of the clutch is provided on the handlebar lever by means of a threaded adjuster and locking. If required, a small amount of adjustment can be obtained here by slackening the locking and turning the adjuster inwards or outwards as required. Recheck the freeplay before retightening the locking.

9 Brake adjustment

The frequency of brake adjustment is dependent largely on the conditions in which the machine is used. It follows that a machine which is ridden hard and covers high mileages will need more frequent attention than its commuting counterpart. It is difficult, therefore, to give precise inspection intervals, but it follows that brake lining wear should be checked, and compensated for, more frequently on a hard-ridden machine.

The front and rear brakes on all models are of the single leading shoe (s/s) drum type. Both front and rear units should be adjusted so that brake operation commences just after the handlebar lever or foot pedal has commenced movement. As a rough guide, Honda recommend that the front brake lever end should move 10 – 20 mm (0.4 – 0.8 in) before the brake begins to take effect. If adjustment is necessary, there are two places on the brake cable where adjustment can be made. Major adjustments should be performed using the adjuster nut at the lower end of the cable at the operating lever on the brake plate. Fine adjustment can be made using the threaded adjuster at the handlebar lever, after the locking has been slackened.

The rear brake should start taking effect when the brake pedal has been depressed by about 20 – 30 mm (0.8 – 1.2 in). Adjustment can be carried out using the adjuster nut at the end of the brake operating rod. If required, the height of the brake pedal in relation to the footrest can be adjusted. A stopper bolt and locknut is fitted to the footrest mounting bar which passes below the crankcase. The head of the stopper bolt bears upon a small integral plate on the brake pedal. The height of the brake pedal can, therefore, be raised or lowered by altering the stopper bolt position, after first slackening the locknut. The brake pedal free play must be readjusted if the pedal height has been altered.

Check also that the stop lamp switch does not need readjusting after brake pedal height or free play adjustment. The stop lamp switch should operate as soon as the brake begins to function; when the brake pedal is approximately 20 mm from its position at rest. Adjust the switch by means of the adjuster nut on the threaded switch body.

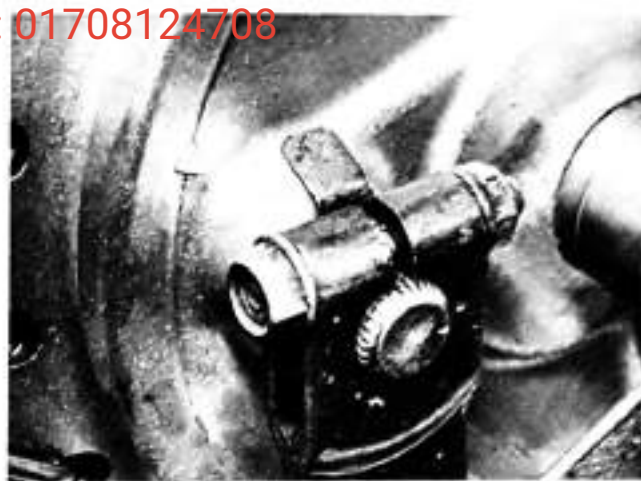
The condition of the brake linings can be ascertained without the need for any dismantling. Incorporated on both the front and rear drum brake plates is an indicator mark. There is also a small metal tag, fitted to the splined end of each brake operating cam, which carries an arrow mark. If, when the brake pedal is depressed, the arrow mark aligns with or passes the indicator mark, then the linings are in need of replacement.

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Front brake wear indicator plate



Rear brake wear indicator plate

Six monthly or every 4500 miles (7200 km)

Carry out the checks listed under the preceding routine maintenance headings and then complete the following:

1 *Cleaning the oil filter screen*

In order that this filter can be cleaned, the engine oil must first be drained. This is no inconvenience, because the oil should be changed at the same time. Remove the drain bolt and drain out the engine oil as described in the two monthly/1500 miles service section. Detach the right-hand exhaust system and the kickstarter and disconnect the clutch cable from the operating arm. Remove the retaining screws and detach the right-hand crankcase cover. Honda recommend the removal of the oil pump complete with the filter screen. Align the three holes in the oil pump internal driven gear with the three holes in the pump outer cover. The three pump retaining screws will now be visible. Remove the screws and lift the complete oil pump unit away. If the screws are very tight an impact driver may be used as long as care is taken not to damage any part of the pump assembly. As an alternative method of filter screen removal, it was found that the screen could be prised away from the pump with careful use of a flat-bladed screwdriver, and then pulled clear from the casing. The disadvantage with this method of screen removal is apparent when refitting the screen. Care must be taken to ensure the screen is installed correctly and that the rubber retainer fits securely over the base of the pump pick-up. This operation can be somewhat 'fiddly' and time-consuming, therefore negating any advantage gained by not removing the oil pump.

Clean the filter screen carefully in clean petrol to remove any residual oil or debris. A small soft-bristled brush such as an old tooth brush is ideal for cleaning the filter screen. If the screen is found to be damaged in any way it must be renewed immediately. Clean out the rubber screen retainer and check it is not damaged.

Refit the screen and retainer to the pick-up extension on the oil pump. Reassemble the components removed and refill the engine with new engine oil. Further details and information on oil pump removal etc will be found in Chapter 2, Section 12.

2 *Checking the steering head bearings*

Place the machine on the centre stand (where fitted) and raise the front wheel clear of the ground by placing blocks under the crankcase. Care should be taken that the machine is supported securely, particularly in the case of CM models which do not have a centre stand.

Check the play in the steering head bearings by standing in front of the front wheel, grasping the bottom of the fork legs and pulling and pushing in a fore and aft direction. Play in the

bearings will be evident as movement between the top and bottom yokes and the frame head lug. Do not confuse play at the steering head with play between the fork stanchions and lower legs.

If adjustment is required, slacken the domed nut which screws onto the centre of the steering stem above the fork top yoke. Using a C-spanner adjust the play by tightening the slotted adjuster nut which lies just below the top yoke. Do not overtighten; it is only too easy to apply a load of several tons to the ball bearings in the steering head bearings. The handlebars should turn smoothly from lock-to-lock with the play just taken up. Apart from causing damage to the bearings, overtightening will cause the machine to weave at low speeds. When adjustment is correct retighten the domed crown nut.

Yearly or every 11 000 miles (18 300 km)

The yearly routine maintenance interval should be considered as an opportunity to carry out what amounts to a minor overhaul, particularly in cases where a winter lay-up gives ample time for detailed attention to the machine. In addition to completing all the tasks in the foregoing sections and also those listed below, give the machine an extensive clean and retouch any chipped or worn paintwork. In addition to this make a close visual inspection of all control assemblies and linkages making repairs where necessary.

1 *Crankcase breather draining – US models only*

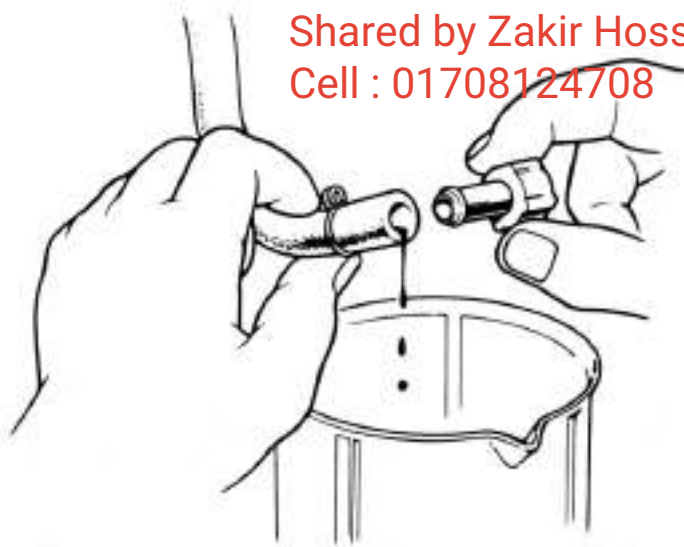
US models are fitted with a crankcase breather which incorporates an oil/water separator. Accumulated water and sludge from the system is stored in the breather tube until the drain plug is removed from the tube end, and any collected fluid is drained. If the machine is ridden in heavy rain or for long periods on full throttle the rate of accumulation will increase; reduce the drainage intervals accordingly. The build-up of deposits can be monitored by viewing the level through the transparent portion of the breather drain tube.

2 *Changing the fork oil*

Drain and replenish the fork oil, working on one leg at a time so that the fork leg remaining undisturbed will support the weight of the machine.

Remove the fork top bolt and then place a drain tray under the drain plug of one fork leg. Remove the plug and allow the oil to drain out. The forks may be pumped up and down a few times to expel any residual oil. Fit and tighten the drain plug, ensuring that the sealing washer is in good condition and is not omitted. Replenish the fork leg with the quantity and grade of fluid specified.

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Draining the crankcase breather tube – US models



Remove fork drain bolt to release fork oil

ENGINE OIL : 1.5 LITRE or 2.6 IMP PINTS

Quick glance maintenance adjustments and capacities

Engine/gearbox oil capacity	1.5 litre (2.6 Imp pint/1.6 US quart)		
Front fork oil capacity (per leg):	CD185/CD200	CM200	CM185
After overhaul	135-140 cc (4.8-4.9 oz)	125 cc (4.2 oz)	125 cc (4.2 oz)
After draining	125-130 cc (4.4-4.6 oz)	115 cc (3.8 oz)	98 cc (3.3 oz)
Contact breaker gap	0.3 - 0.4 mm (0.012 - 0.016 in)		
Sparking plug gap	0.6 - 0.7 mm (0.024 - 0.028 in)		
Tyre pressures (cold):	CD185T/CD200T	CM185T/CM200T	
Up to 90 kg (200 lb) load:			
Front	25 psi (1.75 kg/cm ²)	24 psi (1.75 kg/cm ²)	
Rear	28 psi (2.00 kg/cm ²)	28 psi (2.00 kg/cm ²)	
Over 90 kg (200 lb) load:			
Front	25 psi (1.75 kg/cm ²)	24 psi (1.75 kg/cm ²)	
Rear	40 psi (2.8 kg/cm ²)	32 psi (2.25 kg/cm ²)	
Valve clearances (cold)	0.05 mm (0.002 in)		

Recommended lubricants

Component	Lubricant
Engine/gearbox unit:	
General, all temperature use	SAE 10W/40
-10°C (15°F) to 30°C (85°F)	SAE 20W/50
Below 0°C (32°F)	SAE 10W
-10°C (15°F) to 15°C (60°F)	SAE 20 or 20W
Above 15°C (60°F)	SAE 30
Front forks	Automatic Transmission Fluid (ATF)
Wheel bearings	High melting point grease
Steering head bearings	High melting point grease
Final drive chain	Aerosol chain lubricant or graphite-based chain lubricant such as Linklyfe or Chainguard
Control cables	Engine oil or light machine oil
Pivot points	Multi-purpose grease



Castrol Engine Oils **Castrol Grand Prix**

Castrol Grand Prix 10W/40 four stroke motorcycle oil is a superior quality lubricant designed for air or water cooled four stroke motorcycle engines, operating under all conditions.

Castrol TT Two Stroke Oil

Castrol TT Two Stroke oil is a high quality lubricant specially formulated for both air and water cooled two stroke motorcycle engines. It is readily miscible with fuel and is designed to protect against malfunction of sparking plugs, build up of combustion chamber deposits, seizure and scuffing of pistons and exhaust port blocking.

Castrol R40

Castrol R40 is a castor-based lubricant specially designed for racing and high speed rallying, providing the ultimate in lubrication. Castrol R40 should never be mixed with mineral-based oils, and further additives are unnecessary and undesirable. A specialist oil for limited applications.

Castrol Gear Oils

Castrol Hypoy EP90

An SAE 90 mineral-based extreme pressure multi-purpose gear oil, primarily recommended for the lubrication of conventional hypoid differential units operating under moderate service conditions. Suitable also for some gearbox applications.

Castrol Hypoy Light EP 80W

A mineral-based extreme pressure multi-purpose gear oil with similar applications to Castrol Hypoy but an SAE rating of 80W and suitable where the average ambient tem-

peratures are between 32°F and 10°F. Also recommended for manual transmissions where manufacturers specify an extreme pressure SAE 80 gear oil.

Castrol Hypoy B EP80 and B EP90

Are mineral-based extreme pressure multi-purpose gear oils with similar applications to Castrol Hypoy, operating in average ambient temperatures between 90°F and 32°F. The Castrol Hypoy B range provides added protection for gears operating under very stringent service conditions.

Castrol Greases

Castrol LM Grease

A multi-purpose high melting point lithium-based grease suitable for most automotive applications, including chassis and wheel bearing lubrication.

Castrol MS3 Grease

A high melting point lithium-based grease containing molybdenum disulphide. Suitable for heavy duty chassis application and some CV joints where a lithium-based grease is specified.

Castrol BNS Grease

A bentone-based non melting high temperature grease for ultra severe applications such as race and rally car front wheel bearings.

Other Castrol Products

Castrol Girling Universal Brake and Clutch Fluid

A special high performance brake and clutch fluid with an advanced vapour lock performance. It is the only fluid recommended by

Girling Limited and surpasses the performance requirements of the current SAE J1703 Specification and the United States Federal Motor Vehicle Safety Standard No. 116 DOT 3 Specification.

In addition, Castrol Girling Universal Brake and Clutch fluid fully meets the requirements of the major vehicle manufacturers.

Castrol Fork Oil

A specially formulated fluid for the front forks of motorcycles, providing excellent damping and load carrying properties.

Castrol Chain Lubricant

A specially developed motorcycle chain lubricant containing non-drip, anti corrosion and water resistant additives which afford excellent penetration, lubrication and protection of exposed chains.

Castrol Everyman Oil

A light-bodied machine oil containing anti-corrosion additives for both household use and cycle lubrication.

Castrol DWF

A de-watering fluid which displaces moisture, lubricates and protects against corrosion of all metals. Innumerable uses in both car and home. Available in 400gm and 200gm aerosol cans.

Castrol Easing Fluid

A rust releasing fluid for corroded nuts, locks, hinges and all mechanical joints. Also available in 250ml tins.

Castrol Antifreeze

Contains anti-corrosion additives with ethylene glycol. Recommended for the cooling system of all petrol and diesel engines.

Working conditions and tools

When a major overhaul is contemplated, it is important that a clean, well-lit working space is available, equipped with a workbench and vice, and with space for laying out or storing the dismantled assemblies in an orderly manner where they are unlikely to be disturbed. The use of a good workshop will give the satisfaction of work done in comfort and without haste, where there is little chance of the machine being dismantled and reassembled in anything other than clean surroundings. Unfortunately, these ideal working conditions are not always practicable and under these latter circumstances when improvisation is called for, extra care and time will be needed.

The other essential requirement is a comprehensive set of good quality tools. Quality is of prime importance since cheap tools will prove expensive in the long run if they slip or break and damage the components to which they are applied. A good quality tool will last a long time, and more than justify the cost. The basis of any tool kit is a set of open-ended spanners, which can be used on almost any part of the machine to which there is reasonable access. A set of ring spanners makes a useful addition, since they can be used on nuts that are very tight or where access is restricted. Where the cost has to be kept within reasonable bounds, a compromise can be effected with a set of combination spanners – open-ended at one end and having a ring of the same size on the other end. Socket spanners may also be considered a good investment, a basic $\frac{3}{8}$ in or $\frac{1}{2}$ in drive kit comprising a ratchet handle and a small number of socket heads, if money is limited. Additional sockets can be purchased, as and when they are required. Provided they are slim in profile, sockets will reach nuts or bolts that are deeply recessed. When purchasing spanners of any kind, make sure the correct size standard is purchased. Almost all machines manufactured outside the UK and the USA have metric nuts and bolts, whilst those produced in Britain have BSF or BSW sizes. The standard used in the USA is AF, which is also found on some of the later British machines. Other tools that should be included in the kit are a range of crosshead screwdrivers, a pair of pliers and a hammer.

When considering the purchase of tools, it should be remembered that by carrying out the work oneself, a large proportion of the normal repair cost, made up by labour charges, will be saved. The economy made on even a minor overhaul will go a long way towards the improvement of a tool kit.

In addition to the basic tool kit, certain additional tools can prove invaluable when they are close to hand, to help speed up a multitude of repetitive jobs. For example, an impact screwdriver will ease the removal of screws that have been

tightened by a similar tool during assembly, without risk of damaging the screw heads. And, of course, it can be used again to retighten the screws, to ensure an oil or airtight seal results. Circlip pliers have their uses too, since gear pinions, shafts and similar components are frequently retained by circlips that are not too easily displaced by a screwdriver. There are two types of circlip pliers, one for internal and one for external circlips. They may also have straight or right-angled jaws.

One of the most useful of all tools is the torque wrench, a form of spanner that can be adjusted to slip when a measured amount of force is applied to any bolt or nut. Torque wrench settings are given in almost every modern workshop or service manual, where the extent is given to which a complex component, such as a cylinder head, can be tightened without fear of distortion or leakage. The tightening of bearing caps is yet another example. Overtightening will stretch or even break bolts, necessitating extra work to extract the broken portions.

As may be expected, the more sophisticated the machine, the greater is the number of tools likely to be required if it is to be kept in first class condition by the home mechanic. Unfortunately there are certain jobs which cannot be accomplished successfully without the correct equipment and although there is invariably a specialist who will undertake the work for a fee, the home mechanic will have to dig more deeply in his pocket for the purchase of similar equipment if he does not wish to employ the services of others. Here a word of caution is necessary; since some of these jobs are best left to the expert. Although an electrical multimeter of the Avo type will prove helpful in tracing electrical faults, in inexperienced hands it may irrevocably damage some of the electrical components if a test current is passed through them in the wrong direction. This can apply to the synchronisation of twin or multiple carburettors too, where a certain amount of expertise is needed when setting them up with vacuum gauges. These are, however, exceptions. Some instruments, such as a strobe lamp, are virtually essential when checking the timing of a machine powered by a CDI ignition system. In short, do not purchase any of these special items unless you have the experience to use them correctly.

Although this manual shows how components can be removed and replaced without the use of special service tools (unless absolutely essential), it is worthwhile giving consideration to the purchase of the more commonly used tools if the machine is regarded as a long term purchase. Whilst the alternative methods suggested will remove and replace parts without risk of damage, the use of the special tools recommended and sold by the manufacturer will invariably save time.

Chapter 1 Engine, clutch and gearbox

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Specifications

Engine

Type	
Bore	

185 models
Twin cylinder, air cooled
ohc, four-stroke
53.0 mm (2.09 in)

200 models
Twin cylinder, air cooled
ohc, four-stroke
53.0 mm (2.09 in)

Stroke	41.0 mm (1.61 in)	44.0 mm (1.73 in)
Capacity	181 cc (11.0 cu in)	194 cc (11.9 cu in)
Compression ratio	9.0:1	8.8:1
Lubrication system	Wet sump, force fed by trochoid oil pump	Wet sump, force fed by trochoid oil pump

Cylinder block

Cylinder bore:	All models	
Standard	53.00 – 53.01 mm (2.0866 – 2.0870 in)	
Service limit	53.10 mm (2.0905 in)	
Ovality limit	0.1 mm (0.04 in)	

Pistons

Skirt outside diameter (OD)	52.96 – 52.99 mm (2.0850 – 2.0862 in)
Service limit	52.87 mm (2.0815 in)
Gudgeon pin OD	13.994 – 14.000 mm (0.5509 – 0.5512 in)
Service limit	13.98 mm (0.5504 in)
Gudgeon pin bore ID	14.002 – 14.008 mm (0.5513 – 0.5515 in)
Service limit	14.05 mm (0.5531 in)

Piston rings

Ring to groove clearance:		
Top ring	0.015 – 0.045 mm (0.001 – 0.002 in)	
2nd ring	0.015 – 0.045 mm (0.001 – 0.002 in)	
Service limit:		
Top ring	0.12 mm (0.0047 in)	
2nd ring	0.12 mm (0.0047 in)	
Ring end gap:		
Top ring	0.15 – 0.35 mm (0.001 – 0.014 in)	
2nd ring	0.15 – 0.35 mm (0.001 – 0.014 in)	
Service limit:		
Top ring	0.5 mm (0.020 in)	
2nd ring	0.5 mm (0.020 in)	
Ring thickness:		
Top ring	1.175 – 1.195 mm (0.0463 – 0.0469 in)	
2nd ring	1.175 – 1.195 mm (0.0463 – 0.0469 in)	
Service limit:		
Top ring	1.12 mm (0.0441 in)	
2nd ring	1.12 mm (0.0441 in)	

Valves and valve springs

Valve stem OD:		
Inlet	5.450 – 5.485 mm (0.2146 – 0.2152 in)	
Exhaust	5.430 – 5.445 mm (0.2137 – 0.2144 in)	
Service limit:		
Inlet	5.42 mm (0.2134 in)	
Exhaust	5.40 mm (0.2126 in)	
Valve guide ID:		
Inlet and Exhaust	5.475 – 5.485 mm (0.2156 – 0.2159 in)	
Service limit	5.50 mm (0.2165 in)	
Valve stem to guide clearance:		
Inlet	0.010 – 0.035 mm (0.004 – 0.0014 in)	
Exhaust	0.030 – 0.055 mm (0.0022 – 0.0022 in)	
Service limit:		
Inlet	0.08 mm (0.003 in)	
Exhaust	0.10 mm (0.004 in)	
Valve seat width:		
Inlet and exhaust	1.0 mm (0.04 in)	
Service limit	1.5 mm (0.06 in)	
Valve seat angle	45°	
Valve spring free length:		
Outer	36.45 mm (1.435 in)	
Inner	29.90 mm (1.177 in)	
Service limit:		
Outer	35.5 mm (1.398 in)	
Inner	29.0 mm (1.142 in)	

Valve timing

	CD185T/ CD200T	CM185T ('78)	CM185T ('79)	CM200T
Inlet valve opens at	–	45° BTDC	42° BTDC	7°30'
Inlet valve closes at	–	97° ABDC	100° ABDC	27°30'

Exhaust valve opens at	94° BBDC	98° BBDC	27-30
Exhaust valve closes at	49° ATDC	45° ATDC	7-32
Valve clearance (cold)	All models		
Inlet	0.05 mm (0.002 in)		
Exhaust	0.05 mm (0.002 in)		
Camshaft and rockers			
Cam lobe height:			
Inlet	27.695 mm (1.0904 in)		
Exhaust	27.522 mm (1.0835 in)		
Service limit:			
Inlet	26.6 mm (1.0472 in)		
Exhaust	26.5 mm (1.0433 in)		
Camshaft journal OD	19.967 - 19.980 mm (0.7861 - 0.7866 in)		
Service limit	19.92 mm (0.7843 in)		
Camshaft bushing ID	20.063 - 20.083 mm (0.7898 - 0.7906 in)		
Service limit	20.20 mm (0.7965 in)		
Maximum runout	0.05 mm (0.002 in)		
Rocker arm spindle OD	9.978 - 9.987 mm (0.3929 - 0.3940 in)		
Service limit	9.17 mm (0.3610 in)		
Rocker arm bore ID	10.0 - 10.015 mm (0.3937 - 0.3943 in)		
Service limit	10.1 mm (0.3976 in)		
Crankshaft			
Maximum runout	0.1 mm (0.004 in)		
Connecting rod small end ID	14.016 - 14.034 mm (0.5518 - 0.5525 in)		
Service limit	14.08 mm (0.5543 in)		
Connecting rod big-end clearance:			
Axial	0.10 - 0.40 mm (0.0040 - 0.0157 in)		
Service limit	0.60 mm (0.0236 in)		
Radial	0.004 - 0.012 mm (0.0002 - 0.0005 in)		
Service limit	0.05 mm (0.002 in)		
Cylinder head			
Maximum warpage	0.10 mm (0.004 in)		
Clutch			
Type	Wet, multiplate		
No of plates:			
Plain	4		
Friction	5		
No of springs	4		
Plain plate maximum warpage	0.2 mm (0.008 in)		
Friction plate thickness	3.0 mm (0.1181 in)		
Service limit	2.6 mm (0.1023 in)		
Spring free length:			
185 models	36.2 mm (1.4252 in)		
200 models	37.8 mm (1.489 in)		
Service limit:			
185 models	33.1 mm (1.3031 in)		
200 models	33.1 mm (1.3031 in)		
	34.7 mm (1.3700 in)		
Gearbox			
Type	CD/CM 185, CM 200 (UK)	CD200, CM200 (US)	
Primary reduction ratio	4-speed, constant mesh	4-speed, constant mesh	
Gear ratios:	3.833:1	3.632:1	
1st	2.769:1	2.846:1	
2nd	1.722:1	1.778:1	
3rd	1.272:1	1.273:1	
4th	1.000:1	1.000:1	
Final reduction ratio:			
CD185	2.266:1		
All others	2.333:1		
Selector fork end thickness	5.00 - 5.07 mm (0.1969 - 1.996 in)		
Service limit	4.70 mm (0.1850 in)		
Selector fork bore ID	12.000 - 12.018 mm (0.4724 - 0.4731 in)		
Service limit	12.05 mm (0.4745 in)		
Selector fork shaft OD	11.976 - 11.994 mm (0.4515 - 0.422 in)		
Service limit	11.96 mm (0.4709 in)		

Torque wrench settings:**Cylinder head:**

6 mm bolts (3)	1.0 – 1.4 kgf m (7.2 – 10.1 lbf ft)
Rocker carrier nuts (8)	1.6 – 2.0 kgf m (11.6 – 14.5 lbf ft)
Camshaft sprocket bolts (2)	1.7 – 2.3 kgf m (12.3 – 16.6 lbf ft)
Crankshaft bearing holder:	
6 mm (5)	1.0 – 1.4 kgf m (7.2 – 10.1 lbf ft)
6 mm nut (1)	1.0 – 1.4 kgf m (7.2 – 10.1 lbf ft)
Crankcase bolt, 6 mm (10)	1.0 – 1.4 kgf m (7.2 – 10.1 lbf ft)
Generator centre bolt:	
185 models	4.5 – 5.0 kgf m (32.5 – 36.2 lbf ft)
200 models	5.5 – 6.5 kgf m (40.0 – 47.0 lbf ft)
Oil drain bolt, 12 mm	3.0 – 5.0 kgf m (21.7 – 36.2 lbf ft)

1 General description

The engine fitted to the Honda CD185T/CD200T and CM185T/CM200T models is of the single overhead camshaft, vertical twin, four-stroke type. It is a familiar, well-established, design amongst the smaller Honda motorcycles.

The engine/transmission unit is of all-alloy construction, employing vertically split crankcases which house both the crankshaft assembly and the gear clusters. The cylinder head and cylinder barrel are also of light alloy, the latter incorporating steel liners in which the cylinder bores are machined.

In common with virtually all the machines in the Honda range, and with Japanese motorcycles in general, these models feature an overhead camshaft to operate the valve mechanism. The camshaft actuates each cylinder's pair of valves, one inlet and one exhaust, by means of rockers that bear directly on the camshaft. The camshaft is driven by an endless, double row, roller-type chain. The drive chain lower loop is fitted around a duplex sprocket on the crankshaft. The chain passes up through a tunnel between the cylinders and so to a second duplex sprocket fitted to the overhead camshaft. The camshaft, as its description implies, is located at the top of the cylinder head. The advantage of the chain driven overhead camshaft arrangement to operate the valves, instead of the traditional pushrod system, is that higher engine revolutions may be sustained without risk of damage. This is particularly beneficial in the smaller four-stroke engine, where useful power is obtainable only at high engine speeds.

Lubrication is provided by a small trochoid oil pump feeding the major engine components. The lubricating oil is contained in the lower portion of the crankcase which forms a combined sump and oil bath for the gearbox components. The oil is picked up from the sump through a mesh filter screen attached to the oil pump.

All of these models utilize a traditional constant-mesh 4-speed gearbox. The gearbox mainshaft is driven via spur primary gears on the right-hand end of the crankshaft and the multiplate clutch. The oil pump, fitted outboard of the clutch on the right-hand side of the engine, is driven by the second outermost of the two primary drive gears.

Electrical power is generated by a small 6-volt alternator mounted on the left-hand end of the crankshaft. Fitted outboard of the alternator is the contact breaker points and backplate, access to which is provided by a small, detachable inspection cover. On all models, starting is effected by either the traditional kickstarter or the electric starter motor. The starter motor, mounted at the front of the engine on the left-hand side, turns the crankshaft via a roller clutch by means of a drive sprocket on the motor, a short chain and a crankshaft mounted driven sprocket.

2 Operations with the engine/gearbox unit in the frame

Because of the compactness of the engine in the frame it is not possible to remove either the cylinder head or the barrel or

to work on the gearbox without first removing the engine from the frame. Work can, however, be carried out on the following components without prior removal of the engine from the frame:

- Right and left-hand crankcase covers and final drive sprocket cover*
- Cylinder head cover*
- Camshaft and rocker gear*
- Clutch assembly and gear selector external components*
- Oil pump and filter screen*
- Alternator and starter motor*
- Contact breaker and auto-advance mechanism*
- Carburettor*

3 Operations with the engine/gearbox unit removed from the frame

Certain operations can be accomplished only if the complete engine/gearbox unit is removed from the frame. This is because it is necessary to separate the crankcases to gain access to the parts concerned, or because there is insufficient clearance to withdraw parts after they have been slackened or freed from their normal location. These operations include:

- Removal and replacement of the main bearings*
- Removal and replacement of the crankshaft assembly*
- Removal and replacement of the gear cluster, selectors and gearbox main bearings*
- Removal and replacement of the cylinder head, cylinder barrel and piston*

4 Method of engine/gearbox unit removal

As mentioned previously, the engines and gearbox are built in unit and it is necessary to remove the unit complete in order to gain access to either component. Separation is accomplished after the engine/gearbox unit has been removed from the frame and refitting cannot take place until the crankcases have been reassembled. When the crankcases are separated, the gearbox internals will also be exposed. There is no means of working on the engine without disturbing the gearbox, or on the gearbox without first dismantling the engine.

5 Removing the engine/gearbox unit from the frame

1 If the machine is fitted with a centre stand, place the machine securely on this stand. If only a side stand is fitted (CM models), then place the machine astride a stout wooden box or some similar form of support; a plastic milk bottle crate or a crate of the type used by Moto-cross riders to support their machines, are ideal alternatives, so that it will stand unsupported on level ground. In all cases, ensure that there is no likelihood of the machine falling over during engine removal and damaging both itself and the unfortunate operator. Engine

removal can be made much easier if the machine is raised about two feet by means of a stand. A stout table can be modified for this purpose, or alternatively, a few substantial planks and some concrete blocks will suffice. This procedure is by no means essential, but will greatly ease the discomfort of squatting or kneeling down to work.

2 Place a suitable receptacle below the left-hand crankcase cover, then remove the drain plug from the lower crankcase wall and the filler cap/dipstick, and leave the oil to drain whilst further dismantling is carried out. Draining the oil from the engine is best carried out when the engine is warm; the oil will be thinner and so drain more quickly and more completely. Approximately 1.5 litres (2.6 Imp pint/1.6 US quart) of oil should drain out; select the draining receptacle bearing this in mind. When the oil has finished draining, replace and tighten the drain plug, making sure the sealing washer is in good condition.

3 Turn the petrol tap lever to the 'OFF' position. Release the tension on the fuel pipe by pinching together the 'ears' of the spring retaining clip, and prise off the fuel pipe. A small screwdriver can be used to ease the pipe off without straining it.

4 Release the seat mounting bolts which are situated towards the rear of the seat, and lift it away from the frame. Note that on all the models except the CM185T, the left-hand side bolt also partly retains the grab/lifting handle. On the CM185T model, both bolts secure the rearward section of the passenger grab rail. Place the bolts back in their holes and finger-tighten them to avoid misplacing them.

5 It is always a good policy, whenever possible, to screw back in place, finger-tight, bolts and nuts after a component has been removed. This negates the chance of loss and aids correct reassembly. It is also a good idea to have various clean boxes or containers to hand, into which can be put the numerous small components and their fasteners which have been detached.

6 From the CM models, remove both frame side panels by pulling their locating lugs from their corresponding holes in the rear of the petrol tank and frame. Pull off both side panels from their location on the frame of the CD models. This will expose the air filter assembly on the left-hand side, and the battery and electrical components on the right-hand side of the machine.

7 Remove the single bolt that retains the rear of the petrol tank on the CM200T model. On all other models disengage the lug at the rear of the tank from the retension/support block. The tank can then be pulled upwards and back to release it from the front rubber mounting blocks.

8 Disconnect the battery leads at their terminals to prevent the risk of short circuits. If the machine is to be out of service for an extended length of time, the battery should be removed from the machine and given a trickle charge at approximately one month intervals. To free the battery, detach the securing strap and pull the battery out of the carrier. Disconnect the earth lead and the breather tube, and lift the battery away.

9 Remove the complete exhaust system. The exhaust pipes are each attached to the cylinder head exhaust ports by means of a flange that retains two split collars in place. Slacken and remove the two nuts retaining each flange, remove the flange and catch the collars as they drop free. Each silencer is retained to the frame by a plate fitted to the inside surface of the silencer. Two nuts secure the plate to the silencer and two further nuts retain the plate and silencer to a sub-frame loop. Remove these latter nuts and manoeuvre the complete system clear. The exhaust pipes cannot be separated from the silencers because they are welded-up units.

10 Pull the sparking plug caps from the plugs, and secure the HT leads out of the way over the top frame tube.

11 Pull back the rubber boot from the adjuster at the lower end of the clutch cable. Slacken off the cable adjuster nuts and release the cable from the actuating arm on the rear top of the right-hand crankcase cover. Disengage the cable and let it rest clear of the engine.

12 Prise back the rubber boot from the starter motor terminal,

and disconnect the starter motor cable.

13 Disconnect all the leads at the block connectors and individual snap connectors on the left-hand side of the machine and under the dualseat of the 185 models. Separate the large block connector on the right-hand side of the 200 models.

14 Slacken the single crosshead screw which secures the choke cable clamp to the carburettor. With the cable released, free the inner cable from the operating pulley. Unscrew the screw clip band on the rubber intake hose at the carburettor, and prise it rearwards to clear the carburettor intake. Release the spring securing clip and pull off the crankcase breather hose from its locating stub on the left-hand crankcase. Unscrew the carburettor mixing chamber top and withdraw the throttle valve assembly.

15 Remove the pinch bolts from both the kickstart lever and the gearchange lever and remove both levers.

16 Slacken and remove the three crosshead screws which retain the small gearbox sprocket cover. If they have not been removed previously, the screws will probably be very tight, necessitating the use of an impact driver to free them successfully.

17 On CD models, remove two bolts from the lower forward end and from the lower rearward end, of the full chaincase. The lower half can now be detached. It is not necessary to remove the top half of the chaincase to facilitate engine removal, although its removal will avoid damage. The top half is secured by a further two bolts. Remove the spring clip on the connecting link of the drive chain using a pair of pointed-nose pliers or a small screwdriver. Separate the chain and run it off at the rear sprocket. The chainguard fitted to the CM models can be detached if required to avoid possible damage; it is retained by a single forward bolt and by the nut on the lower, left-hand, rear suspension unit mounting stud.

18 Remove the two bolts from the underside of the crankcase on the right-hand side, and the two long through bolts from the left-hand side, and detach the footrest mounting bar and side stand retaining plate.

19 The engine is now held only by the seven mounting bolts. Remove the four front bolts and place them to one side. It is worthwhile removing the entire engine plate to gain the maximum amount of manoeuvring room. Remove the lower rear mounting bolt having first unscrewed its nut from the right-hand side. Remove the upper bolt which passes through a lug cast in the rear of the cylinder head. It is also worthwhile to remove the other two bolts retaining the head steady/top engine mounting plates, and to remove the plates. Finally, remove the upper rear mounting bolt's nut from the right-hand side, freeing the battery earth lead from below the nut. As the bolt is withdrawn, the unit will drop free, and it is advisable to have an assistant to hand to help with these final stages. Before lifting the engine from position, ensure that all cables, wires and other connections are detached and tucked out of the way so that they do not become snagged. The unit is not particularly heavy, and can be lifted clear of the frame by one person.



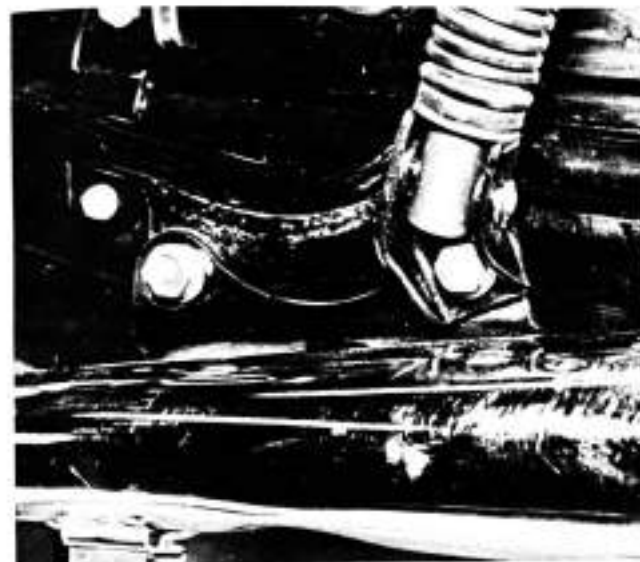
5.4 Release the dualseat after removing the retaining bolts



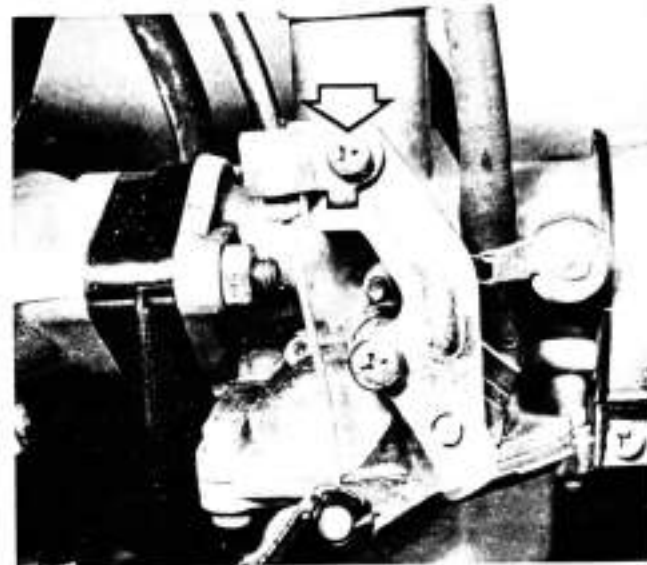
5.7 Disengage rear of tank from retention/support block (not CM200T)



5.9a Exhaust pipes are secured by flange and two nuts



5.9b Silencers are retained by two bolts on sub-frame loops



5.14 Release choke cable from clamp by slackening single screw (arrowed)



5.17 Remove two bolts and detach the chaincase lower half (CD models only)



5.18a Remove these two bolts and bracket from the right-hand underside, and ...

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5.18b ... the two long bolts (arrowed) from the left-hand side, to free the footrest/sidestand assembly

6 Dismantling the engine and gearbox unit: general

1 Before commencing work on the engine unit, the external surfaces should be cleaned thoroughly. A motor-cycle engine has very little protection from road grit and other foreign matter, which will find its way into the dismantled engine if this simple precaution is not taken. One of the proprietary cleaning compounds, such as 'Gunk' or 'Jizer' can be used to good effect, particularly if the compound is permitted to work into the film of oil and grease before it is washed away. Special care is necessary when washing down to prevent water from entering the now exposed parts of the engine unit.

2 Never use undue force to remove any stubborn part unless specific mention is made of this requirement. There is invariably good reason why a part is difficult to remove, often because the dismantling operation has been tackled in the wrong sequence.

3 Mention has already been made of the benefits of owning an impact driver. Most of these tools are equipped with a standard $\frac{1}{2}$ inch drive and an adaptor which can take a variety of screwdriver bits. It will be found that most engine casing screws will need jarring free due to both the effects of assembly by power tools and an inherent tendency for screws to become pinched in alloy castings. If an impact screwdriver is not available, it is often possible to use a crosshead screwdriver fitted with a T-handle as a substitute. An example of this would be the crosshead screwdriver and the box spanner, both from the machine's standard tool kit, fitted together, to form a T-handle attachment.

4 A cursory glance over many machines of only a few years use, will almost invariably reveal an array of well-chewed screw heads. Not only is this unsightly, it can also make emergency repairs impossible. It should also be borne in mind that there are a number of types of crosshead screwdrivers which differ in the angle and design of the driving tangs. To this end, it is always advisable to ensure that the correct tool is available to suit a particular screw.

7 Dismantling the engine/gearbox unit: removing the cylinder head cover, camshaft and rocker gear

1 As noted earlier, it is possible to remove the overhead camshaft and its rocker assembly without removing the engine from the frame. The cylinder head cover and contact breaker points inspection cover should both be removed prior to commencing camshaft/rocker gear removal.

2 Place the engine in an upright position, suitably supported by wooden blocks, so that it remains secure. Loosen and remove the two shouldered bolts that secure the cylinder head cover to the cylinder head. Remove also the two rubber seals fitted to the bolts. Lift the cylinder head cover away, noting the rubber sealing gasket. Unscrew both sparking plugs.

3 If not already removed, unscrew the two screws (185 models) or three screws (200 models) and remove the contact breaker points circular inspection cover and gasket. Apply a spanner to the alternator rotor centre bolt (185 models) or the engine turning hexagon fitted outboard of the automatic timing unit (ATU) (200 models). Using the spanner, rotate the engine in an anti-clockwise (forward) direction until the T1 mark (185 models) or T mark (200 models) on the ATU, visible through the aperture in the contact breaker points baseplate, aligns with the index mark on the left-hand crankcase cover. In this position both pistons will be at TDC and only one camlobe will be engaging a rocker. This will allow removal of the cylinder head without placing undue strain on the camshaft or rocker gear.

4 Slacken the eight nuts which secure the rocker carriers/camshaft holders and the three cylinder head securing bolts. Slacken them evenly, a little at a time, reversing the sequence given in Fig. 1. Following this sequence will help prevent distortion. Note that this operation must only take place when the engine is cold; this too prevents distortion.

5 Remove the eight nuts and their plain washers. Withdraw the three bolts, noting the nut on the single front bolt. Lift the two rocker carriers away individually, complete with the rocker shafts and arms. Note the four hollow dowels, two longer and two shorter, and their respective positions. Place the rocker carrier assemblies to one side for further attention at a later stage; refer to Section 27 of this Chapter for the relevant information.

6 To enable the camshaft to be removed, the driven sprocket must first be released from the shaft flange, where it is retained by two bolts. Using a spanner on the rotor centre bolt (185 models) or the engine turning hexagon (200 models), rotate the engine until one cam sprocket bolt is accessible. In the same manner, prevent engine rotation as the bolt is slackened. Then rotate the engine once more, until the second bolt can be slackened and removed. Rotate the engine again and remove the first bolt. Do not drop the bolts or any foreign matter down the cam chain tunnel, particularly if only a top-end overhaul is to be carried out.

7 With the unusual double row cam chain utilised on these models, there may not be enough lateral 'give' in the chain to enable the sprocket to be pulled clear of the cam flange. Additionally, if the machine is relatively new, there may not be enough slack in the chain to enable sprocket removal. To overcome this, slacken the cam chain tensioner locknut at the rear of the cylinder block, and pull the tensioner slide upwards, tighten the locknut. This will release the tension on the tensioner blade. Rotate the engine, as required, until the cut-out in the cam sprocket is correctly positioned to allow the camshaft journals on the left-hand side of the camshaft to pass through the cut-out. Displace the chain to the right-hand side of the cam sprocket. The camshaft can now be manoeuvred from position, towards the right-hand side of the engine, whilst the sprocket and chain are held. The sprocket can then be wriggled free from the chain. If a top-end overhaul only is envisaged, the cam chain should not be allowed to fall into the cam chain tunnel because retrieval is difficult. To prevent this occurring place a short rod or screwdriver through the chain and across the cylinder head, or secure the chain with a length of stiff wire.

8 It should be noted that in practice this operation was found to be fairly tricky; there is little room for manoeuvring of the components. It was found, however, that if the bushes were removed from each end of the camshaft, then extra clearance could be obtained to facilitate camshaft removal. The right-hand bush can be simply slid off the camshaft end; the left-hand bush is retained by a circlip and thrust washer. Note the locating dowels on the bushes.

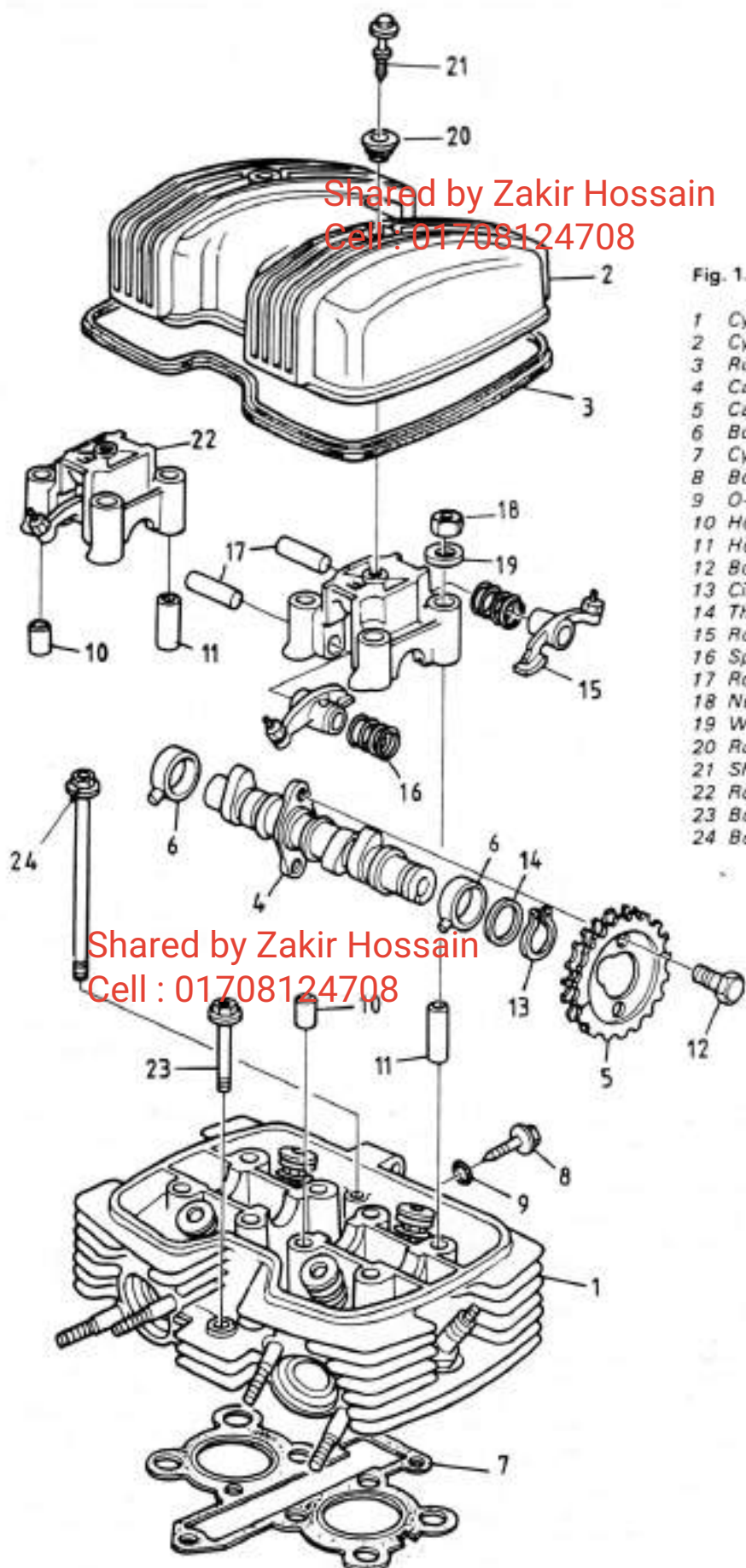
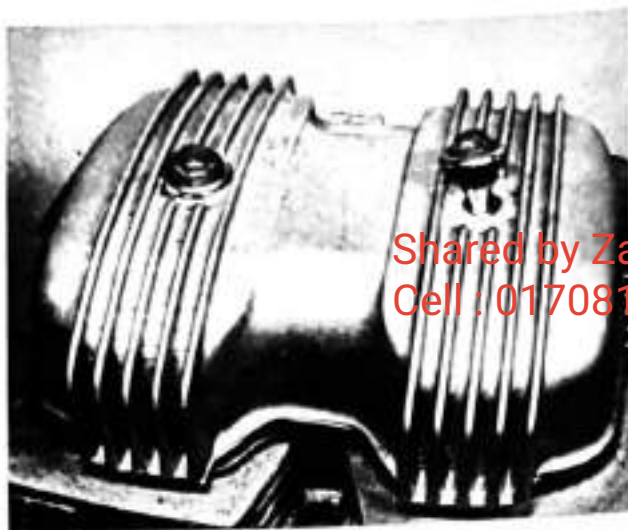


Fig. 1.1 Cylinder head and rocker gear

- 1 Cylinder head
- 2 Cylinder head cover
- 3 Rubber sealing gasket
- 4 Camshaft
- 5 Camshaft sprocket
- 6 Bush – 2 off
- 7 Cylinder head gasket
- 8 Bolt
- 9 O-ring
- 10 Hollow dowel – 2 off
- 11 Hollow dowel – 2 off
- 12 Bolt – 2 off
- 13 Circlip
- 14 Thrust washer
- 15 Rocker arm – 4 off
- 16 Spring – 4 off
- 17 Rocker arm spindle – 4 off
- 18 Nut – 8 off
- 19 Washer – 8 off
- 20 Rubber seal – 2 off
- 21 Shouldered bolt – 2 off
- 22 Rocker carrier/camshaft holder – 2 off
- 23 Bolt
- 24 Bolt – 2 off

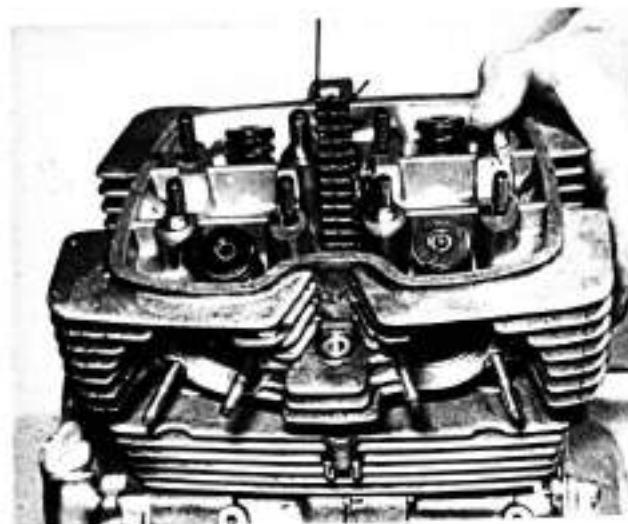
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7.2 Slacken and remove the two head cover retaining bolts



7.6 Remove cam sprocket bolts carefully, one at a time



7.7 Prevent cam chain dropping down cam chain tunnel

8 Dismantling the engine/gearbox unit: removing the cylinder head, cylinder block and pistons

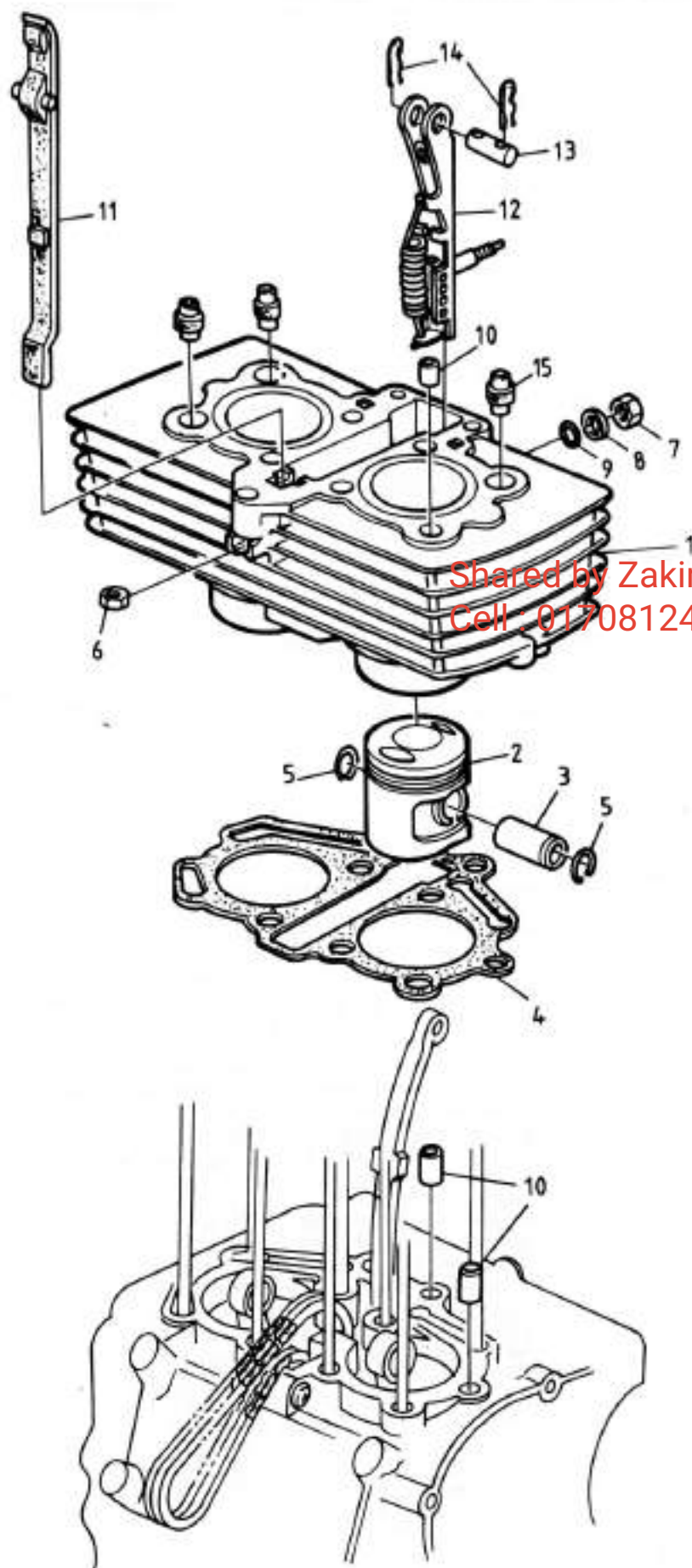
- 1 The nuts which retain the two rocker carriers/camshaft holders are fitted to the upper ends of the cylinder head and cylinder block holding down studs. It follows that once the nuts are removed, and the two rear bolts and single front bolt are removed, no direct restraints secure either the cylinder head or cylinder block.
- 2 Slacken and remove the washer and locknut which secures the cam chain tensioner adjuster if this has not been done previously.
- 3 Remove the four intake manifold securing nuts and pull off the manifold complete with carburettor heat insulator plate. Note the O-rings in each of the inlets in the intake manifold.
- 4 Remove the bolt from the rear of the cylinder head which retains the tensioner blade arm. Note that both this bolt and the adjuster stud are fitted with a sealing O-ring to prevent oil leakage.
- 5 Due to the type of cylinder head gasket used, it is possible that the cylinder head will be stuck firmly to the cylinder block surface. Considerable care should be exercised when trying to separate these two components, and the temptation to use

excessive force should be avoided at all costs. Commence separation by working around the head, using a block of hardwood struck by a hammer. Place the block only on portions of the cylinder head which are well supported by webs, gussets or flanges. Move around the cylinder head progressively a number of times, taking care not to damage the fins. If this method fails to loosen the two components, the application of levers will have to be adopted. Use tyre levers of reasonable length rather than screwdrivers, because they spread the load better and have no sharp edges. Insert the levers only between fins which are well supported and ensure that the areas carrying the direct load are as near as possible to the body of the cylinder/cylinder heads units. Whilst this method of cylinder head removal is not generally to be recommended, there may be no practicable alternative. It follows that extreme caution and control are required. Having accomplished separation, lift the cylinder head up off the cylinder block, preventing the cam chain from falling by placing a bar across the top of the cylinder block, or alternatively, by hooking a length of stout wire under the cam chain. Make a note of the position of the hollow dowels fitted over four of the studs. Three of these are also fitted with oil seals and act as oil feed passages.

6 The tensioner blade is secured to the top of the tensioner arm by a clevis pin and R-clip. Displace the R-clip, withdraw the clevis pin, being careful not to drop either down the cam chain tunnel and remove the tensioner arm. The tensioner arm should be detached at this stage because the clevis pin and R-clip are too wide to pass easily down through the cam chain tunnel. The tensioner blade will remain in position at this stage. If the tensioner arm and blade are left in position, the cylinder block must be manoeuvred around the clevis pin and R-clip as the block is lifted clear.

7 Withdraw the cam chain guide blade from the front of the cam chain tunnel. If the engine is to be dismantled completely, the chain can be allowed to drop into the crankcase. If it is not to be dismantled fully, the cam chain must be prevented from falling into the depths of the crankcase. A length of stiff wire or a metal rod, or a screwdriver blade of suitable length, will suffice.

8 Rotate the engine until the pistons are at TDC, hand feeding the cam chain, if necessary, to prevent it locking. Separate the cylinder block from the crankcase using a soft-headed mallet or a block of wood and a hammer. Once again care should be taken not to damage the fins. In this case levers will not be required and should not be used. Slide the cylinder block up and off the pistons, taking care to support each piston as the cylinder block becomes free. If a top end overhaul only is being carried out, place a clean rag in each crankcase mouth before



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- 1 Cylinder barrel
- 2 Piston – 2 off
- 3 Gudgeon pin – 2 off
- 4 Cylinder base gasket
- 5 Circlip – 4 off
- 6 Nut
- 7 Nut
- 8 Washer
- 9 O-ring
- 10 Hollow dowel
- 11 Static tensioner blade
- 12 Cam chain tensioner
- 13 Clevis pin
- 14 R-clip – 2 off
- 15 Dowel and oil seal – 3 off

Fig. 1.2 Cylinder barrel, piston and camchain tensioner

the lower edge of each cylinder frees the rings. This will preclude any small particles of broken ring falling into the crankcase. Note the positioning, to the side and rear of the left-hand crankcase mouth, of the two hollow dowels. If the dowels are loose, remove them to avoid loss.

9 Prise the outer gudgeon pin circlip of each piston from position, then press the gudgeon pin out of the small-end eye through the piston boss. If the pin is a tight fit, it may be necessary to warm the piston so that the grip on the gudgeon pin is released. A rag soaked in hot water should suffice, if it is placed on the offending piston's crown. This will give the necessary temporary expansion to the piston bosses to allow the gudgeon pin to be pushed out.

10 If the gudgeon pin is still a tight fit after warming the piston, it can be lightly tapped out of position with a hammer and soft metal drift. Do not use excess force and make sure the connecting rod is supported during this operation or there is a risk of its bending.

11 The pistons may be lifted from the connecting rods once the gudgeon pins are clear of the small-end eye. As each piston is removed, scribe the cylinder identification inside the piston skirt. A mark R or L will ensure that the piston is replaced in the correct bore, on reassembly. It is unnecessary to mark the back and front of the piston. On the 185 models there is an arrow mark on the crown indicating the front of the piston, and on the 200 models, the letters IN are stamped on the piston crown indicating the inlet (rearward) side of the piston. There is a cut-out for both valves on the crown of the 185 model's pistons; on the 200 models, there is only one cut-out for the exhaust valve.

12 Each piston is fitted with two compression rings and an oil control ring. It is wise to leave the rings in place on the pistons until the time comes for their examination or renewal in order to avoid confusing their correct order.



8.9 Push out the gudgeon pin to free the piston; note the rag padding

9 Dismantling the engine/gearbox unit: removing the contact breaker, automatic timing unit and alternator

1 These components may all be removed with the engine in the frame. If the alternator is to be removed with the engine in situ, the left-hand exhaust system, the drive sprocket cover and the gearchange lever must be removed. The block connector carrying the leads from the alternator, situated to the rear, left-hand side of the engine (obscured by the left-hand side panel)

must be separated. Also, remove the circular inspection cover from the main left-hand crankcase cover. The inspection cover is held by two crosshead screws (185 models) or three crosshead screws (200 models). The contact breaker baseplate is visible with the inspection cover detached. Disconnect the single lead from the contact breaker points.

2 The alternator of the 185 models incorporates the starter motor clutch on the rear face of the rotor. The stator is fixed around the rotor and is screwed onto the crankcase outer cover. The rotor is a taper fit on the end of the crankshaft. The alternator fitted to the 200 models is of a similar arrangement with the stator screwed inside the crankcase cover but with the rotor fitted outboard of the stator. The rotor on these models also incorporates the starter motor clutch on its rear face.

3 On all models, slacken and remove the three contact breaker baseplate retaining screws. Before disturbing the baseplate mark its position relative to the casing using punch marks or a scribed line. This will aid reassembly and correct ignition timing. On 185 models, the points baseplate can now be lifted clear. On the 200 models, slacken and remove the 8 mm headed ATU retainer bolt and the large engine turning hexagon. To accomplish their removal, stop the engine from rotating by fitting a spanner on the large hexagon, and then remove the 6 mm centre bolt. The contact breaker baseplate and ATU can now be lifted away on the 200 models.

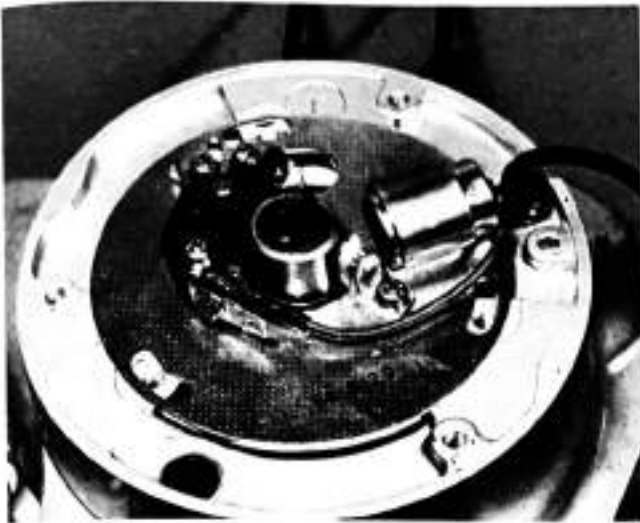
4 Slacken and remove the eight crosshead screws that retain the left-hand side crankcase cover. An impact driver will probably be necessary to avoid mutilation of the cover screws. Note the starter motor cable clamp on the lower part of the cover. Pull the cover clear and place it to one side; the stator is still retained inside the cover, and its lead must be wriggled clear of the machine.

5 In each case, the rotor is now accessible for removal, with the rotor centre bolt, on the 185 models, also securing the ATU. It is now necessary to prevent the crankshaft from rotating whilst the rotor centre bolt is released. If this is to be done with the engine in situ, top gear should be selected (the gear change lever will have to be temporarily refitted) and the rear brake applied, thus locking the engine and transmission. With the engine on the bench, a strap or chain wrench can be used around the rotor, as an alternative to the manufacturer's recommended method of holding the clutch with their clutch outer holding tool (No 07923-1070001). If the cylinder head, barrel and the pistons have been removed, a close-fitting metal bar can be placed through one or both small-end eyes and allowed to bear down on small wooden blocks placed across the crankcase mouth. Do not let the bar bear directly onto the gasket face as damage may result.

6 Slacken and remove the centre bolt. This will allow the ATU to be lifted clear on the 185 models. Access is now gained to an internal thread in the rotor boss. This thread is designed to take the Honda service tool, No. 07733-0020000, which is a T-handled rotor puller. In the absence of this tool, a conventional three-legged puller can be used to good effect.

7 Using the Honda service tool, screw the puller boss into the threaded rotor centre, then gradually tighten the T-handle to place pressure on the crankshaft end. If the rotor proves stubborn, tap the end of the shaft to jar the rotor off its taper. On no account strike the rotor itself, because this can easily damage the unit. If the correct tool is not available, it is possible, as stated earlier, to use a legged puller. When using the latter method, refit the rotor centre bolt prior to tightening down the puller boss; this will prevent damage to the thread in the crankshaft. If, as may well be found, the rotor is very tight and resists removal, place a stout bolt or screw into the rotor centre boss (in place of the rotor bolt) before tightening down the puller boss. The bolt or screw will then bear on the end of crankshaft and will allow rotor removal without risking damage to the standard bolt. Place the rotor to one side to await examination or reassembly, taking care not to drop or bang the unit; this may well dislodge the starter motor clutch rollers at the rear of the unit, as well as causing damage to the rotor itself.

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9.3a Disconnect contact breaker points lead, remove three baseplate screws and (on 200 models as shown), remove large hexagon and 6 mm bolt



9.3b Lift away the ATU from the 200 models at this stage



9.5 The rotor can now be detached (200 model shown)

10 Dismantling the engine/gearbox unit: removing the starter motor, chain and sprockets

1 All the models covered by this manual are fitted with an electric starter motor. The motor is mounted at the front left-hand side of the engine. The drive from the motor is taken from a small drive sprocket on the outboard end of the motor, via a drive chain, to a driven sprocket fitted behind the rotor, and a starter clutch fitted to the rear face of the rotor. Before the starter motor drive gear (two sprockets and the chain) can be removed, it is necessary to remove the alternator rotor and starter clutch as described in Section 9 of this Chapter. It should be noted, however, that the starter motor itself can be detached with the drive gear components in situ.

2 With the rotor and starter clutch removed, slacken and remove the single bolt which retains the driven sprocket guide plate. Remove the guide plate. Slide the driven sprocket off the crankshaft end, followed by the starter motor drive sprocket and the drive chain. Remove the polished spacer plate and the left-hand side crankcase cover gasket, if not previously removed.

3 Pull back the rubber boot on the terminal on the starter motor and disconnect the starter cable. Slacken and remove the four crosshead retaining screws, two at each end, and remove

the starter motor. Note that the screws will probably be tight requiring the use of an impact driver to facilitate removal. Note also the O-ring on the inboard end of the motor.

4 If the starter motor is to be removed with the engine in the frame and the left-hand crankcase cover still in position, then the motor must be moved slightly towards the right-hand side, with the retaining screws removed, to clear the crankcase cover. If the starter cable also requires removal, detach the rubber boot and feed the cable down, behind the crankcase cover, pull back the thicker outer cable to ease passage through the cast guide under the crankcase cover, and pull the cable free.

11 Dismantling the engine/gearbox unit: removing the primary drive cover, oil pump/primary drive gears and clutch

1 If access to the clutch, kickstart components, oil pump and filter screen or primary drive gears is required and the engine is still in the frame, the kickstart lever, and the right-hand exhaust system must be removed. In addition, the clutch cable must be detached at the operating arm on the crankcase cover and the engine oil must be drained.

2 Loosen evenly and remove the ten crosshead screws securing the right-hand crankcase cover (primary drive cover). An impact driver will facilitate their removal. Lift the cover and the gasket away from the engine. Be prepared for a certain amount of residual oil to be released as the cover is detached. Note and remove the two hollow dowels if they are loose.

3 If the engine unit is still in the frame, and it is wished to remove the primary drive assembly, the crankshaft gear pinion retaining nut should be slackened at this stage. Remove the pressure relief valve plunger and spring from the end of the crankshaft. Select top gear and apply the rear brake to prevent the crankshaft from rotating, then slacken the 16 mm nut. Then pull out the securing pin, remove the Belleville washer (note the wording 'OUTSIDE' on the washer) and pull off the oil pump drive gear.

4 If the engine unit is on the workbench, and the cylinder head, barrel and pistons have been removed, then a bar can be passed through one or both connecting rod small-end eyes, as described in paragraph of Section 9 of this Chapter, to 'lock' the crankshaft. Alternatively, a chain or strap wrench around the clutch assembly will prevent crankshaft rotation. The Honda service tool, clutch outer holder (No 07923-1070001), should, of course, be used if it is available. It may occur, as in the case of the machine used for this stripped-down procedure, that none of these 'locking' methods are necessary to release the pinion

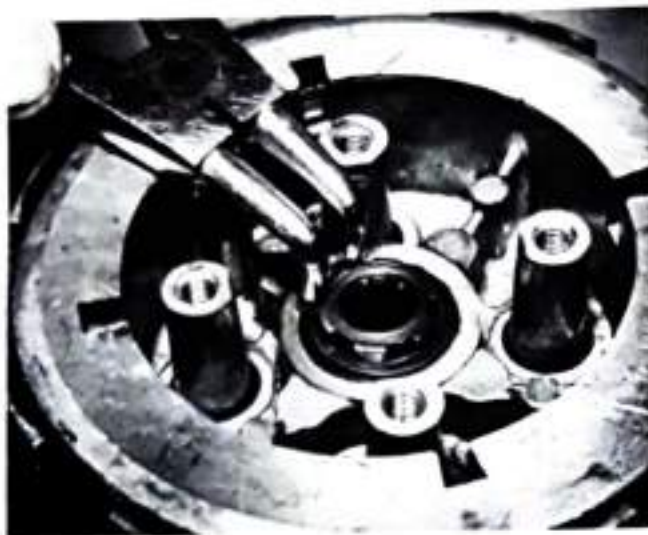
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retaining nut. It was found that a sharp tap on the end of the socket spanner ratchet handle resulted in the desired slackening effect.

5 Lift out the clutch pushrod together with the cup in which it seats, then remove the four bolts which secure the clutch thrust plate, unscrewing them in a diagonal sequence until the clutch spring pressure is released. Lift off the thrust plate and its bearing, lift out the four clutch springs and place them to one side.

6 Remove the circlip which retains the clutch centre, and lift out the centre, followed by the nine plain and friction plates and the pressure plate lifted out together. The plates and centre can be left as a unit for further examination etc later; refer to Section 31 of this Chapter for details.

7 It may now be found that the clutch outer drum, which incorporates the large primary driven gear, will not pull off the gearbox mainshaft end until the small primary drive gear is removed from the crankshaft end. Slide off the primary drive gear. Remove the heavy splined washer from the inside of the clutch outer drum (noting that the rounded side of the washer faces inwards) and pull off the clutch outer drum/primary driven gear.



11.5 Release the circlip to free clutch centre and plates

12 Dismantling the engine/gearbox unit: removing the oil pump and filter screen

1 The oil pump and filter screen can be removed with the engine in the frame if required; refer to Section 12 of Chapter 2, for the relevant information and procedure.

2 With the engine on the workbench, rotate the crankshaft by pulling on the cam chain, until the three holes in the pump outer face align with those in the pump driving gear. Three countersunk crosshead screws will then be visible. If the clutch and oil pump drive gear have been removed, the gear can be temporarily refitted in order to engage with the pump driving gear. Alternatively, a screwdriver can be used to turn the driving gear manually until the holes are aligned.

3 Pass a screwdriver through the holes and release the three countersunk crosshead retaining screws, and lift the pump away. If the screws are particularly tight, an impact driver may have to be used. If an impact driver is to be used, take great care that no damage is inflicted on the pump driving gear or housing etc.

4 The oil filter screen will be lifted away with the oil pump; it is attached to the lower part of the pump by a rubber retainer.

5 Note the O-ring in the lower forward part of the crankcase



11.4 Slacken evenly and remove the four clutch thrust plate bolts

wall, which mates up with the rear face of the lower part of the pump. This O-ring should be removed and renewed during reassembly, to obviate the risk of oil leakage.

13 Dismantling the engine/gearbox unit: removing the gearchange shaft and external mechanism

1 If attention is required to the gearchange shaft and external mechanism components, access may be gained with the engine still in the frame. If the engine is still in situ, detach the right-hand exhaust system and kickstart lever, drain the engine oil, disconnect the clutch cable and remove the right-hand crankcase cover. Remove the clutch and primary drive/oil pump drive gears (see Section 11 of this Chapter) and remove the oil pump (Section 12 of this Chapter). In addition, if the gearchange shaft is to be withdrawn, the gearchange lever must be removed from the left-hand side of the engine.

2 Slacken and remove the shouldered bolt from the centre of the camplate fitted to the end of the selector drum. Lift the camplate away, noting that it is located by the most prominent of the three pins protruding from the end of the selector drum. Remove the three pins to prevent loss; a pair of pliers or a magnetic screwdriver will facilitate removal of the pins.

3 The drum stopper arm (detent mechanism) is similarly retained by a shouldered bolt. Release the tension on the stopper arm return spring gradually by slackening the pivot bolt. Remove the stopper arm and its return spring.

4 The gearchange shaft runs in a bore through the crankcase, emerging on the left-hand side of the engine. The complete unit can be withdrawn from the right-hand side of the unit, once the gearchange pedal has been removed if the engine is still in the frame, with the quadrant centring spring disengaging from its locating post.

14 Dismantling the engine/gearbox unit: removing the kickstart mechanism

1 Should the kickstart mechanism alone require attention, this may be accomplished with the engine in situ. Remove the right-hand crankcase cover as described in the preceding Sections to gain access to the kickstart mechanism.

2 Grasp the return spring end with a pair of pliers at the point where it enters the locating hole in the kickstart shaft. Disengage the spring end in a controlled manner, allowing it to unwind slowly. Displace the kickstart return spring internal guide by using a screwdriver to dislodge it from inside the

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spring coils. Rotate the kickstart shaft and disengage the other end of the spring from its locating lug on the crankcase wall. The complete kickstart mechanism can now be pulled clear of the casing. If examination and/or renovation are required, refer to Section 34 of this Chapter.

3 It should be noted that if specific attention to the kickstart mechanism is not required, the engine dismantling can continue with the mechanism in place: it does not hinder crankcase separation and will remain in the right-hand crankcase half after separation.

15 Dismantling the engine/gearbox unit: removing the neutral indicator switch

1 Although the neutral indicator switch components do not prevent crankcase separation, it is advisable to remove them as this will need to be done prior to reassembly.

2 With the engine lying on its right-hand side, suitably supported on wooden blocks, remove the single shouldered bolt which retains the claw plate which secures the neutral indicator switch body. Remove the claw plate and then pull out the switch body. Note the O-ring on the inner end of the switch body.

16 Dismantling the engine/gearbox unit: separating the crankcase halves

1 With the engine/gearbox unit stripped of all ancillary components, as described in the previous Sections, commence separation of the crankcase halves with the unit lying on its right-hand side. Support the unit with suitable wooden blocks. Note that at this stage the lower end of the cam chain tensioner blade is still retained. Honda recommend complete dismantling of the crankcase halves and removal of the crankshaft before the tensioner blade may be detached. It was found, however, that by employing a length of wire with a suitably hooked end and a steady hand, it was possible to withdraw the R-clip and washer retaining the tensioner blade lower end. The blade can then be displaced sideways to free it from the locating pivot. This method of tensioner blade removal is, of course, only of benefit if the blade is damaged or in need of replacement, and the 'bottom end' of the engine unit does not require attention.

2 With the left-hand side of the engine unit facing upwards, remove two crosshead screws: one from the lower rear section of the engine and the other from the top of the unit, between the crankcase mouths.

3 Turn the engine unit over, so the right-hand crankcase is facing upwards, again supporting the underside of the unit with blocks, and locate the eight 6 mm crankcase joining bolts. It will be noted that the two bolts at the top and one at the bottom of the right-hand crankcase half are deeply recessed; they will require the use of a thin section socket spanner and extension bar to remove them. Slacken the bolts evenly, a little at a time, in a diagonal sequence, to avoid distortion. Remove all the bolts. The right-hand crankcase half can now be lifted away, leaving the crankshaft gear clusters and selector mechanism in the left-hand casing half. The kickstart mechanism will be removed with the right-hand casing half. The crankcase halves will probably separate fairly easily. Should they prove stubborn, a certain amount of persuasion may be employed. Tap around the joint using a soft-headed mallet, or a hammer and a hardwood block interposed between it and the casing. Never use a metal hammer directly on the casing; the soft aluminium alloy is easily damaged by this method. Do not lever the joint apart with screwdrivers or other implements, because this will invariably damage the mating faces of the crankcase halves and lead, inevitably, to oil leakage at a later date.



15.2 Remove the bolt and claw plate to free the neutral indicator switch



16.3a Use a thin section socket spanner and extension bar to reach these two upper and ...



16.3b ... this lower right-hand side crankcase retaining bolts

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17 Dismantling the engine/gearbox unit: gearbox mainshaft and layshaft and selector mechanism

- 1 The gear selection mechanism, the two selector forks and the selector drum, should be removed before the gear shafts and clusters. The selector forks run on a support shaft which is fitted from the right-hand side of the casing.
- 2 Slide out the selector fork support shaft. As the shaft is withdrawn the two forks will be freed; catch them as they are displaced, disengage them from the gear clusters and refit them to the support shaft in their correct positions as an aid to reassembly.
- 3 The selector drum is now free to be slid out of the casing in the same manner as the selector fork support shaft. Before lifting the selector drum out of the casing, check the fit of the three pins in the drum end; if they are loose remove them to avoid loss. Remove the selector drum noting the neutral indicator switch rotor on the lower end.
- 4 Lift out the gearbox layshaft as a complete unit, putting it to one side for examination and, if necessary for further dismantling at a later stage. Similarly remove the gearbox mainshaft. Note the fitting of a plain shim to the right-hand end of the layshaft.

18 Dismantling the engine/gearbox unit: removing the crankshaft assembly

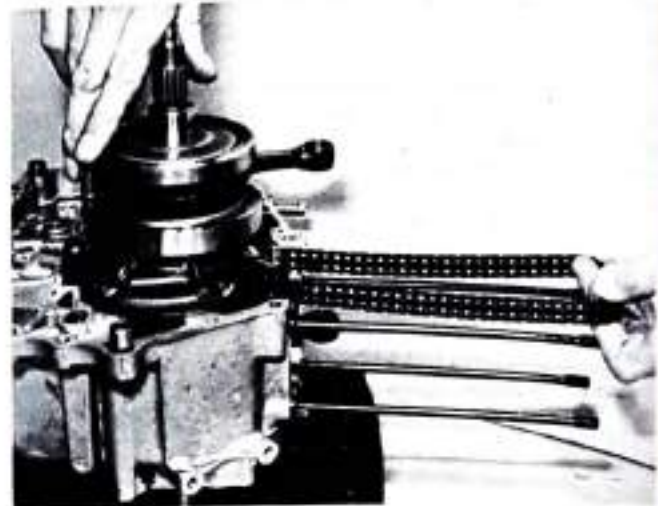
- 1 Before the crankshaft assembly can be removed, the central bearing holder must be released from the left-hand casing half. The cast holder is fitted between the two flywheel assemblies and is secured to the casing by five 6 mm bolts and one 6 mm nut. Remove the retaining bolts and the single nut, grasp the crankshaft firmly and slide it from position complete with cam chain and cam chain tensioner blade. It should be noted that the tensioner blade and connecting rods will have to be manoeuvred until they are in a suitable position to clear the crankshaft mouths.
- 2 The cam chain and the cam chain tensioner blade can now be detached from the crankshaft assembly. The tensioner blade lower end is secured to the central bearing holder by an R-clip and a plain washer. Displace the R-clip and washer and remove the tensioner blade from its locating pin. Due to the inherent rigidity of the double-row cam chain and its subsequent lack of 'twist', the inner weight of the left-hand flywheel assembly must be positioned so that the cam chain can be displaced without being trapped between the flywheel weight and bearing holder. If the flywheel weight is positioned correctly the cam chain can be displaced to either side of the flywheel weight, between the tensioner blade locating pin and the forward guide blade locating slot.

19 Examination and renovation: general

- 1 Before examining the component parts of the dismantled engine/gearbox unit for wear, it is essential that they should be cleaned thoroughly. Use a paraffin/petrol mix to remove all traces of oil and sludge which may have accumulated within the engine.
- 2 Examine the crankcase castings for cracks or other signs of damage. If a crack is discovered, it will require professional attention or in an extreme case, renewal of the casting.
- 3 Examine carefully each part to determine the extent of wear, if necessary checking with the tolerance figures listed in the Specifications Section of this Chapter, or accompanying the text.
- 4 Use a clean, lint-free rag for cleaning and drying the various components, otherwise there is risk of small particles obstructing the internal oilways.
- 5 Should any studs or internal threads require repair, now is the appropriate time to attend to them. Where internal threads



17.4 Lift out the gear shaft assemblies after the selector drum has been removed

18.1 Manoeuvre the crankshaft assembly to facilitate removal
are stripped or badly worn, it is preferable to use a thread insert, rather than tap oversize. Most dealers can provide a thread reclaiming service by the use of Helicoil thread inserts. They enable the original component to be re-used.**20 Crankshaft and gearbox main bearings: removal**

- 1 The crankshaft outer main bearings and gearbox mainshaft and layshaft main bearings will all remain in the crankcase walls as their respective shafts are removed. The crankshaft central main bearing will remain in the bearing holder fitted in between the crankshaft webs. The main bearings are all of the ball journal type.
- 2 If, on examination of the main bearings (refer to Section 21 of this Chapter), they require renewal, the outer bearings can be drifted out of position; they are a press fit in the crankcase casings.
- 3 If the crankshaft left-hand main bearing requires removal and renewal, the removal operation must be preceded by removal of the oil seal and the oil passage plate. Prise the oil seal from its position outboard of the left-hand main bearing; a wide bladed screwdriver will facilitate the removal of the seal. Take care not to bruise the crankcase surface adjacent to the seal housing. Remove the three countersunk crosshead screws from the inside of the left-hand crankcase. Note that an impact driver must be used on these screws to free them; they are tight

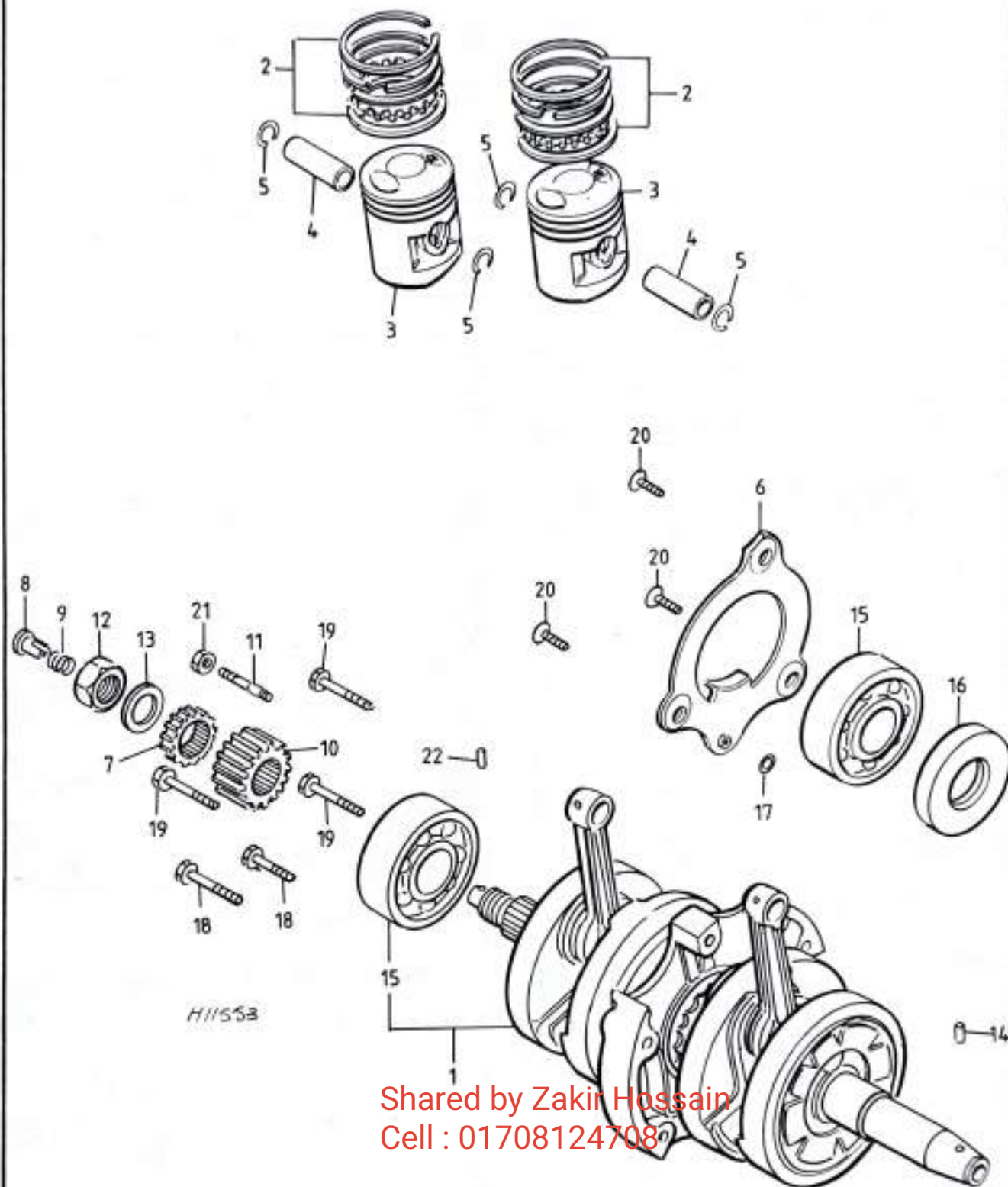


Fig. 1.3 Crankshaft assembly

- | | | |
|---------------------------|-----------------------|------------------|
| 1 Crankshaft assembly | 9 Spring | 16 Oil seal |
| 2 Piston ring set - 2 off | 10 Primary drive gear | 17 O-ring |
| 3 Piston - 2 off | 11 Stud | 18 Bolt |
| 4 Gudgeon pin - 2 off | 12 Nut | 19 Bolt - 4 off |
| 5 Circlip - 4 off | 13 Lock washer | 20 Screw - 3 off |
| 6 Bearing retainer | 14 Pin | 21 Nut |
| 7 Oil pump drive gear | 15 Bearing - 2 off | 22 Pin |
| 8 Oil feed quill | | |

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and have a locking compound applied to their threads. Lift the oil passage plate out of the casing. The bearing can now be removed.

4 The crankshaft centre main bearing is, in effect, an integral part of the pressed-up crankshaft and is not available as a spare part. Because of this, and because Honda do not operate a service exchange scheme, if the centre bearing wears a new crankshaft will be required. It may be possible, however, to find an engineering specialist willing to separate the crankshaft, renew the bearing and then rebuild and true the completed unit.

5 The crankshaft outer main bearings, the mainshaft right-hand bearing and layshaft left-hand bearing may be drifted from position using a suitable tubular drift. A socket spanner of similar outside diameter to that of the bearing is an ideal tool. When tapping the bearings out, care should be taken to keep them square with the bearing housing, and thus prevent them from 'tying' in the housing bores. Should the bearings prove stubborn the cases may be heated in an oven to 100°C. If an oven is not available, boiling water will produce a similar effect, in expanding the casing material which will loosen its grip on the bearing outer races.

6 The mainshaft left-hand bearing and layshaft right-hand bearing are a press fit in blind holes in the casings. Because of

this access cannot be made to drift them out in the normal manner. Bearing removal may be possible after heating the cases by knocking the cases, face downwards, on a wooden block, and so dislodging the bearings under their own weight. If this is not successful it is suggested that the cases be taken to a Honda Service Agent who will have the correct internally expanding bearing puller.

7 New bearings may be drifted into place using a tubular drift in a manner similar to that adopted for removal. Ensure again that the bearings are drifted in carefully without being allowed to twist, and that they are seated correctly in the crankcase bores.

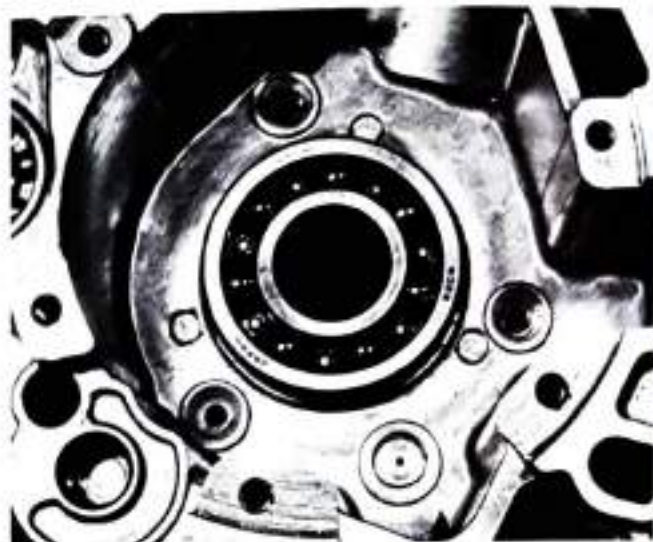
8 The oil seals outboard of the crankshaft and the layshaft left-hand main bearings should be renewed as a matter of course upon reassembly of the crankcase. Refer to Section 22 of this Chapter for more details of oil seal renewal.

9 When refitting the oil passage plate inboard of the crankshaft left-hand main bearing, note the O-ring fitted behind the lower part of the plate in the crankcase wall. Do not omit the O-ring. Apply a locking compound to the threads of the three countersunk crosshead retaining screws and fully tighten them using an impact driver.

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20.3a Use an impact driver to remove the three oil passage plate retaining screws before ...



20.3b ... the crankshaft left-hand main bearing can be drifted out



20.5a Examine crankshaft right-hand and gearbox mainshaft right-hand main bearings and ...



20.5b ... gear shafts left-hand main bearings; drift out those requiring renewal

21 Examination and renovation: big-end and main bearings

1 Big-end failure is invariably indicated by a pronounced knock from the crankcase. The knock will become progressively worse, accompanied by vibration.

2 There should be no vertical play whatsoever in the big-end bearings, after all the oil has been washed out. To check the big-end bearings, arrange the connecting rods in the TDC position. Grasp each rod in turn and pull it firmly upwards and then push downwards, making sure that any end float, which is intentional, is not mistaken for big-end play. Any discernible up-and-down movement will indicate that the bearing is worn and in need of renewal. **Do not** run the machine with a worn big-end bearing, otherwise there is risk of breaking the connecting rod or crankshaft.

3 As stated, a certain amount of side play (end float) is intentional and should be between 0.1 – 0.4 mm (0.004 – 0.016 in) on a new crankshaft. The maximum allowable side play is 0.6 mm (0.024 in). These measurements can be checked with a feeler gauge between the flywheel face and the edge of the connecting rod.

4 The built up nature of the crankshaft assembly precludes the possibility of repair by the home enthusiast/mechanic. Because of the high accuracy required in re-aligning the flywheels after bearing replacement, this operation is beyond the means of the average owner. Big-end repair should, therefore, be entrusted to a Honda agent, who will have the necessary repair or replacement facilities.

5 Wash the main bearings carefully in clean petrol to remove any residual oil, then check for free play, and spin them to check for any signs of roughness or grittiness. If the main bearings do not run freely and smoothly they must be replaced; refer to Section 20 of this Chapter. Warning of main bearing failure is usually given by a characteristic rumble that can be readily heard when the engine is running. Some vibration will also be felt, which will be transmitted via the footrests.

6 Failure of both the big-end bearings and the main bearings may not necessarily occur as the result of high mileage covered, although this is the usual cause of bearing failure. If the machine is only used infrequently, it is possible that condensation within the engine may cause premature bearing failure. The condition of the flywheels is usually the best guide. When condensation troubles have occurred, the flywheels will rust and become discoloured.

22 Examination and renewal: oil seals

1 An oil seal is fitted to the left-hand end of the crankshaft assembly in the left-hand crankcase half, outboard of the crankshaft main bearing. This oil seal prevents oil from entering the alternator and contact breaker assembly. There is also an oil seal fitted outboard of the layshaft left-hand main bearing, behind the gearbox final drive sprocket. Both of these seals should be replaced as a matter of course whenever the engine is fully dismantled.

2 Oil seals tend to lose their effectiveness if they harden with age. It is difficult to give any firm recommendations in this respect except to say that if there is any doubt about the condition of a seal, renew it as a precaution.

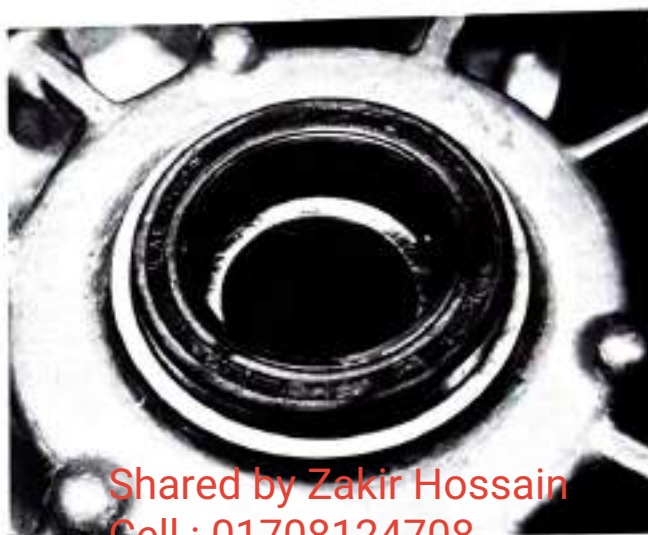
3 New seals may be drifted into position using a tubular drift. Smear the outer surface of the seal with oil to aid fitting.

23 Examination and renovation: cylinder block

1 The usual indication of badly worn cylinder bores and pistons is excessive smoking from the exhausts and piston slap, a metallic rattle that occurs when there is little or no load on the engine. If the top of the cylinder bore is examined carefully, it will be found that there is a ridge at the front and back the depth of which will indicate the amount of wear which has taken



21.1 Check the crankshaft big-end bearings for wear



22.1a Renew the left-hand crankshaft main bearing oil seal and



22.1b ... the oil seal fitted outboard of the layshaft left-hand main bearing

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place. This ridge marks the limit of travel of the top piston ring.

2 Since there is a difference in cylinder wear in different directions, side to side and back to front measurements should be made. Using an internal micrometer, take measurements at three different points down the length of the cylinder bore, starting just below the top piston ring ridge, then about 2 inches below the top of the bore and the last measurement about 1 inch from the bottom of the cylinder bore. Refer to the Specifications for cylinder specifications. If any of the cylinder bore inside diameter measurements exceed the service limit the cylinder must be bored out to take the next size of piston. If there is a difference of more than 0.05 mm (0.002 in) between any two measurements, the cylinder should in any case, be rebored.

3 If an internal micrometer is not available, the amount of cylinder bore wear can be measured by inserting the piston without rings so that it is approximately 20 mm ($\frac{3}{4}$ in) from the top of the bore. If it is possible to insert a feeler gauge thicker than 0.01 mm (0.04 in) between the piston and cylinder wall on the thrust side of the piston, remedial action must be taken.

4 Honda can provide pistons in two oversizes: +0.25 mm (0.010 in) and +0.50 mm (0.020 in) for the CD and CM200 models. For the CD and CM 185 models, Honda can supply pistons in four oversizes: +0.25 mm (0.010 in), +0.50 mm (0.020 in), +0.75 mm (0.030 in), and +1.0 mm (0.040 in).

5 Check that the surface of the cylinder bore is free from score marks or other damage that may have resulted from an earlier engine seizure or a displaced gudgeon pin. A rebore will be necessary to remove any deep scores, irrespective of the amount of bore wear that has taken place, otherwise a compression leak will occur.

6 Make sure the external cooling fins of the cylinder block are not clogged with oil or road dirt which will prevent the free flow of air and cause the engine to overheat.

24 Examination and renovation: pistons and piston rings

1 Attention to the pistons and piston rings can be overlooked if a rebore is necessary, since new components will be fitted.

2 If a rebore is not necessary, examine each piston carefully. Reject pistons that are scored or badly discoloured as the result of exhaust gases by-passing the rings. Remove the piston rings by pushing the ends apart with the thumbs whilst gently easing the ring from its groove. Great care is necessary throughout this operation because the rings are brittle and will break easily, if overstressed. If the rings are gummed in the grooves, three strips of tin can be used to ease them free, as shown in the accompanying illustration.

3 Remove all carbon from the piston crowns, using a blunt scraper which will not damage the surface of the piston. Clean away carbon deposits from the valve cutaways and finish off with metal polish so that a smooth, shining surface is achieved. Carbon will not adhere so readily to a polished surface.

4 Small high spots on the back and front areas of the piston can be carefully eased back with a fine swiss file. Dipping the file in methylated spirit or rubbing its teeth with chalk will prevent the file clogging and eventually scoring the piston. Only very small quantities of material should be removed, and never enough to interfere with the correct tolerances. Never use emery paper or cloth to clean the piston skirt; the fine particles of emery are inclined to embed themselves in the soft aluminium and consequently accelerate the rate of wear between bore and piston.

5 Piston wear usually occurs at the skirt or lower end of the piston and takes the form of vertical streaks or score marks on the thrust side. There may also be some variation in the thickness of the skirt.

6 Measure the outside diameter of the piston about 10 mm (0.4 in) up from the bottom of the skirt, at right angles to the line of the gudgeon pin. If the measurement is under the service limit, the piston must be renewed.

7 The piston ring grooves may also have become enlarged in

use, allowing the piston rings to have greater side float. If the clearance exceeds 0.12 mm (0.0048 in) for either the top ring or the second compression ring, the piston is due for renewal. It is unusual for this amount of wear to occur on its own.

8 Piston ring wear is measured by removing the rings from the piston and inserting them, one at a time, in the cylinder bores. Position each ring, using the piston skirt to push them into place squarely, so that they rest about 10 mm ($\frac{1}{2}$ inch) from the bottom of the bore. Measure the end gap with a feeler gauge, if it exceeds 0.5 mm (0.020 in) the rings require renewal. A replacement set of piston rings is comparatively inexpensive and it is considered good practice to renew them as a matter of course whenever the engine is completely dismantled; this is particularly important if the machine has covered a considerable mileage.

9 Check that there is no build-up of carbon on the inside surface of the rings or in the grooves of the pistons. Any build-up should be removed by careful scraping; a piece of broken old piston ring makes an ideal scraping implement.

10 The piston crowns will show whether the engine has been rebored on some previous occasion. All oversize pistons have the rebore size stamped on the crown. This information is essential when ordering replacement piston rings.

11 If new piston rings are fitted but a rebore has not taken place, the cylinder bores should be 'glaze busted'. This honing operation, as the name suggests, removes the highly polished glazed surface of the bore which has been caused by the countless up and down strokes of the piston and rings. If 'glaze busting' is not carried out, the time required to run-in the new rings will be greatly extended.



Fig. 1.4 Method of freeing gummed piston rings

25 Examination and renovation: gudgeon pins, small-ends and piston bosses

1 The fit of the gudgeon pin in both the small-end eye of the connecting rod, and in the piston bosses should be checked. In the case of the small-end eye, slide the pin into position and check for wear by moving the pin up and down. The pin should be a light sliding fit with no discernible radial play. If play is detected, it will almost certainly be the small-end eye which has worn rather than the gudgeon pin, although in extreme cases the latter may also have become worn. The service limit for the internal diameter of the connecting rod small-ends is 14.08 mm

(0.5543 in). If the gudgeon pin outside diameter is below the service limit of 13.98 mm (0.5504 in), then it must be renewed.

2 The connecting rod is not fitted with a bush type of small-end bearing, and consequently a new connecting rod will have to be fitted if worn. This is not a simple job, as the flywheels must be parted to fit the new component, and is a job for a Honda Service Agent. It should be borne in mind that if the small-end has worn, it is likely that the big-end bearings will require attention. Check the fit of the gudgeon pin in the piston. This is normally a fairly tight fit, and it is not unusual for the piston to be warmed slightly to allow the pin to be inserted and removed. After considerable mileages have been covered, it is possible that the bosses will have become enlarged. If the piston bores have become enlarged beyond the service limit of 14.05 mm (0.5531 in), the piston must be renewed. It is worth noting, as an aid to diagnosis, that wear in the above areas is characterised by a metallic rattle when the engine is running.

26 Examination and renovation: cylinder head, valves, valve seats and valve guides

1 It is best to remove all carbon deposits from the combustion chambers before removing the valves for inspection and grinding-in. Use a blunt ended chisel or scraper so that the surfaces are not damaged. Finish off with a metal polish to achieve a smooth, shining surface. If a mirror finish is required, a high speed felt mop and polishing soap may be used. A chuck attached to a flexible drive will facilitate the polishing operation.

2 Remove each valve in turn, using a valve spring compressor, and place the valves, springs, seats and collet halves in a suitable box or bag marked to denote inlet or exhaust and left and right as appropriate. Alternatively, tie-on labels, suitably inscribed, are ideal for identification purposes. Assemble the valve spring compressor in position on the cylinder head, and gradually tighten the threaded portion to place pressure on the upper spring seat. Do not exert undue force to compress the springs, the tool should be placed under slight load, and then tapped on the end to jar the collet halves free. Continue to compress the springs until the collet halves can be dislodged using a small screwdriver. Note that the valve springs exert considerable force, and care should be taken to avoid the compressed assembly flying apart. To this end, a small magnet is invaluable for retrieving the collet halves, being more delicate than fingers.

3 Before giving the valves and valve seats further attention, check the clearance between each valve stem and the guide in which it operates. Measure also the valve stem diameters. Measure the valve stem at the point of greatest wear and then measure again at right-angles to the first measurement. If the valve stem is below the service limit, it must be renewed. The valve stem/guide clearance can be measured with the use of a dial gauge and a new valve. Place the new valve into the guide and measure the amount of shake with the dial gauge tip resting against the top of the stem. If the amount of wear is greater than the service limit, the guide must be renewed.

4 To remove an old valve guide, place the cylinder head in an oven and heat it to about 150°C. The old guide can now be tapped out from the cylinder side. The correct drift should be shouldered, with the smaller diameter the same size as the valve stem and the larger diameter slightly smaller than the OD of the valve guide. If a suitable drift is not available a plain brass drift may be utilised with great care. **DO NOT** use a blowtorch or other localised form of heat source to heat the cylinder head prior to valve guide removal. Heating in this way may cause irreparable damage due to warping. Each inlet valve guide is fitted with an O-ring to ensure perfect sealing. The O-rings must be renewed with new components. Install new valve guides by reversing the removal procedure, again heating the cylinder head to around 150°C. After installation of a new guide, the seat must be recut to centralise the seat on the guide axis (see paragraph 7).

5 After cleaning the valves to remove all traces of carbon,

examine the heads for signs of pitting and burning. Examine also the valve seats in the cylinder head. The exhaust valve and its seat will probably require the most attention because this is the hotter running of the two. If the pitting is slight, the marks can be removed by grinding the seats and valves together, using fine valve grinding compound. Note that the valve seats must be reground whenever new valve guides are fitted.

6 Valve grinding is a simple task. Commence by smearing a trace of fine valve grinding compound (carborundum paste) on the valve seat and apply a suction tool to the head of the valve. Oil the valve stem and insert the valve in the guide so that the two surfaces to be ground in make contact with one another. With a semi-rotary motion, grind in the valve head to the seat, using a backward and forward action. Lift the valve occasionally so that the grinding compound is distributed evenly. Repeat the application until an unbroken ring of light grey matt finish is obtained on both valve and seat. This denotes the grinding operation is now complete. Before passing to the next valve, make sure that all traces of the valve grinding compound have been removed from both the valve and its seat and that none has entered the valve guide. If this precaution is not observed, rapid wear will take place due to the highly abrasive nature of the carborundum paste.

7 When deeper pit marks are encountered, it will be necessary to use a valve refacing machine and also a valve seat cutter, set to an angle of 45°. This course of action should be resorted to only in an extreme case because there is risk of pocketing the valve and reducing performance. For the same reason, never resort to excessive grinding; reduced engine efficiency will result. If there is any doubt about the condition of a valve or if the valve face width exceeds 1.8 mm (0.07 in) fit a new replacement. If, after recutting a valve seat in the cylinder head, the seat width exceeds 1.5 mm (0.06 in), the width should be restored to 1.0 mm (0.04 in) using 37.5° and 63.5° cutters or grinders. In view of the high cost of these items, and the expertise required for their successful use, it is suggested that the cylinder head be placed in the hands of an experienced agent for this work to be carried out.

8 Examine the condition of the valve collets and the groove on the valve stem in which they seat. If there is any sign of damage, new parts should be fitted. Check that the valve spring collar is not cracked. If the collets work loose or the collar splits whilst the engine is running, a valve could drop into the cylinder and cause excessive damage.

9 Check the free length of each of the valve springs. The springs have reached their serviceable limit when they have compressed to the limit readings given in the Specifications section of this Chapter.

10 Reassemble the valve and valve springs by reversing the dismantling procedure. Fit new oil seals to each valve guide and oil both the valve stem and the valve guide, prior to reassembly. Take special care to ensure the valve guide oil seal is not damaged when the valve is inserted. The springs are of variable pitch. Each must be fitted with the closer wound coils downwards, next to the cylinder head. Check that the split collets are located positively before the spring compressor is released. To ensure that the collets are firmly located, tap the top of each valve stem sharply with a hammer, taking care to strike squarely on the stem and not on the top seat (collar).

11 Check the cylinder head for straightness, especially if it has shown a tendency to leak oil at the cylinder head joint. Laying the cylinder head on a sheet of plate glass will show whether the head is distorted. If the amount of warpage is not too great, it may be machined flat. If the cylinder head is distorted badly, it is advisable to fit a new replacement. Most cases of cylinder head warpage can be traced to unequal tensioning of the cylinder head bolts by tightening them in incorrect sequence.

12 Make sure the cylinder head fins are not clogged with oil or road dirt, otherwise the engine will overheat. If necessary, use a wire brush. Check that no cracks are evident, especially in the vicinity of the holes through which the holding down studs and bolts pass, and near the sparking plug threads.

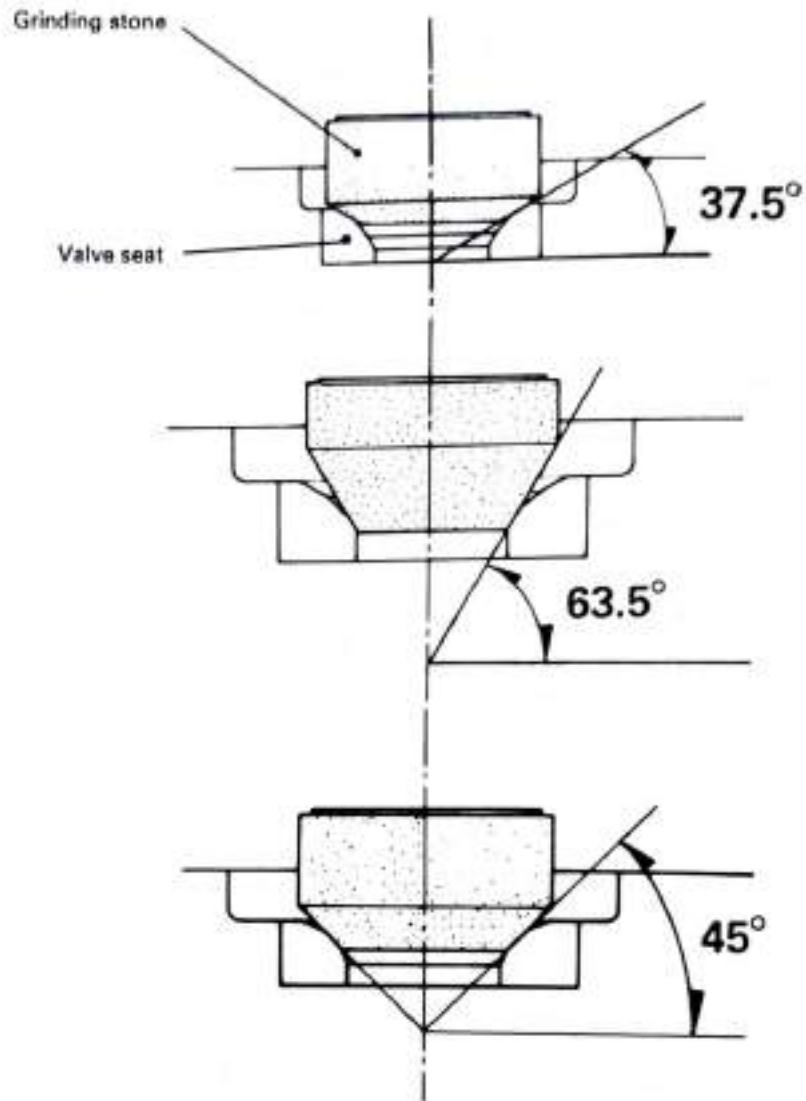


Fig. 1.5 Valve seat grinding angles



26.10a Lubricate valve stem before insertion in the guide



26.10b Fit inner spring lower seat (note new oil seal on guide) and ...

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26.10c ... outer spring lower seat



26.10e Compress valve springs so that collets can be reinstalled

27 Examination and renovation: rocker arms and shafts

- 1 The rocker carriers together with their related components should be considered as two separate assemblies and dismantled separately, to prevent the accidental interchange of components. The various parts of each assembly should be kept in matched sub-assemblies for the same reason.
- 2 It is unlikely that excessive wear will occur in either the rocker arm or the rocker shafts unless the flow of oil has been impeded or the machine has covered a very high mileage. A clicking noise from the rocker area is the usual symptom of wear in the rocker gear, which should not be confused with a somewhat similar noise caused by excessive valve clearances.
- 3 If any shake is present and the rocker arm is loose on its shaft, a new rocker arm and/or shaft should be fitted.
- 4 Dismantling the rocker gear from the rocker carriers is made a simple task on these machines by the use of rocker shafts with one threaded end. Insert a 5 mm screw (a crankcase cover retaining screw will do admirably) into the threaded end of the rocker shaft; the shaft can now be withdrawn from the carrier. This will allow removal of the rocker arms and their end control springs. Take care that each shaft remains with its original rocker arm.
- 5 Check the tip of each rocker arm at the point where the arm



- 26.10d Springs must be fitted with close coils downwards makes contact with the cam. If signs of cracking, scuffing or breakthrough in the case hardened surface are evident, fit a new replacement.
- 6 Check also the thread of the tappet adjusting screw, the thread of the rocker arm into which it fits and the thread of the locknut. The hardened end of the tappet adjuster must also be in good condition.
- 7 Examination of the rockers and their shafts should be carried out on a general basis as already stated. If excessive wear is suspected, refer to the Service limits stated in the Specifications section of this Chapter, and replace components as necessary. Note that during reassembly, which is carried out by reversing the dismantling procedure, the rocker shafts should be lubricated with clean oil prior to being inserted in the carriers. Ensure the shafts are fitted correctly; the threaded end outermost.

28 Examination and renovation: camshaft, camshaft bushes and camshaft drive sprockets

- 1 Examine the camshaft visually for signs of wear. The camshaft lobes should have a smooth surface and be entirely free from scuff marks or indentations. Wear will probably be most evident on the ramps of each cam and where the cam contour changes sharply. It is unlikely that severe wear will be encountered during the normal service life of the machine unless the lubrication system has failed, causing the case hardened surface to wear through. If necessary, check with the Specification given at the beginning of this Chapter and measure the cam height in each case. If any of the cams is below the service limit, the camshaft must be renewed.
- 2 The camshaft runs in two replaceable bushes; one fitted to each end journal. Examine the internal bearing surface of the bushes for any signs of damage or excessive wear. If such defects are found, renewal of the bushes is necessary. If no such signs are discovered, but internal wear is suspected, the bushes may be measured using an internal micrometer. The service limit for the bushes is 20.20 mm (0.7965 in). If the bushes prove to be worn beyond acceptable levels, the camshaft journals should also be examined and measured for wear. The service limit for the journals is 19.92 mm (0.7843 in).
- 3 The sprockets which the camshaft drive chain runs between are of a relatively unusual design. They are of duplex construction featuring a double row of teeth, giving the effect of two sprockets joined together. In consequence, the camshaft drive chain is also a double row item.
- 4 The upper camshaft sprocket is bolted directly to the camshaft and in consequence is easily renewable if the teeth become hooked, worn, chipped or broken. The lower sprocket is

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integral with the crankshaft and if any of these defects are evident, the complete crankshaft assembly must be renewed. Fortunately, this drastic course of action is rarely necessary since the parts concerned are fully enclosed and well lubricated, working under ideal conditions. If sprocket damage is apparent it can often be attributed to the presence of foreign bodies or particles from some other engine component which has broken up.

29 Examination and renovation: camshaft drive chain and chain tensioner

1 Inspect the cam chain for obvious signs of damage, such as cracked, broken or missing rollers or fractured links. Some indication of the amount of wear is given by the extent to which the chain can be bent sideways: if a pronounced curve is evident, the chain should be renewed. Although working in near ideal conditions, being fully lubricated and enclosed, wear will develop after an extended mileage. Therefore, although a low rate of wear may be expected, do not neglect to examine the chain; breakage of the chain will not only necessitate complete engine dismantling for renewal, but may also cause extensive engine damage. Unfortunately, precise cam chain wear figures are not available, but some indication of the extent of chain wear may be gathered from how much adjustment movement remained on the tensioner.

2 The chain tensioner, like the chain, normally lives a trouble-free life. The mechanism should not, however, be neglected. Check the tensioner blade and the guide blade for wear and for separation of the rubber coating from the backing piece. No specifications are laid down for acceptable blade wear, but it is suggested that the components are renewed if wear has reduced the thickness of rubber to less than 50%. If the rubber has begun to separate from the blade, the component should be renewed as a matter of course.

3 Examine also the tensioner arm and its spring for signs of excessive wear. Also examine the tensioner pushrod, ensuring it slides freely in the cylinder block. In cases of doubt, err on the side of safety and renew without question. This advice applies to all components connected with the camshaft drive train.

30 Examination and renovation: starter motor clutch, drive chain and sprockets

1 The power from the electric starter motor is taken from a small drive sprocket on its outer end, via a drive chain, to a larger driven sprocket on the left-hand end of the crankshaft,

behind the alternator rotor. Fitted to the rear face of the alternator rotor is a one-way clutch.

2 The one-way clutch ensures that the starter motor drive is disconnected from the crankshaft immediately the engine starts. It operates on the roller and ramp principle: spring loaded rollers take up the drive until the engine starts and the drive is automatically disconnected. In the normal course of events, it is unlikely that the clutch will malfunction.

3 The only parts likely to require attention are the rollers and their springs, or the bush in the centre of the driven sprocket around which the one-way clutch is fitted.

4 To check whether the starter clutch is operating correctly, fit the driven sprocket to the rear of the clutch and rotor. When rotated in a clockwise direction, as viewed from the sprocket side, the clutch should lock immediately, allowing power to be transmitted from the sprocket to the crankshaft. When rotated anti-clockwise, the sprocket should be free to run smoothly. If the movement is unsatisfactory, remove the sprocket from the clutch. The sprocket boss should be smooth, scoring or damage to the surface indicates that the rollers are similarly marked, and require further inspection.

5 The rollers, springs and plungers may be removed for examination with the clutch still attached to the rear of the rotor. Using a small, flat-bladed screwdriver, carefully push each plunger back against its spring tension until the roller can be removed. Then remove the springs and plungers. Signs of wear will be obvious and will necessitate renewal of the worn or damaged parts.

6 To dismantle the clutch further, the three countersunk crosshead retaining screws must be removed. Using an impact driver, slacken and remove the three screws. The clutch outer unit and the backing plate can now be removed from the rear of the rotor. Note the single locating dowel pin between the rear of the rotor and the clutch outer unit. Examine the clutch outer unit for wear in the form of elongation or scoring of the roller housing. If worn badly, the clutch outer unit must be renewed.

7 Upon reassembly, ensure the dowel pin is correctly located, and apply a small quantity of locking fluid to each of the three retaining screws. Retighten them fully using an impact driver. Reinstall the springs, plungers and the rollers using the same procedure as employed for their removal.

8 Inspect the drive and driven sprockets, looking for chipped, worn, broken or missing teeth. Examine the drive chain for damaged rollers or fractured links. Wear of an excessive nature is extremely unlikely on the sprockets or the chain, due to its fully enclosed situation and limited usage. Replacements should not be required until an extremely high mileage has been covered.



27.4a Use a 5 mm screw to facilitate rocker shaft removal



27.4b With the shaft withdrawn, remove the rocker arm and spring

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29.2 Examine cam chain tensioner blade and tensioner arm (note O-rings on adjuster stud and securing bolt)



30.4 Check sprocket boss for wear; note oil seal which must be renewed if damaged



30.5 If required, the rollers, springs and plungers can be removed



30.6 Remove three screws to separate clutch outer unit from rotor

31 Examination and renovation: clutch assembly

1 After a considerable mileage has been covered, the bonded linings of the clutch friction plates will wear down to or beyond the specified wear limit, allowing the clutch to slip.

2 Give the plain and the friction plates a wash with a petrol/paraffin mix and remove all traces of clutch insert debris. If this precaution is not taken, a gradual build-up of debris will occur and eventually affect clutch action.

3 The degree of wear is measured across the faces of the friction material, using a vernier caliper. When new, each friction plate measures 3.0 mm (0.1181 in). If the plates have worn to 2.6 mm (0.1024 in), they should be renewed as a complete set, even if slipping is not yet apparent.

4 The plain plates should be free from scoring and signs of overheating, which will be apparent in the form of blueing. The plates should also be flat. Check each plain plate for distortion by laying it on a flat surface, such as a sheet of plate glass, and by measuring any detectable gap with feeler gauges. The plates must be less than 0.2 mm (0.008 in) out of true; any more and judder and snatch may result. Replace the plates as a set if warpage is greater than the service limit stated.

5 Measure the uncompressed length of the clutch springs. Should the springs have taken a permanent set, down to a

length of less than that given in the Specifications at the beginning of the Chapter, they should be renewed. Always renew clutch springs as a set.

6 Check the condition of the thrust bearing and the clutch operating pushrod and its seat. Excessive play or wear in these components will cause noise and lead to erratic operation.

7 Check the condition of the slots in the outer surface of the clutch centre and the inner surfaces of the outer drum. In an extreme case, clutch chatter may have caused the tongues of the inserted plates to make indentations in the slots of the outer drum, or the tongues of the plain plates to indent the slots of the clutch centre. These indentations will trap the clutch plates as they are freed, and impair clutch action. This will be shown in the form of clutch drag and slow disengagement during gear changes. If the damage is only slight the indentations can be removed by careful work with a file and the burrs removed from the tongues of the clutch plates in a similar fashion. More extensive damage will necessitate renewal of the parts concerned.

8 The clutch release mechanism attached to the inside of the right-hand outer cover takes the form of a vertical spindle with a cam on the lower end. A light return spring ensures that pressure is taken off the end of the pushrod once the handlebar lever has been released. No attention is normally required, other than greasing prior to reassembly.

32 Examination and renovation: primary drive gears and oil pump drive gear

- 1 The primary drive is by means of a small (18T on the 185 models and 19T on the 200 models) crankshaft-mounted gear driving the clutch by way of a large driven gear on the rear of the clutch outer drum. Examine both gears for chipped, broken or worn teeth. This type of damage cannot be rectified; the component must be renewed. The pinions concerned are unlikely to become damaged unless foreign bodies or particles from some other engine component which has broken up become trapped between the teeth. The primary driven gear is riveted to the clutch outer drum. These two items are not supplied separately as replacements. If one fails the other must be renewed at the same time. If either the primary drive or primary driven gear require renewal, it is preferable to replace them as a pair.
- 2 The oil pump drive gear is fitted outboard of the primary drive gear on the crankshaft end. The examination and renovation procedures detailed for the primary drive gears apply equally to the oil pump drive gear.

33 Examination and renovation: gearbox components

- 1 Examine each of the gear pinions to ensure that there are no chipped or broken teeth and that the dogs on the end of the pinions are not rounded. Gear pinions with any of these defects must be renewed; there is no satisfactory method of reclaiming them.
- 2 It is not normally necessary to dismantle the gear clusters unless the sort of damage described above has occurred or a fault has become apparent on the gearbox shafts.
- 3 If damage or wear warrants dismantling of the clusters, it is advisable that the assemblies are studied prior to dismantling. In order to further eliminate the risk of incorrect reassembly, rough sketches can be made as each pinion, thrust washer, circlip etc is removed. Also strip and rebuild as soon as possible to reduce any confusion which may arise at a later date.
- 4 The accompanying illustrations and photographic sequence show how the clusters are arranged on their shafts. It is imperative that the gear clusters, including the thrust washers and circlips, are assembled in **EXACTLY** the correct sequence, otherwise constant gear selection problems will arise.
- 5 Check the gear selector fork support shaft for straightness by rolling it on a sheet of plate glass. A bent rod will cause

difficulty in selecting gears and will make the gear change particularly heavy.

6 Examine the selector forks carefully, ensuring that there is no scoring or wear where they engage in the gears, and that they are not bent. Damage and wear rarely occur in a gearbox which has been properly used and correctly lubricated, unless very high mileages have been covered.

7 The tracks in the selector drum, with which the selector forks engage, should not show any undue signs of wear unless neglect has led to under-lubrication of the gearbox. Check the tension of the gearchange arm selector lever, gearchange arm and drum stopper arm (detent arm) springs. Weakness in the springs will lead to imprecise gear selection. Check the condition of the gear stopper arm roller and the pins in the change drum end with which it engages. It is unlikely that wear will take place here except after considerable mileage.

8 Examination of the gearchange components should be carried out on the general basis as described. If excessive wear is suspected, refer to the service limits given in the Specification section at the beginning of this Chapter and replace components as necessary.



33.4a Mainshaft 1st gear pinion (13T) remains on shaft



33.4b Fit mainshaft 3rd gear pinion (22T) ...



33.4c ... and secure with splined washer and circlip

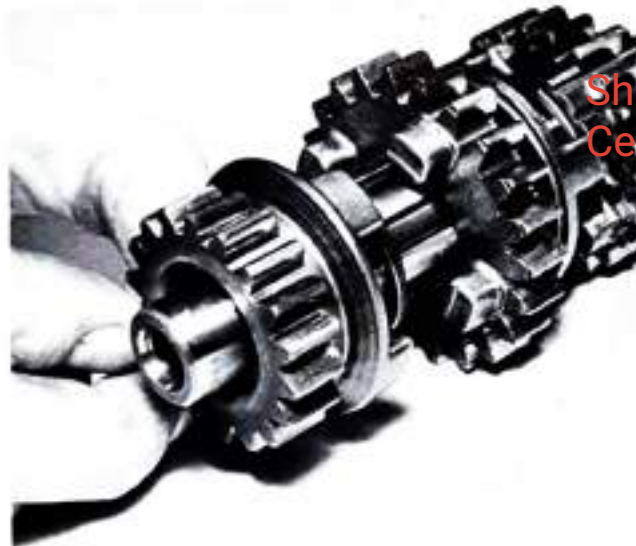
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33.4d Slide mainshaft 4th gear pinion (22T) into position ...



33.4e ... and secure with circlip and splined washer



33.4f Fit 2nd gear pinion (18T) to mainshaft and ...



33.4g ... then the heavy shim and thrust washer

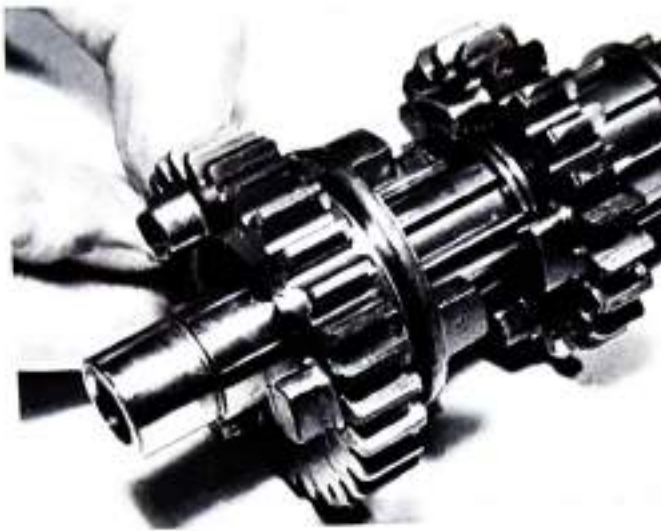


33.4h Slide thrust washer and sleeve gear onto other end of mainshaft



33.4i To layshaft fit 4th gear pinion (22T), splined washer and circlip to one side and circlip on other side

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33.4j Fit layshaft 3rd gear pinion (28T), followed by ...



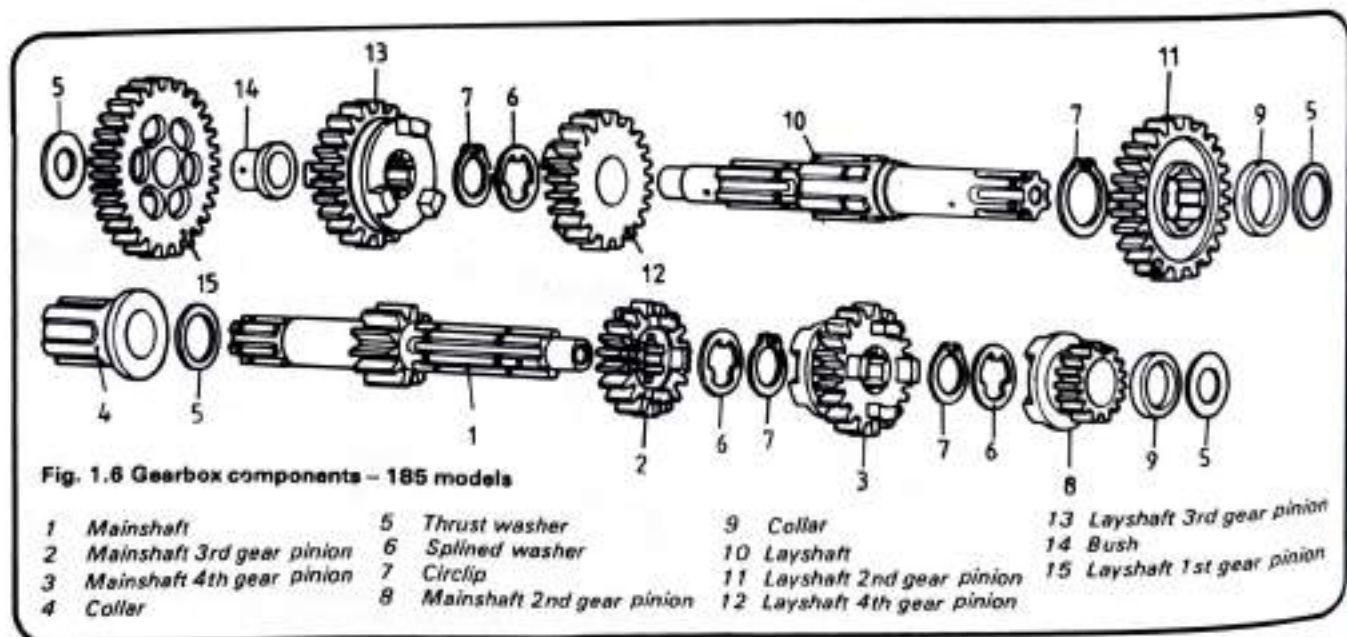
33.4k ... thrust washer, layshaft 1st gear pinion (37T) and shouldered bush



33.4l Slide 2nd gear pinion (32T) onto layshaft followed by ...



33.4m ... the heavy shim and final thrust washer



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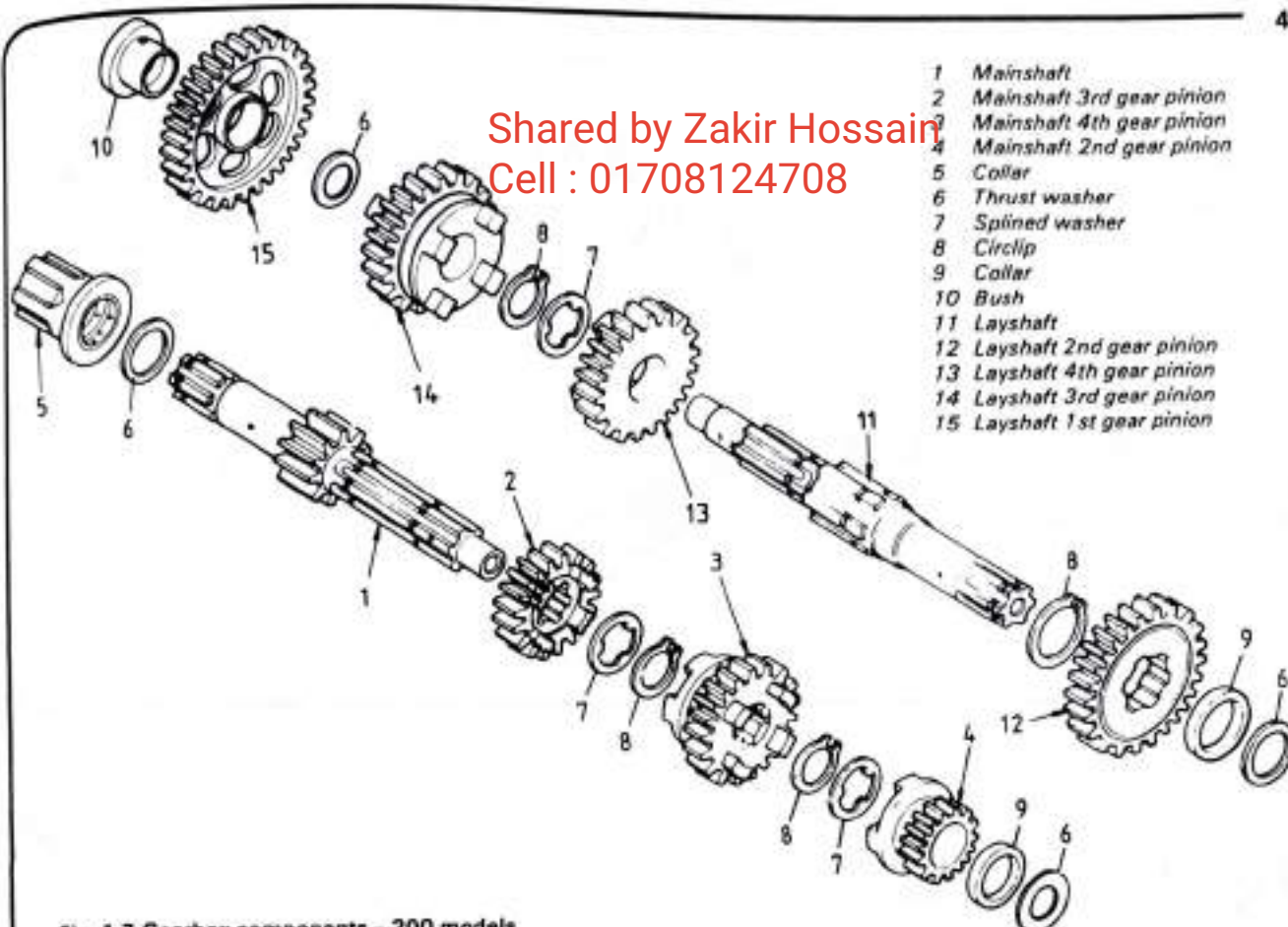


Fig. 1.7 Gearbox components – 200 models

34 Examination and renovation: kickstart components

1 Check the condition of the kickstart components. If slipping has been encountered a worn ratchet and pawl will invariably be traced as the cause. Any other damage or wear to the components will be self-evident. If either the ratchet or pawl is found to be faulty, components must be replaced as a pair. Examine the kickstart return spring, which should be renewed if there is any doubt about its condition.

2 The kickstart shaft assembly will have been partially dismantled when it was removed from the engine during the stripdown procedure. The assembly may be dismantled further into its individual components as follows. Displace the circlip from the inner end of the shaft, having first removed the pawl guide plate. The pawl engagement spring is now free to be slid off the shaft, followed by the pawl piece itself. From the outer end of the shaft, remove the large plain thrust washer the kickstart ratchet pinion, another plain washer and a second circlip.

3 If deemed necessary, the kickstart shaft OD and the ratchet pinion bore ID may be measured; the service limits will be found in the Specifications section at the front of this Chapter.

4 The components may be refitted by reversing the dismantling sequence. Note that when refitting the pawl piece to the shaft, the punch on each should be in alignment with the other. If this alignment is not correct, the kickstart will not operate correctly.

5 In general, kickstart mechanisms are reliable devices and do not wear quickly unless a malfunction occurs, or engine oil level is allowed to fall. Look for chipped or broken teeth and excessive wear, replacing the pawl piece and ratchet pinion if necessary.

35 Examination and renovation: engine cases and covers

1 The aluminium alloy casings and covers are unlikely to suffer damage through ordinary use. However, damage can occur if the machine is dropped, or if sudden mechanical breakages occur, such as the rear chain breaking.

2 Small cracks or holes may be repaired with an epoxy resin adhesive, such as Araldite, as a temporary expedient. Permanent repairs can only be effected by argon-arc welding, and a specialist in this process is in a position to advise on the viability of proposed repair. Often it may be cheaper to buy a new replacement.

3 Damaged threads can be economically reclaimed by using a diamond section wire insert, of the Helicoil type, which is easily fitted after drilling and re-tapping the affected thread. The process is quick and inexpensive, and does not require as much preparation and work as the older method of fitting brass, or similar inserts. Most motorcycle dealers and small engineering firms offer a service of this kind.

4 Sheared studs or screws can usually be removed with screw extractors, which consist of tapered, left-hand thread screws, of very hard steel. These are inserted by screwing anti-clockwise, into a pre-drilled hole in the stud, and usually succeed in dislodging the most stubborn stud or screw. If a problem arises which seems to be beyond your scope, it is worth consulting a professional engineering firm before condemning an otherwise sound casing. Many of these firms advertise regularly in the motorcycle papers.

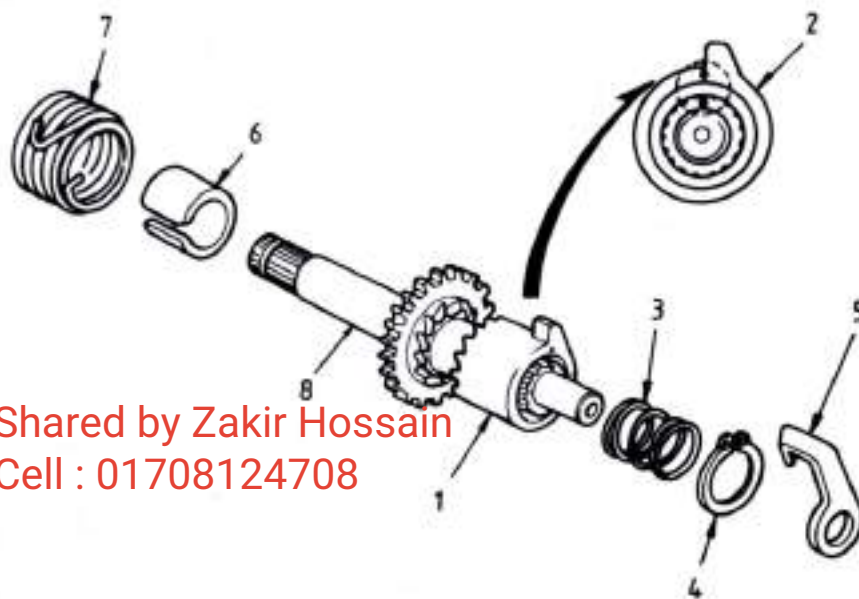


Fig. 1.8 Kickstart mechanism

- 1 Ratchet
- 2 Alignment punch marks
- 3 Spring
- 4 Circlip
- 5 Pawl guide plate
- 6 Spring guide
- 7 Return spring
- 8 Kickstart shaft

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34.4a Check the kickstart shaft splines before reassembly



34.4b Fit circlip as shown, then ...



34.4c ... position pawl and pawl spring on shaft and secure with circlip



34.4d Install the pawl guide plate as shown



34.4e Note alignment of dots on pawl piece and shaft splines



34.4f Fit the plain washer and ratchet pinion, followed by —



34.4g ... the large thrust washer and ...



34.4h ... the spring and spring guide

36 Engine reassembly : general

1 Before reassembly of the engine/gearbox unit is commenced, the various component parts should be cleaned thoroughly and placed on a sheet of clean paper, close to the working area.

2 Make sure all traces of old gaskets have been removed and that the mating surfaces are clean and undamaged. One of the best ways to remove old gasket cement is to apply a rag soaked in methylated spirit. This acts as a solvent and will ensure that the cement is removed without resorting to scraping and the consequent risk of damage. If a gasket becomes bonded to the surface through the effects of heat and age, a new sharp razor blade can be used to effect removal. Old gasket compound can also be removed using a soft brass wire brush of the type used for cleaning suede shoes. A considerable amount of scrubbing can take place without fear of damaging the mating surfaces.

3 Gather together all the necessary tools and have available an oil can filled with clean engine oil. Make sure that all new gaskets and oil seals are to hand, also all replacement parts required. Nothing is more frustrating than having to stop in the middle of a reassembly sequence because a vital gasket or replacement has been overlooked.

4 Make sure that the reassembly area is clean and that there is adequate working space. Refer to the torque and clearance settings wherever they are given. Many of the smaller bolts are easily sheared if over-tightened. Always use the correct size screwdriver bit for the crosshead screws and never an ordinary screwdriver or punch. If the existing screws show evidence of maltreatment in the past, it is advisable to renew them as a complete set.

5 If the purchase of a replacement set of screws is being contemplated, it is worthwhile considering a set of socket or Allen screws. These are invariably much more robust than the originals, and can be obtained in sets for most machines, in either black or nickel plated finishes. The manufacturers of these screw sets advertise regularly in the motorcycle press.

6 In addition it will be necessary to obtain a bottle or tube of thread locking compound such as Loctite.

37 Reassembling the engine/gearbox unit: refitting the crankshaft assembly

1 Place the left-hand crankcase half on blocks on the workbench, checking that a reasonable amount of room is allowed for the shafts to protrude when fitted.

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2 Lubricate the left-hand crankshaft main bearing with clean engine oil. Install a new oil seal outboard of the left-hand main bearing, with the spring side facing inwards. Fit a new O-ring to the oil feed hole in the crankcase, below the left-hand crankshaft main bearing, and refit the oil passage plate. Apply a little locking compound to the threads of the three countersunk crosshead retaining screws, and tighten them fully using an impact driver.

3 Refit the camshaft drive chain to the crankshaft sprocket. As noted during the removal of the chain (Section 18.2 of this Chapter), a certain amount of manoeuvring must take place to facilitate cam chain removal and refitting. Position the flywheel weights so that the cam chain can be passed either side of them between the cam chain tensioner blade locating pin at the rear of the cast retainer, and the guide blade locating slot at the front of the retainer.

4 If the cam chain tensioner blade has been renewed, or was detached for examination, it should be refitted now. Fit the blade lower end over the locating pin and secure it with the plain washer and R-clip. The lower end is distinguishable from the upper end by the letter 'A' marked on the latter.

5 Slide the crankshaft assembly into position in the left-hand crankcase half, aligning the stud at the top of the casing half with the upper hole in the centre bearing holder. Insert the five 6 mm bolts and the single 6 mm nut on the stud, and tighten them evenly, a little at a time, to a torque setting of 1.0 – 1.4 kgf m (7.2 – 10.0 lbf ft).

38 Reassembling the engine/gearbox unit: refitting the gearbox components

1 If the gearshaft assemblies were dismantled for inspection or renovation they must now be assembled as complete units

before fitting into the casing. When reassembling either shaft, refer to the appropriate line drawing and to the photograph sequence to ensure that all pinions, circlips and washers are refitted in **exactly** the correct order. Failure to do this will result in continuous gear selection problems.

2 Install a new oil seal in the crankcase wall outboard of the layshaft left-hand main bearing, with the seal spring side facing inwards. Lubricate all the gearbox bearings with clean engine oil. Similarly, lubricate the gearbox shafts and pinions.

3 Engage the mainshaft and layshaft gear assemblies and lower them into position in the casing together. Ensure the shim and the thrust washer on the end of the mainshaft being inserted and the thrust washer on the layshaft end being inserted, are not omitted. To prevent the thrust washers dropping free as the shafts are inserted, hold them in position with two fingers of one hand.

4 Fit the neutral indicator switch rotor to the left-hand end of the selector drum. Lubricate the selector drum with clean engine oil. Lower the drum, rotor downwards, into the left-hand casing. Align the contact on the rotor with the hole in the crankcase wall through which the neutral switch will pass.

5 Install the two selector forks so that the guide pin of each locates with the selector drum and the fork ends locate with the pinions. Note that each fork has an identification mark stamped on it; fit the lower fork, which engages with the central selector drum track, with the mark facing downwards and fit the upper fork, which engages with the right-hand selector drum track, with the mark facing upwards. Slide the selector fork support shaft into position to secure the forks. When the assembly has been installed correctly, lubricate it with engine oil, and check gearbox operation by turning the change drum so that each gear in turn is engaged. Rotate the mainshaft to aid selection.

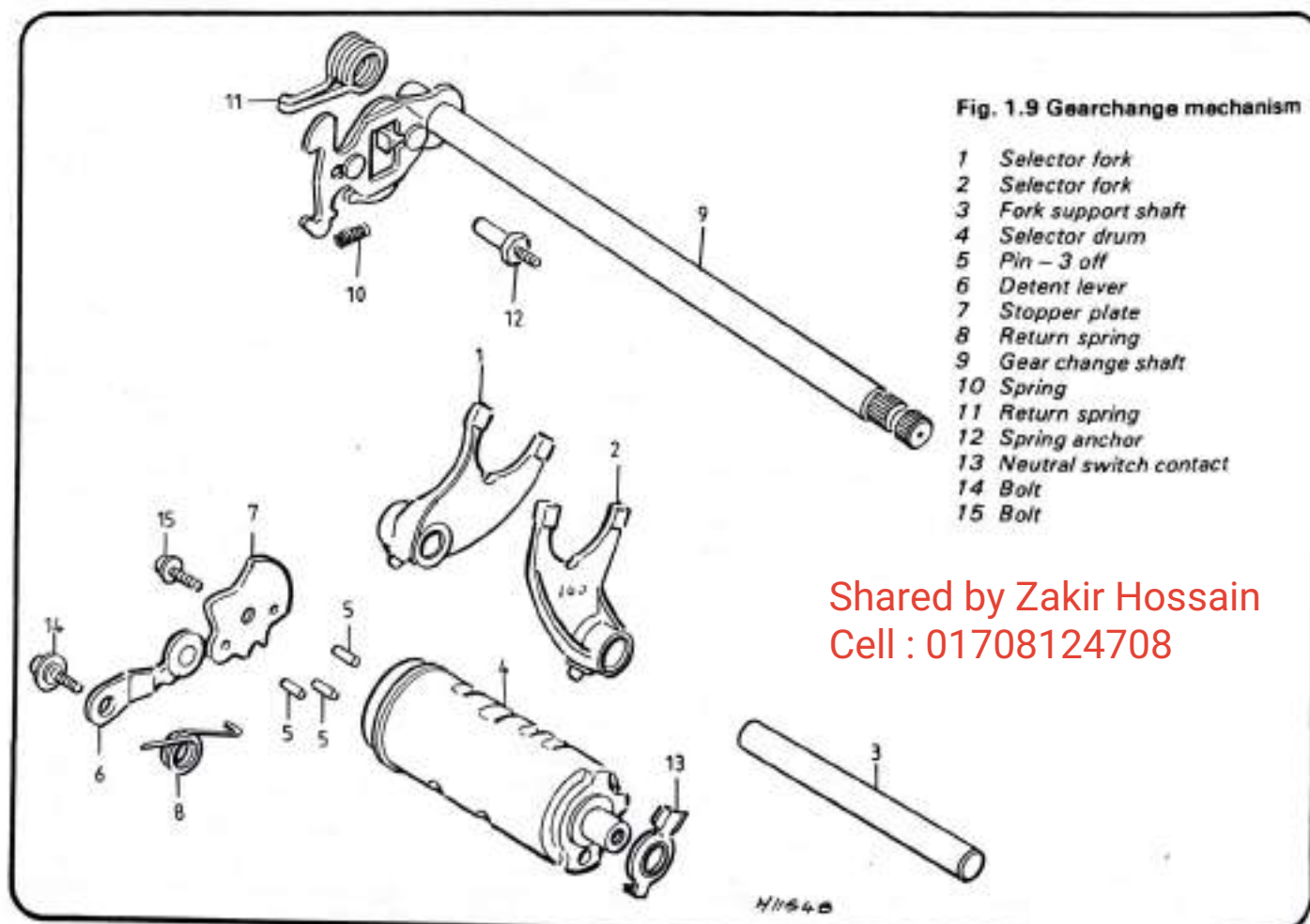


Fig. 1.9 Gearchange mechanism

- 1 Selector fork
- 2 Selector fork
- 3 Fork support shaft
- 4 Selector drum
- 5 Pin - 3 off
- 6 Detent lever
- 7 Stopper plate
- 8 Return spring
- 9 Gear change shaft
- 10 Spring
- 11 Return spring
- 12 Spring anchor
- 13 Neutral switch contact
- 14 Bolt
- 15 Bolt

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37.2 Fit a new O-ring before refitting oil passage plate



38.3 Lower both gear shaft assemblies into casing together; hold shim and washers as shafts are fitted

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38.4a Fit the neutral indicator switch rotor to selector drum as shown, before ...



38.4b ... Inserting the drum (switch rotor downwards) into the casing



38.5a Arrange the two selector forks as shown, then ...



38.5b ... insert support shaft to secure forks

39 Reassembling the engine/gearbox unit: joining the crankcase halves

- 1 Insert the two hollow locating dowels in the left-hand crankcase half. Check that the various moving parts are adequately lubricated with clean engine oil. In the right-hand crankcase half, lubricate the crankshaft main bearing, the mainshaft main bearing and the needle roller bearing in which the blind end of the layshaft runs.
- 2 The jointing faces of each crankcase half should be meticulously cleaned, preferably with methylated spirit, to remove any residual oil or moisture.
- 3 Fit a new crankcase gasket to the left-hand crankcase half. No gasket cement should be necessary. If the jointing surfaces are somewhat suspect, however, a thin coat of gasket cement can be applied to them. Avoiding oil leaks is the priority here, but unsightly, excess amounts of gasket compound should also be avoided.
- 4 Lower the right-hand crankcase half down firmly and squarely onto the left-hand half. Ensure that both the locating dowels engage correctly. Take care to locate the right-hand end of the crankshaft in its main bearing and the gear shafts in their respective bearings. It may be necessary to give the right hand crankcase a few light taps with a soft faced mallet before the jointing surfaces will mate up correctly. **Do not use force.**
- 5 Insert the eight crankcase retaining bolts from the right-hand crankcase half. It will be remembered that three of the bolts, two at the top and one at the bottom, are deeply recessed and require the use of a thin section socket spanner and extension bar to tighten them. Tighten all the bolts in an even and diagonal sequence, to a torque setting of 1.0 – 1.4 kgf m (7.2 – 10.0 lbf ft). Turn the engine around, so that the left-hand crankcase half is uppermost (taking care that the cam chain does not disappear inside the crankcase) and insert the two crosshead screws. One is fitted at the rear of the unit, the second at the top front of the unit, between the crankcase mouths. Tighten the screws fully.
- 6 The operation and free-running of the crankshaft and gearbox assembly should be ascertained at this stage. Any tight spots or resistance must be investigated and rectified before further reassembly takes place.

40 Reassembling the engine/gearbox unit: refitting the kickstart assembly and gearchange mechanism

- 1 With the kickstart assembly reassembled, as described in Section 34 of this Chapter, the device may be refitted to the right-hand crankcase half. Note that the pawl piece must be



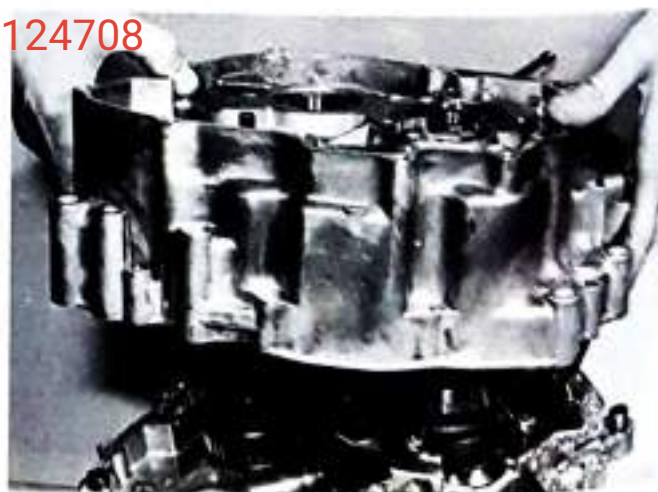
39.3 Fit a new gasket to the left-hand crankcase half and ensure two dowels are fitted

fitted to the kickstart shaft in one position only; that is with the punch mark on each component in alignment. If this is not done incorrect kickstart operation will result.

- 2 Install the kickstart assembly, ensuring that the guide plate on the inner end is correctly located in the groove in the crankcase wall. Slide the return spring onto the shaft and locate the inner turned end in the cross-drilled hole. Grasp the outer turned end with a stout pair of pliers and tension the spring in a clockwise direction until the hooked end can be secured on the anchor lug. With the spring in place, insert the spring guide, ensuring it is pushed fully home.
- 3 Refit the three gearchange pins to the end of the selector drum if not already reinstalled. Install the pin securing stopper plate, noting the locating holes in the rear of the plate which must engage with two of the pins, and fit and tighten the plate securing bolt.
- 4 Refit the change drum stopper arm (detent lever) and its spring in position in the casing and fit and tighten the single retaining bolt. After tightening the bolt ensure that the arm is free to pivot. The gearchange spindle can now be slid into place in its bore in the crankcase, and the operating claw engaged with the selector drum pins behind the stopper plate. Note that there is an oil seal in the left-hand crankcase wall through which the gearchange spindle splined end passes. In order that the splines do not damage the oil seal lip as the shaft is pushed home, grease the splines or wrap tape around them. The large centralising spring on the main claw arm must be positioned correctly, with one spring and either side of the stop pin protruding from the right-hand crankcase wall.
- 5 It is prudent to check the gear selection at this stage. Temporarily refit the gearchange pedal, and attempt to select each gear in turn. Rotating the projecting layshaft will aid engagement. The gears should select up and down the range.

41 Reassembling the engine/gearbox unit: refitting the primary drive pinions, oil pump drive pinion and clutch assembly

- 1 Slide the small primary drive gear pinion onto the right-hand end of the crankshaft.
- 2 Place the clutch outer drum in position on the gearbox mainshaft end, engaging the primary driven gear on the back of the drum with the crankshaft mounted primary drive pinion. Fit the splined thrust washer with its rounded face inwards.
- 3 Place the clutch pressure plate on the workbench so that the inner face is uppermost. Fit the clutch plates to the pressure plate one at a time, starting with a friction plate and then a plain plate, building up the layers alternately until complete. Lightly oil each friction plate as it is installed. Ensure that the internal splines of the plain plates are all aligned and the tangs of the



39.4 Check that all is well, then fit right-hand crankcase half

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friction plates are all in line with each other. If this is not done, it will be impossible to fit the clutch centre. Lower the clutch centre into position on the assembled plates.

4 Moving the assembly of pressure plate, clutch plates and clutch centre to the engine, install the assembly in the outer drum. It will probably be necessary to rotate the pressure plate and centre backwards and forwards to ease the location of the plates in the outer drum. Secure the assembly with the circlip.

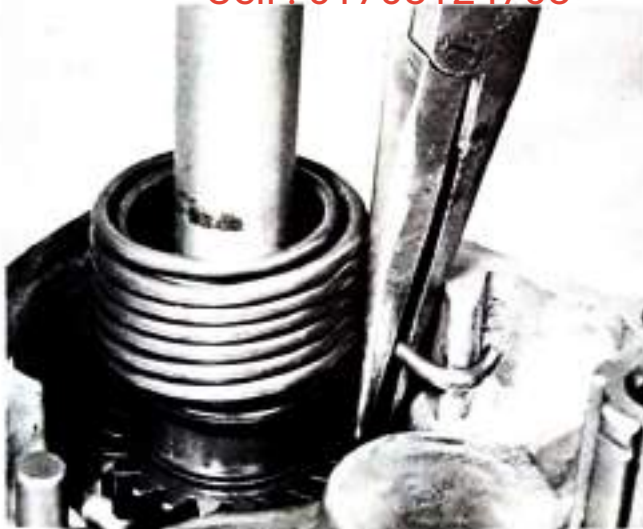
5 Fit the four clutch springs over the projecting threaded pillars. The thrust plate and fitted thrust bearing can now be placed in position, and the four retaining bolts fitted and tightened. Tighten the bolts down evenly in a diagonal sequence, in two or three stages. Install the cup and pushrod in the centre of the thrust bearing.

6 Slide the oil pump drive gear pinion into position on the crankshaft end. Then fit the Belleville washer; note the wording 'OUTSIDE' on one face of the washer and fit it accordingly. Refit the locking pin in the end of the crankshaft and the 16 mm retaining nut. Tighten the nut using the same procedure as used for its removal (see Section 11.3 and 11.4 of this Chapter). Refit the oil pressure relief valve plunger and spring in the end of the crankshaft.

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40.2a Insert the assembled kickstart mechanism



40.2b Secure the outer end of the spring around the casing lug



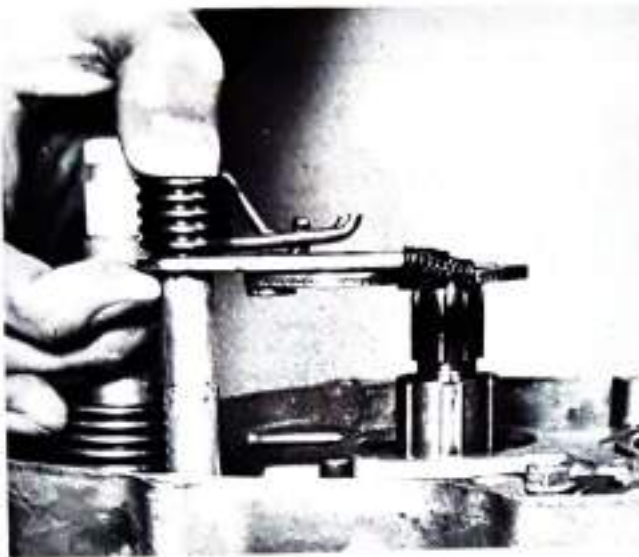
40.3a Refit the pin securing plate, noting the locating holes, and ...



40.3b ... secure the stopper plate by tightening the centre bolt



40.4a Fit and secure the detent lever ensuring it is free to pivot



40.4b Insert the complete gearchange spindle assembly into the casing, ensuring that ...



40.4c ... the operating claw engages with the pins and the centraliser spring ends locate with the stop pin (arrowed)

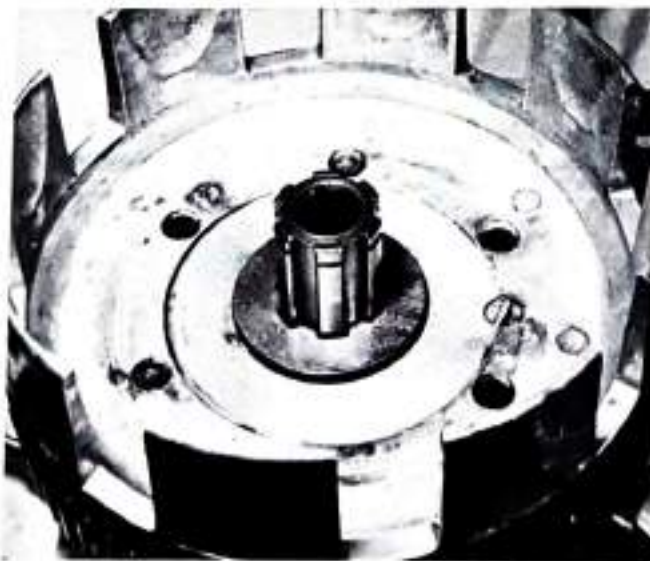
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41.1 Fit primary drive pinion to crankshaft end



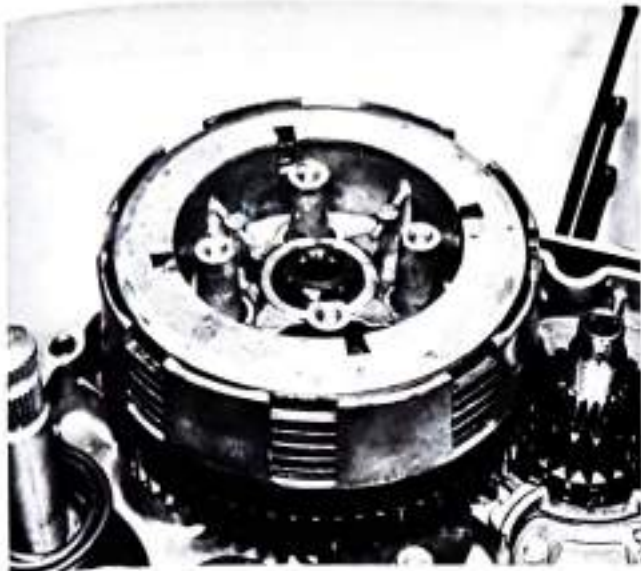
41.2a Place clutch outer drum in position ...



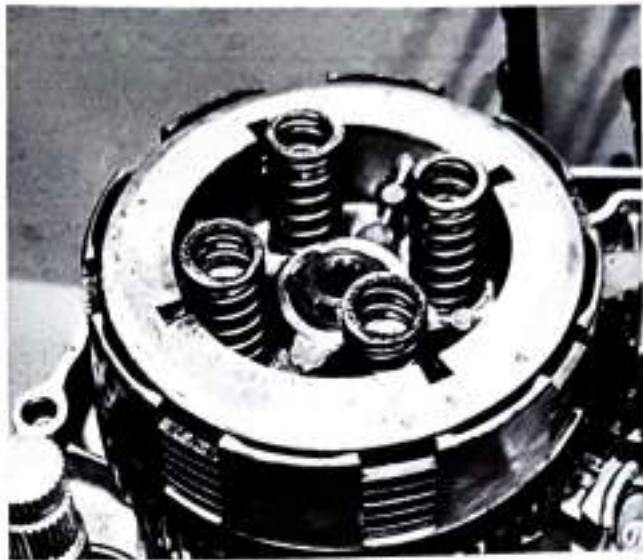
41.2b ... and fit splined thrust washer as shown



41.3 With the clutch plates assembled on the pressure plate, fit the clutch centre



41.4 Fit the assembly to the outer drum



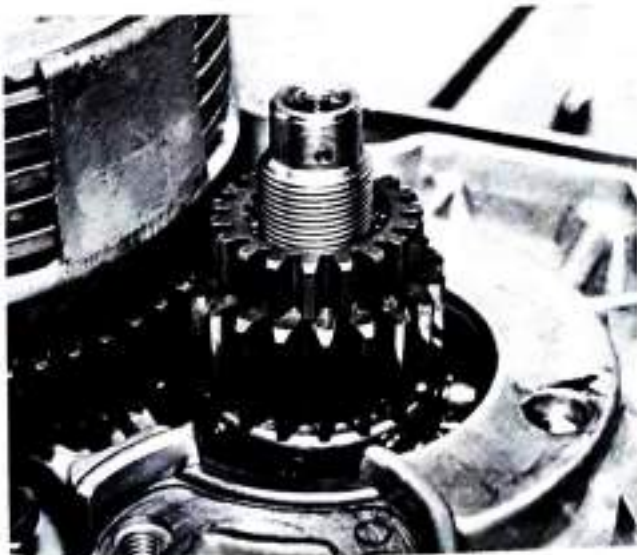
41.5a Fit the four clutch springs over the pressure plate pillars



41.5b With the thrust plate and bearing refitted, install the centre cup ...



41.5c ... and the clutch pushrod



41.6a Slide the oil pump drive pinion into position then ...



41.6b ... fit the Belville washer accordingly

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41.6c Install the locking pin in the crankshaft end then ...



41.6d ... refit the retaining nut



41.6e Reinstall the oil pressure relief valve assembly

42 Reassembling the engine/gearbox unit: refitting the oil pump and filter screen

- 1 Install a new O-ring in the recess in the lower part of the right-hand crankcase wall. Fit the filter screen and rubber retainer to the lower part of the oil pump.
- 2 Place the oil pump in position, engaging the pump driven gear with the crankshaft-mounted drive gear pinion. Align the three holes in the pump outer cover with those in the pump driven gear, so that the three countersunk crosshead retaining screws can be refitted. Tighten the screws evenly and fully.
- 3 With the pump installed, check that the driven gear is free to rotate with no binding or tight spots.

43 Reassembling the engine/gearbox unit: refitting the right-hand crankcase cover (primary drive cover)

- 1 Make a final check to ensure all the components are correctly fitted and tightened and the mating surfaces are clean.
- 2 Ensure the clutch operating mechanism is refitted, with the clutch pushrod engaging properly.
- 3 Lightly grease the clutch operating mechanism in the outer cover.

- 4 Place a new gasket on the casing and lower the casing slowly down over the kickstart shaft, taking great care not to damage the oil seal. An oil leak will result if the seal is damaged; light greasing of the kickstart shaft splines or wrapping the splines in tape, will lessen the chance of damage to the oil seal. Locate the casing on the two dowels and if necessary tap it gently into position with a soft-headed mallet.
- 5 If the casing resists seating, do not use excess force, but raise it slightly and discover the cause of the trouble. A potential cause of the difficulty is the kickstart shaft being pulled slightly out of alignment by the return spring.
- 6 Refit and tighten the ten crosshead retaining screws.

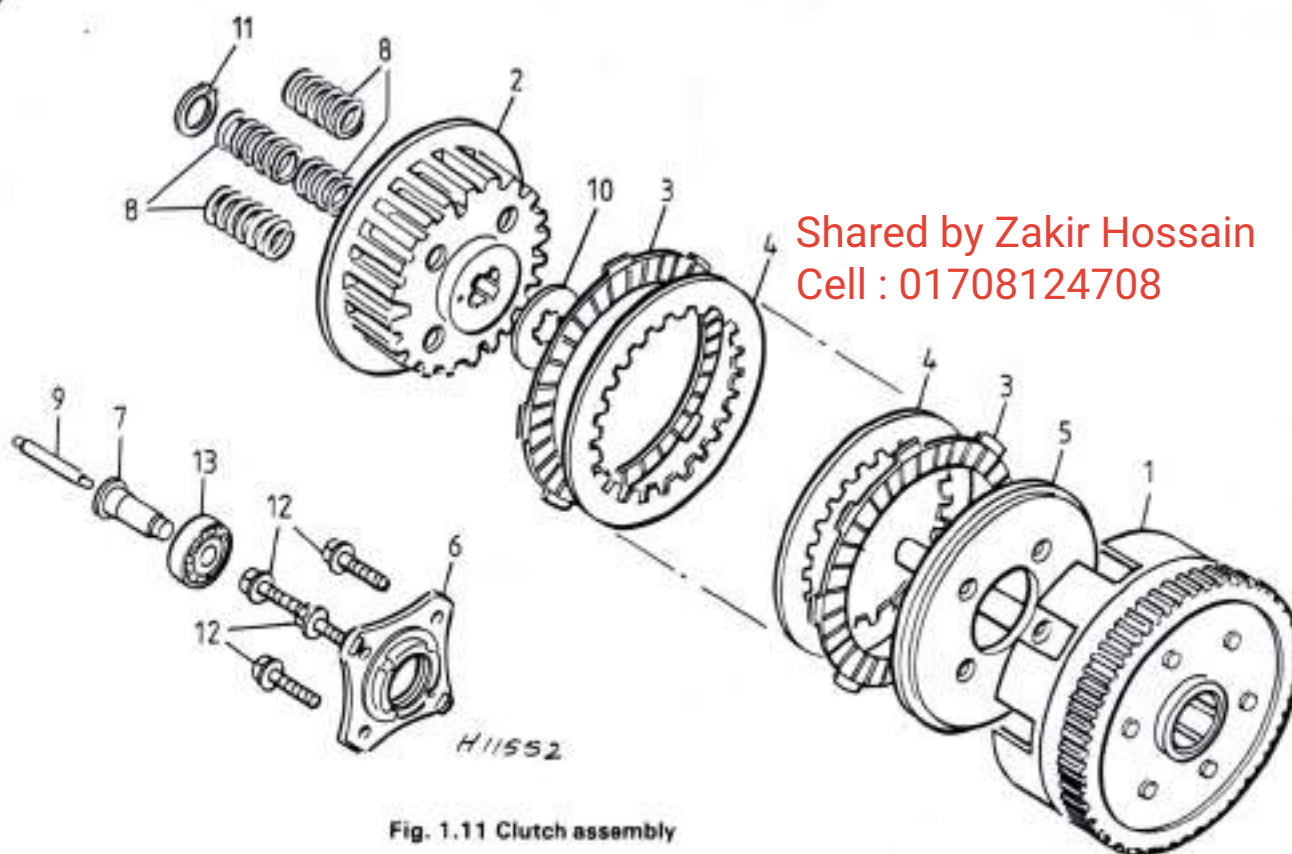
44 Reassembling the engine/gearbox unit: refitting the neutral indicator switch and gearbox final drive sprocket

- 1 Turn the unit over and support it on wooden blocks with the left-hand side uppermost.
- 2 Refit the neutral indicator switch in its bore in the crankcase wall; it is a simple push fit. Note the O-ring on the inner end of the switch body; if the O-ring is suspect it should be renewed or an oil leak may occur. Position the claw plate which secures the switch body and refit and tighten the single shouldered bolt which retains the claw plate. Feed the neutral switch lead up through the rubber grommet in the casing wall.
- 3 Place the final drive sprocket on the splined gearbox output shaft and then fit the securing plate. Rotate the plate so that the two bolt holes align with those in the sprocket and then fit and tighten the two bolts. Note that if the rear chain was removed together with the sprocket, this stage should be left until the engine has been installed in the frame.

Fig. 1.10 Section view of splined washer when fitted to the clutch



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Fig. 1.11 Clutch assembly

- | | |
|------------------|-------------------|
| 1 Outer drum | 8 Spring - 4 off |
| 2 Clutch centre | 9 Pushrod |
| 3 Friction plate | 10 Splined washer |
| 4 Plain plate | 11 Circlip |
| 5 Pressure plate | 12 Bolt - 4 off |
| 6 Thrust plate | 13 Thrust bearing |
| 7 Pushrod seat | |



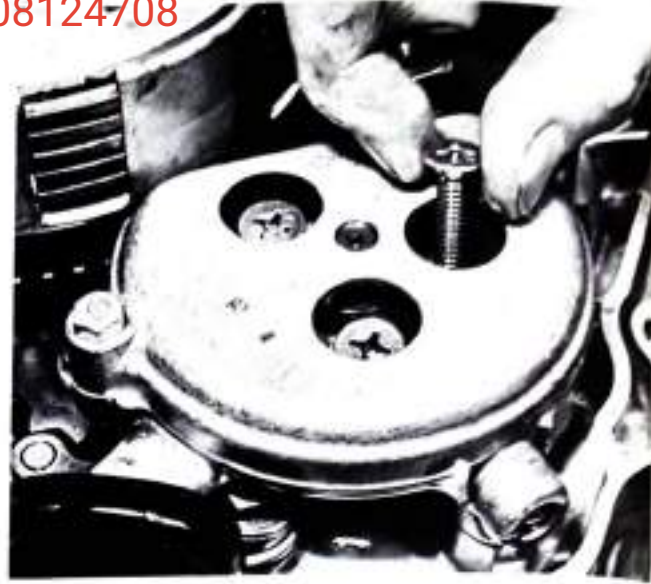
42.1a Install a new O-ring in this recess



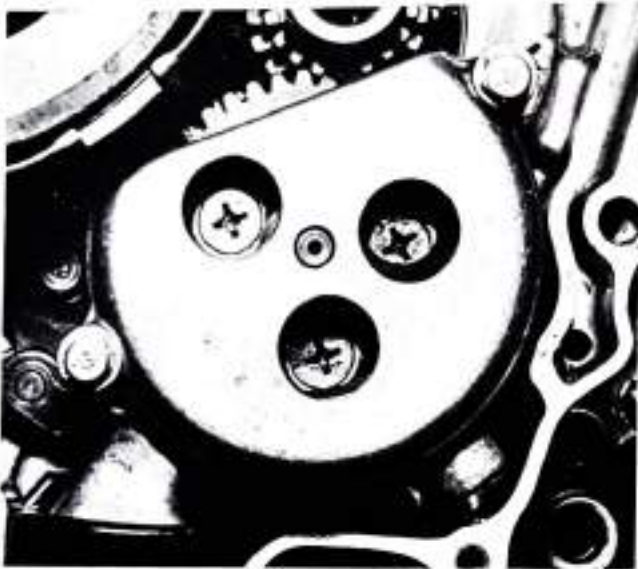
42.1b Clean and refit the filter screen and rubber retainer



42.2a Lower pump assembly into position then ...



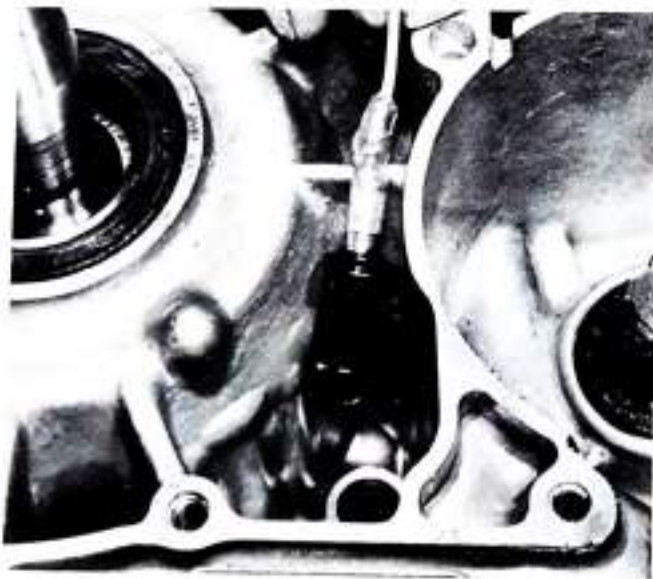
42.2b ... reinstall the three retaining screws ...



42.2c ... tightening them carefully with an impact driver



43.3 Lightly grease clutch operating mechanism



44.2 Refit neutral indicator switch; note O-ring on switch body



44.3a Refit gearbox sprocket and retaining plate and ...



44.3b ... secure the plate with the two bolts

45 Reassembling the engine/gearbox unit: refitting the starter motor, drive chain and sprockets and the alternator

- 1 Place a new gasket against the left-hand side and refit the polished spacer plate to the crankcase. Refit the two locating dowels and position a new gasket on the outside of the spacer plate.
- 2 Check the condition of the O-ring on the starter motor boss; if suspect the O-ring should be renewed. Lubricate the O-ring before sliding the motor into position and securing it with the four crosshead screws at the front of the crankcase.
- 3 Refit the starter motor driven sprocket to the rear of the rotor, and slide the assembly onto the crankshaft end. Line up the internal slot of the rotor with the locating pin on the tapered crankshaft end. Push the rotor into position. Tap the rotor gently with a soft-headed mallet until it is correctly seated. **Do not use excessive force to seat the rotor; the starter clutch rollers may be displaced necessitating removal of the rotor again to refit them.** Note that the driven sprocket must be fitted with the longer boss facing inwards. Secure the driven sprocket by refitting the sprocket guide plate, and fitting and tightening the guide plate retaining bolt. Loop the starter motor drive chain over the rotor, engaging it with the driven sprocket and fit the small drive sprocket to the chain. Fit the drive sprocket, with its boss facing inwards, to the splined end of the starter motor. Ensure that the starter motor drive is functioning correctly by



45.1 Refit polished spacer plate. Do not omit two dowels

rotating the crankshaft anti-clockwise. Make sure the camchain does not disappear into the depths of the crankcase.

- 4 Whilst there is still room for manoeuvring, the heavy duty starter motor cable should be routed up behind the spacer plate and reconnected to the terminal on the starter motor.

5 On the 185 models, refit the ATU, aligning the pin on the rear face of the ATU with the locating hole in the end of the rotor internal boss. Fit the rotor retaining centre bolt and tighten the bolt to a torque setting of 4.5 – 5.0 kgf m (32.5 – 36.2 lbf ft). On the 200 models, insert the rotor retaining centre bolt and tighten it to a torque setting of 5.5 – 6.5 kgf m (40 – 47 lbf ft). During tightening of the centre bolt, the crankshaft must be prevented from rotating once again. Use one of the methods described in the removal sequence (see Section 9.5 of this Chapter).

46 Reassembling the engine/gearbox unit: refitting the contact breaker assembly and left-hand crankcase cover

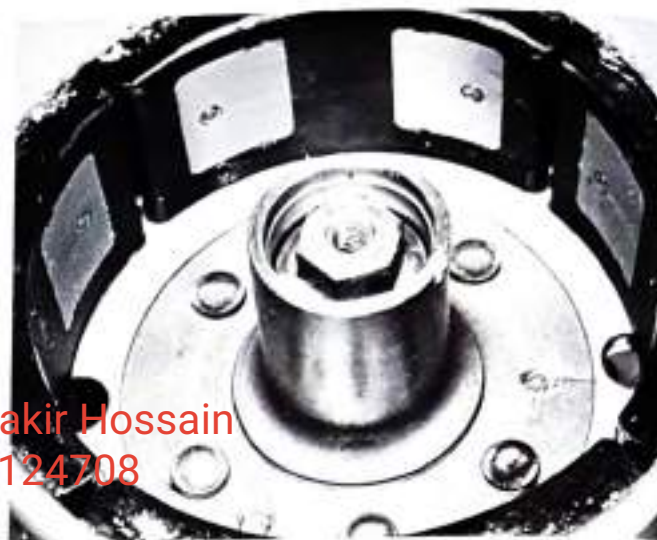
- 1 On the 185 models, the left-hand crankcase cover can now be refitted. The stator, retained on the inside of the cover by three crosshead screws, will also be refitted at this stage. Route the leads from the stator through the rubber grommet in the cover rear wall. Ensure the starter motor cable is fitted below the cast guide on the underside of the cover. Insert and tighten the eight crosshead cover retaining screws, noting the alternator cable guide secured by the lower rear screw. The contact breaker baseplate can now be refitted and secured lightly by the three crosshead screws. Do not tighten the screws until the ignition timing has been checked.
- 2 On the 200 models, the left-hand cover can now be refitted. Locate the ATU on the end of the rotor centre boss, engaging the locating pin on the rear face of the ATU with the notch in the end of the rotor centre boss. Refit the contact breaker baseplate and secure it lightly with three crosshead screws. Finally, refit the large engine turning hexagon and the ATU retaining bolt.
- 3 On all models, unless the position of the baseplate in relation to the crankcase was marked prior to dismantling, the ignition timing will require adjustment. It should, in any case, be checked before running the rebuilt engine. Similarly, the contact breaker points gap must also be checked and if necessary adjusted prior to checking the ignition timing. The checking and adjustment procedure is described fully under the four monthly/3000 mile service heading in Routine Maintenance at the beginning of the manual. In view of the necessity to check and adjust the ignition timing and points gap, and the need for timing marks to be visible when setting the valve timing, the circular inspection cover should not be refitted yet.



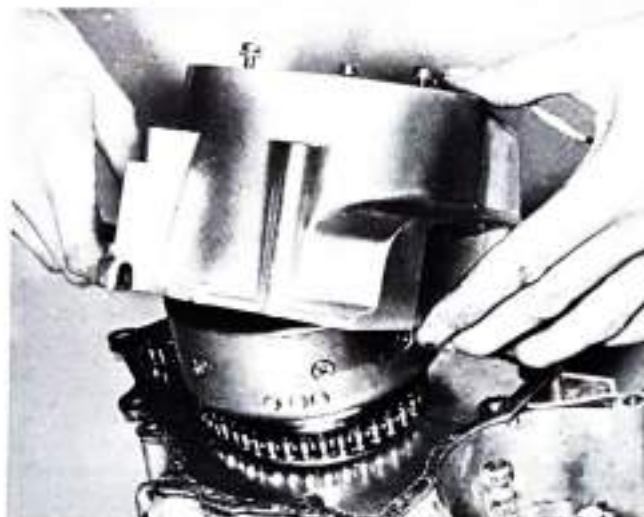
45.3a Locate the driven sprocket/rotor assembly over crankshaft end (200 model shown)



45.3b Refit guide plate and retaining bolt as shown



45.5 Refit and tighten rotor centre bolt (200 model shown)



46.2a Lower left-hand crankcase cover into position (200 model shown)



46.2b Pin on ATU must locate with slot in rotor centre boss

47 Reassembling the engine/gearbox unit: refitting the pistons, cylinder block and cylinder head

- 1 Place the crankcase unit upright on the workbench, taking care that the camshaft drive chain does not slide back into the crankcase.
- 2 Before replacing the pistons it is advisable to pad temporarily the crankcase mouth. If a circlip is misplaced, it may fall into the crankcase and necessitate a further engine strip for its retrieval. Leave the rag in position until the cylinder block is lowered into position since a similar catastrophe can occur should the piston rings break as they enter the cylinder bore.
- 3 Before refitting the pistons, the piston rings must be refitted. The three-piece oil control ring should be fitted first, one piece at a time, starting with a side rail, then the centre expander and then the final side rail. Ensure the end gaps of the side rails are positioned at least 20 mm (0.8 in) away from each other. These components may be fitted either way up. The oil control ring must be fitted to the lowest of the three grooves on the pistons. The remaining rings should be fitted with the identification markings uppermost. If any difficulty is encountered in identifying the two top rings, it should be noted that the 2nd ring has a slightly tapered section and the top ring a parallel section, with slightly chamfered edges on the outer face.

Arrange the ring end gaps away from each other as shown in the accompanying illustration.

- 4 Fit the pistons to their original connecting rods (unless new pistons are being fitted of course), with the arrow mark embossed on each 185 model's piston crown facing forwards. Fit the pistons on the 200 models with the 'IN' mark on the crowns facing rearwards, towards the inlet port. The gudgeon pins should be a light sliding fit. If they prove a tight fit, warm the pistons first, to expand the metal. Do not forget to lubricate the gudgeon pin, small-end eye and the piston bosses before reassembly.

- 5 Use new circlips. **NEVER** re-use old circlips. Check that each circlip has located correctly in its groove and is positioned with the circlip ends away from the cut-out in the boss in the 6 o'clock position. An incorrectly located circlip may become displaced in use, causing damage to the bore and piston.

- 6 Fit a new cylinder base gasket over the holding studs; gasket compound must not be used. Refit the two hollow locating dowels to the rear and left-hand side of the left-hand crankcase mouth. If required piston ring clamps can be fitted to each piston to aid fitting of the cylinder block over the pistons. Ring clamps will aid this operation but are by no means essential items to successfully refit the block. Position the pistons at TDC.

- 7 The cylinder block replacement operation requires two

persons; one to hold the pistons and help feed the pistons into the bores, and the other to support the cylinder block as it is lowered into position. Lubricate the cylinder bores and the pistons rings thoroughly. Position the cylinder block above the pistons and, with a length of suitably bent wire, hook the camshaft chain up through the chain tunnel between the cylinder bores. Run a screwdriver or rod through the chain across the top of the barrel to prevent the camshaft chain falling free.

8 Carefully slide the cylinder block down until the pistons enter the cylinder bores. Keeping the pistons square to the bores gently ease the block down until the rings are all engaged in their bores. Remove the ring clamps if they were used. Remove the rag padding from the crankcase mouths and push the block fully down onto the base gasket and dowels.

9 If the cam chain tensioner blade moving arm has not already been refitted, it should now be reinstalled at the rear of the cam chain tunnel. Prior to refitting the arm, fit a new O-ring to the stud which projects from the rear of the arm. Install the arm in the rear of the cam chain tunnel and fit the plain washer and lock nut. Do not tighten the nut at this stage. Reconnect the tensioner blade with the moving tensioner arm by inserting the clevis pin and fitting the R-clip. Take care not to drop any components down the cam chain tunnel. Pull the moving arm upwards so that it is extended fully and lock it in this position

by tightening the adjuster locknut. With the arm so placed, the tensioner blade has the flattest profile; this facilitates cam chain and camshaft reassembly.

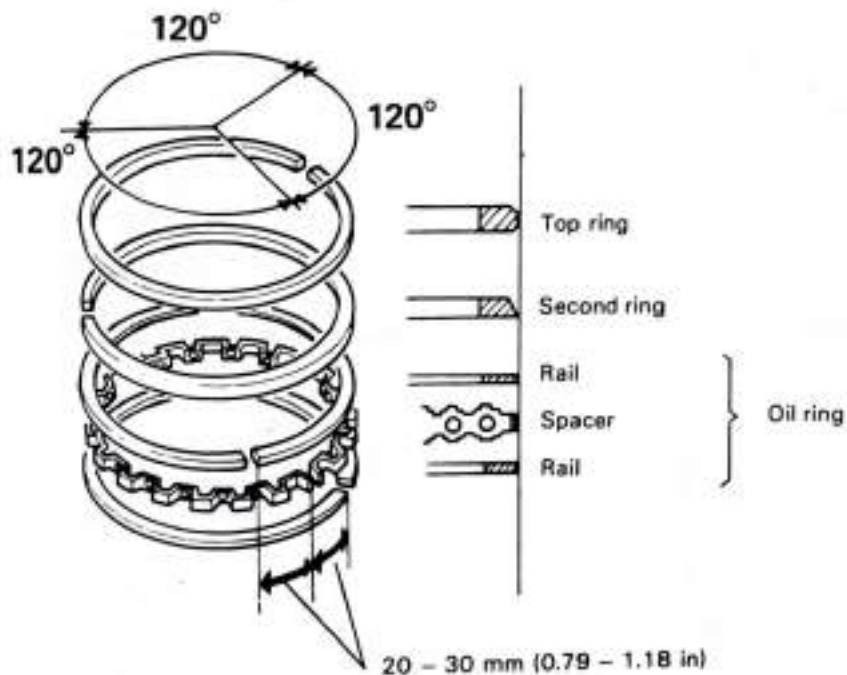
10 Install the cam chain guide blade in the front of the cylinder block, ensuring that the lower end locates correctly in the casing.

11 Fit the four hollow dowels to the outer holding down studs. Note that although the dowels are all of the same length, those fitted to the rear left-hand and forward right-hand studs sit deeper in the cylinder block top mating surface. Note also that three of the dowels are fitted with O-ring seals which must be renewed at this stage. The dowel not fitted with an O-ring is the forward, left-hand dowel. Place a new cylinder head gasket in position over the dowels.

12 Position the cylinder head above the engine and using the hooked wire, pull the camshaft drive chain up through the chain tunnel in the cylinder head and fit the head onto the dowels. Place a screwdriver through the chain again, across the cylinder head.

13 Insert and tighten the chain tensioner upper bolt; do not omit the O-ring using a new item to obviate oil leaks.

14 Insert but do not yet tighten the two long retaining bolts at the rear of the cylinder head and the single shorter retaining bolt at the front of the head. Also refit the captive nut to the forward retaining bolt.



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Fig. 1.12 Piston ring gap positions



47.4a Fit pistons with 'IN' mark (200 models) towards rear



47.4b Lubricate small end eye and gudgeon pin before reassembly



47.5 Use new circlips to avoid risk of failure

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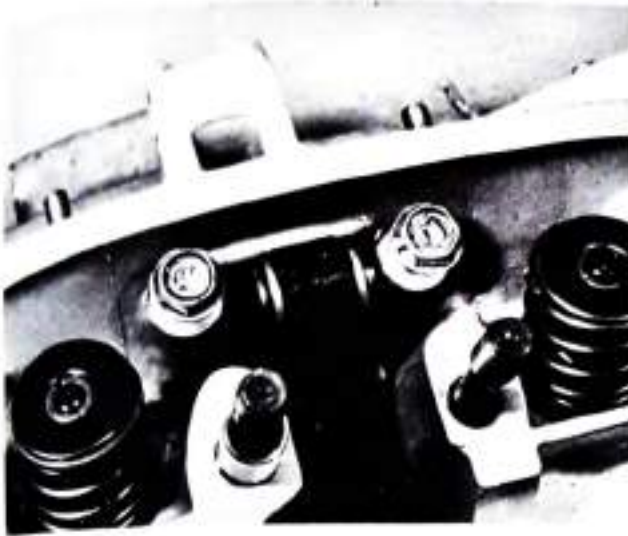
47.10 Install cam chain guide blade at front of cam chain tunnel



47.11 Fit all dowels and O-rings and a new cylinder head gasket



47.13 Use a new O-ring on the cam chain tensioner bolt



47.14a Do not yet tighten these two rear head bolts



47.14b Lightly fit the single front bolt; note the captive nut

48 Reassembling the engine/gearbox unit: refitting the camshaft, timing the valves and fitting the rocker gear

1 Before reassembling the camshaft and sprocket in the cylinder head, refit the bushes to each camshaft and journal. The right-hand bush is a simple sliding fit on the end of the camshaft. The left-hand bush is retained by a thrust washer and circlip. If, as was found during dismantling, the bushes hinder camshaft movement, so making camshaft removal and refitting more difficult, the bushes can be refitted after camshaft installation. In practice it was found that camshaft reinstallation was a far easier affair than camshaft removal, and could be carried out with the bushes in place.

2 Secure the cam chain, using a length of thin wire so that during camshaft replacement the chain will not fall into the tunnel. Position the camshaft sprocket to the left of the chain and insert the camshaft from the right-hand side. The timing marks in the form of two opposed dots scribed on the sprocket face, must face towards the left-hand side. To allow the camshaft centre boss to enter the sprocket centre, the sprocket must be positioned with the offset cutaway approximately vertical. With the sprocket positioned against the flange, rotate the camshaft until the slot machined in the left-hand end is in the 3 o'clock position, flush with the cylinder head mating surface. Rotate the sprocket so that the opposed dot marks are parallel with the cylinder head. Apply a spanner to the alternator rotor centre bolt on the 185 models, or the engine turning hexagon on the 200 models, and turn the rotor in a forward (anti-clockwise) direction until the 'T1' mark (185 models), or 'T' mark (200 models) on the ATU is in exact alignment with the raised index mark on the casing. Locate the dowels on each camshaft bush with the milled recesses in the cylinder head. Now lift the cam chain onto the sprocket and align the bolt holes in the sprocket with those in the camshaft flange.

3 Fit and finger-tighten the first camshaft sprocket bolt. If the timing is correct, the 'T1' or 'T' mark on the ATU must be in line with the raised index mark and the two dots on the sprocket must be parallel with the cylinder head mating surface. In addition, the milled slot in the camshaft left-hand end must be in the 3 o'clock position again flush with the cylinder head mating surface; refer to the accompanying illustration and photographs. If the marks do not align exactly, repeat the timing sequence and re-check. Absolute accuracy is essential.

4 Apply non-permanent locking fluid to the sprocket bolts and tighten them to a torque setting of 1.7 – 2.3 kgf m (12.3 – 16.6

lbf ft). The crankshaft must be rotated to insert the second bolt, and the bolts must be tightened evenly.

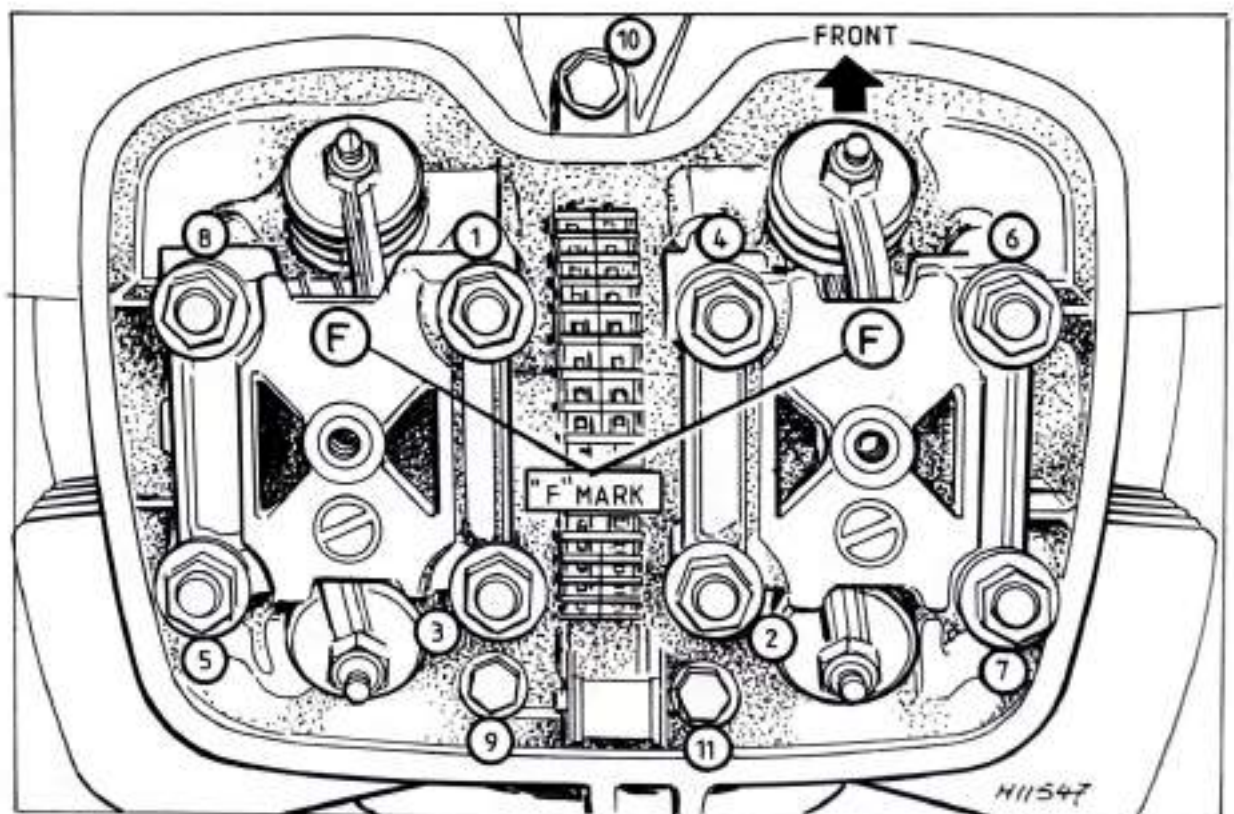
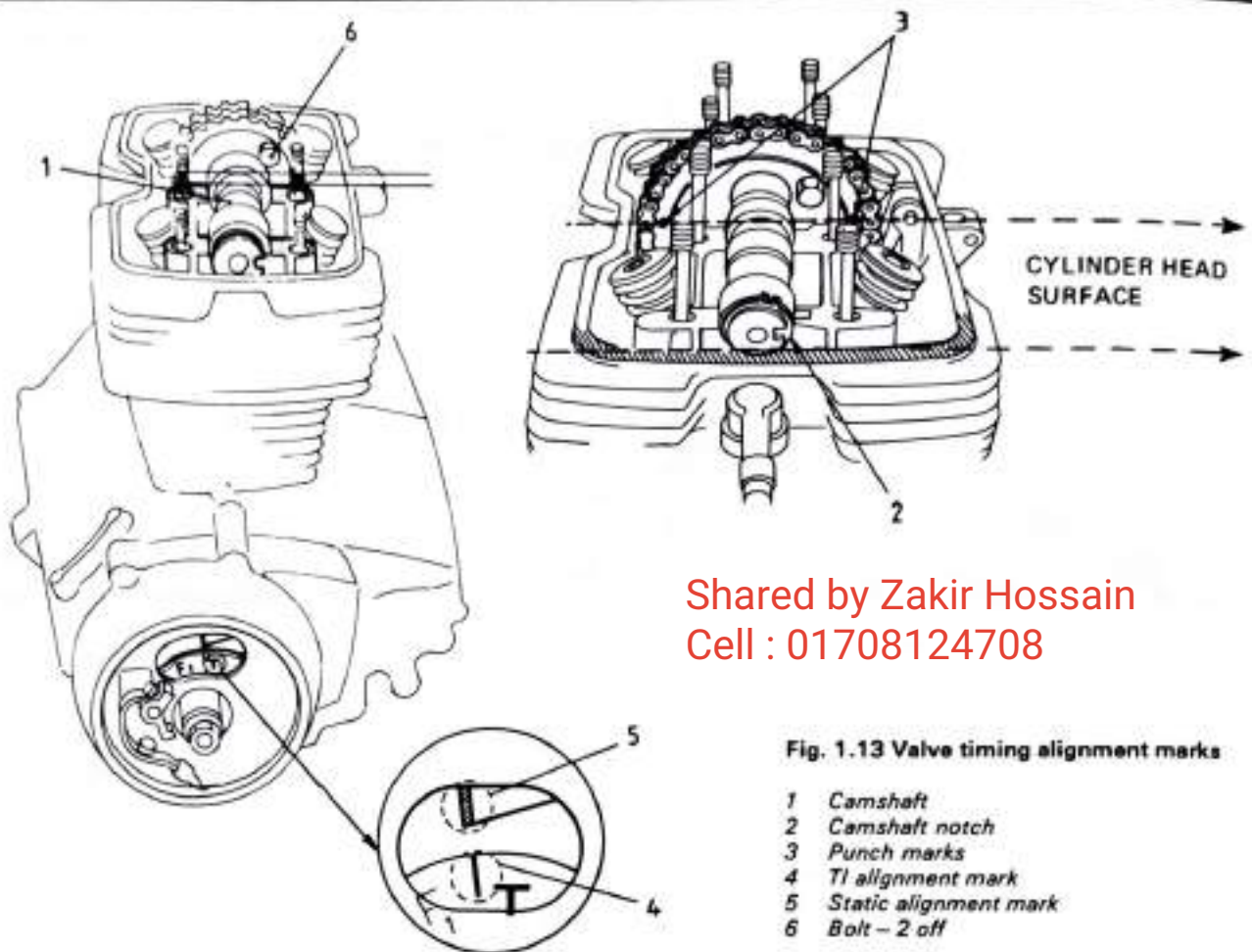
5 Lubricate thoroughly the camshaft journals, bushes and lobes and fill the troughs below each lobe with engine oil. Similarly, lubricate the valve stems and springs. Fit the four hollow dowels to the holding down studs; one dowel is fitted to each of the studs whose nuts are numbered 1, 2, 5 and 6 on the accompanying illustration. Note that two of the dowels are longer; these should be fitted to the studs corresponding to the numbers 1 and 6.

6 Before refitting the assembled rocker carriers, slacken off the rocker adjustment of all four rocker arms. Place the rocker carriers in position, noting that the 'F' mark must be fitted towards the front of the engine, and refit the eight securing nuts and their washers on the holding down studs. Tighten down the nuts, and the three previously fitted bolts, evenly, in a number of steps, in the sequence given in the accompanying illustration. Tighten the nuts to a torque setting of 1.6 – 2.0 kgf m (11.6 to 14.5 lbf ft) and the bolts to 1.0 – 1.4 kgf m (7.2 – 10.0 lbf ft).

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48.1 Left-hand bushing is retained by thrust washer and circlip





48.2b Slot in camshaft end must be positioned as shown



48.6 Refit the assembled rocker carriers, noting the 'F' mark on each

the rocker arm concerned and by turning the adjuster until the correct setting is obtained. Hold the adjuster still and tighten the locknut. Then recheck the setting before passing to the next valve.

- 4 Rotate the engine one full turn (360°) by turning the rotor in an anti-clockwise direction, until the pistons are again at TDC on the compression stroke (the T1' or T' mark again aligned with the index mark). The valves of the second cylinder should now be closed allowing checking and readjustment of the clearances. Use the same procedure as for the first cylinder to adjust the tappet settings.

5 Make sure that the cylinder head cover sealing ring is in good condition and correctly positioned and refit the cylinder head cover. Secure the cover by means of the special bolts and rubber washers. If the condition of the washers is suspect, they should be renewed. Tighten the two special bolts to a torque setting of 0.8 – 1.2 kJ m (5.8 – 7.8 lbf ft).

- 6 Check that the engine is still positioned with the 'T-1' or 'T' mark against the index mark and adjust the cam chain tension. Do this by slackening the locknut on the adjuster and then tightening it again. The chain will tension automatically. Hold the adjuster screw to prevent it rotating, whilst tightening the locknut.



49.3 Feeler gauge should be a light sliding fit



49.5 Refit head cover, ensuring sealing ring is not damaged

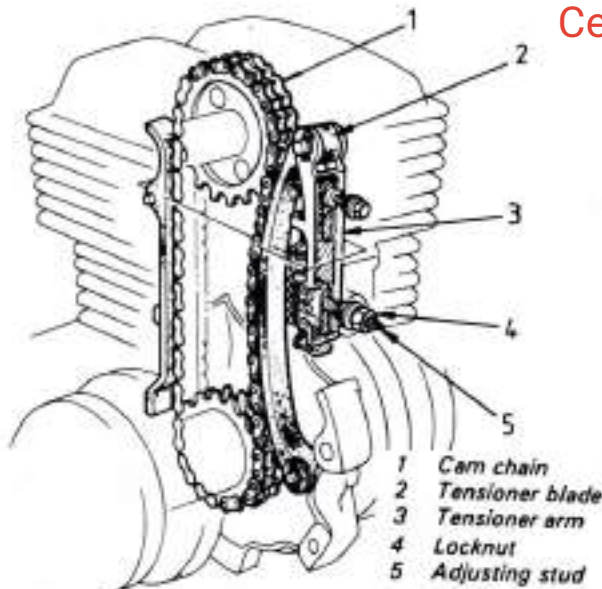
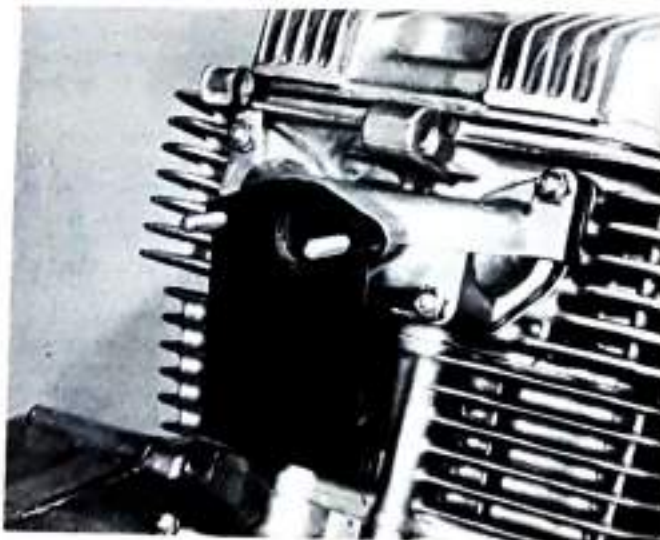


Fig. 1.15 Cam chain tensioner assembly



50.2 Position and secure manifold and heat insulator plate; note guide clamp



50.3 Refit the carburettor

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50 Reassembling the engine/gearbox unit; refitting the intake manifold and carburettor

1 It is probably easier to refit the intake manifold and the carburettor body at this stage rather than after the engine has been refitted in the frame.

2 Check that the O-ring fitted in each inlet tract on the intake manifold at the cylinder head/manifold joint is in good condition and seated correctly in the locating groove. If the O-rings are perished, compacted or damaged in any other way, they must be renewed or an air leak will result. Place the intake manifold, complete with carburettor heat insulator plate, in position on the mounting studs and secure it with the four retaining nuts. Note that the upper left-hand retaining nut also serves to retain the guide clamp for the left-hand cylinder's HT lead.

3 Check the condition of the O-ring in the carburettor mating flange; if damaged in any way it must be renewed to prevent the chance of air leaks. Offer up the carburettor, locating the flange with the rubber insulator block at the rear of the intake manifold. Secure the carburettor with the two retaining nuts, noting the right-hand nut also secures a guide clamp for the right-hand cylinder's HT lead.

51 Refitting the engine/gearbox unit in the frame

- 1 It is worth checking at this stage that nothing has been omitted during the rebuilding sequences. It is better to discover any left-over components at this stage rather than just before the rebuilt engine is started.
- 2 It is helpful to enlist the aid of a convenient bystander at this stage, because this will ease considerably the early stages of installation. If essential, the unit can be lifted by one person without incurring permanent damage to the machine or owner. The unit is best fitted from the right-hand side, taking care not to topple the machine in the process.
- 3 With the engine between the frame confines, insert the two 10 mm rear bolts from the left-hand side of the machine. With these two bolts inserted, the engine will be supported sufficiently for it to be released from your grip. Fit the nuts to the end of each bolt, ensuring that the earth lead is secured below the nut on the upper rear mounting bolt. Do not yet tighten the nuts on the rear bolts. Refit the engine front mounting bracket to the base of the front downtube and, lifting the front of the engine slightly, insert the four bolts from the left-hand side of the machine and secure them with the four nuts. As before, do not tighten the bolts. Finally, reassemble the two head steady plates and refit the three 8 mm bolts, from the left-hand side, and the retaining nuts.
- 4 When all the bolts are in position, tighten the 8 mm bolts to a torque setting of 2.0 – 2.5 kgf m (14.5 – 18.1 lbf ft) and the 10 mm bolts to a torque setting of 5.5 – 7.0 kgf m (39.8 – 50.8 lbf ft).
- 5 Reconnect the alternator output leads at the white block connector fitted on the left-hand side of the machine. On the 185 models reconnect also the block connector at the rear of the engine and the single lead for the condenser at its snap connector. Reconnect the large white block connector on the right-hand side, outboard of the regulator/rectifier unit, on the 200 models.
- 6 The intake hose from the air filter box can now be reconnected to the carburettor intake, and secured with the screw clip. Refit the throttle valve assembly which will have been left attached to the throttle cable, and screw down the mixing chamber top.
- 7 Push the crankcase breather pipe back onto its stub on the crankcase, and secure it with the spring clip.
- 8 Reconnect the choke operating cable, locating the inner cable with the operating pulley, after securing the outer cable under the retaining clamp by tightening the single crosshead screw on the clamp.
- 9 Loop the final drive chain over the engine sprocket and mesh the two ends onto the rear wheel sprocket. Replace the master link making certain that the spring link is replaced the correct way round. That is, with the closed end facing the direction of travel.
- 10 Manoeuvre the footrest assembly into position beneath the crankcase, and fit the two retaining bolts on the right-hand underside. On the left-hand side, the side stand is retained by the two long bolts which also secure the left-hand part of the footrest assembly and pass under the crankcase to retain the right-hand footrest mounting bracket. Tighten the footrest mounting bracket bolts to a torque setting of 2.0 – 2.5 kgf m (14.5 – 18.1 lbf ft). Note that the footrest mounting bracket will fit in one position only, if turned round, the bolt holes will not align. Note also that the two nuts on the long through bolts are held captive on the mounting bracket.
- 11 On the CD185T and CD200T models, the two halves of the full chain enclosure can now be refitted. Fit the upper half first and secure it with one forward and one rear bolt. Then position and secure, with two further bolts, the lower chaincase half. The small sprocket cover can now be refitted. Secure it with the three crosshead screws.
- 12 Refit the gearchange lever and its retaining pinch bolt. Similarly, refit the kickstart lever, and its single retaining pinch

bolt. Check the operating angle of both levers before tightening the pinch bolts. Tighten the gear change lever pinch bolt to a torque setting of 0.8 – 1.2 kgf m (5.8 – 9.7 lbf ft) and the kick start lever pinch bolt to a torque setting of 1.0 – 1.5 kgf m (7.2 – 10.8 lbf ft).

13 Reconnect the clutch cable to the operating arm on the right-hand crankcase cover.

14 Place a new exhaust port sealing ring in each exhaust port. Offer up the exhaust pipe/silencer assemblies one at a time. Position the split collars around the flange of the pipe as it is held against the exhaust port, and fit the retaining flange. Secure each retaining flange with two nuts. Secure the silencers to the pillion footrest mounting sub frames. The pillion footrests will be refitted at the same time, and secured by the rearward of the two silencer retaining bolts.

15 Refit the fuel tank (not forgetting the retaining bolt on the CM200T model) and reconnect the fuel pipe from the petrol tap to the carburettor. Secure the pipe with a spring clip at each end. Refit the dualseat and the side panels.

16 Reconnect the HT leads to their respective plugs. Remake the connections to the battery, taking great care to observe the correct polarity; negative (-) earth. The red lead must be reconnected to the positive (+) terminal. Make a final check around the machine to ensure that no connections have been missed. Check around the engine to make certain that all components have been fitted and are functioning correctly, and that all bolts, nuts and screws are correctly tightened. It will be noted that the circular inspection cover has still not yet been refitted to the left-hand main crankcase cover; reference to the following Section will explain why this is so.

52 Final adjustments and preparation

- 1 Check the free play in the clutch cable, and reset it, if necessary, by means of the adjuster and locknut at the crankcase end to give 10–20 mm (0.4–0.8 in) free play at the lever end. Fine adjustment can be carried out at the lever end.
- 2 Check that the engine oil drain plug has been reinstalled and fully tightened to a torque setting of 3.0 – 5.0 kgf m (21.7 – 36.2 lbf ft). Do not overtighten the drain bolt to compensate for a worn sealing washer; renew the washer to obviate risk of oil leakage. Fill the crankcase with SAE 10W/40 engine oil to the upper level on the combined filler plug/dipstick when it is rested on the crankcase hole into which it screws. The crankcase will accept approximately 1.5 litres (2.6 imp pint/1.6 US quart) of oil. Note that the machine must be upright when checking the oil level on the dipstick. The CM models must, therefore, be held in an upright position as the oil level is checked; if this precaution is not observed, an incorrect reading on the dipstick will occur, resulting in either a lack of or an excess of oil in the crankcase.
- 3 Check the free play in the final drive chain. If free play, when checked at a mid-way point on the lower run (CM models) or through the chaincase inspection aperture (CD models), is not within the range 15 – 25 mm (0.6 – 1.0 in), adjustment is required. Make the necessary adjustment using the procedure described in Section 14 of Chapter 5.
- 4 Check, and if necessary adjust, the contact breaker gap and then the ignition timing. If a scribe mark was made on the contact breaker baseplate prior to dismantling then this should be aligned to provide a datum setting for the timing check. The correct procedure for checking the contact breaker gap and ignition timing is given in Chapter 3, Sections 4 and 6 respectively. If a stroboscopic lamp is to be used, then checking can only take place after the engine has been started.

53 Starting and running the rebuilt engine

- 1 Open the petrol tap to allow fuel to flow to the carburettors, and close the choke. Kick the engine over rapidly with the ignition switch turned off, to help prime the oilways. Start the

engine using either the kickstarter or the electric starter. Raise the choke as soon as the engine will run evenly and keep it running at a low speed for a few minutes to allow oil pressure to build up and the oil to circulate.

2 Bear in mind that the engine parts should have been liberally coated with oil during assembly, so the engine will tend to smoke heavily through the exhausts initially. The excess of oil should gradually burn away as the engine settles down. Do not despair if the engine will not fire up at first, as it is quite likely that the excess oil will foul the sparking plugs necessitating their removal and cleaning. When the engine does start, listen carefully for any unusual noises, and if present, establish, and if necessary rectify, the cause. Check around the engine for any signs of leaking gaskets, or oil leaks.

3 Make sure each gear engages correctly and that all controls function effectively, particularly the brakes. This is an essential last check before taking the machine on the road.

4 After the initial few minutes running, stop the engine and recheck the oil level. If necessary, add oil to bring the level back up to the upper level on the dipstick.

54 Taking the rebuilt machine on the road

- 1 Any rebuilt machine will need time to settle down, even if



51.3a Refit upper rear mounting bolt (note earth lead below nut) and then ...



51.3c Refit head steady plates and three bolts

parts have been replaced in their original order. For this reason it is highly advisable to treat the machine gently for the first few miles to ensure oil has circulated throughout the lubrication system and that any new parts fitted have begun to bed down.

2 Even greater care is necessary if the engine has been rebored or if a new crankshaft has been fitted. In the case of a rebore, the engine will have to be run-in again, as if the machine were new. This means greater use of the gearbox and a restraining hand on the throttle until at least 500 miles have been covered. There is no point in keeping to any set speed limit; the main requirement is to keep a light loading on the engine and to gradually work up performance until the 500 mile mark is reached. These recommendations can be lessened to an extent when only a new crankshaft is fitted. Experience is the best guide since it is easy to tell when an engine is running freely.

3 If at any time a lubrication failure is suspected, stop the engine immediately, and investigate the cause. If an engine is run without oil, even for a short period, irreparable engine damage is inevitable.

4 When the engine has cooled down completely after the initial run, recheck the various settings, especially the valve clearances. During the run most of the engine components will have settled into their normal working locations.

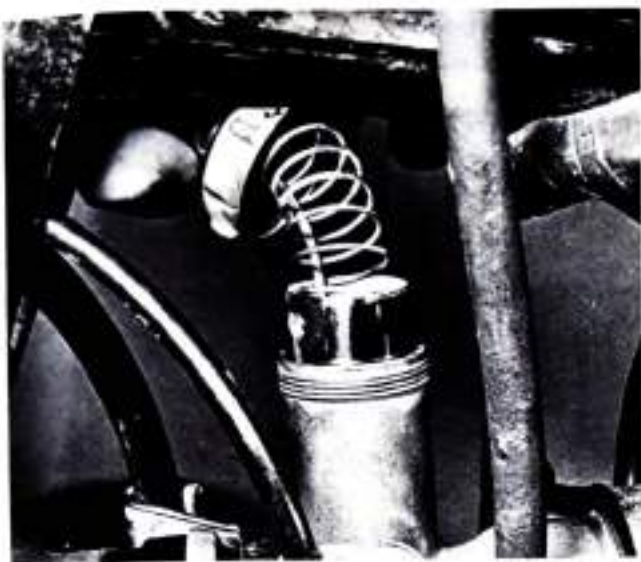


51.3b ... lower rear mounting bolt



51.6a Tighten screw clip to secure intake hose

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51.6b Carefully refit throttle valve assembly and screw down top



51.7 Reconnect the breather pipe to the crankcase stub



51.9 Refit spring link correctly, with the closed end facing normal direction of travel



51.11 Refit sprocket cover



51.12a Fit gearchange lever and retaining pinch bolt and ...

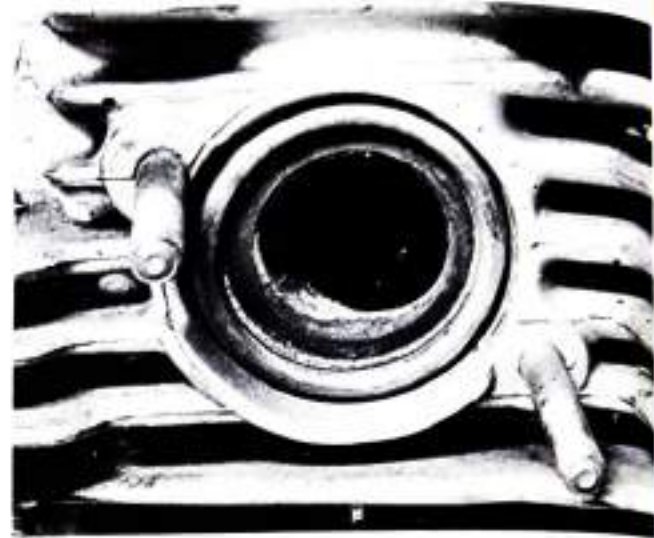


51.12b ... kickstart lever and pinch bolt

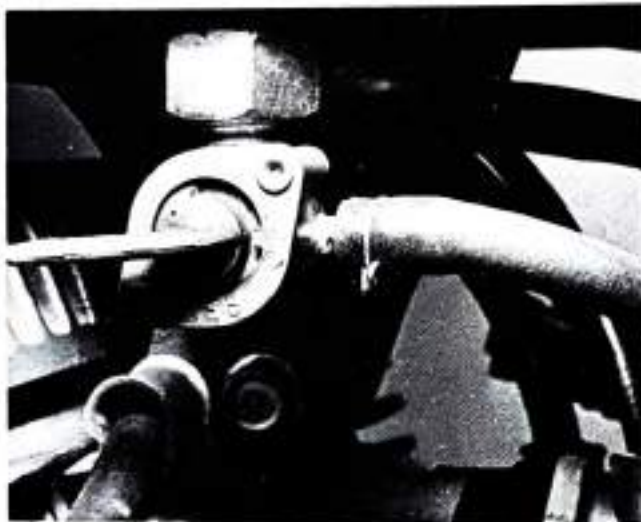
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51.13 Reconnect and adjust, if necessary, the clutch cable



51.14 Fit a new sealing ring in each exhaust port



51.15 Reconnect the fuel pipe and secure with spring clip



52.2 Refill the crankcase with the correct quantity of SAE 10W/40 oil

55 Fault diagnosis: engine

Symptom	Cause	Remedy
Engine starting difficulty or does not start	Lack of compression	
	(1) Valve stuck open	Adjust tappet clearance.
	(2) Worn valve guides	Renew.
	(3) Valve timing incorrect	Check and adjust.
	(4) Worn piston rings	Renew.
	(5) Worn cylinder bore	Rebore.
	No spark at plug	
	(1) Fouled or wet sparking plugs	Clean.
	(2) Fouled contact breaker points	Clean.
	(3) Incorrect points gap	Adjust.
	(4) Incorrect ignition timing	Check and adjust.
	(5) Open or short circuit in ignition	Check and renew fuse.
	No fuel flowing to carburettor	
	(1) Blocked fuel tank cap vent hole	Clean.
	(2) Blocked fuel tap	Clean.
	(3) Faulty carburettor float valve	Renew.
	(4) Blocked fuel pipe	Clean.

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Engine stalls whilst running	Fouled sparking plug or contact breaker points	Clean.
	Ignition timing incorrect	Adjust.
	Blocked fuel line or carburettor jets	Clean.
Noisy engine	Tappet noise:	
	Excessive tappet clearance	Check and reset.
	Weakened or broken valve spring	Renew springs
	Knocking noise from piston:	
	Worn piston and cylinder noise	Rebore cylinder block and fit oversize pistons.
	Carbon in combustion chambers	Decoke engine.
	Worn gudgeon pin or connecting rod small end	Renew.
	Cam chain noise:	
Engine noise	Stretched cam chain	Renew chain.
	Worn cam sprocket or timing sprocket	Renew sprockets
	Excessive run-out of crankshaft	Renew.
	Worn crankshaft bearings	Renew.
	Worn connecting rods	Renew flywheel assembly.
	Worn transmission splines	Renew.
	Worn or binding transmission gear teeth	Renew gear pinions.
Smoking exhaust	Too much engine oil	Check oil level and adjust as necessary.
	Worn cylinder bores and piston rings	Rebore and fit oversize pistons and rings.
	Worn valve guide	Renew
	Damaged cylinder bores	Rebore cylinder block and renew pistons
Insufficient power	Valve stuck open or incorrect tappet adjustment	Re-adjust
	Weak valve springs	Renew.
	Valve timing incorrect	Check and reset.
	Worn cylinder bores and piston rings	Rebore and fit oversize pistons and rings.
	Poor valve seating	Grind in valves.
	Ignition timing incorrect	Check and adjust
	Defective plug cap(s)	Fit replacement(s).
	Dirty contact breaker points	Clean or renew.
Overheating	Accumulation of carbon on cylinder head	Decoke engine.
	Insufficient oil	Refill to specified level.
	Faulty oil pump and/or blocked oil passage	Strip and clean.
	Ignition timing too advanced	Re-adjust

56 Fault diagnosis: clutch

Symptom	Cause	Remedy
Engine speed increases but machine does not respond	Clutch slip	Check clutch adjustment for free play at handlebar lever. Check thickness of inserted plates.
Difficulty in engaging gears. Gear changes jerky and machine creeps forward when clutch is withdrawn. Difficulty in selecting neutral	Clutch drag	Check clutch adjustment for too much free play. Check clutch drums for indentations in slots and clutch plates for burrs on tongues. Dress with file if damage not too great.
Clutch operation stiff	Damaged, trapped or frayed control cable	Check cable and renew if necessary. Make sure cable is lubricated and has no sharp bends.

57 Fault diagnosis: gearbox

Symptom	Cause	Remedy
Difficulty in engaging gears	Selector forks bent Gear clusters not assembled correctly	Renew. Check gear cluster arrangement and position of thrust washers.
Machine jumps out of gear	Worn dogs on ends of gear pinions Stopper arms not seating correctly	Renew worn pinions. Remove right-hand crankcase, cover and check stopper arm action.
Gearchange lever does not return to original position	Broken return spring	Renew spring.
Kickstart does not return when engine is turned over or started	Broken or poorly tensioned return spring	Renew spring or re-tension.
Kickstart slips	Ratchet assembly worn	Renew all worn parts

Chapter 2 Fuel system and lubrication

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Specifications

		CD185/CD200/CM200 (UK)		CM185/CM200 (US)		
Fuel tank						
Overall capacity		10.5 lit (2.7 US/2.3 imp gal)		10.5 lit (2.7 US/2.3 imp gal)		
Reserve capacity		3.5 lit (1.0 US/0.8 imp gal)		3.0 lit (0.8 US/0.6 imp gal)		
Carburettor		CD200	CD185	CM200 (UK)	CM185	CM200 (US)
Make	—	Keihin	Keihin	Keihin	Keihin	Keihin
Type	—	PD9913	PD33A	PD98A	PB23A	PB23A
Main jet	—	105	110	105	105	105
Slow jet	—	42	150	—	—	—
		All models				
Jet needle clip position		2nd groove from top				
Pilot screw setting		2 turns out				
Float height:						
CM200 models		12.5 mm (0.49 in)				
All others		14.5 mm (0.57 in)				
Idle speed		1200 ± 100 rpm				
Twist grip free play		2-6 mm (0.08 – 0.24 in)				
Air cleaner						
Element type		Oiled, polyurethane foam				
Element oil type		SAE 80 or 90, gear oil				
Engine lubrication						
System type		Wet sump, force fed				
Oil capacity		1.5 litre (2.8 imp pint/1.6 US quart)				
Oil filter		Gauze screen fitted to oil pump				
Oil pump						
Type		Trochoid				
Inner rotor/outer rotor clearance		0.15 mm (0.006 in)				
Service limit		0.2 mm (0.008 in)				
Outer rotor/housing clearance		0.15 – 0.18 mm (0.006 – 0.007 in)				
Service limit		0.25 mm (0.010 in)				
End clearance		0.01 – 0.07 mm (0.0004 – 0.003 in)				
Service limit		0.12 mm (0.005 in)				

1 General description

The fuel system comprises a petrol tank from which petrol is fed by gravity to the float chamber of the single Keihin carburettor. The fuel flow is controlled by a petrol tap with a built-in filter, located beneath the tank on the left-hand side. The tap has three positions controlled by an integral lever: 'Off', 'On' and 'Reserve', the latter providing a reserve supply of fuel when the main supply is exhausted.

The use of a single carburettor to serve two cylinders by means of an intake manifold, is a somewhat unusual arrangement by present day, multi-carburettor standards. It has very practical advantages, however, in simplicity and economy. The carburettor itself is of conventional concentric design, the float chamber being integral with the lower part of the carburettor body. For cold starting, the carburettor has a choke, manually operated by a knob below the instrument panel and connected to the carburettor by cable. The choke mechanism enables the mixture to be richened temporarily for easier cold starts. The machine should be allowed to run on 'Choke' for the least amount of time.

Air entering the carburettor passes through a moulded plastic air cleaner casing, which contains a detachable oil-impregnated foam filter element. This effectively removes any airborne dust and dirt, which would otherwise enter the engine and cause premature wear. The air cleaner casing is mounted on the left-hand side of the machine and is attached directly to the carburettor intake by a short rubber hose.

The exhaust system takes the form of twin integral exhaust pipes and silencers.

Lubrication is by the wet sump, pressure principle, in which oil is delivered under pressure from the sump, through a mechanical pump, to the working parts of the engine. The pump is of the trochoid type and is driven from the right-hand end of the crankshaft via a small drive gear pinion. Incorporated in the pump is the mesh-type filter screen. The engine oil is also shared by the primary drive, clutch and gearbox components.

2 Petrol tank: removal and replacement

1 Before the petrol tank is removed, the dualseat must be detached by unscrewing the two bolts which pass through seat base lugs and into the frame. Note that on the CD200T model, the bolt on the left-hand side also partly retains the lifting handle used to aid raising the machine onto its centre stand, and on the CM185T model, both bolts secure the rearward section of the passenger grab rail.

2 Also remove, from the CM models, the two frame side panels. Their forward end locating spigots engage with rubber grommets inserted into holes at the rear of each side of the petrol tank.

3 The fuel tank is retained at the forward end by two guide channels, one of which is placed each side of the underside of the tank, which locate with two rubber buffers on the frame top tube. The rear of the tank rests on a small rubber block placed across the frame top tube. On the CD185T/CD200T, and the CM185T models, this 'saddle' also acts as a retention strap retaining the rear of the tank. The strap secures the tank by means of the lug projecting from the rear. On the CM200T model, the rear of the tank is retained by a single bolt passing through the lug.

4 With the dualseat removed, remove the single rear bolt (CM200T only), or release the rubber retaining block to free the rear of the tank. Turn the petrol tap lever to the 'Off' position and ease off the rubber fuel feed pipe. Release the pressure on the fuel line by pinching together the ears of the securing spring clip. There is no need to drain the tank. Raise the rear end of the tank and pull it backwards, moving it gently from side to side in order to force the guide channels from the rubber buffers at the nose.

5 Store the tank in an upright position in a safe place whilst

it is removed from the machine, well away from any naked lights or flames. It will otherwise present a considerable fire or explosion hazard. Check that the tap is not leaking and that it cannot be accidentally knocked into the 'On' position.

6 To replace the tank, reverse the procedure adopted for its removal. Move it from side to side before it is fully home, so that the rubber buffers engage with the guide channels correctly. If the difficulty is encountered in engaging the front of the tank with the rubber buffers, apply a small amount of petrol to the buffers to ease location.

3 Petrol tap: removal, examination and replacement

1 Before the petrol tap can be removed, it is first necessary to drain the tank. This is easily accomplished by removing the feed pipe from the carburettor float chamber and allowing the contents of the tank to drain into a clean receptacle, with the tap turned to the 'Reserve' position. Alternatively, if the tank is only partially full, it can be removed and placed on one side, so that the fuel level is below the tap outlet; take care not to damage the tank paintwork if the latter method is used.

2 The tap unit is retained by a gland nut to the threaded stub on the underside of the tank. To remove the tap, unscrew the gland nut fully and withdraw the tap unit. The filter column may be slid off the stand pipe, for cleaning. This should be done using a soft brush and clean petrol. If the filter screen has become perforated, it should be renewed.

3 Check the condition of the gland nut to tank, sealing gasket. This will probably remain in position on the tank bottom as the tap unit is removed. If leakage had been apparent, prior to dismantling, renew the gasket.

4 If the tap lever leaks, it will be necessary to renew it as a complete unit. It is not possible to dismantle the tap for repair.

5 When reassembling the tap, reverse the procedure for dismantling.

6 Check that the feed pipe from the tap to the carburettor is in good condition and that the push-on joints are a good fit, irrespective of the retaining wire clips. If particles of rubber are found in the filter, replace the pipe, since this is an indication that the internal bore is breaking up.

4 Petrol feed pipe: examination

1 A thin-walled synthetic rubber feed pipe is used to convey fuel from the petrol tap to the float chamber union. It is only necessary to replace the pipe if it becomes hard and cracks or starts to split. Check that the wire retaining clips, fitted at each end of the pipe, have not worn through the pipe surface. It is unlikely that the retaining clips themselves should need renewing due to fatigue, because the main seal between the pipe and the unions is effected by an interference fit.

2 Do not replace a broken pipe with one of natural rubber, even temporarily. Petrol causes natural rubber to swell very rapidly and disintegrate, with the result that minute particles of rubber would easily pass into the carburettors and cause blockages of internal passageways. Plastic pipe of the correct bore size can be used as a temporary substitute but it should be replaced with the correct type of tubing as soon as possible since it will not have the same degree of flexibility.

5 Carburettor: removal

1 To improve access to the carburettor, remove the dualseat and petrol tank, as described in Section 2 of this Chapter, and detach the frame side panels. Ensure the petrol tap lever is in the 'Off' position before removing the tank.

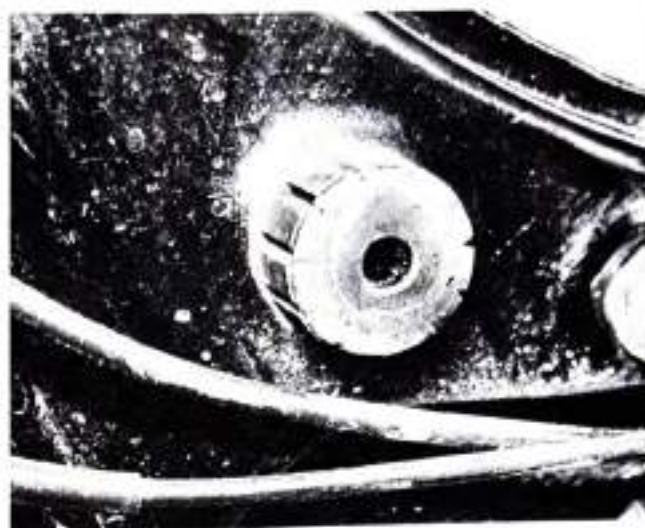
2 Before the carburettor can be removed from the intake manifold stub, it is necessary to detach the petrol feed pipe, the air cleaner hose, and the carburettor top and throttle valve assembly. The choke cable must also be disconnected.

3 Prise off the rubber air hose from the carburettor intake, having slackened fully the screw clip retaining band. It should not be necessary to remove the air cleaner assembly unless a complete overhaul is contemplated.

4 Unscrew the carburettor top, and withdraw the throttle valve assembly. If the valve or needle require attention, they can be detached by compressing the return spring against the underside of the top, and disengaging the cable end from its recess in the valve. The needle is held by a spring clip, which is itself positioned by a second clip inside the valve. Unless this assembly is obviously worn, it is normally advisable to leave it undisturbed. The assembly should be positioned out of harm's way by tying or taping it to the frame top tube.

5 Slacken the crosshead screw on the choke cable retaining guide clip; this will release the cable from tension, allowing its lower end to be freed from the operating pulley arm.

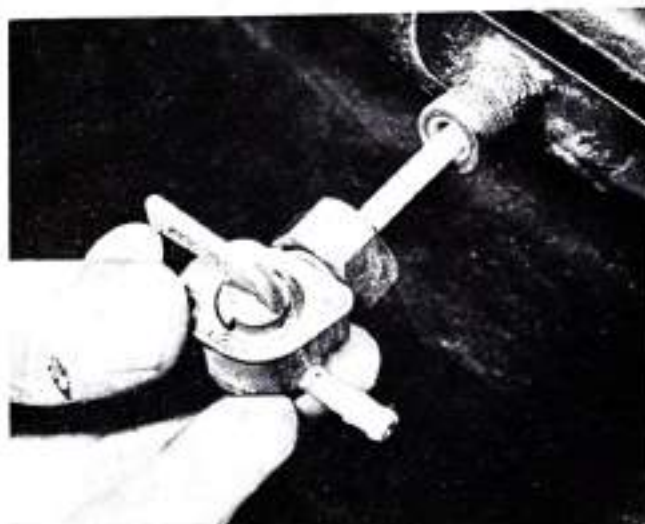
6 Slacken and remove the two manifold flange mounting nuts, and pull the carburettor body rearwards, clear of the mounting studs. Note that the right-hand nut also retains a guide for the HT cable. Do not detach the separate heat shield unless necessary. The carburettor components can now be dismantled for examination if required.



2.3a Rubber buffers secure forward end of tank



2.3b Retention strap/mounting block secures rear of tank (except CM200T)



3.2a Detach fuel pipe, slacken gland nut and withdraw tap complete



3.2b Check filter column for dirt; renew if damaged

6 Carburettor: dismantling, examination and reassembly

1 Before stripping the carburettor it is advisable to lay clean paper on the workbench as there are small components in the carburettor that may be lost.

2 Detach the float chamber by inverting the carburettor and removing the three screws and spring washers that retain it to the main body. There is a sealing gasket around the edge of the float chamber, which will remain either with the float chamber or the main body of the carburettor.

3 With a pair of pointed-nose pliers, pull out the pivot pin from the twin float assembly. This will free the floats and the float needle. Lift away the floats and the float needle together, being careful that the small float needle does not drop free and become lost. Pull the float assembly and needle safely aside for examination at a later stage.

4 The main jet is located in the centre of the mixing chamber, between the twin floats. It is threaded into the base of the needle jet holder (in the central column). When unscrewing any jet, a close fitting screwdriver must be used to prevent damage to the slot in the soft jet material. Unscrew and remove the

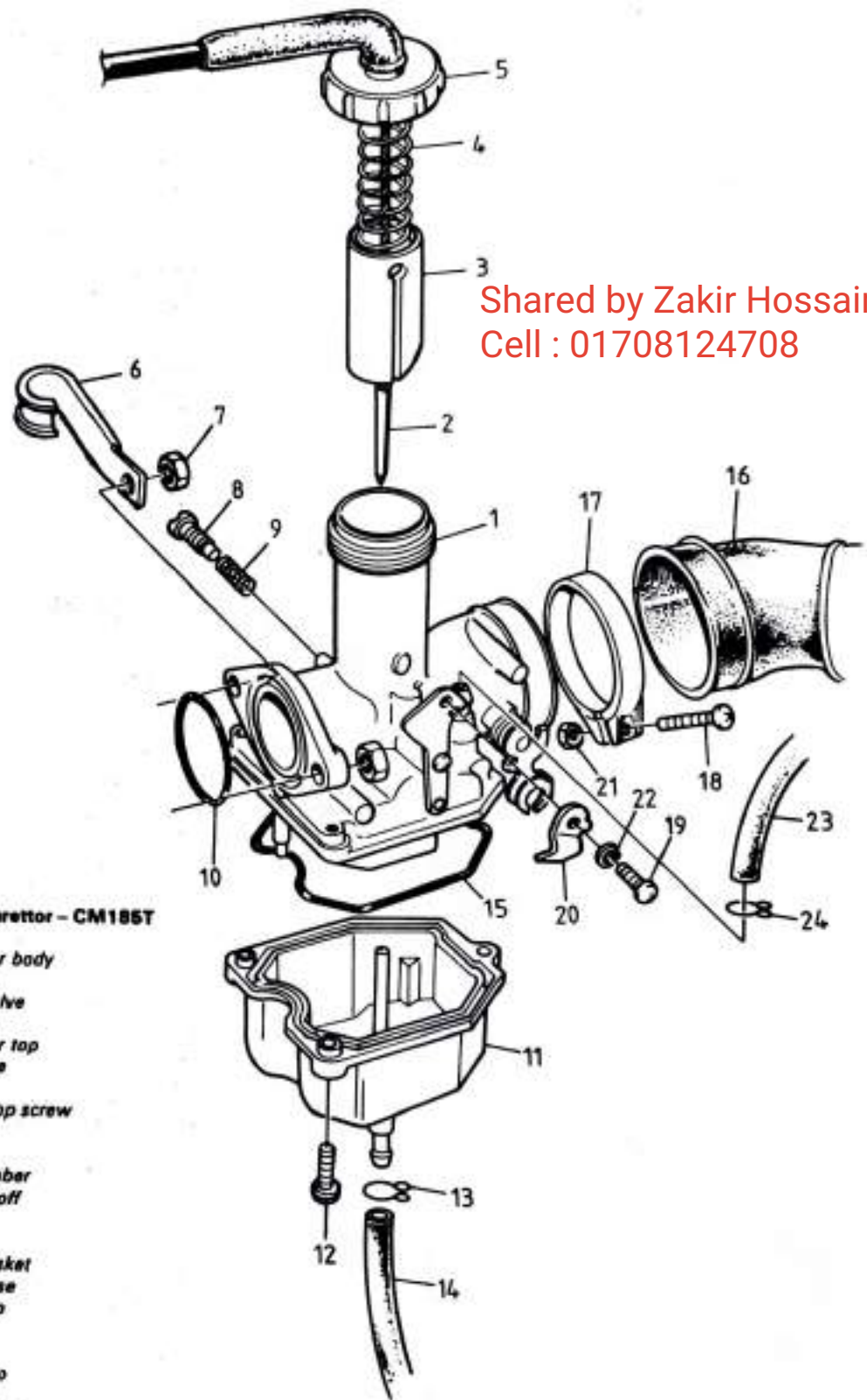


Fig. 2.1 Carburettor - CM185T

- 1 Carburettor body
- 2 Jet needle
- 3 Throttle valve
- 4 Spring
- 5 Carburettor top
- 6 Cable guide
- 7 Nut
- 8 Throttle stop screw
- 9 Spring
- 10 O-ring
- 11 Float chamber
- 12 Screw - 3 off
- 13 Clip
- 14 Drain pipe
- 15 Sealing gasket
- 16 Air inlet hose
- 17 Hose clamp
- 18 Screw
- 19 Screw
- 20 Cable clamp
- 21 Nut
- 22 Spring washer
- 23 Fuel pipe
- 24 Clip

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main jet. Using a small close fitting spanner, unscrew and remove the needle jet holder from below (when inverted) the main jet.

5 Return the carburettor to its upright position and allow the needle jet to fall out.

6 Note that the slow jet, situated beside and forward of the central column, cannot be removed for cleaning; it is a press fit in the carburettor body.

7 The pilot air screw, situated at the rear of the mixing chamber, may be unscrewed and removed together with the spring, washer and O-ring.

8 Although the carburettor top has been unscrewed, further dismantling should not prove necessary unless wear has occurred. After an extended period of service, the throttle slide will wear and may produce a clicking sound within the carburettor body. Wear will be evident from inspection, usually at the base of the slide and in the locating groove. A worn slide should be replaced as soon as possible because it will give rise to air leaks which will upset carburation.

9 The throttle needle is suspended from the slide, where it is retained by a spring clip. The needle is normally suspended from the 2nd groove down, but other grooves are provided as a means of adjustment so that the mixture strength can be either increased or decreased by raising or lowering the needle. It is unlikely the position of the needle will need to be changed in normal usage.

10 The cable operated cold starting choke should not require attention as it is of the simple spring loaded plunger type. Wear is unlikely to occur unless the machine has covered a very high mileage, in which case the whole carburettor will then require renewal.

11 Check that the floats are in good order and not punctured. If either float is punctured, it will produce the wrong petrol level in the float chamber, leading to flooding and an over-rich mixture. Because the floats are moulded from a plastic material, it is not possible to effect a permanent repair. In consequence, a new replacement must be fitted if damage is found. Flooding may be caused by an incorrect float height, rather than a damaged float. If this is the case refer to Section 7 for the adjustment procedure.

12 The float needle and needle seat will wear after lengthy service and should be closely examined with a magnifying glass. Wear usually take the form of a ridge or groove, which will cause the float needle to seat imperfectly. If damage to the seat has occurred the carburettor body must be renewed because the seat is not removable.

13 Before the carburettor is reassembled, by reversing the dismantling procedure, it should be cleaned out thoroughly using compressed air. Avoid using a piece of rag since there is always risk of particles of lint obstructing the internal passages or the jet orifices.

14 Never use a piece of wire or any pointed metal object to clear a blocked jet. It is only too easy to enlarge the jet under these circumstances and increase the rate of petrol consumption. If the compressed air is not available, a blast of air from a tyre pump will usually suffice. As a last resort, a fine nylon bristle may be used.

15 Do not use excessive force when reassembling a carburettor because it is easy to shear a jet or some of the smaller screws. Furthermore, the carburettor is cast in a zinc-based alloy which itself does not have a high tensile strength. If any of the castings are damaged during reassembly, they will almost certainly have to be renewed.

16 Take care when reinstalling the throttle valve that the cut-out on the slide is on the right-hand side, so that it will align correctly with the throttle stop screw in the carburettor body. Also take care when replacing the carburettor top because the needle is easily bent if it does not locate with the needle jet. Before reattaching the air cleaner hose, check that the throttle slide rises and falls when the throttle is operated.

17 Install the pilot air screw, positioning it 2 turns out from the lightly seated position. The carburettor should then be adjusted following the procedure in Section 7, paragraphs 2 - 4.

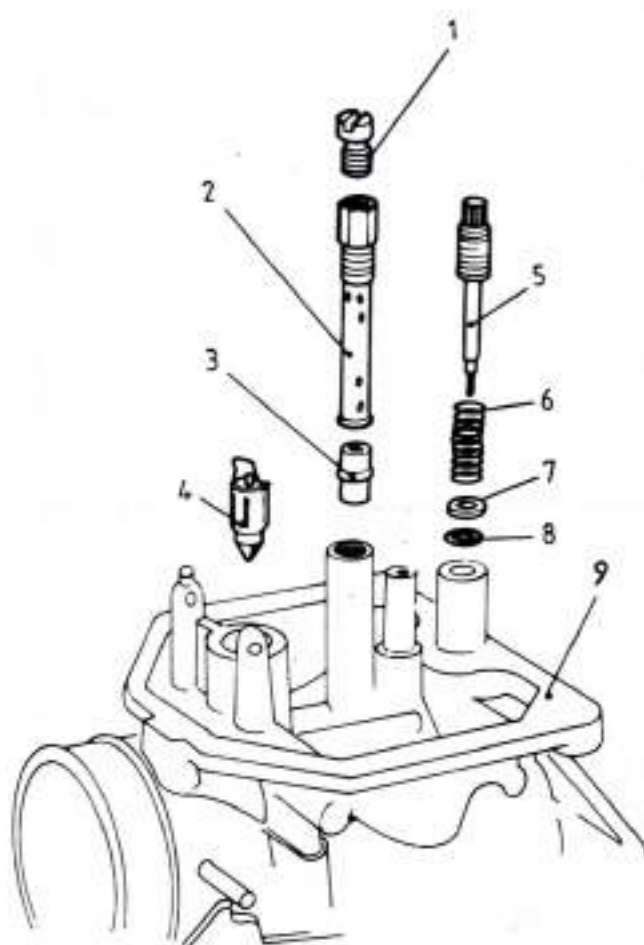


Fig. 2.2 Carburettor jet arrangement

- | | |
|---------------------|--------------------|
| 1 Main jet | 6 Spring |
| 2 Needle jet holder | 7 Washer |
| 3 Needle jet | 8 O-ring |
| 4 Float needle | 9 Carburettor body |
| 5 Pilot air screw | |

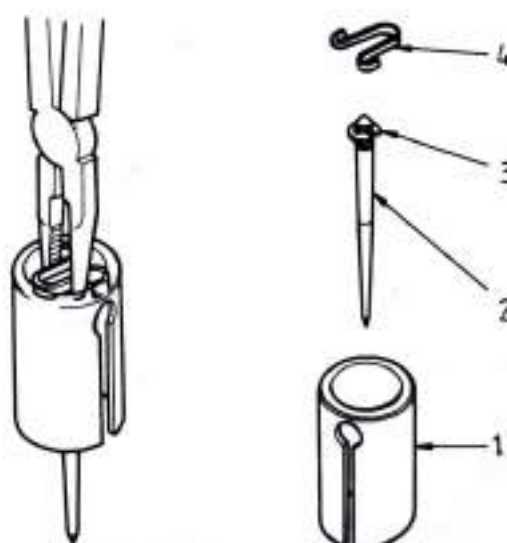


Fig. 2.3 Carburettor throttle valve components

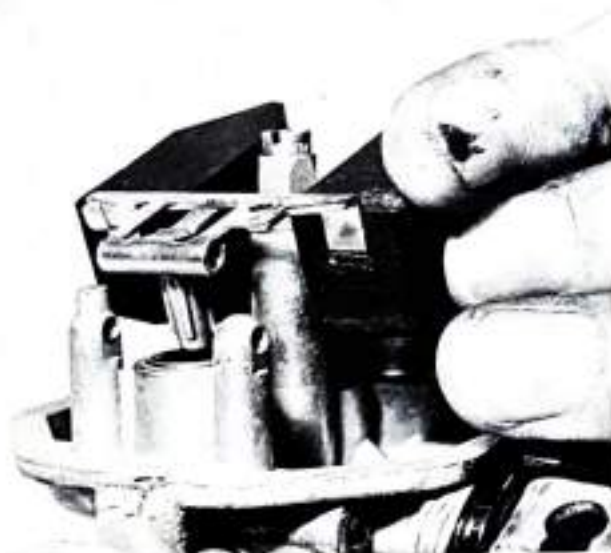
- | | |
|------------------|-------------------------|
| 1 Throttle valve | 4 Retaining spring clip |
| 2 Jet needle | |
| 3 Needle clip | |



6.2 Lift off float chamber after removing three retaining screws



6.3a Displace the float pivot pin ...

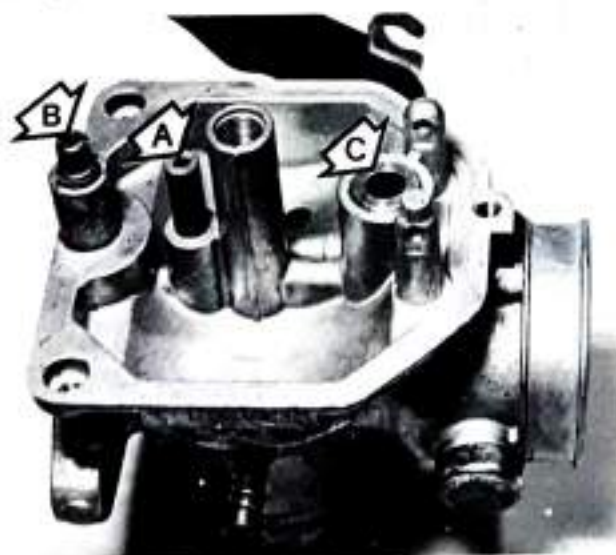


6.3b ... to free the float assembly; note float needle attached to assembly



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6.4 Main jet and needle jet holder can be unscrewed and separated for cleaning etc



6.6 Slow jet (A) cannot be removed; unscrew and remove pilot screw (B); examine float needle seat (C)



6.16a Reinstall the needle jet and ...



6.16b ... ensure it is positioned as shown, then ...



6.16c ... refit needle jet holder



6.16d Check the condition of the float chamber O-ring and ...



6.16e ... the intake flange O-ring, prior to reassembly

7 Carburettor: adjustment

1 If flooding of the carburettor or excessive weakness (ie fuel starvation) has been experienced, it is wise to start operations by checking the float level. This will involve detaching the carburettor, if not already removed, inverting it and removing the float chamber bowl. If the float level is correct, the height of the top of the float assembly above the flange of the mixing chamber, when the needle is lightly seated, should be as given in the Specifications. To adjust the setting, bend the tang on the float arm carefully in the direction required. See the accompanying diagram.

2 Replace the carburettor and, with the engine at normal operating temperature turn the pilot jet screw, located in the mixing chamber, inwards until the engine misfires or decreases in speed. Note the position of the screw, then turn it outwards until similar symptoms are observed. The screw should be set exactly between these two positions. This position is normally about 2 turns out from a lightly seated position. Note that in normal usage, the pilot air screw should not require adjustment, it being set correctly on factory assembly. If the carburettor has

been completely dismantled, as in Section 6, the pilot air screw must be reset upon reassembly. It must also be noted here that, due to stringent Federal emissions regulations, as laid down by the Environmental Protection Agency (EPA) in the USA, all models sold on the US market, which have been manufactured after December 31st 1977, have their pilot air screws pre-set. Therefore, unless the carburettor is completely overhauled, it is recommended that the pilot air screw should not be disturbed. Failure to observe this recommendation could, theoretically, result in the operator being prosecuted and fined for a breach of the emissions regulations.

3 If, with the pilot air screw adjusted correctly, the idling speed is now too low or too high, adjustment must be made using the idling speed (throttle stop) screw. Adjust the screw until the engine will tick over at the correct engine speed; 1200 ± 100 rpm. The engine speed will have to be approximated, there being no tachometer (rev-counter) fitted to these machines.

4 Note that these adjustments should always be made with the engine at normal operating temperature and with the air cleaner connected, otherwise a false setting will be obtained.



7.2 Pilot jet screw must be set correctly after dismantling



7.3 Use throttle stop screw to obtain correct idling speed

Chapter 3 will show how the condition of the sparking plugs can be interpreted with some experience as a reliable guide to carburettor mixture strength.

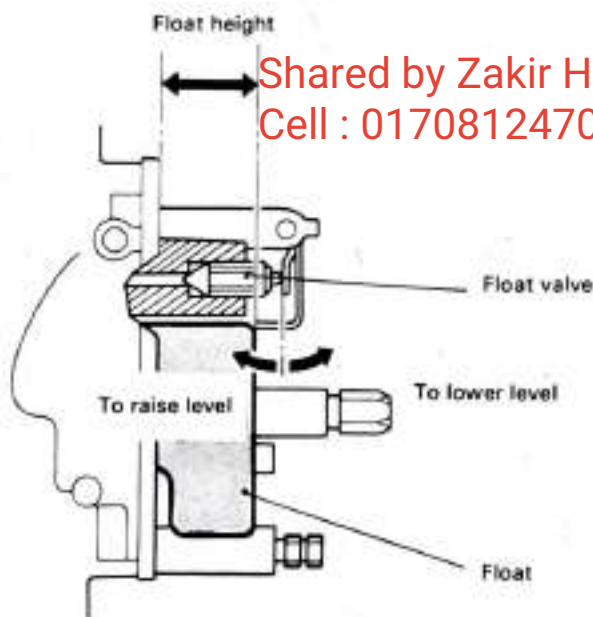


Fig. 2.4 Float level adjustment

8 Carburettor settings

1 Some of the carburettor settings, such as the sizes of the needle jets, main jets and needle positions, etc., are predetermined by the manufacturer. Under normal circumstances it is unlikely that these settings will require modification, even though there is provision made. If a change appears necessary, it can often be attributed to a developing engine fault.

2 As an approximate guide, the pilot jet setting controls engine speed up to $\frac{1}{2}$ throttle. The throttle slide cutaway controls engine speed from $\frac{1}{2}$ to $\frac{3}{4}$ throttle. The size of the main jet is responsible for engine speed at the final $\frac{1}{4}$ to full throttle. It should be added, however, that these are only guidelines. There is no clearly defined demarcation line due to a certain amount of overlap that occurs between the carburettor components involved.

3 Always err slightly on the side of a rich mixture, since a weak mixture will cause the engine to overheat. Reference to

9 Air cleaner: dismantling, servicing and reassembly

1 Pull off the left-hand frame side panel to gain access to the air cleaner case. Remove the two securing nuts and release the lid of the air cleaner case. Note that the lower nut also retains a small plate which, at its lower end, forms an anchor point for the tool kit retaining strap. Pull out the leaf spring at the rear of the filter element and remove the element.

2 The foam element sleeve can be separated from its central support core, and slid off. Note that on the CD185T/CD200T models the element consists of two separate sleeves fitted together on the support core. With the element separated from its core, cleaning and re-oiling of the element can take place.

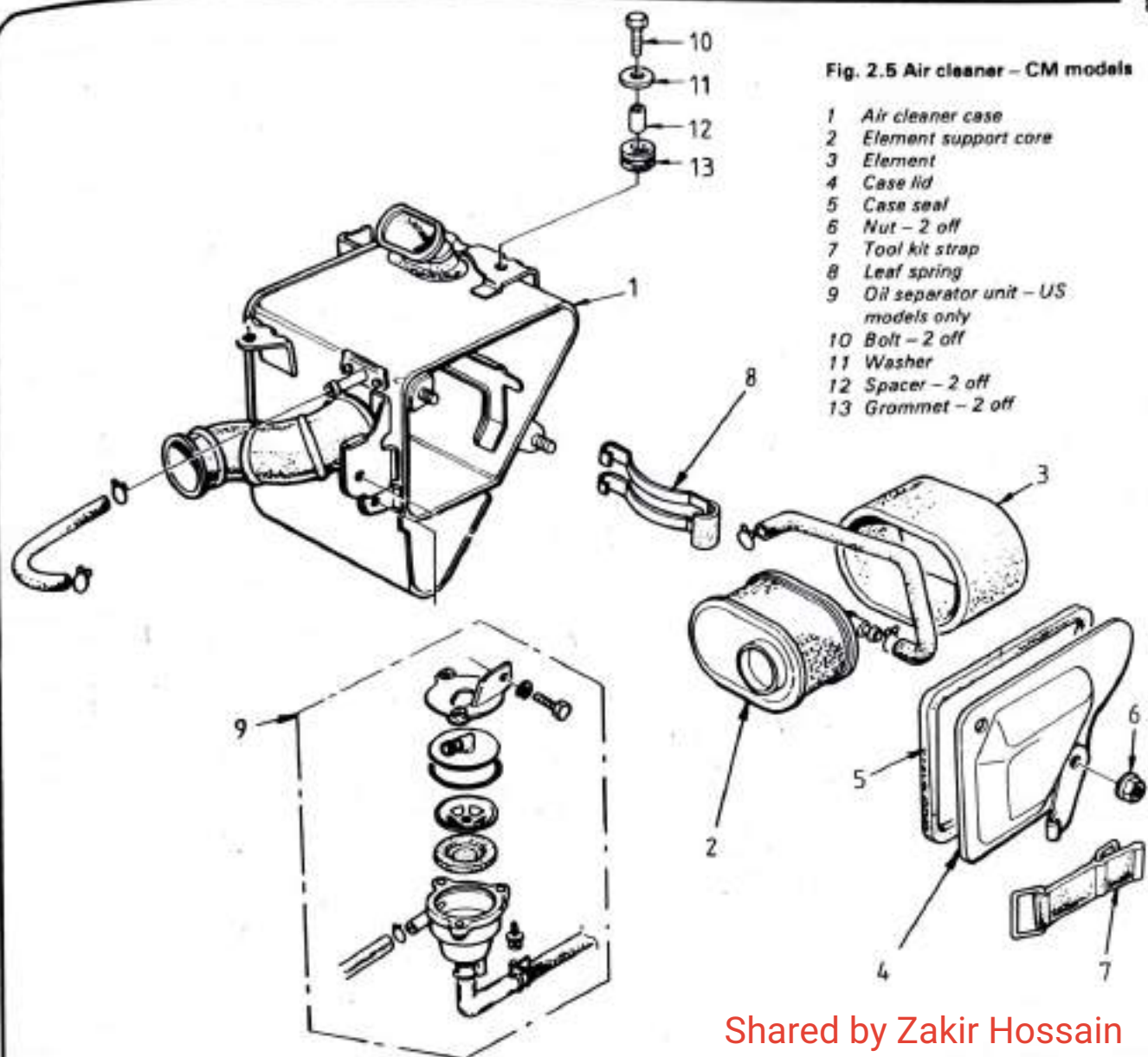
3 The element is of synthetic foam impregnated with oil. Wash the element in special solvent (available from Honda service agents) or alternatively, wash thoroughly in denatured alcohol or methylated spirit. Do not use petrol. Squeeze the element gently to dislodge any accumulated debris; do not attempt to wring the foam element out as it is easily torn. The solvent should be removed by squeezing the foam carefully in some clean rag. Allow the element to dry thoroughly allowing any residual solvent to evaporate.

4 Check the cleaned element carefully, looking for holes or splits, which if found, will necessitate the element's renewal. Immerse the element in clean engine or gear oil (Honda recommend gear oil SAE 80 or 90) until it is saturated. Squeeze the element gently to remove any excess oil. The object is to leave the element moist, but not dripping, with oil. Slide the element sleeve back onto its support core, and refit the unit in the air cleaner case.

5 Ensure the element is correctly seated and retained properly with the spring clip. If it is not seated correctly, air leakage may occur. Refit the case lid, ensuring the sealing gasket is in good condition and is refitted, and secure the lid with the two nuts. Do not forget the tool kit retainer strap's anchor plate below the lower nut.

6 The air cleaner element should receive attention in this manner every four months or 3000 miles (4800 km), whichever is sooner. In dusty conditions, it is advisable to increase the frequency of cleaning and re-impregnating.

7 If the foam element begins to harden or is torn or perforated, it should be replaced without question. Never run without the element connected to the carburettor because the carburettor is jetted to compensate for the addition of this component.



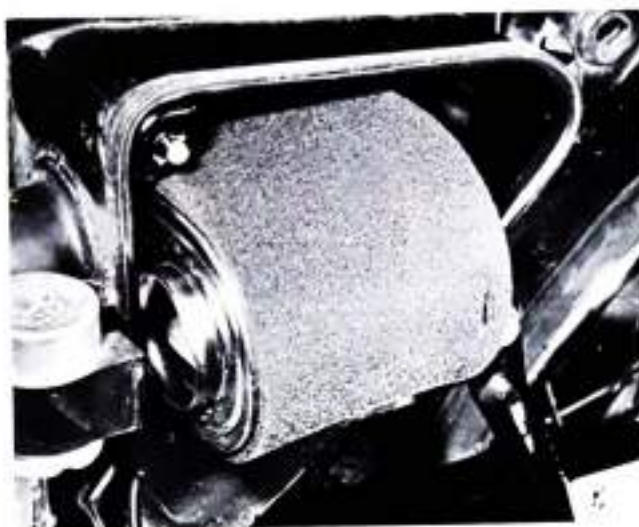
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9.1a Remove two nuts and detach air filter case lid ...



9.1b ... and displace spring retaining clip ...



9.1c ... then remove the element complete

10 Exhaust system: general

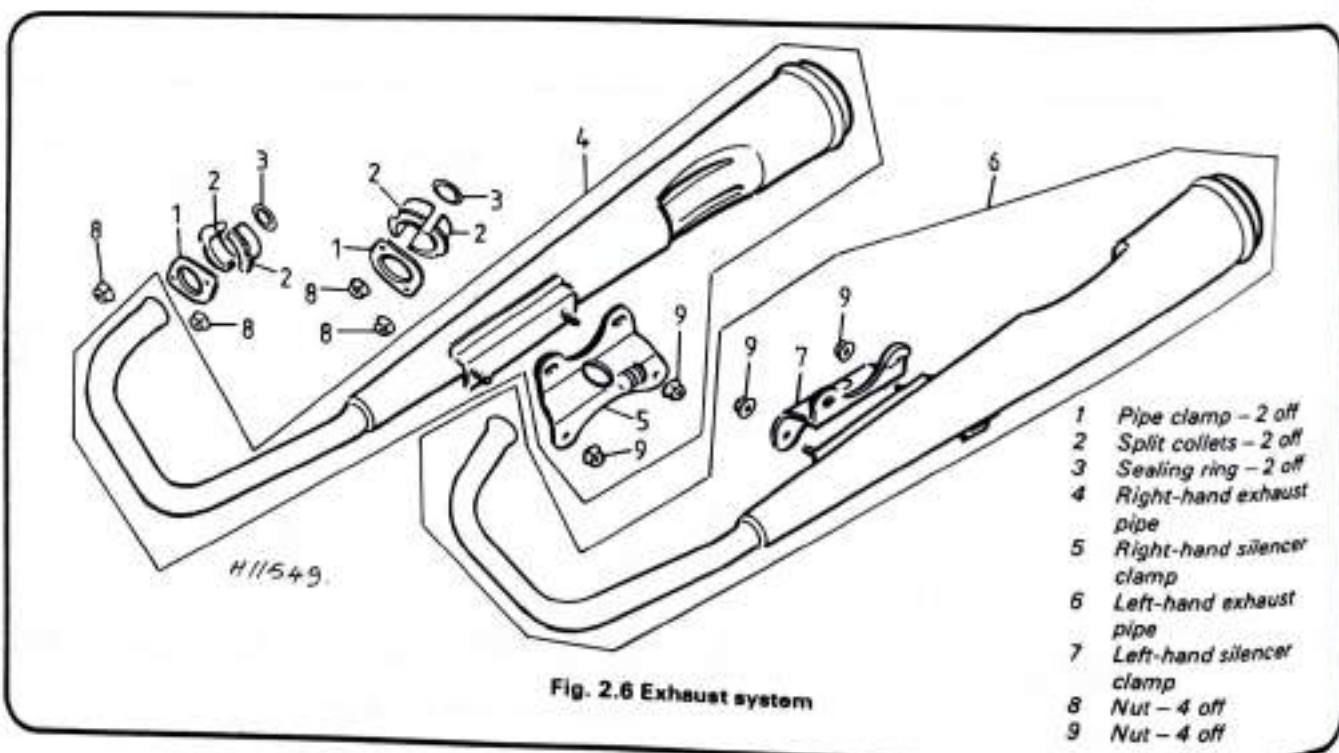
- 1 The exhaust system on a four-stroke motor-cycle will require very little attention, as, unlike two-stroke machines, it is not prone to the accumulation of carbon. The only points requiring attention are the general condition of the system, including mountings and the chromium plating, and ensuring that the system is kept airtight, particularly at the exhaust port.
- 2 Air leaks at the exhaust port, ie the joint between the exhaust pipe and the cylinder head, will cause mysterious backfiring when the machine is on overrun, as air will be drawn in causing residual gases to be ignited in the exhaust pipe. To this end, make sure that the composite sealing ring is renewed each time the system is removed.
- 3 There is no means of detaching the baffles from the silencer to facilitate their cleaning; usually the silencer will have reached the point of visible deterioration long before there is any chance

of obstruction through carbon build-up.

4 The silencer, as its name implies, effectively reduces the exhaust noise to an acceptable level without having any adverse effects on engine performance. Do not tamper with or remove the baffles from within the silencer. Although a much louder exhaust note may give the impression of greater speed, this is rarely the case in practice. Tampering with the standard system, designed to give optimum performance with a low noise level, will upset the balance and cause reduced performance, even though the changed exhaust note creates the illusion of speed.

11 Engine lubrication

- 1 The engine oil, which is common also to the primary drive and transmission, is contained within a crankcase compartment formed by the right-hand crankcase cover and the sump. This compartment holds 1.5 litres (2.6 Imp pint/1.6 US quart) of oil, and has a dipstick integral with the filler cap.
- 2 The lubrication system works on the wet sump, pressure fed, principle. A rotating vane type oil pump, driven from the outermost of a pair of primary drive gears on the right-hand end of the crankshaft, delivers oil to most parts of the engine. The oil is picked up in the sump through a filter screen attached to the lower part of the oil pump. The filter screen protects the oil pump, and thus the rest of the engine, from impurities which could otherwise cause damage. The clean oil is then pressure fed, by the oil pump, up through the right-hand crankcase cover, where it is diverted into two routes. In one direction it goes through a passage in the right-hand crankcase cover, and through another passage across the top of the complete crankcase, to both ends of the crankshaft. The other direction takes the oil through a passage via two cylinder head studs to the rocker arms and camshaft. The transmission also receives oil under pressure, relying upon this simple yet efficient system. The remaining few components within the engine/transmission unit, those subject to less stress, which are not pressure fed with oil, are lubricated on the splash principle.
- 3 The need to change the engine oil at the prescribed regular intervals (1500 miles/2400 km), and to clean out the filter screen at the recommended half-yearly or 5000 miles/8000 km intervals, is paramount to the long-term reliability of these machines. If the machine is used only for short journeys



especially in the lower temperature range. It is prudent to make the oil changes more frequent. Low running temperatures encourage condensation within the engine and if the lubricating oil becomes contaminated with water, it will emulsify, leading to severe corrosion of many of the working parts.

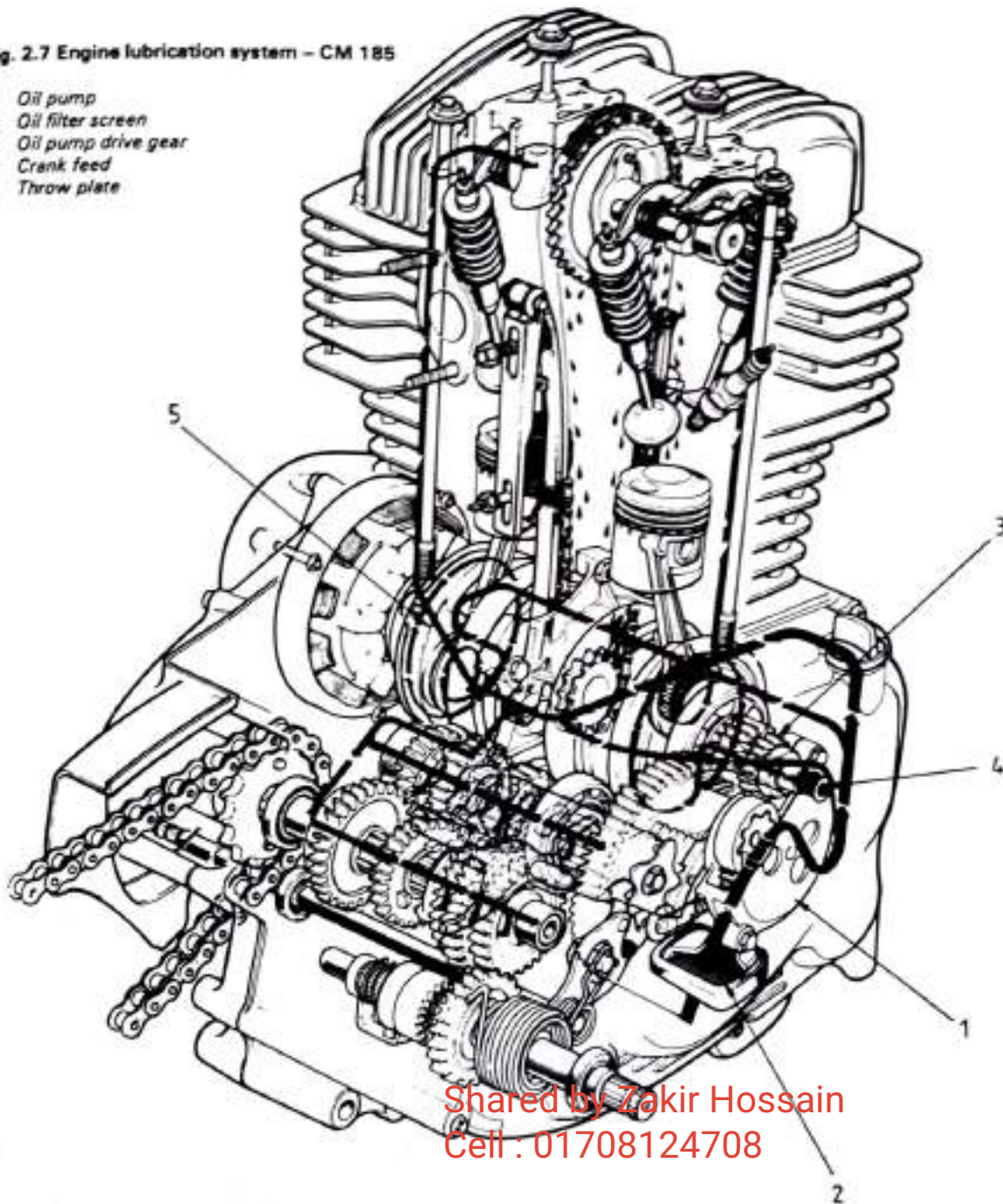
4 The oil pump is unlikely to give trouble unless particles of metal find their way into the rotor assembly and cause scoring. This will usually only occur if some engine component has broken up during service, and underlines the necessity to

completely strip and clean the engine if any such unfortunate incident occurs. Impurities in the oil will normally be trapped by the filter screen, before they enter the pump and cause any harm.

5 Note that checking the oil level must be done before the engine has been run, because the oil would otherwise be somewhat dispersed around the engine. Also note that to check correctly the oil level, the dipstick/filler cap should be rested on the edge of the filler orifice and not screwed home.

Fig. 2.7 Engine lubrication system – CM 185

- 1 Oil pump
- 2 Oil filter screen
- 3 Oil pump drive gear
- 4 Crank feed
- 5 Throw plate



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12 Oil pump: removal, examination and renovation

1 There is no necessity to strip the engine in order to gain access to the oil pump, although it should be stressed that this component should not be removed and dismantled unless really essential. It will give good service during the normal service life of the machine without attention and is likely to give trouble only if metallic particles or other foreign bodies contaminate the oil and score the pump rotors.

2 To remove the pump without dismantling the engine, first drain off the engine oil by means of the crankcase drain plug on the lower left-hand crankcase side. Disconnect the clutch cable at the operating lever on the rear top of the right-hand crankcase cover. Remove the kick start lever pinch bolt and remove the lever. Remove the right-hand exhaust pipe and silencer assembly. Slacken evenly and remove the ten retaining screws from the right-hand crankcase cover, and detach the cover; be prepared for some residual oil to be released.

3 The oil pump driving gear housing will be seen to have three circular holes in its outer face. Turn the engine over until the holes in the driving gear are in line, then pass a screwdriver through to release the three countersunk retaining screws. The pump can now be lifted away. If the three countersunk crosshead screws are particularly tight, an impact driver may have to be used. Take great care in that case, that no damage is inflicted on the pump drive gear or housing etc. Note that the oil filter screen will lift away with the oil pump; the screen should be separated for cleaning purposes. Also note the O-ring in the lower forward part of the crankcase wall.

4 Remove the end cover from the back of the pump unit; it is retained by a single crosshead screw. Note the hole in the cover and the corresponding locating pin in the rear of the pump body; these must align during reassembly. Remove also the end plate gasket. Shake out the inner and outer rotors.

5 If desired, the driving gear housing can now be separated. Slacken and remove the two 5 mm bolts, separate the two halves of the gear cover and remove the rotor drive shaft and the pump drive gear.

6 Wash all the oil pump components with petrol and allow them to dry before carrying out a full examination. Before part reassembling the pump for the various measurements to be made, check the castings for cracks or other damage, especially the pump end cover. Examine all the components, especially the rotors, for signs of scuffing and wear, or for signs of scoring, chipping or other surface damage which will occur if metallic particles find their way into the oil pump assembly. Renewal of the affected parts is the only remedy under these circum-

stances, bearing in mind that the rotors must always be renewed as a matched pair.

7 Reassemble the pump rotors and measure the clearance between the outer rotor and the pump body, using a feeler gauge. If the measurement exceeds the service limit of 0.25 mm (0.010 in) the rotor or the body must be renewed, whichever is worn. Measure the clearance between the outer rotor and the inner rotor, using a feeler gauge. If the clearance exceeds 0.2 mm (0.008 in) the rotors must be renewed as a set. With the pump rotors installed in the pump body, lay a straight edge across the mating surface of the pump body. Again with a feeler gauge measure the clearance between the rotor faces and the straight edge. If the clearance exceeds 0.12 mm (0.005 in) the rotors should be renewed as a set.

8 Reassemble the rest of the pump components by reversing the dismantling procedure. The components parts must be absolutely clean upon reassembly, or damage will result. Lubricate the rotors thoroughly, removing them again if necessary, before refitting the end cover and gasket. Note that the locating hole in the end cover must correctly locate with the protruding pin on the pump body. Note also that there are two protruding pins on the pump drive shaft. There are similar flats in the flats on the pump drive gear pinion and one flat in the centre of the inner rotor. With the pump drive shaft inserted correctly, all the flats will align.

9 The oil pump filter screen should now be cleaned, prior to refitting the pump to the engine. The screen should be carefully washed out in clean petrol, and any residual oil or debris which may have accumulated, carefully removed. A small, soft bristled brush (ie an old tooth brush) is ideal for cleaning the filter screen. If the screen is damaged in any way it must be renewed immediately. Refit the filter screen to the pump pick-up extension.

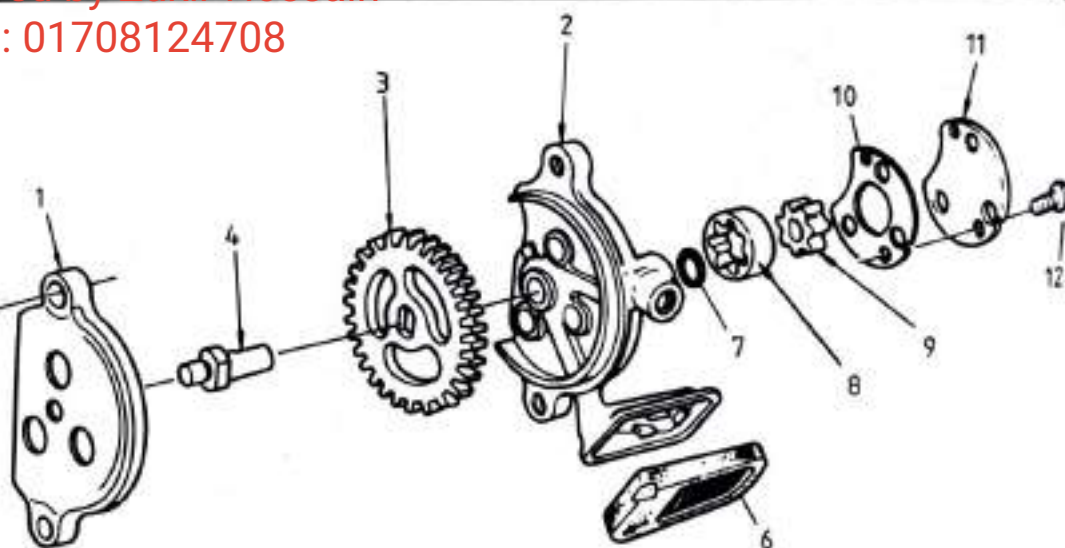
10 Fit a new O-ring to the recess in the casing lower wall, and position the pump against the wall. Fit and tighten the three screws evenly. Check that the rotors are not binding during tightening, by stopping and rotating the drive gear, and check again after tightening, in the same manner. A stiff pump is almost invariably due to dirty rotors.

13 Oil filter screen: location and cleaning

1 The engine is equipped with a detachable filter screen attached to the pick-up extension on the lower part of the oil pump. The oil pump and screen are located behind the right-hand crankcase cover. Removal and cleaning of the filter screen is covered in the appropriate section in the Routine Maintenance Chapter at the beginning of this Manual.

Fig. 2.8 Oil pump

- 1 Outer gear housing
- 2 Gear housing
- 3 Drive gear
- 4 Rotor drive shaft
- 5 Bolt - 2 off
- 6 Screen
- 7 O-ring
- 8 Outer rotor
- 9 Inner rotor
- 10 End cover gasket
- 11 End cover
- 12 Screw



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12.4a Lift off end cover from rear of pump; note locating pin on pump body (arrowed)

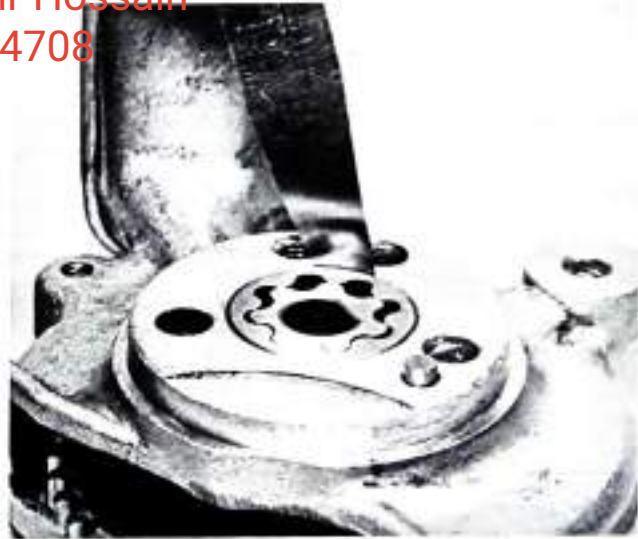


12.4b Remove the inner and outer rotors

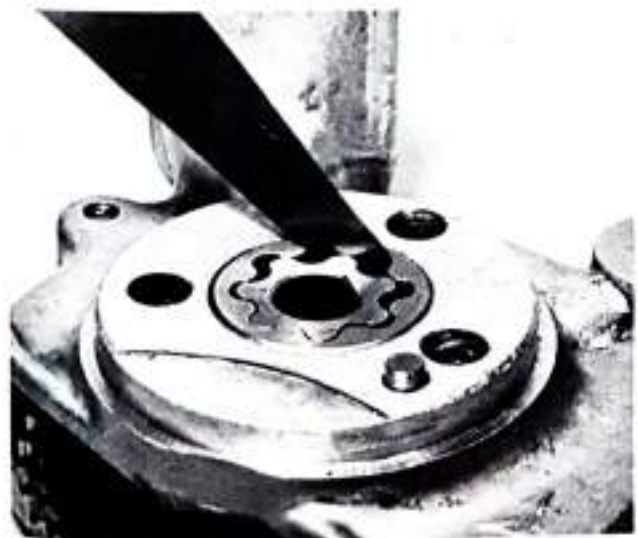
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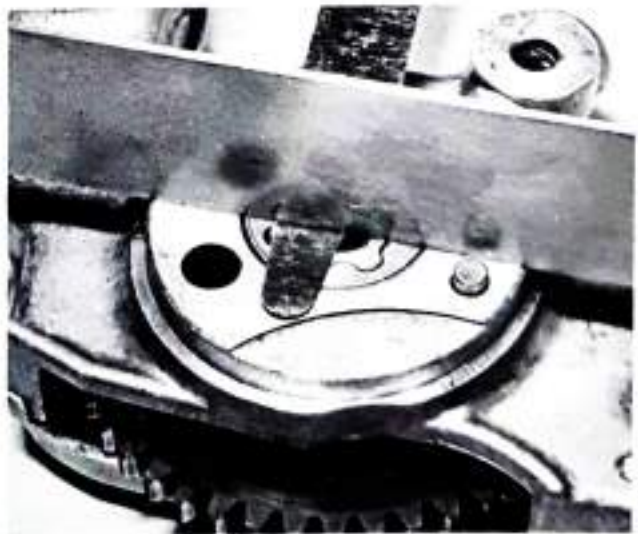
12.5 If required, remove two bolts and separate driving gear housing



12.7a Measure the clearance between outer rotor and pump body ...



12.7b ... the clearance between inner and outer rotors and ...



12.7c ... the clearance between the rotor faces and a straight edge

14 Fault diagnosis: fuel system and lubrication

Symptom	Cause	Remedy
Excessive fuel consumption	Air cleaner choked or restricted Fuel leaking from carburettor. Float sticking Badly worn or distorted carburettor Jet needle setting too high Main jet too large or loose Carburettor flooding	Clean or renew. Check all unions and gaskets. Float needle seat needs cleaning. Replace. Adjust as figure given in Specifications. Fit correct jet or tighten if necessary. Check float valve and replace if worn.
Idling speed too high	Throttle stop screw in too far Carburettor top loose Pilot jet incorrectly adjusted Throttle cable sticking	Adjust screw. Tighten top. Refer to relevant paragraph in this Chapter. Disconnect and lubricate or replace.
Engine dies after running for a short while	Blocked air hole in filler cap Dirt or water in carburettor	Clean. Remove and clean out.
General lack of performance	Weak mixture; float needle stuck in seat Air leak at carburettor joint	Remove float chamber or float and clean. Check joint to eliminate leakage, and fit new O-ring.
Engine does not respond to throttle	Throttle cable sticking Petrol octane rating too low	See above. Use higher grade (star rating) petrol.
Engine runs hot and is noisy	Lubrication failure	Stop engine immediately and investigate cause. Do not restart until cause is found and rectified

Chapter 3 Ignition system

Contents

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Specifications

Alternator	CD185T	CD200T	CM185T	CM200T
Make	Hitachi or Nippon Denso	Hitachi	Hitachi	Hitachi
Type	6-coil stator	8-coil stator	6-coil stator	8-coil stator
Output (6V)	0.086 kw #5000 rpm	0.080 kw #5000 rpm	0.086 kw #5000 rpm	0.080 kw #5000 rpm
Ignition	Battery, ignition coil and contact breaker points			
Type				
Ignition timing	15° BTDC #1200 rpm			
Retarded	1800 rpm			
Advance starts				
Advance finishes:	37° - 40° BTDC #3100 rpm			
CD185/CM185	32° - 35° BTDC #2800 rpm			
CD200/CM200				
Contact breaker	0.3 - 0.4 mm (0.012 - 0.016 in)			
Gap				
Condenser	0.22 - 0.26 Microfarads			
Capacity				
Sparking plugs	NGK or Nippondenso			
Make				
Type:	CR7HS or U22FSR-L			
UK models	C7HS or U22FS			
US models	0.6 - 0.7 mm (0.024 - 0.028 in)			
Gap				

1 General description

The spark which is necessary to ignite the petrol/air mixture in the combustion chambers is derived from an ignition coil mounted on the frame and an alternator mounted on the left-hand end of the crankshaft. Current is produced immediately the engine is rotated and this is fed to the ignition coil via the contact breaker assembly. When the contact breaker points separate, the primary circuit is broken, causing a high tension voltage to be developed in the secondary windings of the ignition coil. This voltage jumps across the sparking plug gap, to create the spark that ignites the mixture.

2 Crankshaft alternator: checking the output

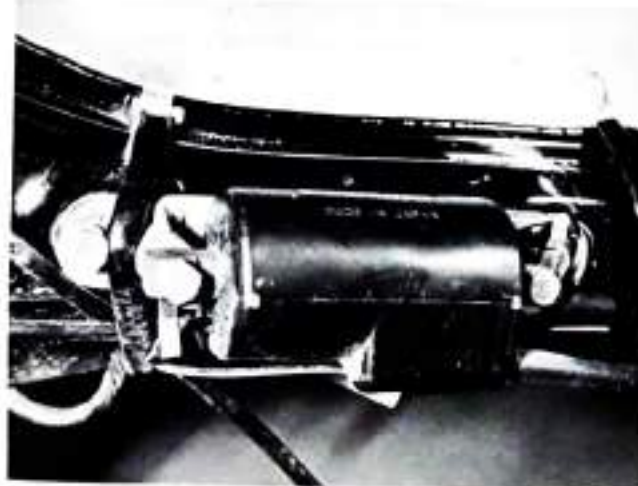
1 The alternator is instrumental in creating the power in the ignition system, and any failure or malfunction will affect the operation of the ignition system. If the machine will not start and there is no evidence of a spark at the sparking plug, a check should first be made to ensure there is no fault at either the contact breaker assembly or in the ignition itself. Refer to Sections 3 and 4 of this Chapter.

2 If the above checks have failed to show any faults, the output from the alternator itself should be suspected, assuming the wiring is not broken or disconnected at any point. Reference should be made, therefore, to Chapter 6, Section 2 for output checking procedures by basic test methods.

3 Ignition coil : checking

1 The ignition coil is a sealed unit and designed to give long service without need for attention. The coil is located beneath the petrol tank, and is mounted on the right-hand side of the frame top tube. It follows, therefore, that the tank must be removed to gain access to the coil. Refer to Section 2 of the previous Chapter for the tank removal procedure.

2 If a weak spark and difficult starting causes the performance of the coil to be suspect, it should, in general, be tested by a Honda service agent or an auto-electrical expert. They will have the necessary appropriate test equipment. It is, however, possible to perform a number of basic tests, using a test lamp and battery arrangement, or a multimeter, connected as shown in the accompanying diagram. In both checks shown, continuity should be found in the primary windings and the secondary windings. Note that when performing these tests, the plug caps should be removed from the ends of the secondary (HT) leads.



3.1 Ignition coil is sealed unit; secured by two bolts

3 Should any of these checks not produce the expected result, the coil should then be taken to a Honda service agent or auto-electrician for a more thorough check. If the coil is found to be faulty, it must be replaced; it is not possible to effect a satisfactory repair.

4 A defective condenser in the contact breaker circuit can give the illusion of a defective coil and for this reason it is advisable to investigate the condition of the condenser before condemning the ignition coil. Refer to Section 7 of this Chapter for the appropriate details.

4 Contact breaker : adjustment

1 To gain access to the contact breaker assembly, it is necessary to detach the circular inspection cover, retained on the 185 models by two crosshead screws, and on the 200 models by three crosshead screws, from the left-hand crankcase cover. Note the paper gasket on the cover. Remove the sparking plugs to make rotation of the engine easier.

2 Rotate the engine slowly by means of the ATU centre bolt (185T models) or by the engine turning hexagon (200T models only) until the points are in the fully open position. Examine the contact faces. If they are dirty, burnt or pitted, they should be removed for renewal as described in Section 5 of this Chapter. If the points are in sound condition, proceed with adjustment as follows:

3 Adjustment is carried out by slackening the two screws which retain the fixed contact breaker point and inserting a screwdriver on the adjusting lugs provided. Turn in the appropriate direction until the gap is within the range 0.3 to 0.4 mm (0.012 to 0.016 in) and then retighten the two retaining screws. It is imperative that the points are open FULLY while this adjustment is made, or a false reading will result. Recheck the setting when the screws have been tightened; it is not unknown for the setting to move slightly during the retightening operation.

4 If adjustment has proved necessary, it is likely that the ignition timing will have been affected, and this should be checked. See Section 6 of this Chapter for the relevant procedure.

5 Before replacing the cover and gasket, place a slight smear of grease on the contact breaker cam and a few drops of thin oil on the felt wick which lubricates the surface of the cam. Do not over-lubricate or there is a chance of oil working onto the contact breaker points faces. Ensure that the cover gasket is in good condition; if it is torn or damaged in some way, the ingress of moisture is made easier.



4.1 Remove three screws (200 models) and detach the inspection cover



Spark plug maintenance: Checking plug gap with feeler gauges



Altering the plug gap. Note use of correct tool



Spark plug conditions: A brown, tan or grey firing end is indicative of correct engine running conditions and the selection of the appropriate heat rating plug



White deposits have accumulated from excessive amounts of oil in the combustion chamber or through the use of low quality oil. Remove deposits or a hot spot may form



Black sooty deposits indicate an over-rich fuel/air mixture, or a malfunctioning ignition system. If no improvement is obtained, try one grade hotter plug



Wet, oily carbon deposits form an electrical leakage path along the insulator nose, resulting in a misfire. The cause may be a badly worn engine or a malfunctioning ignition system



A blistered white insulator or melted electrode indicates over-advanced ignition timing or a malfunctioning cooling system. If correction does not prove effective, try a colder grade plug



A worn spark plug not only wastes fuel but also overloads the whole ignition system because the increased gap requires higher voltage to initiate the spark. This condition can also affect air pollution

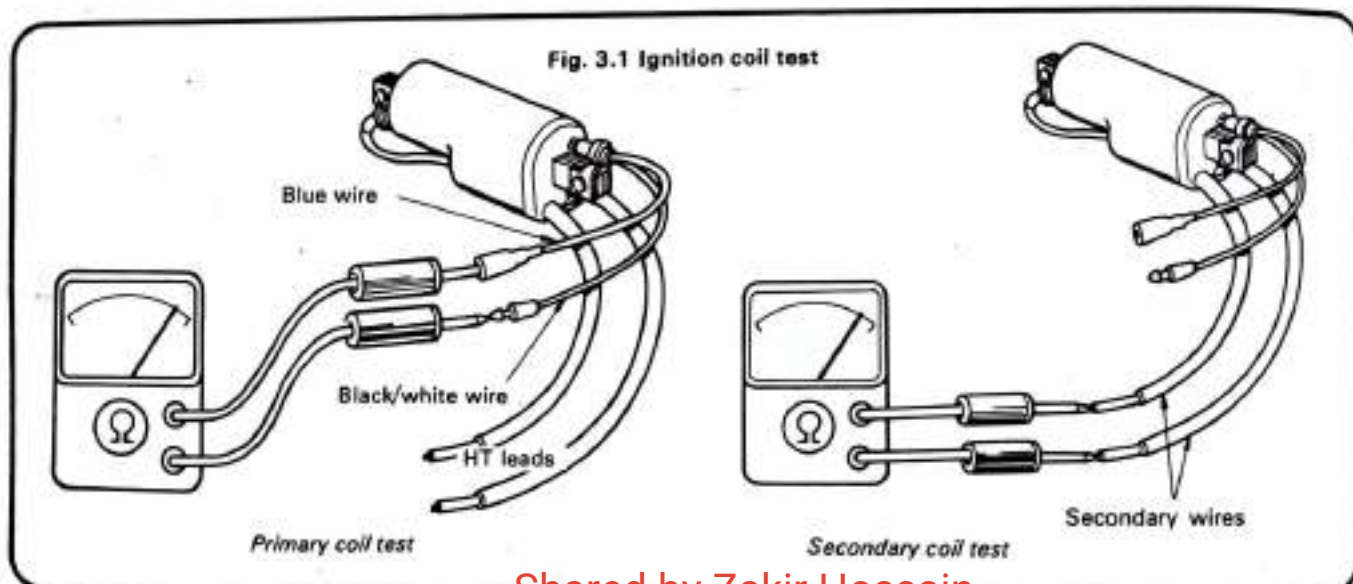
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4.3a Check points gap with correct size feeler gauge



4.3b Slacken two screws (arrowed) and adjust the points gap



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5 Contact breaker points: removal and renewal

- 1 If the contact breaker points are burned, pitted or badly worn, they should be removed for dressing. If it is necessary to remove a substantial amount of material before the faces can be restored, the points should be replaced without question.
- 2 To remove the contact breaker points, detach the circlip which secures the moving contact to the pin on which it pivots. Remove the nut and bolt that secures the flexible lead wire to the end of the contact return spring, noting the arrangement of the insulating washers so that they are replaced in their correct order during reassembly. Lift the moving contact off the pivot, away from the assembly.
- 3 The fixed contact is removed by unscrewing the two screws which retain the contact to the contact breaker baseplate.
- 4 If the points are only slightly pitted, it is possible to restore them or dress them. The points should be dressed with an oilstone or fine emery cloth. Keep them absolutely square throughout the dressing operation, otherwise they will make angular contact on reassembly and rapidly burn away.
- 5 Replace the contacts by reversing the dismantling procedure making sure that the insulating washers are replaced in the correct order. It is advantageous to apply a thin smear of

grease to the pivot pin, prior to the replacement of the moving contact arm.

- 6 Check, and if necessary, re-adjust the contact breaker gap when the points are fully opened, and then check the ignition timing following the procedure given in the next Section.

6 Ignition timing : checking and resetting

- 1 In order to check the accuracy of the ignition timing, it is first necessary to remove the circular inspection cover from the left-hand crankcase cover. Remove the two (185T models) or three (200T models) crosshead retaining screws, and remove the cover and gasket. This gives access to the contact breaker points and also allows the timing marks inscribed on the automatic timing unit (ATU) to be viewed through the aperture in the points baseplate. It will also be noted that the left-hand crankcase cover carries a fixed index mark which can be viewed through the baseplate aperture. On 185 models, two sets of marks are inscribed on the ATU: 'F1', 'T1' and two parallel lines, and also 'F2', 'T2' and a further two parallel lines. The second set should be disregarded; these are used on other machines fitted with the ATU. On 200 models one set only of marks is shown on the ATU; 'F', 'T' and the two parallel lines. In both

cases the two parallel lines indicate the point of full advance when they align with the fixed index mark.

2 Note that the contact breaker gap must be correct before the accuracy of the ignition timing is verified. Adjustment of the contact breaker gap will always alter the ignition timing.

3 If the ignition timing is correct, the 'F1' line (185 models) or 'F' line (200 models) will coincide exactly with the fixed pointer line, when the contact breaker points are just commencing to separate. The moment at which the contact breaker points separate can be determined using a bulb with two leads soldered to its contacts. Connect one lead to the contact on the contact breaker assembly. The other lead should be connected to a good earthing point on the engine. Note that if the lead ends are fitted with crocodile clips, connection and disconnection will be made more convenient. If a multimeter is available, this may be used in preference to the battery and bulb arrangement. Set the multimeter to 0-10 volts dc scale, connect the negative lead to a good earthing point on the engine, and the positive lead to the moving contact of the contact breaker assembly. With the ignition switched on it will be found that the bulb will light as the points open. In the case of the multimeter, a corresponding needle deflection will occur.

4 To adjust the position of the contact breaker points relative to the points cam, slacken the three crosshead screws at the periphery of the contact breaker baseplate. The whole baseplate can then be rotated until the points commence to separate as the timing marks on the ATU coincide with the fixed pointer. If the timing is too early (advanced), the baseplate should be rotated anti-clockwise; if the timing is too late (retarded), rotate the baseplate clockwise to remedy the situation.

5 It cannot be overstressed that optimum performance depends on the accuracy with which the ignition timing is set. Even a small error can cause a marked reduction in performance and the possibility of engine damage, as the result of overheating.

6 Although the base of the contact breaker cam forms part of the automatic ignition advance mechanism (ATU) the ignition is timed when the unit is in the fully retarded position (balance weights unextended).

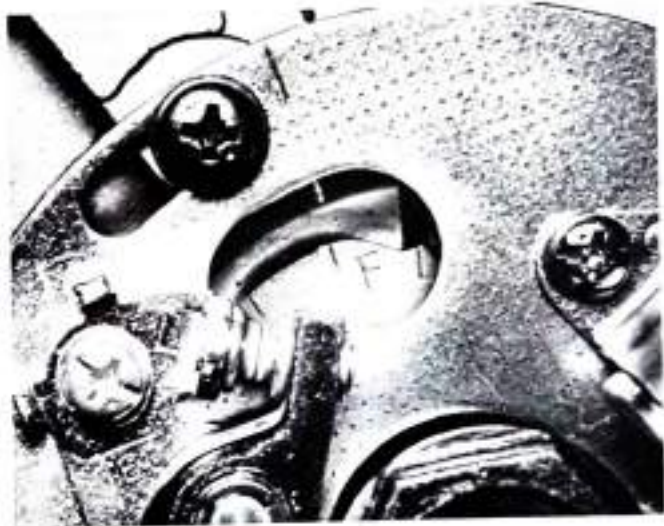
7 An alternative method of checking the ignition timing can be adopted, whilst the engine is running, using a stroboscopic lamp. When the light from the lamp is aimed at the timing marks it has the effect of 'freezing' the moving marks on the ATU in one position, and thus the accuracy of the timing can be seen. An additional advantage is also provided in that the correct and smooth functioning of the automatic advance-retard unit (ATU) can be determined.

8 The stroboscopic lamp should be connected up as recommended by the lamp's manufacturer and the engine started.

In order for this test to be carried out correctly, the engine speed (rpm) must be known. Because these machines are not fitted with a tachometer (rev-counter), a test tachometer is necessary if testing of the ignition timing by the stroboscopic method is to be successful. It should be noted that the 'well tuned ear' may be able to deduce when the engine is at its correct idling speed ($1200 \text{ rpm} \pm 100 \text{ rpm}$) but would be very unlikely to gauge accurately when full advance has been reached.

9 Up to about 1300 rpm the 'F1' mark (185 models) or 'F' mark (200 models) should be in line with the fixed index mark. As the speed is increased the ATU should be seen to rotate and the two parallel advance marks should move closer to the fixed index mark. On reaching 3100 rpm (185 models), or 2800 rpm (200 models) full advance should be reached and the two parallel advance marks should be in alignment with the fixed index mark. Alterations to the ignition timing should be made in the normal way as described in the first paragraphs of this Section.

10 If it is found that the static ignition timing is correct but that full advance cannot be reached, or that progression to full advance is erratic, there is some indication that the mechanical automatic timing unit is not functioning correctly. Refer to Section 8 of this Chapter for detail relating to examination of this component.



6.3 The 'F1' mark must be aligned with fixed index mark as points commence to open



6.4 Slacken the three baseplate retaining screws (arrowed) to adjust timing



6.5 Fixed index mark should be between twin vertical marks at full advance

7 Condenser: location, removal and replacement

- 1 A condenser is included in the contact breaker circuitry to prevent arcing across the contact breaker points as they separate. The condenser is connected in parallel with the points and if a fault develops, ignition failure is liable to occur.
- 2 If the engine proves difficult to start, or misfiring occurs, it is possible that the condenser is at fault. To check, separate the contact breaker points by hand when the ignition is switched on. If a spark occurs across the points and they have a blackened and burnt appearance, the condenser can be regarded as unserviceable.
- 3 It is not possible to check the condenser without the appropriate equipment. In view of the low cost involved, it is preferable to fit a new replacement and observe the effect on engine performance.
- 4 The condenser on the 200T models is located on the contact breaker baseplate. The condenser has an integral bracket and is attached to the baseplate by two crosshead screws; removal and replacement is therefore simple. On all the remaining models, the condenser is located below the dualseat, just rearward of the petrol tank locating block. It is secured by a single bolt passing through its integral bracket.
- 5 To gain access to the condenser on the CD200 model, simply remove the three crosshead retaining screws and detach the circular inspection cover from the left-hand crankcase cover. On the other models, access to the condenser is gained after the dualseat has been removed.
- 6 Ensure the mounting bracket on each condenser makes a good contact and is tightened fully, since this fitting provides the earth connection of the condenser.

8 Automatic timing unit (ATU): removal, examination and replacement

- 1 Fixed ignition timing is of little advantage as the engine speed increases, and it is necessary to incorporate a method of advancing the timing by centrifugal means. A balance weight assembly located behind the points; linked to the points cam, is used on these models. A centre fixing bolt holds the unit secure, permitting the cam to move by means of this linkage and so advance the ignition timing.
- 2 To remove the ATU from the engine, remove the circular contact breaker points cover from the left-hand crankcase cover, and also the points baseplate complete. Slacken and remove the centre retaining bolt, and the large engine turning

hexagon on the 200 models, and pull off the ATU. Note that during removal of the centre bolt, the engine must be prevented from rotating; this is easily accomplished by selecting top gear and applying the rear brake.

3 The unit comprises spring loaded balance weights, which move outwards against the spring tension as centrifugal force increases. At idling speed the force cannot overcome spring pressure. As the engine speed rises the force increases until it is sufficient to throw the balance weights to the limit of their travel, ie full advance. The balance weights must move freely on their pivots and be rust-free. The tension springs must also be in good condition. Keep the pivots lubricated and make sure the balance weights move easily, without binding. Most problems arise as a result of condensation within the engine, which causes the unit to rust and balance weight movement to be restricted.

4 If any malfunction or breakage has occurred, renew the complete unit; no component parts are available separately. If the unit is in good condition, lightly oil it and slide it back into position, ensuring the locating pin on its rear face correctly locates with the slot in the rotor centre (185 models) or the rotor centre mounting boss (200 models). Refit the centre bolt and washer (185 models) and the engine turning hexagon and centre bolt (200 models); check that the large hexagon engages properly with the end of the points cam. It will be noticed that the ATU centre bolt on the 185 models is also the rotor securing centre bolt and as such should be tightened to a recommended torque setting of 4.5 - 5.0 kgf m (32.5 - 36.2 lbf ft). The smaller dimensioned separate ATU retaining bolt of the 200 models should be tightened fully to a torque setting of 1.0 - 1.4 kgf m (7 - 10 lbf ft). Reassemble the remaining components by reversing the dismantling sequence.

9 Sparking plugs: examination, checking and resetting the gaps

- 1 Both the CD models are fitted with the two NGK CR7HS or Nippon Denso U22 FSR-L sparking plugs as standard. The CM 200 model as available in the UK is also fitted with a pair of this grade of plugs. The two CM models in the USA are equipped with two NGK C7HS or Nippon Denso U22 FS sparking plugs as standard. All models have their plugs gapped to within the range 0.6 - 0.7 mm (0.024 - 0.028 in). Although operating conditions may, at certain times, indicate a change in sparking plug grade is required, the type recommended by the manufacturer gives the best, all round service.
- 2 Check the gap of the plug points during every 3000 mile



7.4 Condenser bracket is retained by two screws to base-plate (200T models only)



8.3 Check balance weights, pivots and springs on ATU

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service. To reset the gap, bend the outer electrode to bring it closer to the centre electrode and check that a 0.6 - 0.7 mm (0.024 - 0.028 in) feeler gauge can be inserted. Never bend the central electrode or the insulator will crack, causing engine damage if the particles fall in whilst the engine is running.

3 With some experience, the condition of the sparking plug electrodes and insulator can be used as a reliable guide to engine operating conditions. See the accompanying colour photographs.

4 Always carry a spare sparking plug of the recommended grade. In the rare event of plug failure, it will enable the engine to be restarted.

5 Beware of overtightening the sparking plug, otherwise there is a risk of stripping the threads from the aluminium alloy cylinder head. The plug should be sufficiently tight to seat firmly

on its copper sealing washer, and no more. Use a spanner which is a good fit to prevent the spanner from slipping and breaking the insulator.

6 If the threads in the cylinder head strip as a result of overtightening the sparking plugs, it is possible to reclaim the head by the use of a Helicoil thread insert. This is a cheap and convenient method of replacing the threads; most motorcycle dealers operate a service of this nature at an economic price.

7 Make sure the plug insulating caps are a good fit and have their rubber seals. They should also be kept clean to prevent tracking. These caps contain the suppressors that eliminate both radio and TV interference.

8 Before refitting a sparking plug in the cylinder head, coat the threads sparingly with a graphited grease to aid future removal.

10 Fault diagnosis: ignition system

Symptom	Cause	Remedy
Engine will not start	Defective ignition coil Dirty points Poor points contact	Renew Clean or renew and check gap. Check for arcing. Renew condenser.
Engine starts but runs erratically	Plug gap incorrect or has whiskered Incorrect ignition timing Weak points arm return spring Defective ignition coil Low output from flywheel generator Plug lead insulation breaking down	Adjust and clean or renew. Check timing and points gap Renew. Renew after expert testing. Have tested and renew. Check and renew.
Engine misfires	Faulty condenser in ignition circuit Fouled sparking plug Poor spark due to generator failure	Replace condenser and retest. Replace plug and have original cleaned. Check output from generator.
Engine lacks power and overheats	<u>Retarded</u> ignition timing	Check timing and also contact breaker gap. Check whether auto advance mechanism has jammed.
Engine 'fades' when under load	Pre-ignition	Check grade of plugs fitted; use recommended grades only. Verify whether lubrication system has pressure.

12. MARCH 2000 UNABLE TO ALIGN F1 CORRECTLY. GAP RE-ADJUSTED TO 18 - 20 THOU. THIS IMPROVED TIMING, POINTS PLATE STILL FULLY C/W - NO ADJUSTMENT LEFT. PROBABLY CAUSED BY WORN CAM ON ATU - NOT AVAILABLE SEPARATELY.

126980 - ATU REPL. ADJUSTMENT NOW POSSIBLE.
(24.4.00.) NEW SPARK PLUGS.

Chapter 4 Frame and forks

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Specifications

Frame

Type	Open diamond, engine used as a stressed member
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Front forks

	CD185T/CD200T	CM185T	CM200T
Type	Oil dampened, telescopic		
Travel	115 mm (4.5 in)	115 mm (4.5 in)	115 mm (4.5 in)
Oil capacity (per leg)	135 – 140 cc (4.8 – 4.9 oz)	125 cc (4.2 oz)	125 cc (4.2 oz)
Oil type	Automatic transmission fluid (ATF)		
Fork spring free length	482.0 mm (18.97 in)	504.6 mm (19.9 in)	504.6 mm (19.9 in)
Service limit	472.6 mm (18.60 in)	495 mm (19.5 in)	495 mm (19.5 in)
Fork stanchion OD	All models 30.915 – 30.950 mm (1.217 – 1.218 mm)		
Service limit	30.870 mm (1.214 in)		
Lower leg ID	31.042 – 31.104 mm (1.222 – 1.224 in)		
Service limit	31.165 mm (1.227 in)		

Rear suspension

Type	Swinging arm fork, two oil filled rear suspension units		
Travel	CD185T/CD200T 64 mm (2.5 in)	CM185T 64 mm (2.5 in)	CM200T 64 mm (2.5 in)
Spring free length	182.6 mm (7.19 in)	184.1 mm (7.25 in)	182.1 mm (7.17 in)
Service limit	180.0 mm (7.09 in)	175.6 mm (6.92 in)	173.5 mm (6.83 in)

Torque settings

Steering stem nut (22 mm)	6.0 – 7.0 kgf m (43.4 – 50.6 lbf ft)
Top yoke pinch bolts, 2 (8 mm)	2.0 – 2.5 kgf m (14.5 – 18.1 lbf ft)
Bottom yoke pinch bolts, 2 (8 mm)	2.0 – 2.5 kgf m (14.5 – 18.1 lbf ft)
Swinging arm pivot bolt (14 mm)	4.0 – 5.0 kgf m (29.0 – 36.2 lbf ft)
Driven sprocket bolts, 4 (10 mm)	5.5 – 6.5 kgf m (39.8 – 47.0 lbf ft)
Rear suspension unit nuts, 4 (10 mm)	3.0 – 4.0 kgf m (21.7 – 29.0 lbf ft)
Footrest mounting bar bolts, 4 (8 mm)	2.0 – 2.5 kgf m (14.5 – 18.1 lbf ft)

1 General description

The models covered by this manual utilise a frame of open diamond type, being a composite structure of tubular construction with a monocoque spine. The engine unit is bolted between the spine and the downtube to form a structural part of the frame. The front forks are of traditional telescopic design, being controlled by hydraulic dampers. Rear suspension is provided by a swinging arm fork, pivoting on replaceable bushes, and controlled by two hydraulically dampened, suspension units.

2 Front forks: removal – general

1 It is unlikely that the forks will require removal from the frame together with the fork yoke and steering stem, unless accident damage has been sustained. Although feasible, this method is both unwieldy and time consuming. It is far easier to remove the individual fork legs, then the steering head assembly, where necessary, as described in Section 4 of this Chapter. If attention to the fork legs only is required, refer to the following Section.

2 Before any dismantling work can be undertaken, the machine should be placed on the centre stand (CD models) and blocked securely so that the front wheel is held off the ground. On the CM models, which are not fitted with a centre stand as standard equipment, an alternative method of support must be found. A plastic milk crate, or a crate of the type used by Moto-cross riders to support their machines, are ideal for this purpose. Alternatively, a stout wooden box will suffice, provided that it is large enough to raise the front wheel well clear of the ground. Ensure that there is no likelihood of the machine toppling over sideways or forwards when positioned astride the support.

3 Detach the speedometer drive cable at the wheel by unscrewing the single crosshead screw which retains it, and pulling the cable clear. If it has not been removed previously, this screw may prove to be very tight, necessitating the use of an impact driver to free it.

4 Slacken and remove the brake cable adjusting nut and disconnect the cable from the trunnion in the operating arm. Refit the adjusting nut to the cable end to avoid loss.

5 Withdraw the split pin, remove the wheel spindle nut, and

withdraw the spindle, noting the spacer on the right-hand side. The wheel can now be lowered clear of the forks and put to one side.

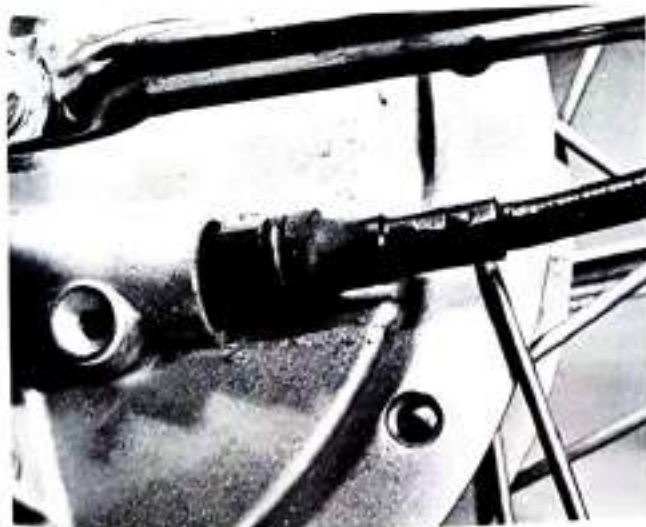
6 To avoid damage to the painted or chromium plated surface, it is a good idea to remove the front mudguard at this stage, irrespective of whether removal will be necessary later, as in the case of individual fork leg removal. The mudguard on all the models except the CM200T, is retained by a total of six bolts. Four of the bolts secure the mudguard integral mounting brackets to the inside of the fork lower legs and the two remaining bolts secure the mudguard stays to their locating points at the rear of the lower legs. The front mudguard fitted to the CM200T model is retained by four bolts and is not fitted with retaining stays.

3 Front forks: removing the fork legs from the yokes

1 Slacken the pinch bolts on the upper and lower yokes to allow the fork legs to be pulled down and clear of the frame. In order to make individual fork leg dismantling as easy as possible, the large chromed fork top bolts should be slackened before the fork legs are detached. Lower the fork leg until the top of the leg is roughly halfway between the two yokes, and then retighten the lower yoke pinch bolt. The fork top bolt can now be slackened without the fork legs rotating. If the legs prove reluctant to move, the fork top bolts can be tapped with a soft-headed mallet to jar the leg out of position. If, as in the case of the machine used in the preparation of this manual, serious corrosion has set-in behind the painted metal stanchion shrouds (fitted to the CD models only), the fork legs may prove very difficult to remove. Positioning a length of hardwood above the top yoke so it rests on the fork top bolt and then striking the hardwood with a hammer proved to be a crude but effective method of driving the fork legs out of the yokes. The painted shrouds can be left in position unless the fork yokes are to be removed.

2 If fork leg dismantling is now required, refer to Section 6 of this Chapter.

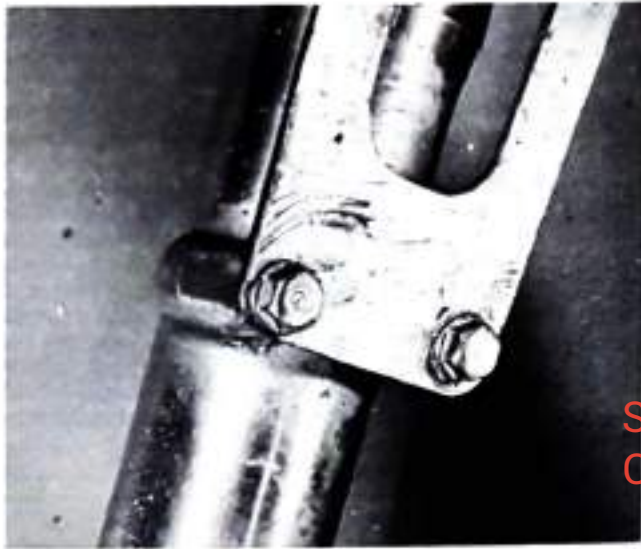
3 Reassembly of the forks is tackled in the reverse order of that given for dismantling; refer to Section 8 for reassembly and refitting procedures.



2.3 Remove retaining screw and pull speedometer cable clear



2.5 Withdraw the front wheel from the forks



2.6a Front mudguard is held by two bolts on each lower leg and ...



2.6b ... on all models (except CM200T), two bolts at bottom of leg



3.1a Slacken the pinch bolts in the top yoke, and ...



3.1b ... slacken the lower yoke pinch bolts



3.1c If fork legs show this type of corrosion they will be difficult to remove



3.1d Withdraw fork legs separately from yokes

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4 Front forks: removing the front yokes and steering head assembly from the frame

- 1 The preceding Section describes the procedure for removal of the fork legs without having to disturb the steering head bearings or yokes. Further dismantling of the fork assembly may be accomplished as follows, after carrying out fork leg removal.
- 2 Before the yokes and steering head assembly can be detached, it will be necessary to remove the handlebars, complete with control levers, the headlamp unit and speedometer/warning light unit. Start by covering the petrol tank with an old blanket, thick rags, or similar, to protect the paintwork from damage. Disconnect the battery positive (+) lead at the terminal to isolate the electrics.
- 3 Slacken and remove the two crosshead screws from the underside of each switch cluster on the handlebars. Free the switchgear leads from their retaining clips (two on each side of the handlebar) and slide off the twistgrip and right-hand switch cluster, and separate and detach the left-hand switch cluster. Disconnect the clutch cable at the handlebar lever. Pull off the plastic cover above the handlebar mounting plate. Unscrew the knurled ring which secures the speedometer cable to the instrument head and lift the speedometer/warning light console away, detaching the bulb holders from the base as this is done.
- 4 It will be noted on 185 models that the handlebar mounting plate, which incorporates the top half of the retaining clamps, also forms a mounting point for the choke operating knob and the ignition switch. Remove the two crosshead screws that secure the ignition switch. Remove the choke operating knob. Slacken and remove the four 8 mm mounting plate/handlebar clamp retaining bolts. Turn the ignition switch body so that it does not hinder the mounting plate as it is lifted clear. Lift up the handlebars and control levers and rest the assembly across the top of the protected tank, clear of the steering head. The 200 models are fitted with conventional handlebar clamps held by two bolts each. The front bolt of each also secures the choke

knob mounting bracket mounted forward of the handles. Unscrew the clamp bolts, displace the clamps and mounting bracket and lift the handlebar rearwards to rest across the petrol tank.

- 5 On CD models release the headlamp rim/reflector assembly by removing the two screws that pass through the shell in the 8 o'clock and 4 o'clock positions and into the assembly. Disconnect the bulb leads and place the reflector unit to one side. Disconnect the horn leads, remove the single retaining bolt and remove the horn. On the CM models, the headlamp, horn and front flashing indicators are all fitted to one mounting bracket. Release the two headlamp shell mounting bolts, disconnect the leads from the rear of the shell and lift the headlamp unit away. Remove the 8 mm bolt that retains the horn and one end of its mounting plate to the mounting bracket and lift the horn away after disconnecting it. Slacken and remove the two 6 mm bolts, that retain the mounting bracket to the top yoke. Remove also the two collars and rubber spacers. Disconnect the indicator leads. The mounting bracket can now be lifted clear of the steering head, disengaging the control cables and wiring leads from the guides on the bracket.

- 6 Slacken and remove the large chromium plated domed nut (crown nut) which retains the top yoke to the top of the steering head column. Holding the headlamp shell steady (CD models), lift off the upper yoke. If it proves stubborn, it may be tapped free using a soft-headed mallet. The headlamp shell (CD models) can be lifted off the lower yoke complete with the two fork shroud/headlamp brackets. Note that it will be necessary to release the front indicator lamp leads from the inside of the headlamp shell.

- 7 Before proceeding further, it should be noted that the steering head bearings are of the uncaged ball type and these will drop free as the steering head and lower yokes are removed. Some provision must, therefore, be made to catch them; a plastic box or lipped tray should suffice. It is also a good idea to position some clean rag under the steering head area to contain any errant steel balls.

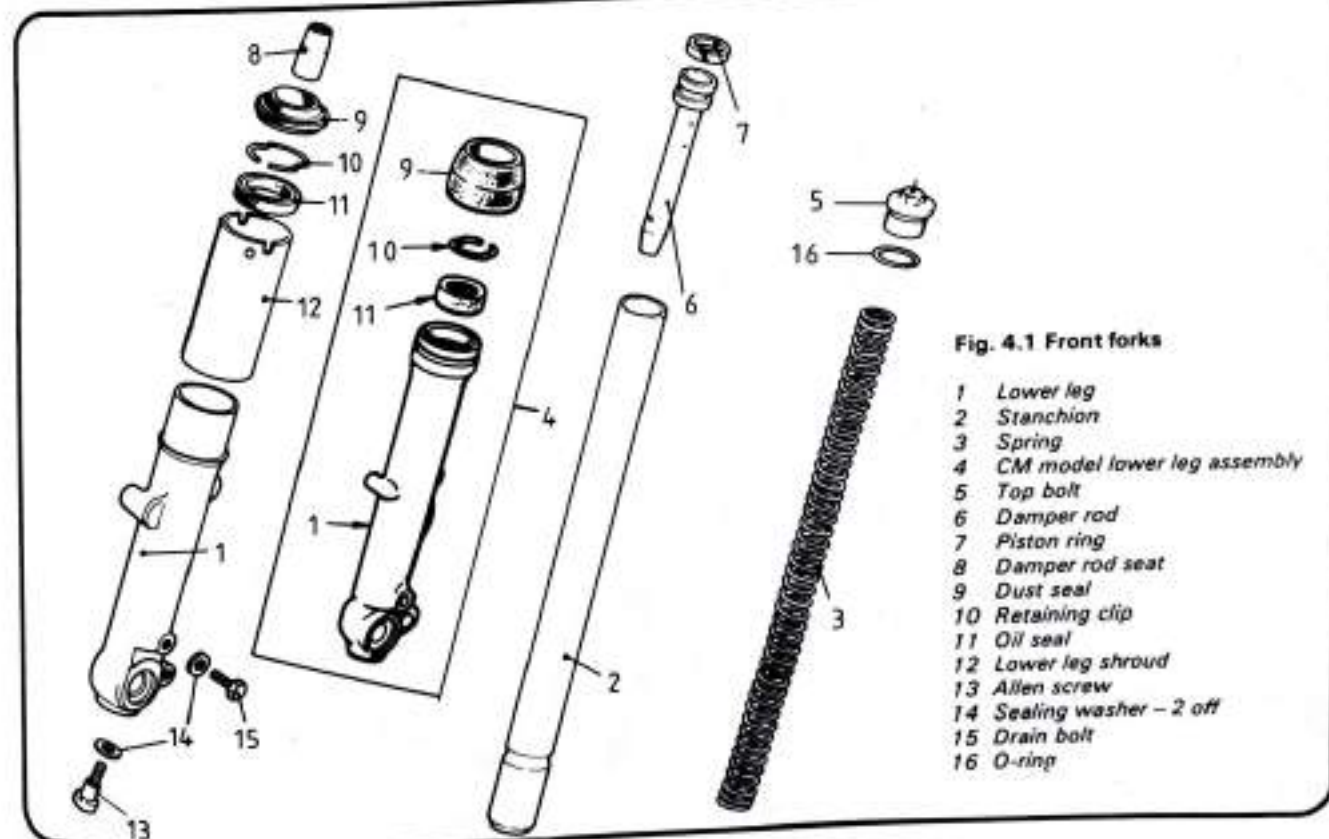


Fig. 4.1 Front forks

- 1 Lower leg
- 2 Stanchion
- 3 Spring
- 4 CM model lower leg assembly
- 5 Top bolt
- 6 Damper rod
- 7 Piston ring
- 8 Damper rod seat
- 9 Dust seal
- 10 Retaining clip
- 11 Oil seal
- 12 Lower leg shroud
- 13 Allen screw
- 14 Sealing washer - 2 off
- 15 Drain bolt
- 16 O-ring

- 8 To release the lower yoke and the steering head stem, unscrew the adjuster ring at the top with a suitable C-spanner. If such a spanner is not available, a brass drift can be used to loosen the ring. With the adjuster ring removed, carefully lower the yoke and steering stem, catching the balls from the lower race as they drop free. Make sure none of the balls are left clinging to the lower race; there should be 21 steel balls in the race. The bearings in the upper race will almost certainly stay in place. These can be dislodged once the lower yoke is out of the way. There should also be 21 steel balls in the upper race.
- 9 The steering head assembly should be reassembled in the reverse order of that given for dismantling. Before reassembly takes place, refer to the following Section.

5 Steering head bearings: examination and renovation

- 1 Before reassembly of the forks is commenced, examine the steering head races. The ball bearing tracks of the respective cup and cone bearings should be polished and free from indentations or cracks. If wear or damage is evident, the cups

and cones must be renewed as a complete set. In order for the straight line steering on any motorcycle to be consistently good, the steering head bearings must be absolutely perfect. Even the smallest amount of wear on the cups and cones may cause steering wobble at high speeds and judder during heavy front wheel braking. The cups and cones are an interference fit on their respective seatings and can be tapped from position with a suitable drift.

2 Note that when each head race bearing has its full complement of ball bearings they are not packed tightly and that sufficient room is left to accommodate one extra ball. This spacing is essential because if the bearings are packed tightly against each other, they will skid rather than roll, thus greatly accelerating the rate of wear. There is a total number of 42 ball bearings, in the head race assembly, each of $\frac{3}{4}$ in diameter, and with 21 being fitted in each race.

3 Ball bearings are cheap; if the originals are marked or discoloured, they should be renewed. To hold the steel balls in position during reassembly, pack the bearings with grease.

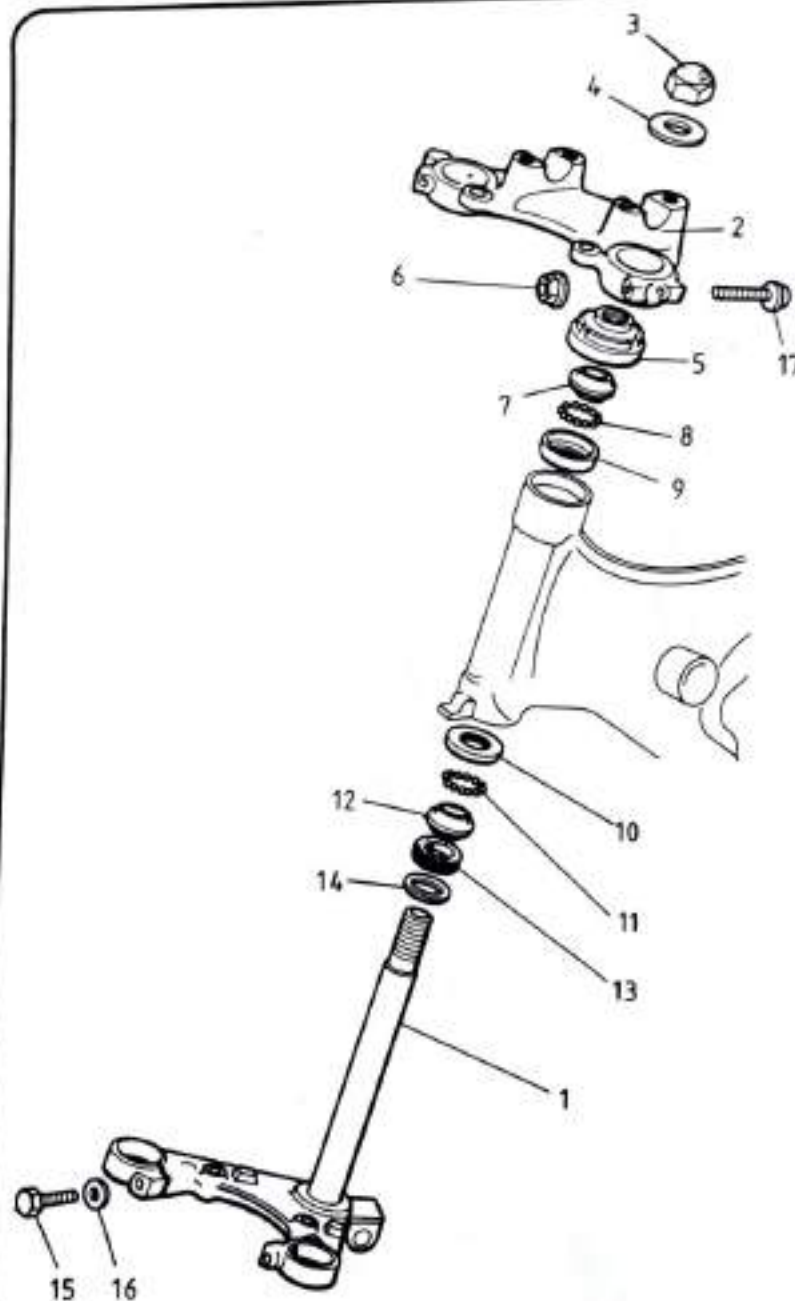


Fig. 4.2 Steering head assembly

- 1 Steering stem
- 2 Upper yoke
- 3 Crown nut
- 4 Washer
- 5 Bearing adjusting ring
- 6 Nut - 2 off
- 7 Upper bearing cone
- 8 Steel ball - 21 off
- 9 Upper bearing cup
- 10 Lower bearing cup
- 11 Steel ball - 21 off
- 12 Lower bearing cone
- 13 Dust seal
- 14 Washer
- 15 Pinch bolt - 2 off
- 16 Washer - 2 off
- 17 Pinch bolt - 2 off

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6 Front fork legs: dismantling

- 1 It is advisable to dismantle each fork leg separately, using an identical procedure. There is less chance of unwittingly exchanging parts if this approach is adopted. With the forks removed, as described in Section 2, unscrew fully and remove the large fork leg top bolt. Note the O-ring on the bolt. Pull out the long fork spring. Invert the legs over a suitable drain tray or bowl, and allow the oil to drain. This process can be accelerated by pumping the oil out by telescoping the legs.
- 2 From the fork legs of the CM models, prise up the rubber dust seals and slide them off the stanchions. This will reveal the oil seals at the top of the lower legs and the wire retaining clips.
- 3 From the fork legs of the CD models, the chromed lower leg shroud must be removed prior to gaining access to the dust seal etc. Due to the somewhat delicate construction of these shrouds, care must be taken when removing them. Use a length of hardwood, preferably with a wedge shaped or tapered end, and a hammer, to tap off the shrouds as shown in the accompanying photograph. Do not use a metal implement, such as a screwdriver, against the edge of the shroud as it will almost certainly result in damage to the chromed finish, if not the shroud itself. With the shrouds removed, prise out the dust

seals.

4 On all models, the oil seal can be removed after the fork stanchion and damper assembly have been withdrawn.

5 Clamp the fork lower leg in a vice fitted with soft jaws, or wrap a length of rubber inner tube or heavy cloth, around the leg to prevent damage. Unscrew the socket head screw, recessed into the housing which carries the front wheel spindle. The fork stanchion (upper tube) can now be drawn out of the lower leg. Invert the stanchion and allow the damper rod and its seat to drop out. Note the piston ring fitted to the top of the damper rod.

6 The fork seal is retained inside the top of the lower leg by a wire clip. Remove the clip and lever the old seal out of position with a screwdriver, taking care not to mark the surrounding surface. The oil seal may need a considerable amount of persuasion to come out of the fork leg, particularly if the seal is damaged. If the screwdriver proves to be ineffective, try using a tyre lever, preferably with the levering end somewhat curled to act as a drift. The oil seal must only be disturbed if it is worn or damaged, indicated by oil leakage, as the seal will almost certainly be damaged when being prised from position. It is considered good practice to renew the seals whenever the forks are dismantled.



6.1a Remove top bolt from fork leg; note O-ring



6.1b Withdraw the long fork spring from the stanchion



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6.3 Remove the fork shrouds (CD models) using a 'soft' drift as shown



6.5a Remove the socket head bolt; note copper sealing washer



6.5b Withdraw the damper rod assembly; note the piston ring



6.6 Oil seal can be prised out with wire retaining clip removed

7 Front forks: examination, renovation and reassembly

1 The front forks are not provided with bushes. The fork legs slide directly against the hard chrome surface of the stanchions and if wear occurs, the stanchion and/or the lower fork will have to be renewed. Wear normally occurs only after a very considerable mileage has been covered and can be detected by a juddering sensation when the front brake is applied. A slack steering head assembly will give the same effect, so this should always be checked first and adjusted if necessary, before condemning the forks. If excessive wear is suspected, refer to the service limits for the internal diameter of the lower leg and the outside diameter of the fork stanchion as given in the Specifications section at the front of this Chapter.

2 Wear is often visually apparent in the form of scuffing or break-through of the chrome surface of the fork stanchions. Check the stanchions in particular for scoring over the length which enters the oil seal. Bad scoring here will damage the oil seal and lead to fluid leakage. If evidence of damage of this nature is apparent, the stanchion in question must be renewed. In extreme cases the fork lower leg will have to be renewed as well.

3 If damping action is lost, the piston ring around the damper piston should be renewed; it is catalogued as a separate item. Check that the small holes in the damper tube are not blocked and if no substantial improvement is shown when the forks are reassembled and refilled, renew the complete damper assembly.

4 It is advisable to renew the oil seals when the forks are dismantled, even if they appear to be in good condition. This will save a strip down of the forks at a later date if oil leakage occurs. Use the procedure given in Section 6.6 of this Chapter to effect removal. When fitting the new oil seal, use a suitably sized socket as a drift to ensure that the seal is fitted squarely into its recess. Take care not to damage the seal when it is being fitted. Ensure the wire retaining clip is refitted securely. Before refitting the stanchion, lubricate the new seal face liberally with damping fluid (ATF).

5 Check that the dust excluder rubbers are not split or worn where they bear on the fork stanchions. A worn excluder will allow the ingress of dust and water which will damage the oil seal and eventually cause wear of the fork stanchions.

6 It is rarely possible to straighten forks which have been badly damaged in an accident, especially if the correct jigs are not available. It is always best to err on the side of safety and fit new replacements, especially since there is no easy means of checking to what extent forks have been overstressed. The fork

stanchions can be checked for straightness by rolling them on a flat surface. Any misalignment will immediately be obvious.

7 The fork springs will take a permanent set, that is, become somewhat compressed, after lengthy service. In this case, the fork action will become spongy, necessitating renewal of the springs. The correct spring lengths are given in the Specifications section of this Chapter. Note that the fork springs have closer coiled springing at one end. Upon refitting, the end with the closer pitches must be at the top of the stanchion. When renewing the fork springs, it should be remembered that they must always be renewed as a pair, not singly.

8 Reassemble the forks by reversing the dismantling procedure given in Section 6 of this Chapter.

8 Refitting the forks in the frame

1 If it has been necessary to remove the fork assembly completely from the frame, refitting is accomplished by following the dismantling procedure in reverse. Check that none of the balls are misplaced whilst the steering head stem is passed through the head set. It has been known for a ball to be displaced, drop down and wear a groove or even jam the steering, so be extremely careful in this respect.

2 Take particular care when adjusting the steering head bearings; the adjusting ring should be tightened down until it is a 'snug' fit against the top race and then slackened off approximately $\frac{1}{4}$ of a turn. In other words, the adjusting ring should be tight but not so tight as to damage the bearings by restricting movement. Final adjustment of the bearings must take place when the forks are fully reassembled. With the top yoke reinstalled, refit the large domed nut to the top of the steering stem, and partially tighten it down.

3 If the fork stanchions prove difficult to re-locate through the fork yokes, make sure their outer surfaces are clean and polished so that they will slide more easily, and apply some lubricant to the internal surfaces of the yoke 'eyes'. It is often advantageous to use a screwdriver blade to open up the clamps as the stanchions are pushed upwards into position. Ensure that the fork legs are fitted correctly, with the rim of each stanchion flush with the top of the yoke eyes. With the stanchions so placed, partially tighten the lower yoke pinch bolts.

4 Difficulty may also be experienced in raising the fork stanchions so that the end locates in the top yoke. If a stanchion puller is not available use a piece of wooden dowel with a tapered end, 'screw' it in the thread and use it to pull the stanchion into place.

5 The fork legs must be filled with the correct quantity and grade of damping fluid before the top bolts are refitted. Refer to the Specification section at the beginning of this Chapter for the relevant information on the damping fluid. Refilling may be done either during reassembly of the fork legs or after they have been installed in the yokes. In either case, check that the socket bolts and the copper sealing washers in the bottom of the lower legs have been refitted and tightened before refilling the legs with oil. Apply a little locking compound to the threads of the sealing bolts prior to their refitting. If the fork drain bolts were removed, they too must be reinstalled (not forgetting their copper washers) prior to refilling. With the forks refilled, check the condition of the O-rings on the top bolts and then refit the bolts. Tighten them to a torque setting of 2.5 – 3.0 kgf m (18.1 – 21.7 lbf ft).

6 When replacing the front wheel, first make sure that the speedometer drive pegs inside the front hub have not ridden up out of their location. After checking this, insert the wheel spindle (not forgetting the spacer on the right-hand side). Push through the spindle and fit the washer and the castellated nut. Tighten the nut and fit a new split pin. Refit the speedometer cable and its retaining screw.

7 Before final tightening of the pinch bolts and steering stem top nut, check the adjustment of the steering head bearings. If

they are too slack, judder will occur. There should be no detectable play in the head races when the handlebars are pulled and pushed with the front brake applied hard.

8 Overtight head races are equally undesirable. It is possible to unwittingly apply a loading of several tons on the head bearings by overtightening, even though the handlebars appear to turn quite freely. Overtight bearings will cause the machine to roll at low speeds and give generally imprecise handling, with a tendency to weave. Adjustment is correct if there is no perceptible play in the bearings and the handlebars will swing to full lock in either direction, with the front wheel clear of the ground. Only a slight tap should cause the handlebars to swing.

9 Make any adjustment required using the C-spanner applied to the slotted adjuster ring below the domed top nut. With adjustment correct, tighten the top nut to a torque setting of 6.0 – 7.0 kgf m (43.4 – 50.6 lbf ft).

10 When everything is in place, with the forks fitted loosely, bounce the forks a few times, so that the various components settle into their natural position. Finally tighten the lower pinch bolts and then the upper pinch bolts. Note that this operation must be done last, only after the front wheel has been replaced and the steering head bearings adjusted. The pinch bolts in both yokes should be tightened to a torque setting of 2.0 – 2.5 kgf m (14.5 – 18.1 lbf ft).



7.2 Damage of this type must be rectified before reassembly



7.3 Check piston ring for wear and damper rod holes for blockage



7.5 Check condition of dust excluders/seals; renew them if split or perished etc



7.8a Refit the damper rod to the stanchion



7.8b With shroud (CD models only) and dust seal/excluder refitted ...



7.8c ... refit stanchion to lower leg, not forgetting the damper rod seat



8.4 This hole must face rearwards when forks are refitted on CD models



8.5 Refill fork legs with correct quantity of fluid



8.6a Refit front wheel, ensuring lug on lower leg engages with slot on brake plate



8.6b Refit spindle, not forgetting the right-hand side spacer

9 Steering head lock

- 1 The steering head lock is fitted in the underside of the lower yoke of the forks, where it is secured by two crosshead screws. When in the locked position, a lug extends from the body of the lock when the forks are on full left-hand lock and abuts against a portion of the steering head lug.
- 2 If the lock malfunctions it must be renewed. A repair is impracticable. When the lock is changed a new key must of course be used to match the new lock.
- 3 To remove the lock, unscrew the two crosshead screws and pull away the lock complete. The retaining screws will almost certainly be tight and require the use of an impact driver to release them.

10 Frame assembly: examination and renovation

- 1 The frame is unlikely to require attention unless accident damage has occurred. In some cases, replacement of the frame is the only satisfactory course of action if it is badly out of alignment. Only a few frame repair specialists have the jigs and mandrels necessary for resetting the frame to the required standard of accuracy and even then there is no easy means of assessing to what extent the frame may have been overstressed. Repair work of this nature can prove expensive and it is always worthwhile checking whether a good replacement frame of identical type can be obtained through any form of Service Exchange Scheme, or through a reputable breaker's yard. Remember that in these cases, the necessary documentation must also be obtained with the replacement frame.
- 2 After the machine has covered a considerable mileage, it is advisable to examine the frame closely for signs of cracking or splitting at the welded joints. Rust can also cause weakness at

these joints. Minor damage can be repaired by welding or brazing, depending on the extent and nature of the damage.

3 If the machine is stripped for a complete overhaul, this affords a good opportunity to inspect the frame for the type of damage described above. Prime areas to check are the front downtube immediately behind the steering head and the top tube immediately behind the steering head; the two points where fractures are most likely to occur. The straightness of the tubes concerned will show whether the machine has been involved in a previous accident.



9.2 Steering lock must be renewed if it malfunctions

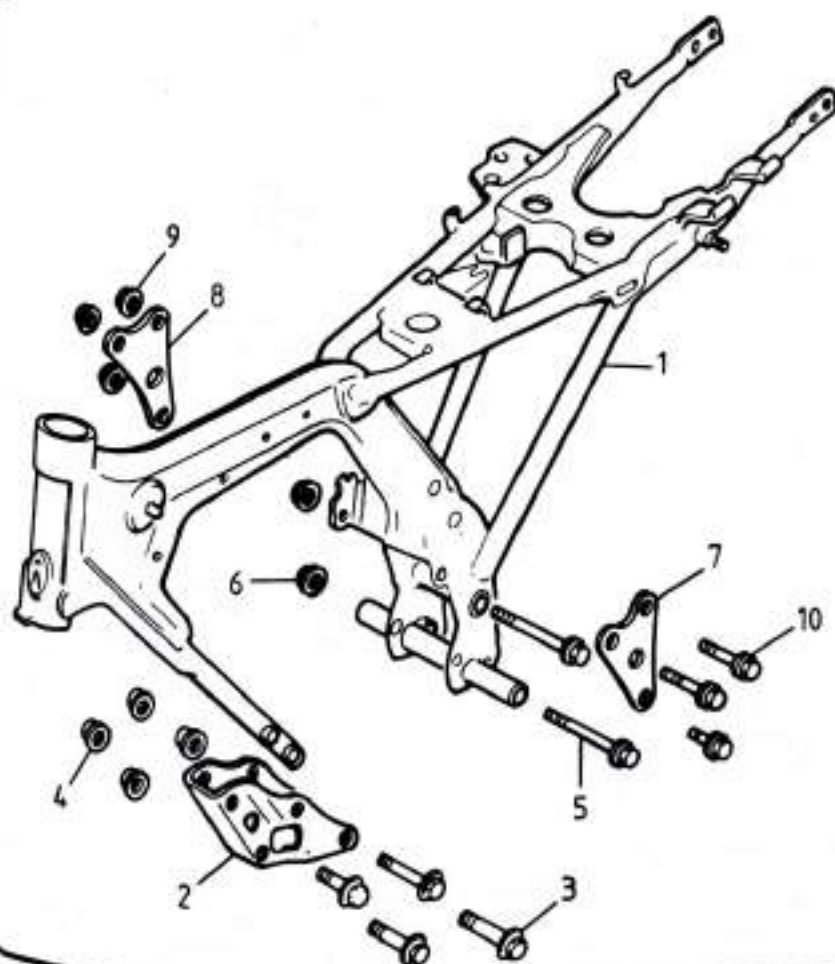


Fig. 4.3 Frame

- 1 Frame
- 2 Front engine mounting plate
- 3 Bolt - 4 off
- 4 Nut - 4 off
- 5 Bolt - 2 off
- 6 Nut - 2 off
- 7 Left-hand mounting plate
- 8 Right-hand mounting plate
- 9 Nut - 3 off
- 10 Bolt - 3 off

11 Swinging arm: examination and renovation

1 The rear fork assembly pivots on shouldered bushes (CM models) or bonded rubber bushes (CD models) pressed into the cross-member at the end of each fork arm, which are supported by a pivot shaft running through the frame down tube box-section. Worn swinging arm pivot bearings will give imprecise handling, with a tendency for the rear end of the machine to twitch or hop. The play can be detected by placing the machine on its centre stand (CD models) or on a suitable crate or wooden box (CM models), and with the rear wheel clear of the ground, pulling and pushing on the fork ends in a horizontal direction. Any play will be magnified by the leverage effect. In the UK, excess play will cause the machine to fail the DOT test.

2 To remove the rear swinging arm fork, first position the machine on the centre stand (CD models only). The CM models must be supported by an alternative method: place a plastic milk crate, or a crate of the type used by Moto-cross riders to support their machines, or alternatively, a stout wooden box, below the crankcases. In all cases, ensure the machine is resting firmly and securely.

3 It should be noted at this stage that whilst not strictly necessary, removal of the exhaust system will make removal of the swinging arm assembly easier. Detaching the two exhaust pipe/silencer units also removes the risk of damage to what are expensive items to replace.

4 From the CD models, remove the four retaining bolts and detach the two halves of the chaincase. From the CM models, remove the forward retaining bolt, washer and nut from the chainguard; the guard cannot yet be detached. On all models, remove the final drive chain by detaching the master link and running the chain off the rear wheel sprocket.

5 Detach the brake torque arm from the lug on the brake plate; it is secured by a retaining nut and split pin. Unscrew the adjuster nut on the brake operating arm and pull the rod through the trunnion on the brake arm. Replace the adjuster nut to avoid loss of the trunnion and brake rod spring.

6 Remove the split pin from the end of the wheel spindle and remove the castellated nut. Slacken and remove the large sleeve nut fitted on the CD models only. The wheel spindle can now be pulled out to the right. As the spindle is withdrawn, the spacer fitted between the brake plate and the right-hand fork leg will drop free. The rear wheel can now be lifted out of the frame.

7 Remove the nuts and washers from the top and bottom rear suspension mounting studs, and detach the two units. The chainguard on the CM models can now be removed; its rear end is located on the lower, left-hand rear suspension unit mounting stud.

8 Remove the locknut and washer from the end of the pivot shaft, which can then be tapped out from the right-hand side. Note that the lower ends of the silencer/pillion footrest mounting, sub-frame loops are secured by the pivot shaft. Working the swinging arm fork up and down will aid removal of the shaft. If the shaft resists removal it is almost certainly due to corrosion causing it to stick in the bushes. Corrosion of this type is more likely to occur in CD models, which utilise bonded rubber bushes in the swinging arm, and, as can be seen from the accompanying photograph, corrosion can reach an advanced stage even at relatively low mileages. If necessary drive out the swinging arm pivot shaft using a suitable drift. Care should be taken not to damage the threaded shaft end. After removal of the shaft the swinging arm fork can be pulled from position between the frame lugs and removed from the machine.

CD models

9 The CD models are fitted with a swinging arm fork mounted on two bonded rubber bushes which are an interference fit in the fork bosses. Wear in the bushes themselves is unlikely to occur until a large mileage has been covered, but wear between the bush inner sleeves and pivot shaft may occur if the shaft nut

has become loose. In normal service there should be no relative movement between the inner bushes and shaft; the fork movement is allowed by flexing of the bonded rubber.

10 Inspect the bush rubber for damage or separation from both inner and outer sleeves. Check also that the inner sleeves have not been moving on the shaft. If damage is evident the bushes must be renewed. Driving the bushes from position is unlikely to prove successful, particularly if they have been in position for a long time and corrosion has taken place. Removal is accomplished most easily using a fabricated puller as shown in the accompanying illustration. The puller sleeve should have an internal diameter slightly greater than the outside diameter of the bush outer sleeve. If possible use a high tensile nut and bolt because these will be better able to take the strain during use. New bushes may be drawn into place using the same puller. If difficulty is encountered in removing old bushes it is recommended that the swinging arm fork be returned to a Honda Service Agent, who can bring his expertise to bear on the problem.

11 After fitting new bushes the swinging arm fork may be refitted to the machine by reversing the dismantling procedure. No lubricant should be used on the bushes because they are made of rubber. The shaft should be lubricated to prevent corrosion between the shaft, bush inner bearings and frame members. A thick waterproof grease is recommended. On final assembly, ensure that the pivot shaft nut is tightened to a torque setting of 5.0 – 6.0 kgf m (36 – 43 lbf ft).

CM models

12 Remove the dust excluder caps from each end of the fork crossmember. Note the presence of the lipped sealing rings in the caps. Push out the two inner bushes.

13 Wash the inner and outer bushes carefully in petrol or another solvent. Do not remove the outer bushes from position in the fork crossmember unless they need renewal as they are made of a brittle material that will probably fracture while being drifted out. When drifting in new bushes, ensure that they enter their housing squarely. Use a soft wood or hard rubber pad between the bush and drive, to prevent chipping. Check the pivot shaft for straightness by rolling it on the edge of a dead flat surface. If the shaft is bent it must be renewed or straightened.

14 Reassemble the swinging arm fork by reversing the dismantling procedure. Grease the pivot shaft and bearings liberally before reassembly and check that the sealing rings in the dust caps are in good condition. Tighten the pivot shaft nut to a torque setting of 5.0 – 6.0 kgf m (36 – 43 lbf ft).

12 Rear suspension units: removal, examination and renovation

1 The swinging arm rear fork assembly is supported by two suspension units, of the hydraulically damped spring type. Each unit consists of a hydraulic damper, effective primarily on rebound, and a concentric, chromed spring. It is mounted by means of rubber-bushed lugs at top and bottom.

2 There is no means of draining the units or topping up, because the dampers are built as a sealed unit. If the damping falls or the units begin to leak, the complete damper assembly must be renewed. This applies equally if the damper rod has become bent. It is, however, possible to renew the springs independently of the sealed damper units. Removal of the spring however, does entail the use of a special jig in which the spring is tensioned. This jig, listed as a spring compressor (No. 07959-3290001) by Honda, also requires the use of a further Honda service item; the spring holding clamps (07967-11050100). Purchase of such items would be somewhat prohibitive to the average owner whose use of such equipment would be limited. It is suggested, therefore, that this job is entrusted to a reputable Honda Service Agent. Note that the units must always be dealt with as a pair, never individually.

Fig. 4.4 Swinging arm assembly – CD models

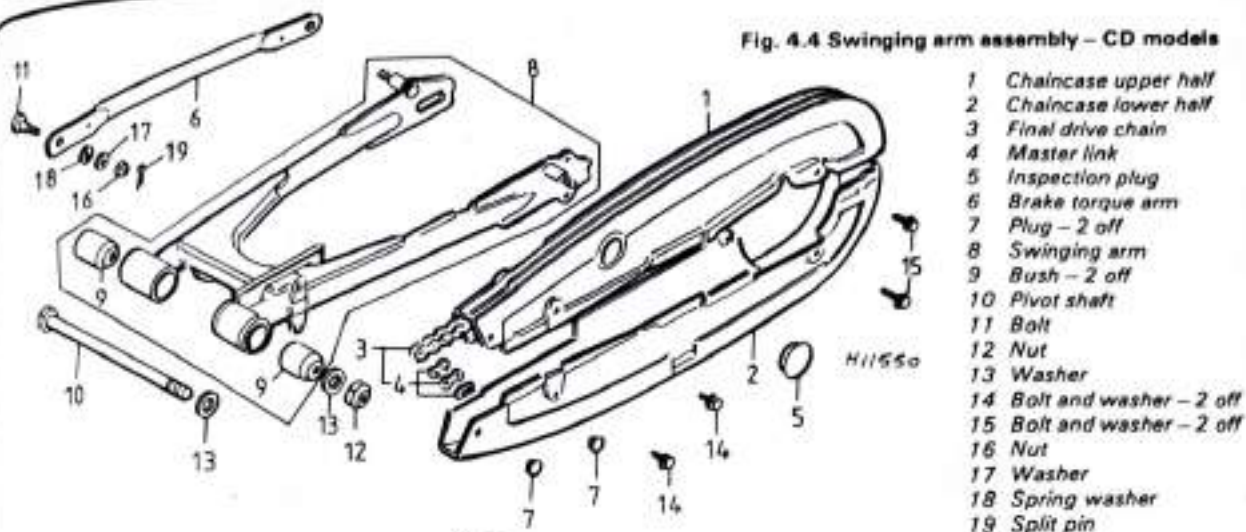
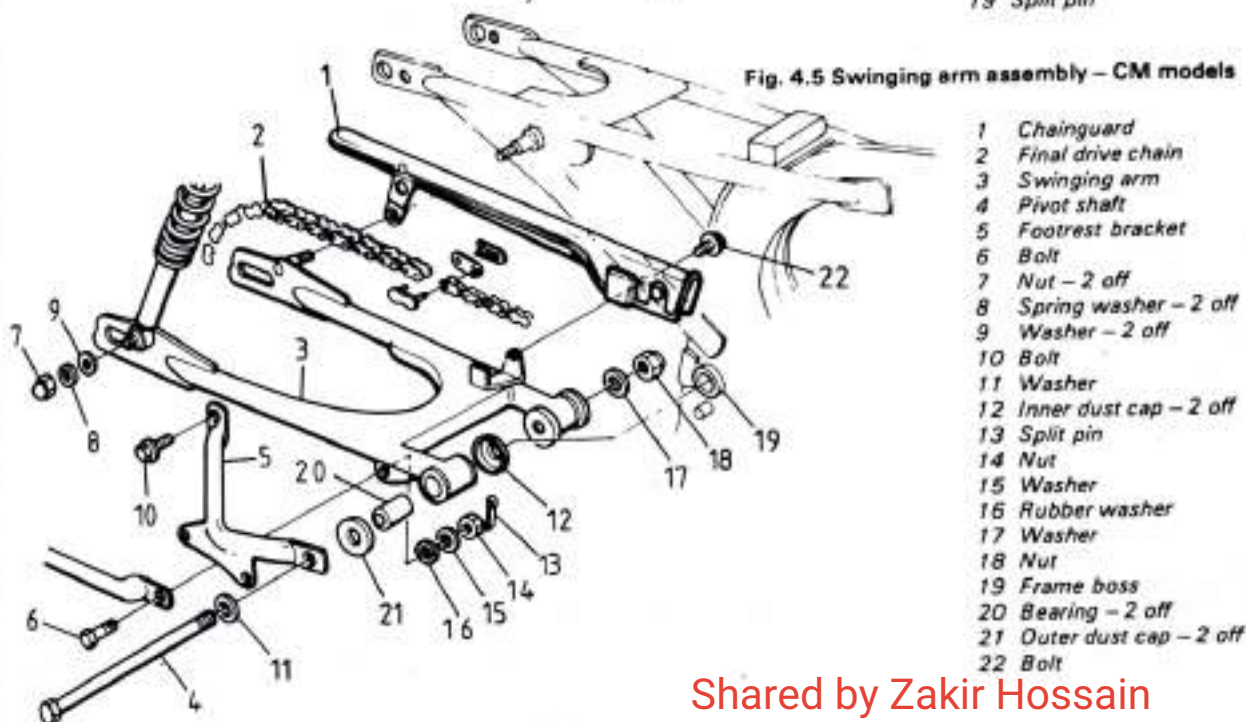
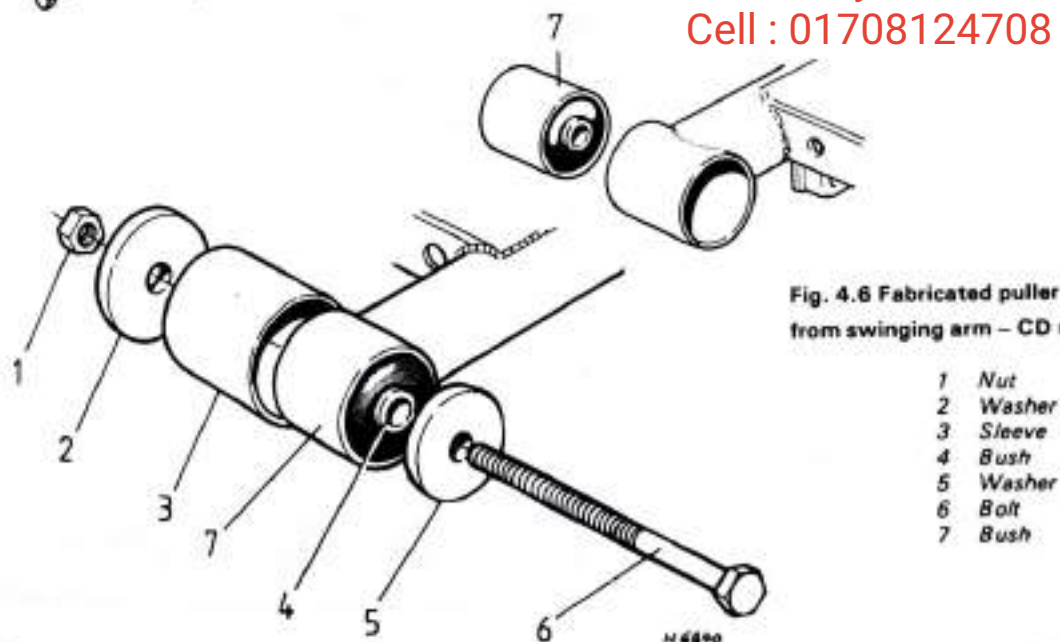


Fig. 4.5 Swinging arm assembly – CM models



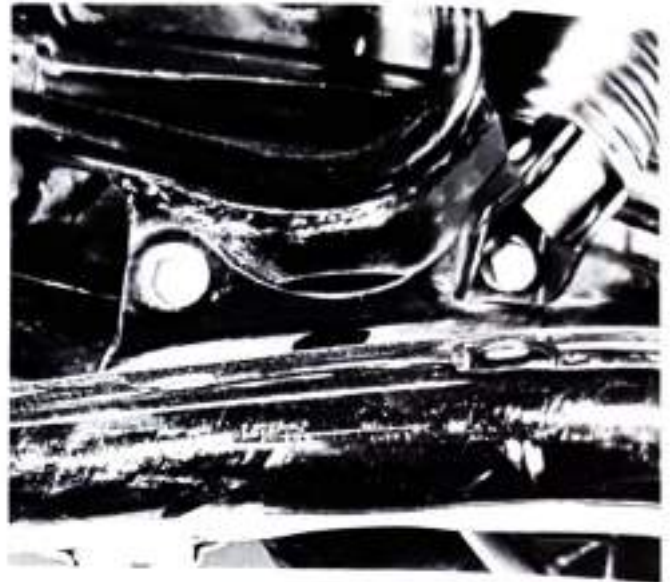
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Fig. 4.6 Fabricated puller for bush extraction from swinging arm – CD models





11.3a Remove two retaining nuts and remove the exhaust pipe from here and ...



11.3b ... remove two bolts to remove silencer from here, to facilitate swinging arm removal



11.4 Remove the chaincase from CD models



11.5 With the split pin withdrawn, remove torque arm nut



11.6 Remove brake plate and rear wheel



11.7 Rear suspension units are secured by nuts and washers

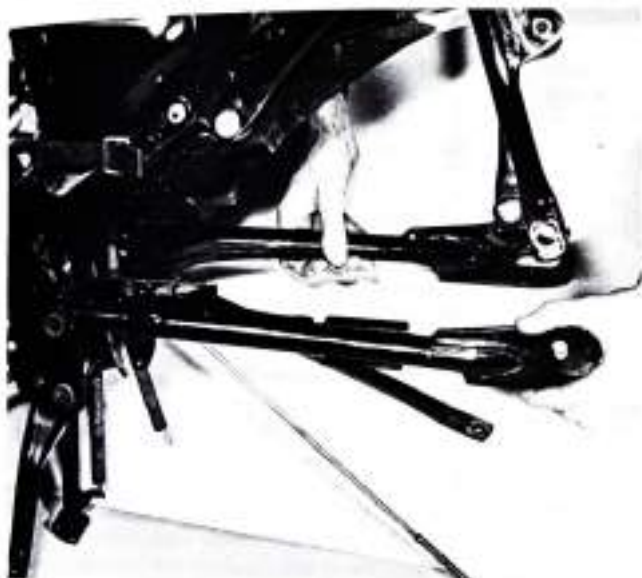
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11.8a Remove pivot shaft locknut and washer



11.8b Corrosion like this (due to lack of lubrication) will make pivot shaft removal difficult



11.8c Remove swinging arm from frame



11.10 Bushes must be drifted out if worn



11.11a Check that torque arm nut is tight and split pin is secure



11.11b Relocate sub-frame loop with swinging arm pivot shaft

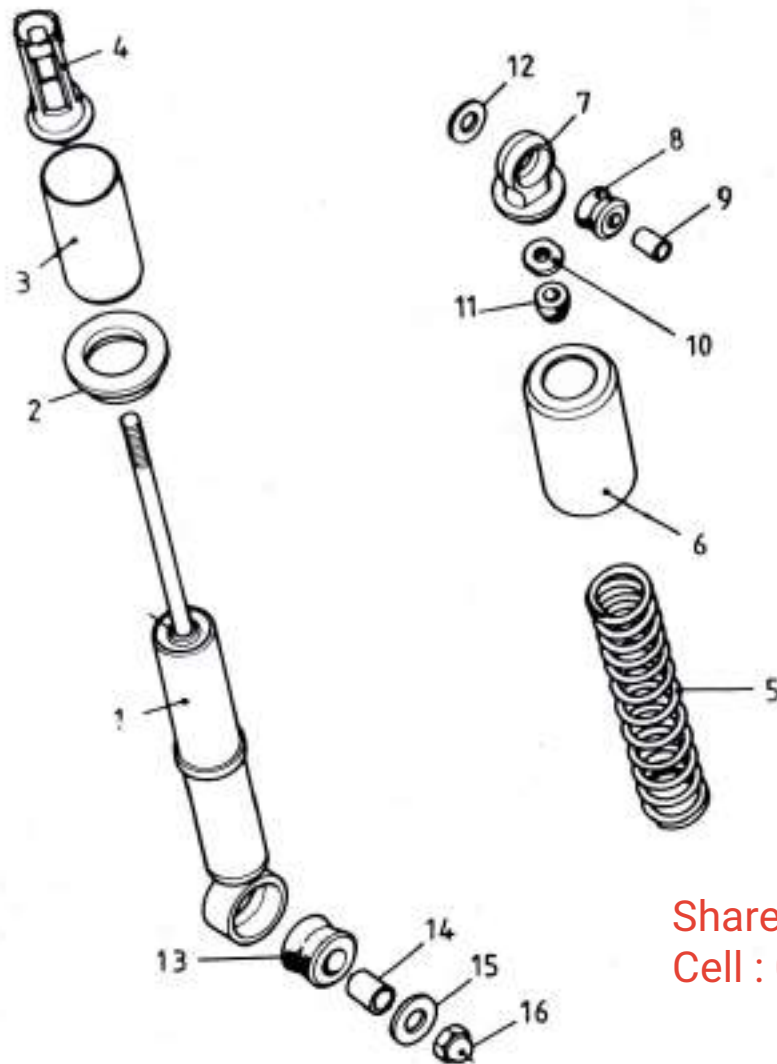


Fig. 4.7 Rear suspension unit – CD models

- 1 Hydraulic damper
- 2 Adjusting ring
- 3 Lower shroud
- 4 Spring guide
- 5 Spring
- 6 Upper shroud
- 7 Upper mounting lug
- 8 Rubber bush
- 9 Spacer
- 10 Nut
- 11 Rubber bump stop
- 12 Washer
- 13 Rubber bush
- 14 Spacer
- 15 Washer
- 16 Nut

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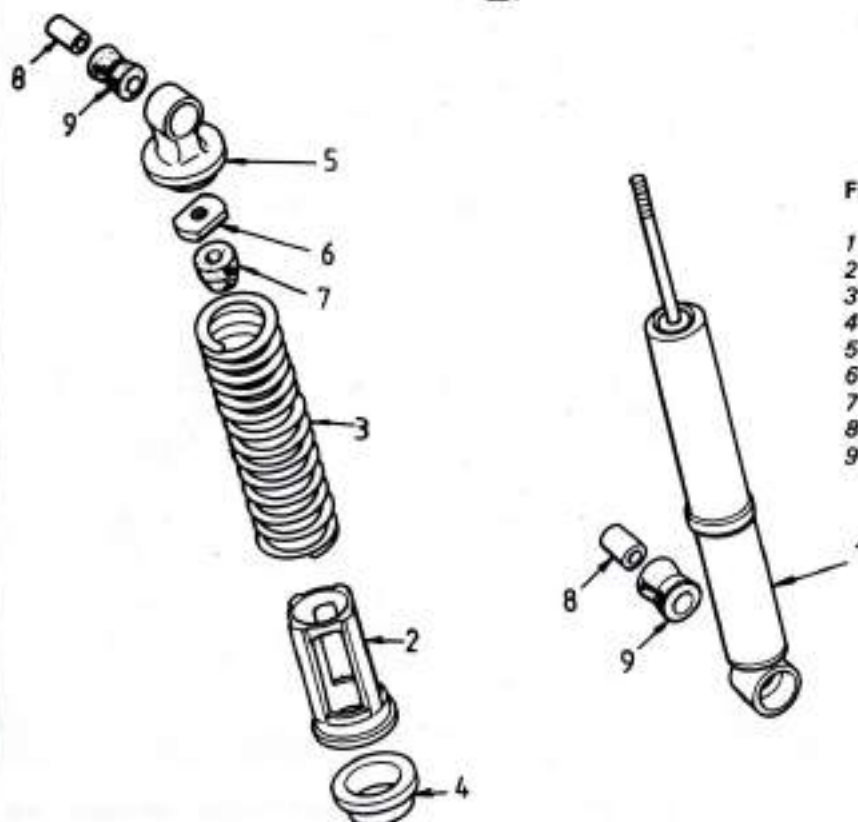


Fig. 4.8 Rear suspension unit – CM models

- 1 Hydraulic damper
- 2 Spring guide
- 3 Spring
- 4 Adjusting ring
- 5 Upper mounting lug
- 6 Nut
- 7 Rubber bump stop
- 8 Spacer – 2 off
- 9 Rubber bush – 2 off

13 Centre stand: examination – CD models only

- 1 The centre stand shares a common pivot pin with the rear brake pedal, the right-hand end of the pin being fitted with a split pin. A strong return spring, attached to a small lug on the left-hand side of the stand, retracts and holds the stand in position when not in use.
- 2 The pivot assemblies on centre stands are often neglected with regard to lubrication and this will eventually lead to wear. It is prudent to remove the pivot shaft from time to time and grease it thoroughly. This will prolong the effective life of the stand.
- 3 Be especially careful to check the condition and correct location of the return spring. If this fails, the stand will fall onto the road in use, and may unseat the rider if it catches in a drain cover or similar obstacle.

14 Side stand: examination

- 1 The side stand is attached to a plate welded to the main footrest mounting bar, on the left-hand side of the machine. Two long bolts secure the left-hand end of the main footrest mounting bar and the side stand, and then pass under the crankcases to locate with their nuts held captive on the right-hand side footrest bar mounting bracket.
- 2 Check that the extension spring is in good condition and not overstretched. An accident is almost certain if the stand extends whilst the machine is on the move. Like that of the centre stand, the propstand pivot should be lubricated occasionally to prevent wear. Check also, that the pivot bolt remains tight.

15 Footrests: examination and renovation

- 1 The rider's footrests form an assembly mounted below the engine, and can be detached as a complete unit along with the side stand which is retained to the left-hand end of the assembly. The assembly is retained by two short 8 mm bolts on the right-hand side which secure a small mounting bracket. From the left-hand side, two longer 8 mm bolts secure the side stand and then pass under the engine to locate with their nuts held captive on the footrest bar mounting bracket.
- 2 The footrests are secured to the mounting bar by a clevis pin and split pin arrangement. The footrests pivot upwards on their clevis pins, and are spring loaded, to keep them in their horizontal position. If an obstacle is struck, they will fold up, reducing the risk of damage to the footrest or to the main frame.
- 3 If the footrests are damaged in an accident, it is possible to dismantle the assembly into its component parts. Detach each footrest from the mounting bar by withdrawing the split pin and displacing the spring. Then remove the clevis pin. Slight bending can be rectified by stripping the assembly to the bare footrest bar, and straightening the bends by clamping the bar in a vice. A blowlamp should be applied to the affected area prior to bending to avoid setting up stresses in the material, which may lead to subsequent fracturing. Do not attempt to straighten the footrests while they are attached to the frame.
- 4 If heat is applied to the footrest piece during any straightening operation it follows that the footrest rubber must be removed temporarily.

16 Rear brake pedal: examination and renovation

- 1 The rear brake pedal is mounted on a spindle, which is also the pivot for the centre stand. It is retained, together with its return spring and the stop lamp switch spring, by a washer and split pin.
- 2 If the brake pedal is bent or twisted it can be removed after disconnecting the return spring and the brake rod, and withdrawing the split pin. The pedal should be straightened by

adopting the same method as recommended for bent footrests. It should be borne in mind that heating the pedal will almost certainly destroy the chrome plate with which the component is finished, therefore if the cosmetic appearance of the machine is important, the part should be renewed. If the brake pedal is severely distorted, err on the side of safety, and fit a replacement pedal; it is precisely when the brake is needed most and applied hard that the lever is most likely to fail.

- 3 The rear brake pedal is returned to its normal position by an extension spring. This should be checked to ensure that it is not stretched, and pulls the brake off cleanly.

17 Speedometer head: removal and replacement

- 1 The speedometer head on the 185 models cannot be removed from the machine individually; it must be removed as a complete unit with the warning light console. The speedometer/warning light console is secured by a single domed nut. To remove the console, unscrew and remove the nut, disconnect the speedometer cable by unscrewing the knurled ring and lift the unit from position, detaching the bulb holders from the base of the console as it is lifted clear.
- 2 The speedometer head on the 200 models may be removed from the machine as a separate item from the warning light console. The speedometer head is retained to the combined speedometer/warning light console mounting bracket, by two domed nuts. To remove the speedometer head, simply disconnect the drive cable, by unscrewing the knurled retainer ring, and remove the two nuts. As the speedometer is lifted up from the mounting bracket, detach the single bulb holder. The speedometer head can now be lifted clear.
- 3 Apart from defects in either the drive or drive cable, a speedometer which malfunctions is difficult to repair. Fit a replacement or alternatively entrust the repair to a competent instrument repair specialist. A further alternative is to obtain a speedometer from a crash repair specialist who may have perfectly good instruments with scratched cases and glass etc.
- 4 Remember that a speedometer in correct working order is a statutory requirement in the UK. Apart from this legal necessity, reference to the odometer reading is the most satisfactory means of keeping pace with the maintenance schedules.

18 Speedometer drive cable: examination and maintenance

- 1 It is advisable to detach the speedometer drive cable from time to time in order to check whether it is adequately lubricated and whether the outer cable is compressed or damaged at any point along its run. A jerky or sluggish movement at the instrument head can often be attributed to a cable fault.
- 2 To grease the cable, uncouple both ends and withdraw the inner cable. After removing the old grease, clean with a petrol soaked rag and examine the cable for broken strands or other damage. On US models the cable may be lubricated with oil using the oiling procedure given for control cables in the Routine Maintenance Section.
- 3 Regrease the cable with high melting point grease, taking care not to grease the last six inches closest to the instrument head. If this precaution is not observed, grease will work into the instrument and immobilise the sensitive movement.
- 4 If the speedometer cable breaks on one of the models imported into the UK (the CD185T/200T and the CM200T) it is possible to renew the inner cable alone, provided the outer cable is not damaged or compressed at any point along its run. On the CM185T and CM200T models (as sold in the US), the complete cable must be renewed if it has failed. Before inserting the new inner cable, it should be greased in accordance with the instructions given in the preceding paragraph. Try to avoid tight bends in the run of the cable because this will accelerate wear and make the instrument movement sluggish.



16.1 Rear brake pedal retained by split pin



18.1 Speedometer drive cable retained by knurled ring; instrument head held by two nuts

19 Speedometer drive: location and examination

1 The speedometer drive gearbox is fitted on the left-hand side of the front wheel hub. The drive rarely gives trouble provided it is kept properly lubricated. Lubrication should take place whenever the front wheel is removed for wheel bearing inspection or replacement.

20 Dualseat: removal

1 The dualseat is fixed to the rear of the machine by a bracket below the seat bolted to the rear part of the frame assembly. It is not positively located at the front; it merely rests on a rubber block.

2 To remove the seat, remove the two 8 mm retaining bolts. Note that on all models except the CM185T, the left-hand bolt also serves to retain the rearward end of the grab handle/lifting handle. On the CM185T, the rearward left-hand bolt also serves to retain the passenger grab rail. With the two bolts removed, hold the seat firmly at the rear, pull gently and the seat will then become detached. Replace by reversing this procedure.

21 Cleaning the machine

1 After removing all surface dirt with a rag or sponge which is washed frequently in clean water, the machine should be allowed to dry thoroughly. Application of car polish or wax to the cycle parts will give a good finish, particularly if the machine received this attention at regular intervals.

2 The plated parts should require only a wipe with a damp rag, but if they are badly corroded, as may occur during the winter when the roads are salted, it is permissible to use one of the proprietary chrome cleaners. These often have an oily base which will help prevent corrosion from recurring.

3 If the engine parts are particularly oily, use a cleaning compound such as Gunk or Jizer. Apply the compound whilst the parts are dry and work it in with a brush so that it has an opportunity to penetrate and soak into the film of oil and grease. Finish off by washing down liberally, taking care that water does not enter the carburetors, air cleaners or the electrics.

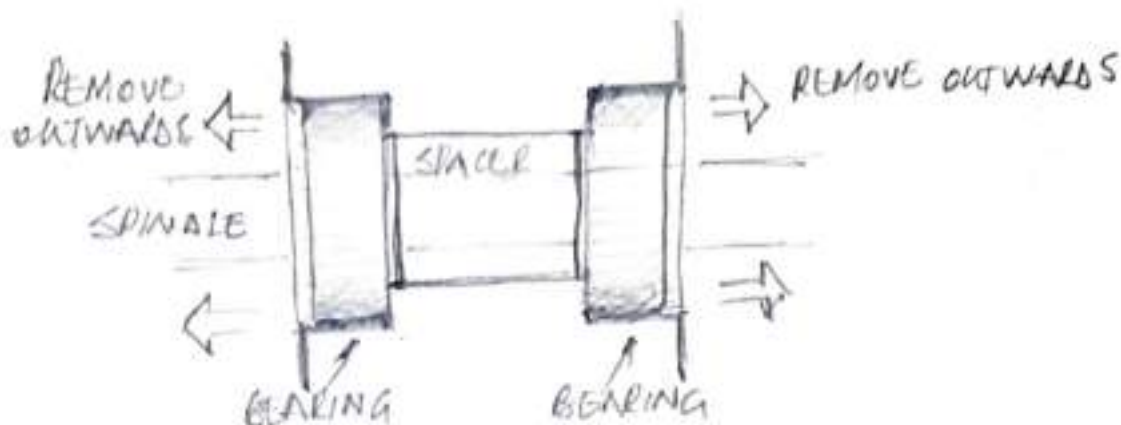
4 If possible, the machine should be wiped down immediately after it has been used in the wet, so that it is not garaged under damp conditions that will promote rusting. On those models not equipped with a full enclosure chaincase, make sure the chain is wiped and re-oiled, to prevent water from entering the rollers and causing harshness with an accompanying rapid rate of wear. Remember there is less chance of water entering the control cables and causing stiffness if they are lubricated regularly as described in the Routine Maintenance section.



19.1 Speedometer drive gear should be kept greased

22 Fault diagnosis: frame and forks

Symptom	Cause	Remedy
Machine veers either to the left or the right with hands off handlebars	Bent frame Twisted forks Wheels out of alignment	Check, and renew. Check, and renew. Check and realign.
Machine rolls at low speed	Overtight steering head bearings	Slacken until adjustment is correct.
Machine judders when front brake is applied	Slack steering head bearings Worn fork stanchion or lower legs	Tighten, until adjustment is correct. Dismantle forks and renew worn components.
Machine pitches on uneven surfaces	Ineffective fork dampers Ineffective rear suspension units	Check oil content of front forks. Check whether units still have damping action.
Fork action stiff	Fork legs out of alignment (twisted in yokes)	Slacken yoke clamps, and fork top bolts. Pump fork several times then retighten from bottom upwards.
Machine wanders. Steering imprecise. Rear wheel tends to hop	Worn swinging arm pivot	Dismantle and renew bushes and pivot shaft.



Chapter 5 Wheels, brakes and tyres

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Specifications

Wheels

Type	Chromed steel rims and steel spokes	
Size	CD185/CD200	CM185/CM200
	17 in front, 17 in rear	17 in front, 16 in rear

Brakes

Front:	
Type	Single leading shoe 140 mm (5.5 in) drum
Brake lining thickness	4.0 mm (0.1575 in)
Service limit	2.0 mm (0.0787 in)
Brake drum ID	140 mm (5.51 in)
Service limit	141 mm (5.551 in)
Rear:	
Type	Single leading shoe 130 mm (5.1 in) drum
Brake lining thickness	4.0 mm (0.1575 in)
Service limit	2.0 mm (0.0787 in)
Brake drum ID	130 mm (5.118 in)
Service limit	131 mm (5.1575 in)

Tyres

Size:	CD185/CD200	CM185/CM200
Front	3.00 x 17-4PR	3.00 x 17-4PR
Rear	3.00 x 17-8PR	3.50 x 16-4PR

Tyre pressures (cold)

Up to 90 kg (200 lb) load:		
Front	25 psi (1.75 kg/cm ²)	24 psi (1.75 kg/cm ²)
Rear	32 psi (2.25 kg/cm ²)	28 psi (2.0 kg/cm ²)
Over 90 kg (200 lb) load:		
Front	25 psi (1.75 kg/cm ²)	24 psi (1.75 kg/cm ²)
Rear	40 psi (2.8 kg/cm ²)	32 psi (2.25 kg/cm ²)

Torque settings

Front spindle nut (14 mm)	4.0 - 5.0 kgf m (29.0 - 36.2 lbf ft)
Rear spindle nut (14 mm)	4.0 - 5.0 kgf m (29.0 - 36.2 lbf ft)
Rear spindle lock nut (22 mm)	6.0 - 7.0 kgf m (43.4 - 50.6 lbf ft)
Brake torque arm bolts (8 mm)	2.0 - 2.5 kgf m (14.5 - 18.1 lbf ft)

1 General description

All the models covered in this manual are fitted with 17 inch diameter front wheels. Both of the CD models are also fitted with 17 inch diameter rear wheels. The CM models both have 16 inch diameter rear wheels as standard equipment. All models utilise a traditional type of wheel; chromed steel rims laced to the hub by steel spokes.

All the models feature a traditional single leading shoe (sls) drum brake on each wheel.

The front wheel on all models carries a 3.00 inch section 4 ply rating tyre as standard equipment. The CD models utilise the same size tyre on the rear wheel, but of a block rather than a ribbed tread pattern. The tyres on the CM models are both of a block tread pattern. The rear fitment on the CM models is of 3.50 inch section and 4 ply rating.

2 Front wheel: examination and renovation

1 Place the machine on the centre stand (CD models) or place a suitable support, a plastic crate or stout wooden box, under the engine (CM models) so that the front wheel is raised clear of the ground. Spin the wheel and check the rim alignment. Small irregularities can be corrected by tightening the spokes in the affected area although a certain amount of experience is necessary to prevent over-correction. Any flats in the wheel rim will be evident at the same time. These are more difficult to remove and in most cases it will be necessary to have the wheel rebuilt on a new rim. Apart from the effect on stability, a flat will expose the tyre bead and walls to greater risk of damage if the machine is run with a deformed wheel.

2 Check for loose and broken spokes. Tapping the spokes is the best guide to tension. A loose spoke will produce a quite different sound and should be tightened by turning the nipple in an anti-clockwise direction. Always check for run out by spinning the wheel again. If the spokes have to be tightened by an excessive amount, it is advisable to remove the tyre and tube as detailed in Section 15 of this Chapter. This will enable the protruding ends of the spokes to be ground off, thus preventing them from chafing the inner tube and causing punctures.

3 Front wheel: removal all models

1 With the machine supported on the centre stand (CD models) or suitably supported on a plastic crate or a stout wooden box (CM models), ensure the machine is secure and raise the front wheel clear of the ground.

2 Detach the speedometer drive cable at the wheel by unscrewing the single crosshead retaining screw. If not previously removed, the screw may have to be released by the use of an impact driver.

3 Slacken and remove the brake cable adjusting nut and disconnect the cable from the trunnion in the operating arm. Refit the adjusting nut to the end of the cable to avoid loss.

4 Remove the split pin from the left-hand end of the wheel spindle, slacken and remove the wheel nut and washer, and withdraw the spindle from the right-hand side. Note the spacer fitted between the inside of the right-hand fork leg and the brake drum; the spacer will drop free as the wheel is lowered clear of the forks.

4 Front brake: examination and renovation

1 Remove the front wheel from the forks in the manner described in the preceding Section. With the wheel removed, the brake backplate assembly can be lifted away for examination.

2 Examine the drum surface for signs of scoring or oil contamination. Both of these conditions will impair braking

efficiency. Remove all traces of dust, preferably using a brass wire brush, taking care not to inhale any of it, as it is of an asbestos nature, and consequently dangerous. Remove oil or grease deposits, using a petrol soaked rag.

3 If deep scoring is evident, due to the linings having worn through to the shoe at some time, the drum must be skimmed on a lathe, or renewed. Whilst there are firms who will undertake to skim a drum whilst fitted to the wheel, it should be borne in mind that excessive skimming will change the radius of the drum in relation to the brake shoes, therefore reducing the friction area until extensive bedding in has taken place. Also full adjustment of the shoes may not be possible. If in doubt about this point, the advice of one of the specialist engineering firms who undertake this work should be sought.

4 If fork oil or grease from the wheel bearings has badly contaminated the linings, they should be renewed. There is no satisfactory way of degreasing the lining material, which in any case is relatively cheap to replace. It is false economy to try to cut corners with brake components; the whole safety of both machine and rider being dependent on their condition.

5 The linings are bonded to the shoes, and the shoes must be renewed complete with the new linings. Examine the linings to determine whether they are thin or unevenly worn; if this is the case, they should be renewed. The linings are approximately 4 mm (0.158 in) thick when new and should be renewed when reduced to the service limit of 2.0 mm (0.079 in).

6 To remove the brake shoes, first remove the brake operating arm and its return spring, from the splined end of the brake operating cam. Straighten out and remove the split pin, and remove the plain washer which it retains. The split pin and washer retain the brake shoes to the pivot post on the brake backplate. Using a stout pair of pliers, release the brake shoe return springs. The two brake shoes can now be lifted clear of the brake backplate. An alternative method of removal, more useful if the original shoes are to be refitted, is described below.

7 Remove the split pin and plain washer. Turn the brake operating arm so that the brake is in the fully on position. Fold the shoes together until the spring tension is relaxed, and then lift the shoes and springs off the brake backplate.

8 Fitting new shoes is a direct reversal of the procedure given in paragraph 6. Note that before refitting existing shoes, roughen the lining surface sufficiently to break the glaze which will have formed in use.

9 The brake operating cam can be displaced for greasing prior to reassembly, after removing the actuating arm. No further attention is normally required. Note the alignment marks on the cam splined end, and on the arm. These must align on reassembly. Note also the lining wear indicator which fits on the splined end of the cam prior to the operating arm. The internal splines of the wear indicator are interrupted at one point by a flat; this flat must engage with the cut-out in the splines on the cam end.

5 Front wheel bearings: examination and replacement

1 To gain access to the front wheel bearings it is first necessary to remove the front wheel (see Section 3 of this Chapter), the brake backplate and the speedometer drive gearbox. With the latter removed, the oil seal can be prised carefully out of position.

2 There are two bearings in the front wheel. If the wheel has any side play when fitted to the machine or any roughness, the wheel bearings need to be renewed.

3 Using a small flat ended drift, place it inside the hub against one of the wheel bearings and tap the bearing out of position. Remove the central spacer and then tap out the other bearing from the other side. Drive the bearings outwards in each case. Careful and even tapping will prevent the bearing 'tying' and damage to the races.

4 Remove all the old grease from the hub and bearings, giving the latter a final wash in petrol. When the bearings are clean lubricate them sparingly with a very light oil. Check the bearings

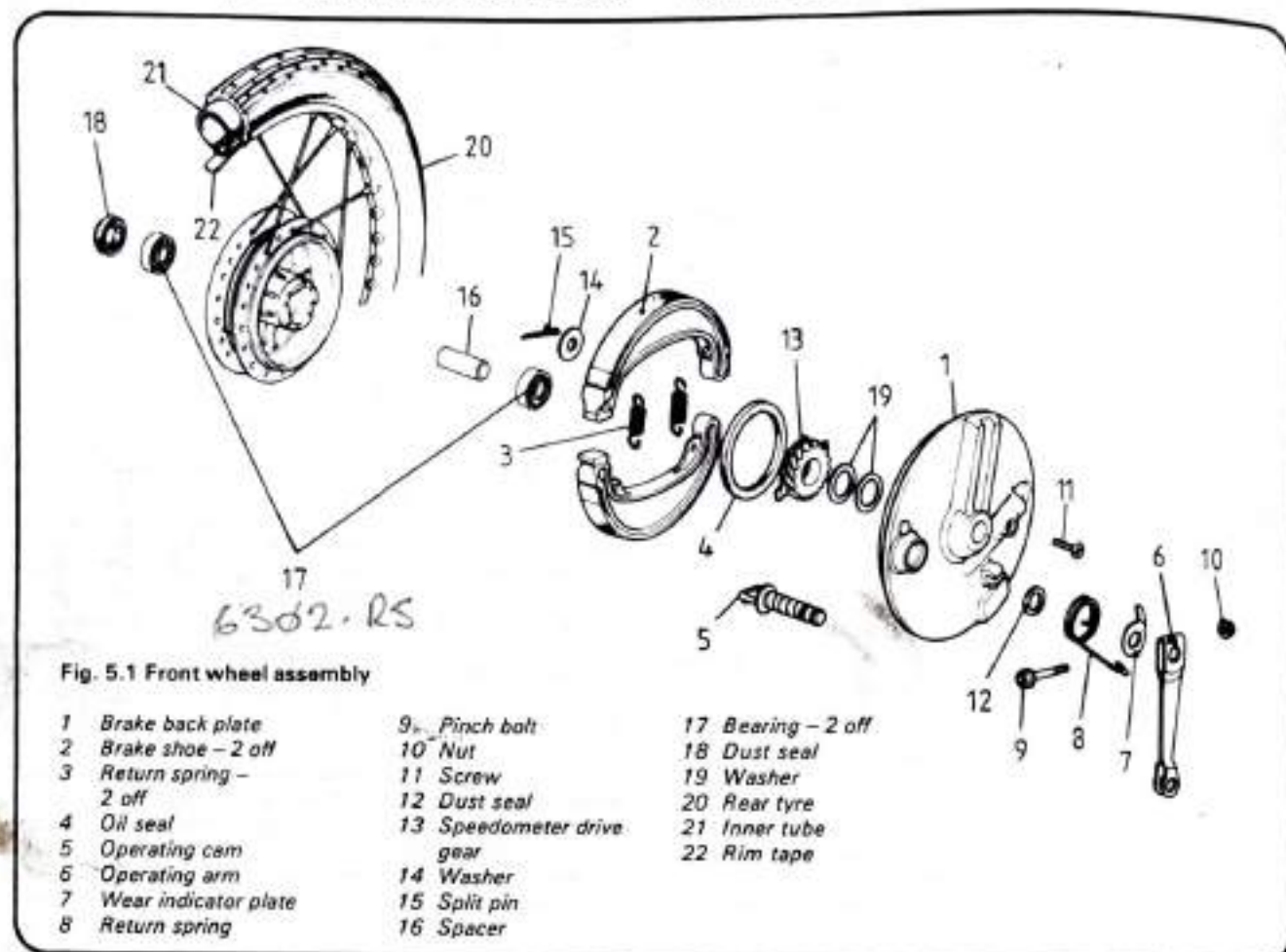
for play and roughness when they are spun by hand. All used bearings will emit a small amount of noise when spun but they should not chatter or sound rough. If there is any doubt about the condition of the bearings they should be renewed.

5 Before replacing the bearings pack them with high melting point grease. Do not overfill the hub centre with grease as it will expand when hot and may find its way past the oil seals. The hub space should be about 2/3 full of grease.

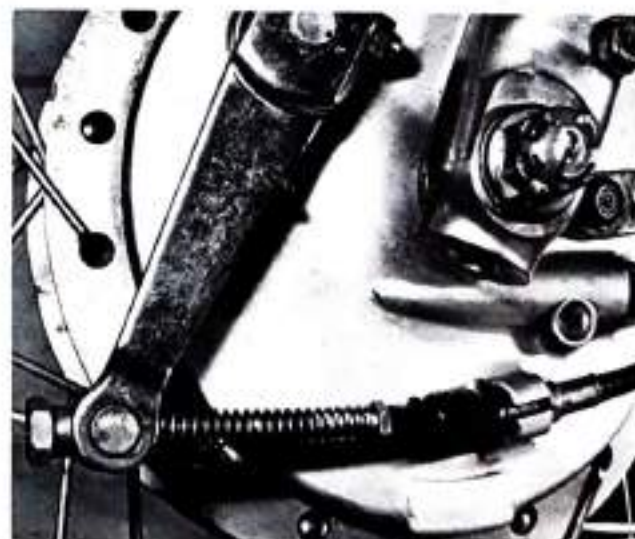
6 Tap in one bearing, turn the wheel over, insert the central

spacer and then tap in the other bearing. Use a soft drift on the outside ring of the bearing to drift them in. Do not drive the centre ring of the bearing or damage will be incurred. Take great care to ensure that the bearings enter the housings perfectly squarely without 'tying' or twisting.

7 Refit the oil seals carefully. Note that the seals must be renewed if damage is evident. Drift the seals into place with a thick walled tube of approximately the same dimension as the seal. A large socket spanner is ideal.



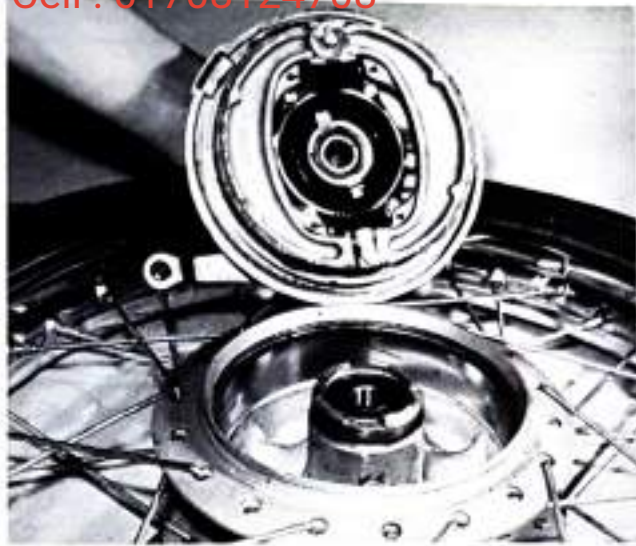
3.2 Remove retaining screw and detach speedometer cable



3.3 Disconnect brake cable with adjuster nut slackened and removed



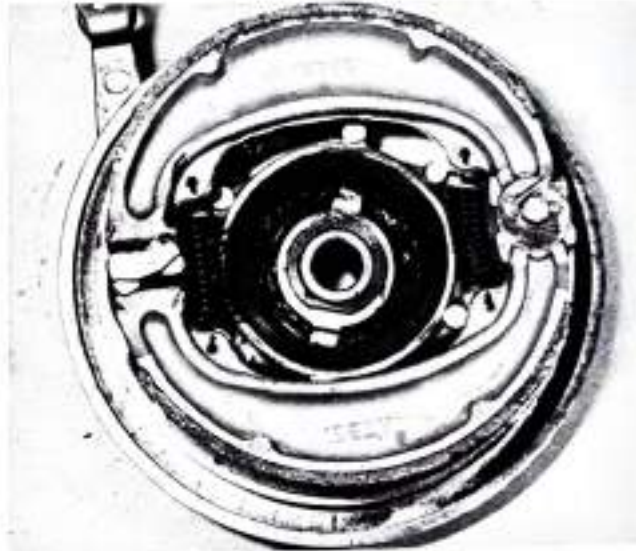
3.4 Remove split pin and castellated wheel spindle nut



4.1 Lift brake plate clear of wheel for examination etc



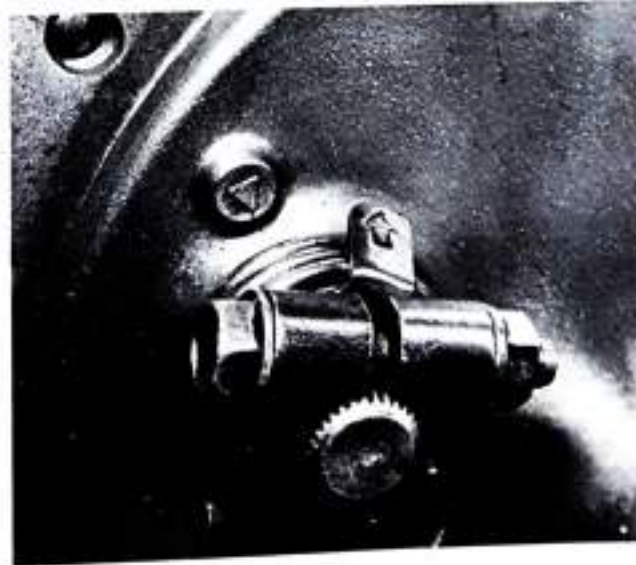
4.2 Clean brake drum thoroughly; note maximum diameter figure



4.4 Examine shoes for signs of contamination and/or excessive wear



4.9a Grease operating cam before refitting brake plate to wheel



4.9b Note punch marks on cam splined end and operating arm; also note wear indicator



5.3 Wheel bearings may be drifted from position if worn



5.7 Refit the oil seal ensuring it is not damaged

6 Front wheel: refitting

- 1 Locate the speedometer drive tabs with the slots in the hub.
- 2 Reinstall the brake backplate on the wheel and roll the wheel into position in the forks. Locate the spacer on the right-hand side, between the fork leg and the brake drum.
- 3 Hold the wheel in place and push the spindle through from the right-hand side. Push the spindle through until it is right home. Fit the plain washer and castellated nut, tightening the latter to a torque setting of 4.0 – 5.0 kgf m (28.9 – 36.2 lbf ft), and secure the assembly with a new split pin of the correct size.
- 4 Ensure the lug cast onto the inside of the left-hand fork lower leg engages with the groove in the brake backplate. If the lug and groove do not engage, the backplate will be free to rotate. Needless to say, the results of this occurring as the brake is applied would probably be disastrous. Refit the brake adjusting nut and adjust the brake operation as necessary.
- 5 Relocate the speedometer drive cable and secure it with the crosshead retaining screw.

7 Rear wheel: examination

- 1 Place the machine on the centre stand (CD models) or on a suitable plastic crate or stout wooden box (CM models), so that



5.6 Tap in new bearings carefully

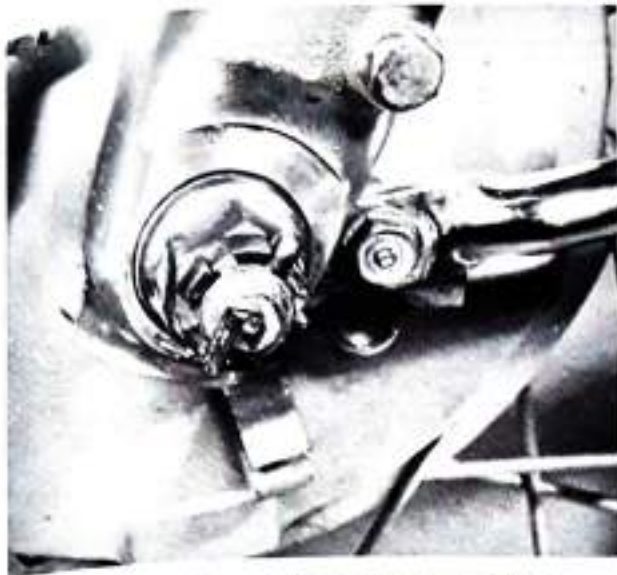
the rear wheel is clear of the ground. Follow the procedure described in Section 2 of this Chapter for checking the front wheel for alignment of the rim, loose or broken spokes or any other defects.

8 Rear wheel: removal

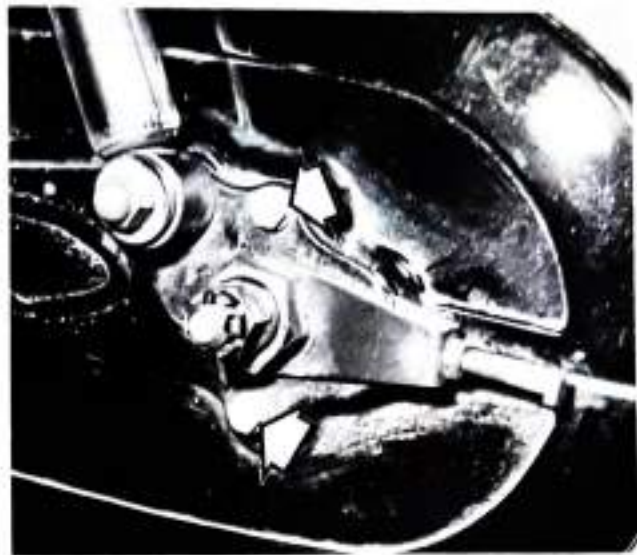
- 1 Place the machine on the centre stand (where fitted) or support the machine on a suitable crate or blocks so that the rear wheel is clear of the ground.
- 2 On CM models, separate the rear chain at the master link by displacing the spring with a screwdriver. Run the chain off the rear sprocket and refit the master link to one end of the chain to prevent its loss. The CD models are fitted with a QD wheel which allows wheel removal without any need to disturb the chain or the chaincase.
- 3 Unscrew the adjuster nut from the end of the brake operating rod and then depress the brake pedal so that the rod leaves the trunnion in the operating arm. Push out the trunnion and fit it, together with the adjuster nut, to the rod, to avoid loss. Detach the brake torque arm from the back plate by removing the securing split pin and unscrewing the nut.
- 4 Remove the split pin from the wheel spindle and unscrew the castellated wheel spindle nut. Support the weight of the wheel and push out the spindle. On CM models the wheel can now be eased backwards from between the fork ends. On CD models, displace the wheel spacer from between the right-hand side of the hub and the fork end. Ease the wheel over to the right to clear the cush drive unit, which remains attached to the left-hand fork end, and lift the wheel out towards the rear.
- 5 If attention to the chain, sprocket or cush drive unit is required on CD models proceed as follows. Detach first the upper half, and then the lower half of the full chain enclosure. Each section is secured by two bolts. Unscrew the large sleeve nut which fits on the cush drive hub spindle and lift the unit away from the fork end, still meshed with the chain. Disengage the sprocket from the chain. The chain can be separated, if required, by displacing the spring clip and master link.

9 Rear brake: examination and renovation

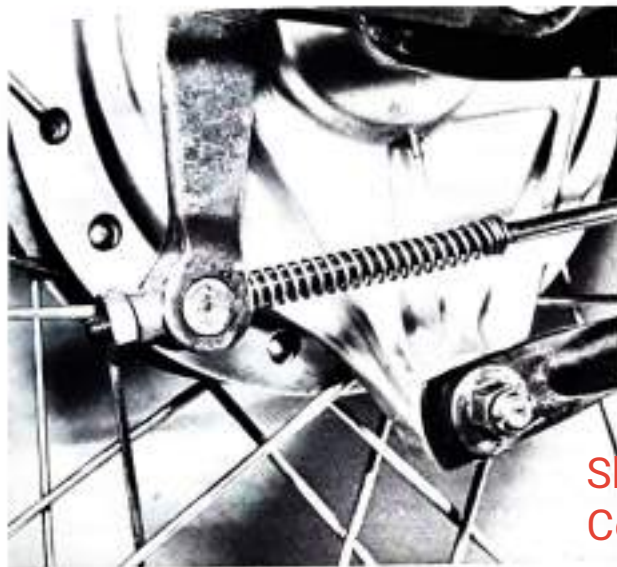
- 1 The procedure for examining and, if necessary, overhauling the rear drum brake is identical to that described for the front brake in Section 4 of this Chapter.
- 2 It should be noted that if attention to the brake plate only is required, the rear wheel may be left in situ, the brake cable and torque arm disconnected, and the brake plate lifted away separately.



6.3 Secure spindle with castellated nut and new split pin



6.2 Remove these two rear bolts, and two forward bolts, and remove chainguard (CD models)



8.3 Unscrew and remove the brake adjuster nut and detach the torque arm



8.4a Pull the spindle out towards the right-hand side



8.4b Displace wheel to right-hand side and remove; sprocket carrier will remain (CD models)



9.1a Check drum for corrosion and/or scoring; note maximum diameter figure

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9.1b Examine shoes carefully. Split pin and washer retain shoe ends to pivot post



9.1c Note alignment punch marks and wear indicator



9.2 Brake plate can be removed separately ...

10 Rear wheel bearings: examination and replacement

- 1 To gain access to the rear wheel bearings, the wheel must be removed from the machine and the brake back plate withdrawn from the brake drum.
- 2 Knock out the bearings, using the same procedure as described for the front wheel in Section 5 of this Chapter. The information contained therein with regard to bearing testing, regreasing and refitting apply similarly to the rear wheel. When refitting or renewing bearings, install the right-hand bearing first, then the central spacer and then the left-hand bearing.
- 3 Note that there is an additional outrigger bearing and oil seal contained in the sprocket carrier assembly on the CD models. The bearing should not be omitted when checking the rear bearings (see Section 11 of this Chapter).

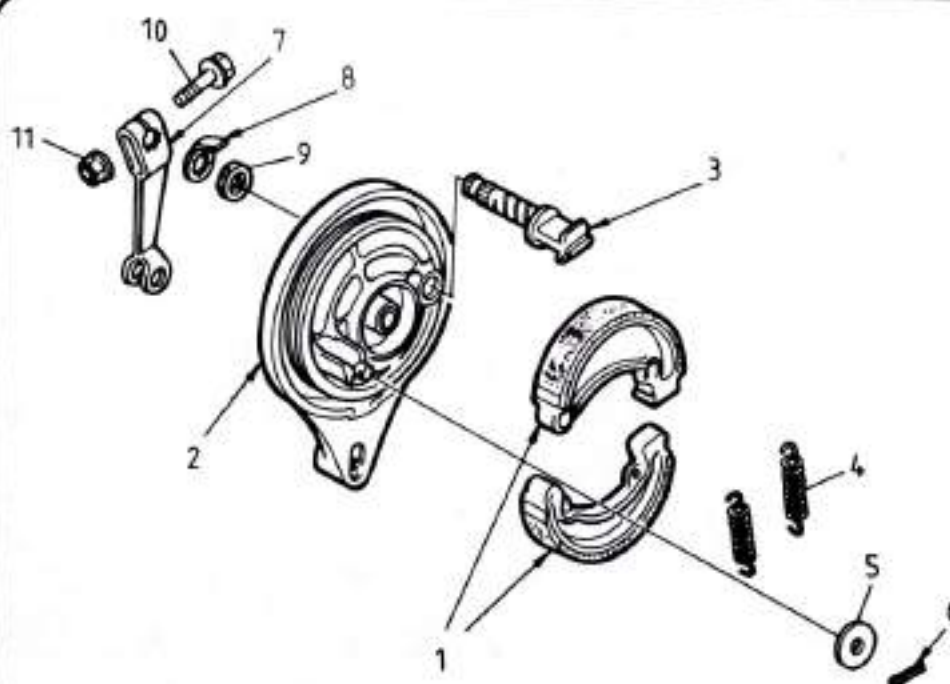


Fig. 5.2 Rear brake

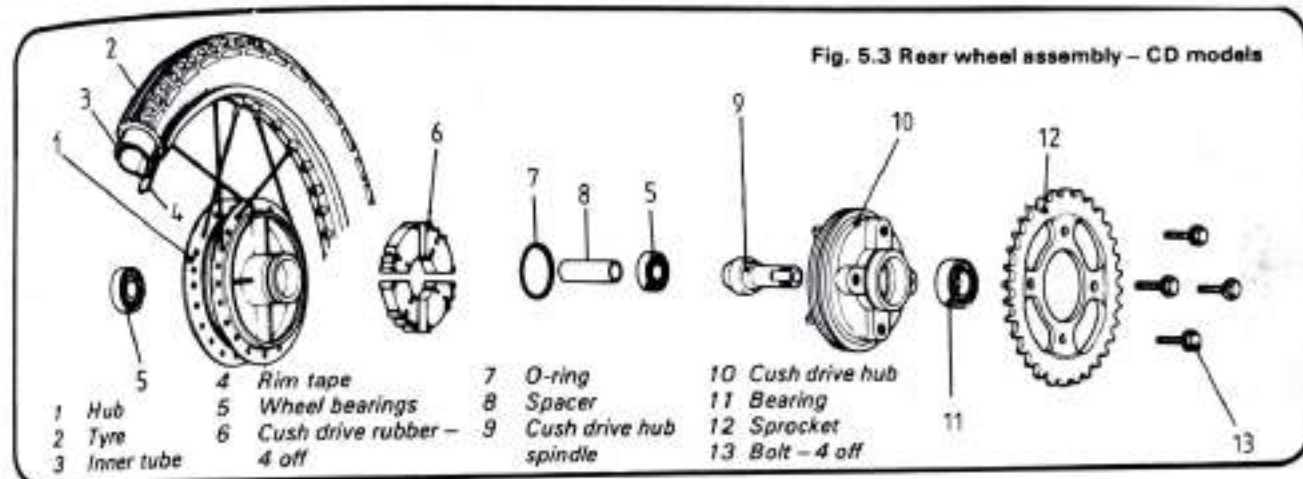
- 1 Brake shoe
- 2 Brake backplate
- 3 Operating cam
- 4 Spring - 2 off
- 5 Washer
- 6 Split pin
- 7 Operating arm
- 8 Wear indicator plate
- 9 Dust seal
- 10 Pinch bolt
- 11 Nut



10.2a Wheel bearings on both sides of hub ...



10.2b ... have integral oil seals



11 Rear wheel sprocket: removal, examination and replacement

1 The rear wheel sprocket assembly (on the CD models only) can be removed as a separate unit after the rear wheel has been separated and detached from the frame as described in Section 8 of this Chapter. Alternatively, it can be removed still attached to the rear wheel if the chain enclosure and final drive chain are detached. The sprocket on these models is secured to a separate sprocket carrier by four hexagon headed bolts. The sprocket carrier contains an outrigger bearing with an integral oil seal in its centre, and four cast vanes on its rear face. The vanes engage with four rubber blocks in the wheel hub to form the cush drive unit (see the following Section). Through the centre of the sprocket assembly is fitted a sleeve, threaded at its outer end, through which the wheel spindle passes. The sprocket assembly is secured by the large nut which fits on the outer end of the spindle sleeve. Detaching the sprocket assembly is therefore a simple operation, involving the removal of the sleeve nut followed by removal of the complete assembly towards the right and then rearwards.

2 The rear wheel sprocket on the CM models is retained in an altogether more simple way. The sprocket is retained on the left-hand side of the hub by a large circlip, and is located by four pins which pass into the cush drive in the hub and are retained by hexagonal nuts.

3 To remove the sprocket, detach the large circlip and pull the sprocket off the hub, together with the four drive pins. Note that

on older machines, it is possible that the pins may become jammed in place in the bonded bush centres, in which case, the four nuts may be removed to allow the sprocket to be detached. 4 On all models, examine the sprocket condition, having cleaned it thoroughly first. In particular, pay attention to the condition of the teeth.

5 The sprocket should be renewed if the teeth are hooked, chipped, broken or badly worn. It is considered bad practice to renew one sprocket on its own; both final drive sprockets should be renewed as a pair, preferably with a new final drive chain. If this recommendation is not observed, rapid wear resulting from the running of old and new parts together will necessitate even earlier replacement on the next occasion.

6 As noted in paragraph 1 of this Section, the CD models have an additional bearing located in the sprocket assembly which supports the sprocket sleeve through which the wheel spindle fits. In common with the wheel bearings, this bearing is of the ball journal type. When wear occurs in the bearing, the sprocket will give the appearance of being loose on its mounting bolts. The bearing is a tight press fit in the sprocket assembly centre and must be drifted from position in the same way as described for the wheel bearings in Section 5 of this Chapter. Wash out the bearing to remove all traces of old grease and then examine it carefully. If the bearing has any play or runs roughly, it must be renewed. If the bearing is not to be renewed, it should be repacked with high melting point grease and refitted in its housing. Ensure it is fitted with the integral oil seal facing outwards.

7 The sprocket may be refitted by reversing the dismantling

procedure. On the CM models it is important that the recesses in the rear of the sprocket are engaged correctly by the milled flats on each cush drive pin. If it was found that the sprocket retaining nuts had to be removed, they must be refitted and tightened to a torque setting of 5.5 – 6.5 kgf m (39.8 – 47.0 lbf ft).

8 Refit the rear wheel assembly (unless the cush drive unit is to receive attention) by reversing the method adopted for removal.

12 Rear wheel cush drive: examination and renovation

1 All models are fitted with a cush drive arrangement in the rear hub which allows a small amount of movement between the hub and the drive sprocket to help damp out transmission shock loads.

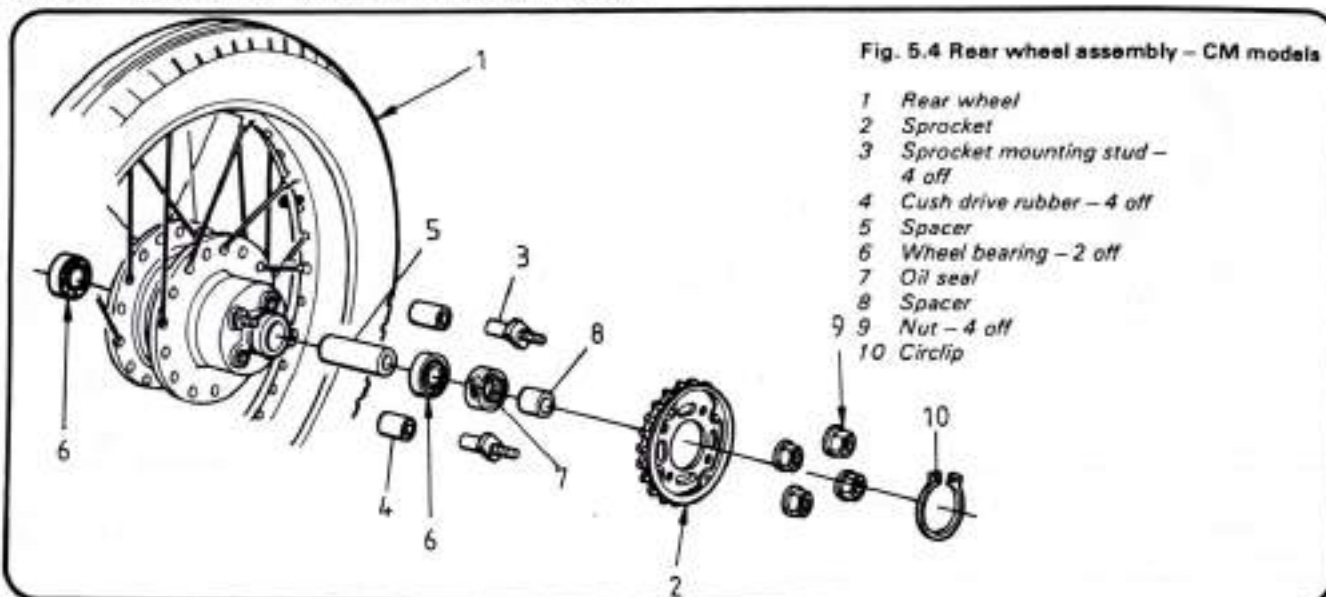
2 On the CD models, the cush drive unit comprises a set of synthetic rubber buffers housed within a series of vanes cast in the hub shell. The sprocket carrier on which the rear wheel sprocket is mounted, has four cast-in vanes which engage with slots in the rubbers when the wheel is in position in the frame.

3 On CM models the pins at the rear of the sprocket pass into individual bonded rubber bushes in the wheel hub. After

removal of the large circlip lift the sprocket away to gain access to the bushes. If the pins have become corroded in place the application of a releasing fluid may facilitate removal.

4 The rubber blocks of the CD models' cush drive arrangement should be examined for signs of damage or general deterioration. Pay particular attention to the blocks' condition if there has been harshness or roughness in the transmission, especially when pulling away from a standstill or opening and closing the throttle. Renew the rubbers as a set if there is any doubt about their condition; there should be no difficulty in removing or refitting them.

5 The bonded rubber bushes of the cush drive unit fitted to the CM models are an extremely tight fit in the hub. If they have become compacted or the bonded bush centres are damaged (as shown by the locating pins jamming in position), they should be renewed. It is extremely unlikely that this operation can be performed at home due to the tight fit on the bushes. In all probability, any attempt to dislodge them will result in the inner metal sleeve tearing out of the rubber, making subsequent removal difficult. For this reason it is suggested that the wheel should be taken to a Honda Service Agent who will have the equipment necessary to extract the old bushes, and fit the new ones.



11.1a Remove large sleeve nut and ...



11.1b ... displace the sprocket carrier assembly (CD models)



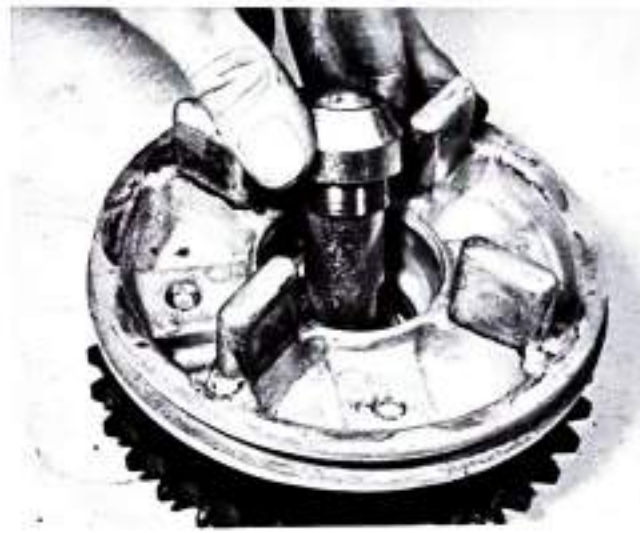
11.3 Examine sprocket carefully; this sprocket is in good condition. Note number of teeth (34) stamped on sprocket



11.5a Sprocket carrier bearing must be packed with grease



11.5b Bearing must be fitted as shown; integral oil seal outwards



11.6 Fit centre sleeve (CD models) as shown, upon reassembly



12.4 Check cush drive rubbers for wear (CD models)

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13 Rear wheel: refitting

- 1 Refitting of the rear wheel is accomplished by a reversal of the removal procedure given in Section 8 of this Chapter.
- 2 Do not omit the chain adjusters (if they were removed) when inserting the wheel spindle. If the chain was detached, it must be refitted and secured with the spring clip correctly positioned, that is, with the closed end facing the normal direction of travel. Check that the wheel is correctly aligned and chain tension is correct (see Section 14). Do not forget to secure the brake torque arm nut by means of the split pin. The wheel spindle nut should be tightened to a torque setting of 4.0 – 5.0 kgf m (28.9 – 36.2 lbf ft) and the sleeve nut, on the CD models only, to a setting of 6.0 – 7.0 kgf m (43.4 – 50.6 lbf ft). Secure the spindle nut with a new split pin.



13.2a Refit the torque arm and refit the nut and ...



13.2b ... secure it with a new split pin



13.2c Spring link must be fitted correctly as shown



13.2d Use a new split pin to secure castellated nut

14 Final drive chain: examination, adjustment and lubrication

1 The CD models are fitted with a full enclosure chaincase. Owners of these models should reap the benefits of such a device in the form of longer service from the chain and sprockets, less frequent lubrication and adjustment of the chain, and a cleaner rear wheel assembly. The final drive of the CM models is fully exposed, with only a light chainguard over the top run. If not properly maintained the chain will have a short life. A worn chain will cause rapid wear of the sprockets and they too will need replacement.

2 The chain tension will need adjustment at regular intervals, to compensate for wear. This is accomplished by firstly placing the machine on its centre stand (CD models only) or by supporting the machine on a suitable plastic crate or stout wooden box (CM models). Slacken the castellated wheel nut, after removing the split pin, and then slacken the large nut securing the sprocket assembly (CD models only).

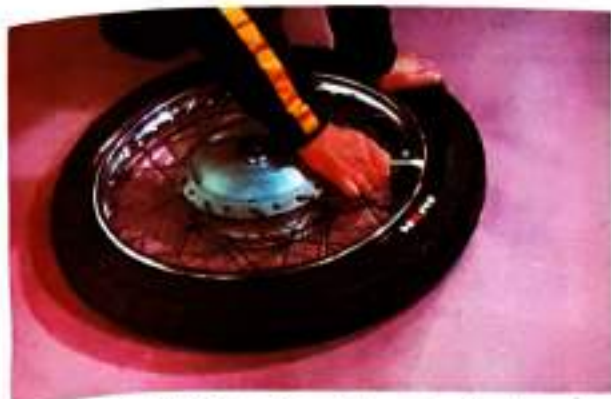
3 Slacken the locknuts on the chain adjusters on the fork ends. Screw the adjusters inwards an equal amount to tighten the chain. The tension is correct if there is 15–25 mm ($\frac{1}{2}$ in – 1 in) up and down movement in the centre of the lower chain

run (CM models) or when checked with a finger through the inspection aperture in the chaincase. Always check the chain when it is at its tightest point; a chain rarely wears evenly. This may be accomplished by turning the wheel whilst applying a finger to the lower chain run, or through the chaincase inspection aperture.

4 Always adjust the drawn bolts an even amount so that correct wheel alignment is preserved. The fork ends are marked with a series of vertical lines to provide a visual check. If desired, wheel alignment can be checked by running a plank of wood parallel to the machine so that it touches both walls of the rear tyre. If wheel alignment is correct, it should be equidistant from either side of the front wheel tyre when tested on both sides of the rear wheel; it will touch the front tyre if both tyres have the same cross-section. See the accompanying diagram.

5 Do not run the chain overtight to compensate for uneven wear. A tight chain will place excessive stresses on the gearbox and rear wheel bearings leading to their early failure. It will also absorb a surprising amount of power.

6 After a period of running the chain will provide lubrication. Lack of oil will accelerate the rate of wear of both chain and sprockets and will lead to harsh transmission. The application of engine oil will act as a temporary expedient, but it is preferable to use an aerosol chain lubricant whilst the chain is in situ on



Tyre removal: Deflate inner tube and insert lever in close proximity to tyre valve



Use two levers to work bead over the edge of rim



When first bead is clear, remove tyre as shown



Tyre fitting: Inflate inner tube and insert in tyre



Lay tyre on rim and feed valve through hole in rim



Work first bead over rim, using lever in final section



Use similar technique for second bead, finish at tyre valve position



Push valve and tube up into tyre when fitting final section, to avoid trapping

the machine and then, from time to time, remove the chain and immerse it in a molten lubricant such as 'Linklyfe' or 'Chainguard' after it has been cleaned in a paraffin bath. These latter lubricants achieve better penetration of the chain links and rollers and are less likely to be thrown off when the chain is in motion. On CD models always keep the chaincase drain holes free of obstruction. Failure to do this will encourage accumulation of rainwater which will promote corrosion and wear.

7 To check whether the chain is due for replacement, lay it lengthwise in a straight line and compress it endwise until all play is taken up. Anchor one end, then pull in the opposite direction to take up the play which develops. If the chain extends by more than $\frac{1}{4}$ inch per foot, it should be renewed, in conjunction with the sprockets. Note that this check should ALWAYS be made after the chain has been washed out, but before any lubricant is applied, otherwise the lubricant may take up some of the play.

8 When fitting the chain on the machine, make sure the spring link is positioned correctly with the closed end facing the direction of travel.

9 When ordering a new chain, always quote the size, the number of chain links and the type of machine to which the chain is to be fitted.



14.6 Lubricate the chain regularly; an aerosol lubricant is recommended

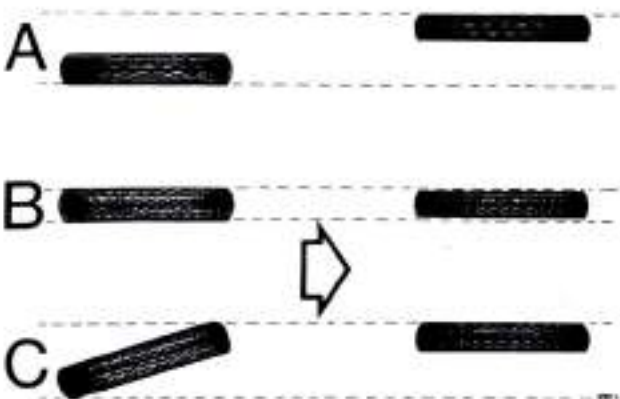


Fig. 5.5 Method of wheel alignment

B - Correct
A and C - Incorrect



14.4 Marks must be aligned equally each side

15 Tyres - removal and replacement

1 At some time or other the need will arise to remove and replace the tyres, either as the result of a puncture or because a replacement is required to offset wear. To the inexperienced, tyre changing represents a formidable task yet if a few simple rules are observed and the technique learned, the whole operation is surprisingly simple.

2 To remove the tyre from either wheel, first detach the wheel from the machine by following the procedure in Section 3 or Section 8 of this Chapter, depending on whether the front or rear wheel is involved. Deflate the tyre by removing the valve insert and when it is fully deflated, push the bead of the tyre away from the wheel rim on both sides so that the bead enters the centre well of the rim. Remove the locking cap and push the tyre valve into the tyre itself.

3 Insert a tyre lever close to the valve and lever the edge of the tyre over the outside of the wheel rim. Very little force should be necessary; if resistance is encountered it is probably due to the fact that the tyre beads have not entered the well of the wheel rim all the way round the tyre.

4 Once the tyre has been edged over the wheel rim, it is easy to work around the wheel rim so that the tyre is completely free on one side. At this stage, the inner tube can be removed.

5 Working from the other side of the wheel, ease the other edge of the tyre over the outside of the wheel rim that is furthest away. Continue to work around the rim until the tyre is free completely from the rim.

6 If a puncture has necessitated the removal of the tyre, reinflate the inner tube and immerse it in a bowl of water to trace the source of the leak. Mark its position and deflate the tube. Dry the tube and clean the area around the puncture with a petrol soaked rag. When the surface has dried, apply the rubber solution and allow this to dry before removing the backing from the patch and applying the patch to the surface.

7 It is best to use a patch of self-vulcanising type which will form a very permanent repair. Note that it may be necessary to remove a protective covering from the top surface of the patch, after it has sealed in position. Inner tubes made from synthetic rubber may require a special type of patch and adhesive if a satisfactory bond is to be achieved.

8 Before replacing the tyre, check the inside to make sure the agent which caused the puncture is not trapped. Check also the outside of the tyre, particularly the tread area, to make sure nothing is trapped that may cause a further puncture.

9 If the inner tube has been patched on a number of past occasions, or if there is a tear or large hole, it is preferable to

discard it and fit a replacement. Sudden deflation may cause an accident, particularly if it occurs with the front wheel.

10 To replace the tyre, inflate the inner tube sufficiently for it to assume a circular shape but only just. Then push it into the tyre so that it is enclosed completely. Lay the tyre on the wheel at an angle and insert the valve captive in its correct location.

11 Starting at the point furthest from the valve, push the tyre bead over the edge of the wheel rim until it is located in the central well. Continue to work around the tyre in this fashion until the whole of one side of the tyre is on the rim. It may be necessary to use a tyre lever during the final stages.

12 Make sure there is no pull on the tyre valve and again commencing with the area furthest from the valve, ease the other bead of the tyre over the edge of the rim. Finish with the area close to the valve, pushing the valve up into the tyre until the locking cap touches the rim. This will ensure the inner tube is not trapped when the last section of the bead is edged over the rim with a tyre lever.

13 Check that the inner tube is not trapped at any point. Reinflate the inner tube, and check that the tyre is seating correctly around the wheel rim. There should be a thin rib moulded around the wall of the tyre on both sides which should be equidistant from the wheel rim at all points. If the tyre is unevenly located on the rim, try bouncing the wheel when the tyre is at the recommended pressure. It is probable that one of the beads has not pulled clear of the centre well.

14 Always run the tyres at the recommended pressures and never under or over-inflate. See Specifications for recommended pressures.

15 Tyre replacement is aided by dusting the side walls, particularly in the vicinity of the beads, with a liberal coating of French chalk. Washing up liquid can also be used to good effect, but this has the disadvantage of causing the inner surfaces of the wheel rim to rust.

16 Never replace the inner tube and tyre without the rim tape in position. If this precaution is overlooked there is a good chance of the ends of the spoke nipples chafing the inner tube and causing a crop of punctures.

17 Never fit a tyre which has a damaged tread or side walls. Apart from the legal aspects there is a very great risk of a blowout, which can have serious consequences on any two wheel vehicle.

18 Valve cores and dust caps

1 Valve cores seldom give trouble, but do not last indefinitely. Dirt under the seating will cause a puzzling slow-puncture. Check that they are not leaking by applying spittle to the end of the valve and watching for air bubbles.

2 Tyre valve dust caps are often left off when a tyre has been replaced, despite the fact that they serve an important two-fold function. Firstly, they prevent dirt or other foreign matter from entering the valve and causing the valve to stick open when the tyre pump is next applied. Secondly, they form an effective second seal so that in the event of the tyre valve sticking, air will not suddenly be lost.

17 Fault diagnosis: wheels, brakes and tyres

Symptom	Cause	Remedy
Ineffective brake	Worn brake linings Foreign bodies on brake lining surface Incorrect engagement of brake arm serration Worn brake cam	Replace. Clean. Reset correctly. Replace.
Handlebars oscillate at low speeds	Buckle or flat in wheel rim, most likely front wheel Tyre not straight on rim	Check rim alignment by spinning wheel. Correct by retensioning spokes or building on new rim. Check tyre alignment.
Machine lacks power, poor acceleration	Brakes binding	Warm brake drum provides best evidence. Re-adjust brakes.
Brakes grab when applied gently	Ends of brake shoes not chamfered Elliptical brake drum	Chamfer with file. Lightly skim on lathe.
Brake pull-off spongy	Brake cam binding in housing Weak brake shoe springs	Free and grease. Renew if springs have not become displaced.
Harsh transmission	Worn or badly adjusted final drive chain Hooked or badly worn sprockets Loose rear sprocket Worn damper rubbers	Adjust or renew. Renew as a pair. Check bolts. Renew rubber inserts.
Tyres wear more rapidly in middle of tread	Over-inflation	Check pressures and run at recommended settings.
Tyres wear more rapidly at outer edges of tread	Under-inflation	Check pressures and run at recommended settings.

Chapter 6 Electrical system

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Specifications

Battery

Make	Yuasa
Type:	
185 models	6N12A-2D
200 models	6N12A-2C
Voltage	6 volts
Capacity	12 amp hr (12 Ah)
Earth	Negative

Alternator

Make	185 models	200 models
Type	Nippon Denso or Hitachi	Hitachi
Output	6-coil stator	8-coil stator
	0.086 kw	0.080 kw
	± 5000 rpm	± 5000 rpm

Regulator

Type:	
200 models	Combined with rectifier, integrated circuit
185 models	—

Rectifier

Type	Silicon diode
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Fuse

Rating	20 A
--------------	------

Starter motor

Brush length	12.2 – 12.4 mm (0.4803 – 0.4882 in)
Service limit	7.0 mm (0.0276 in)
Mica depth	0.5 – 0.8 mm (0.0197 – 0.0315 in)
Service limit	0.2 mm (0.008 in)

Bulbs	UK models	US models
Headlamp	35/35W	35/25W (178 CM185T) 35/36.5W
Tail/stop lamp	5/21W	3/32 CP
Direction indicators	21W	21/21 CP
Pilot lamp	4W	Not fitted
Speedometer light	3W	3W (2 CP)
Neutral indicator	3W	3W (2 CP)
Indicator warning	1.7W	1.7W (1 CP)
High beam indicator	1.7W	1.7W (1 CP)

All bulbs rated at 6 volts

1 General description

The models covered in this manual are fitted with a 6 volt electrical system. The circuit comprises a battery, which receives charge from an alternator mounted on the left-hand end of the crankshaft, a solid-state rectifier on the 185 models, and a combined regulator/rectifier unit on the 200 models.

The alternator takes two different forms, dependent on which model is under consideration. The alternator fitted to the 185 models consists of a rotary magnet revolving at engine speed within six radially placed coils. The coils, which do not actually touch the magnet, are secured to the inside of the left-hand crankcase cover. The rotating central magnet is referred to as the rotor, whilst the outer ring of six coils is known as the stator. The alternator utilised on the 200 models takes the form of a rotary magnet (the rotor) revolving at engine speed around eight radially placed coils (the stator). The stator coils are secured inside the left-hand crankcase cover.

The rectifier converts the alternating current (ac), produced by the alternator, to direct current (dc) to enable the lights and ancillary equipment to be used and the battery to be charged. The regulator (as fitted to the 200 models only) maintains the output to within a specified limit to prevent overcharging. The alternator output from 185 models is self regulating in that the output is matched to the requirements of the electrical system.

2 Crankshaft alternator: checking the output

1 If the battery appears to be consistently over-charged, indicated by a frequent need to top up the battery, or under-charged, the charging system should be checked.

2 In order for the output to be tested, certain items of test equipment must be available. An ammeter and a voltmeter of the centre zero type and a multimeter will be required. The voltmeter should have a range of at least 0-10 volts and the ammeter at least 0-8 amps. It is unlikely that the average owner/rider will have access to this equipment or instruction in its use. In consequence, if the performance of the alternator is in any way suspect, it should be checked by a Honda Service Agent or an auto-electrical specialist.

3 If the necessary test equipment is available, a general check on the alternator can be made as follows. Note that this check must be made with a fully charged battery and after the engine has been allowed to reach normal working temperature.

4 Disconnect the lead from the positive (+) terminal of the battery and reconnect it to the positive terminal of the ammeter. Connect the lead from the positive terminal of the voltmeter to the positive terminal of the battery. Connect both the negative lead from the ammeter and the positive lead from the voltmeter to the positive terminal of the battery. The negative (earth) lead from the battery must remain attached to its earthing point on the frame.

5 Start the engine using the kickstarter. Do not start the engine with the electric starter motor when the ammeter is connected. If this precaution is not observed, a large current will pass into the ammeter and burn it out. Ensure that the positive lead from the ammeter is securely attached to the positive terminal lead of the battery during the test. If the leads were to be disconnected, the current would flow directly to the lights

and burn them out.

6 Measure the battery voltage and the charging current, gradually raising the engine speed to gain the various readings. Needless to say, the engine speed should be held at the higher speeds (5000 rpm and 10 000 rpm) for the shortest duration possible, and certainly for not more than a few seconds. The measurements gained should correspond with the figures given in the tables below. Note that there is only one set of figures for the US market machines. This is because the lights on these machines are wired to operate whenever the machine is running. If the voltage and current is correct, it may be assumed that the system is functioning properly.

Charging specifications	CD200/CM200
Charging starts at	1450 rpm
Charging rate at	5000 rpm, 5.0A min, 7.9V
Charging rate at	10 000 rpm, 8.5A max, 8.1V

Note: These figures apply only when lights are on.

Charging specifications	CD185/CM185
-------------------------	-------------

Main switch	Lighting switch	Charging starts at:	5,000rpm	10,000rpm
ON	OFF [-]	1,200rpm V _B = 6.3V	Min. 2.2A V _B = 7.0V	Max. 4.0A V _B = 8.3V
	ON [H]	2,200rpm V _B = 6.3V	Min. 3.3A V _B = 7.0V	Max. 6.0A V _B = 8.3V

Note: read off lowest line when checking CM185, whose lights are always in 'On' position.

If the results of the test are not similar to those given in the tables it is probable that the alternator coils have failed or a loose connection has occurred between the alternator and battery. The connections should be checked visually and cleaned if corrosion is evident. The condition of the coils in the stator may be checked, by determining whether continuity exists, without removal of the stator from the machine.

7 Remove the left-hand side panel from the frame so that access may be gained to the white 6-pin block connector carrying the alternator, neutral indicator switch and ignition coils leads. Separate the block connector. Using a multimeter set to the resistance function, check that continuity (low resistance or no resistance) exists between the following pairs of wires: Yellow and White, Pink and White, Yellow and Pink. If no continuity exists between any two wires there is evidence that a broken circuit has occurred in one of the stator coils. A further check should be carried out between an earth point on the engine and the following wires: Pink, Yellow, White. In this case, no continuity should exist (infinitely high resistance). If continuity is found here, there is evidence that one of the stator coils is shorting to earth.

8 If such defects are present, a repair is impractical, and the stator must, therefore, be renewed. Before consigning to the

scrap bin what is an expensive item to replace, it is worth placing the machine in the hands of a Honda Service Agent or an auto-electrician for further investigation.

9 If the tests on the alternator current and voltage output were unsatisfactory, yet the tests on the stator proved to be correct, the rectifier or regulator/rectifier unit should be checked as described in Sections 3 and 4 of this Chapter.

3 Rectifier: testing – 185 models only

1 The rectifier on the 185 models is bolted to the left-hand side of the machine, ahead of the air cleaner assembly. Its function is to convert the alternating current from the alternator to direct current which can then be used to charge the battery and operate the electrical system. It takes the form of a small sealed unit mounted in a rubber sleeve at the front of the air cleaner case.

2 The rectifier is deliberately placed in this location so that it is not exposed directly to water or oil and yet has free circulation of air to permit cooling. It should be kept clean and dry.

3 This component is a solid state unit. If damage occurs, no repair is possible; the complete assembly must be renewed. The performance of the rectifier can only be tested using special equipment. However, a continuity check may be carried out on the unit as a whole, giving some indication of its condition.

4 If a pocket multimeter is possessed, then a simple test can be performed to determine whether or not the unit is malfunctioning. If a multimeter is not possessed, and the performance of the unit is suspect, a Honda Service Agent or an auto-electrical expert should be consulted.

5 To test the rectifier, using a multimeter set to the resistance testing function, disconnect the wires leading from the rectifier at the block connector. Test the continuity across the terminals shown in the accompanying table, placing the positive probe on those terminals shown in the left-hand column, and the negative probe on those in the right-hand column. In these tests continuity should exist. Repeat the test but this time reversing the probes. In this test non-continuity should exist. If any test is incorrect, the rectifier must be renewed.

TESTER	+	-
CORD COLOR	RED/WHITE	PINK GREEN YELLOW
	PINK	GREEN
	YELLOW	GREEN

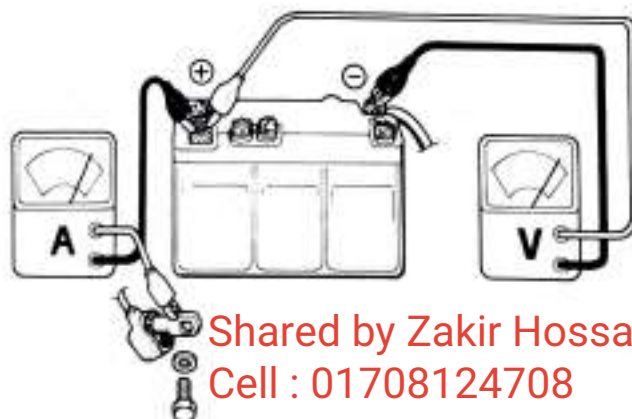


Fig. 6.1 Ammeter/voltmeter test connections

4 Regulator/rectifier: testing – 200 models only

1 The regulator/rectifier is a combined solid-state unit. If damage occurs, no repair is possible; the complete assembly must be renewed. The performance of the regulator can only be tested using special equipment. However, a continuity check may be carried out on the unit as a whole, giving some indication of its condition.

2 The regulator/rectifier unit on these models is mounted behind the right-hand frame side panel, to the rear of the battery. The unit is secured to the rear of the starter motor relay mounting plate.

5 Battery: examination and maintenance

1 A conventional lead-acid type battery of 6 volt, 12 Ah capacity is fitted to these models.

2 The transparent plastic case of the battery permits the upper and lower levels of the electrolyte to be observed without the need to remove the battery from the machine. Simply pull off the right-hand side panel, and the battery is exposed. Maintenance is normally limited to keeping the electrolyte level between the prescribed upper and lower limits and by making sure the vent pipe is not blocked. The lead plates and their separators can be seen through the transparent case, a further guide to the general condition of the battery.

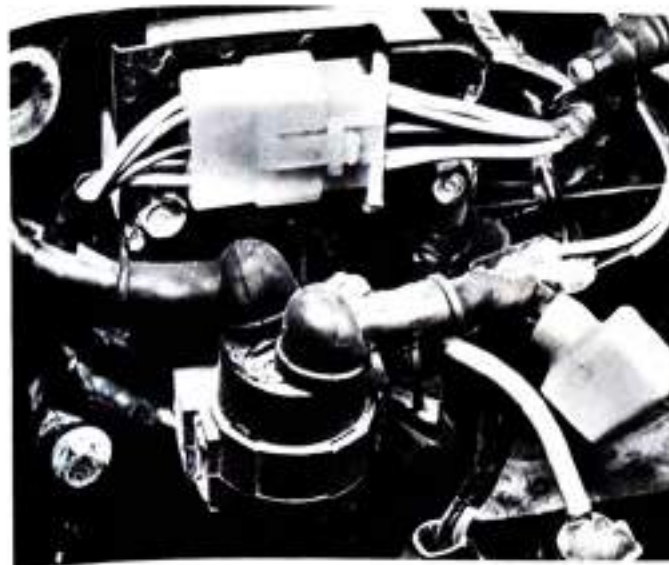
3 Unless acid is spilt, as may occur if the machine falls over, the electrolyte should always be topped up with distilled water, to restore the correct level. If acid is spilt on any of the machine, it should be neutralised with an alkali such as washing soda and washed away with plenty of water, otherwise serious corrosion will occur. Top up with sulphuric acid of the correct specific gravity (1.260 – 1.280) only when spillage has occurred. Check that the vent pipe is well clear of the frame tubes or any of the other cycle parts, for obvious reasons.

4 If the machine is laid up for any period of time, the battery should be removed and given a 'refresher' charge every six weeks or so, in order to maintain it in good condition. If the battery is permitted to discharge completely, the plates will sulphate and render the battery useless. For charging instructions, refer to the following Section.

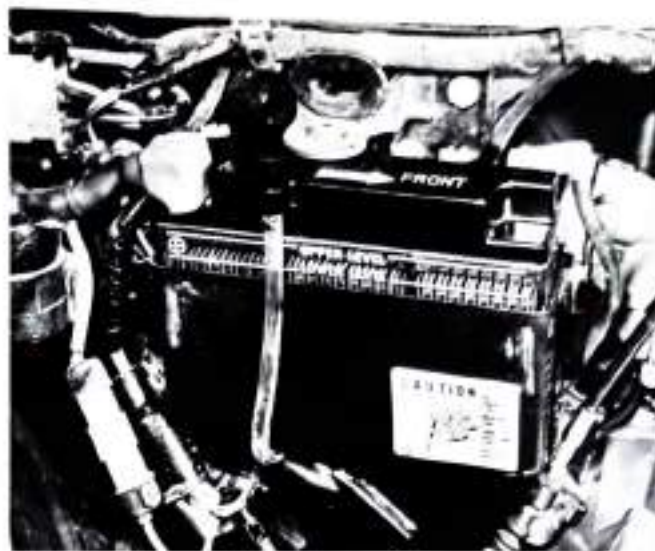


2.7 Stator is retained inside left-hand crankcase cover by three screws (200T model shown)

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4.2 Regulator/rectifier unit (200T models only) and its leads connector



5.2 Electrolyte level can be checked without removing the battery

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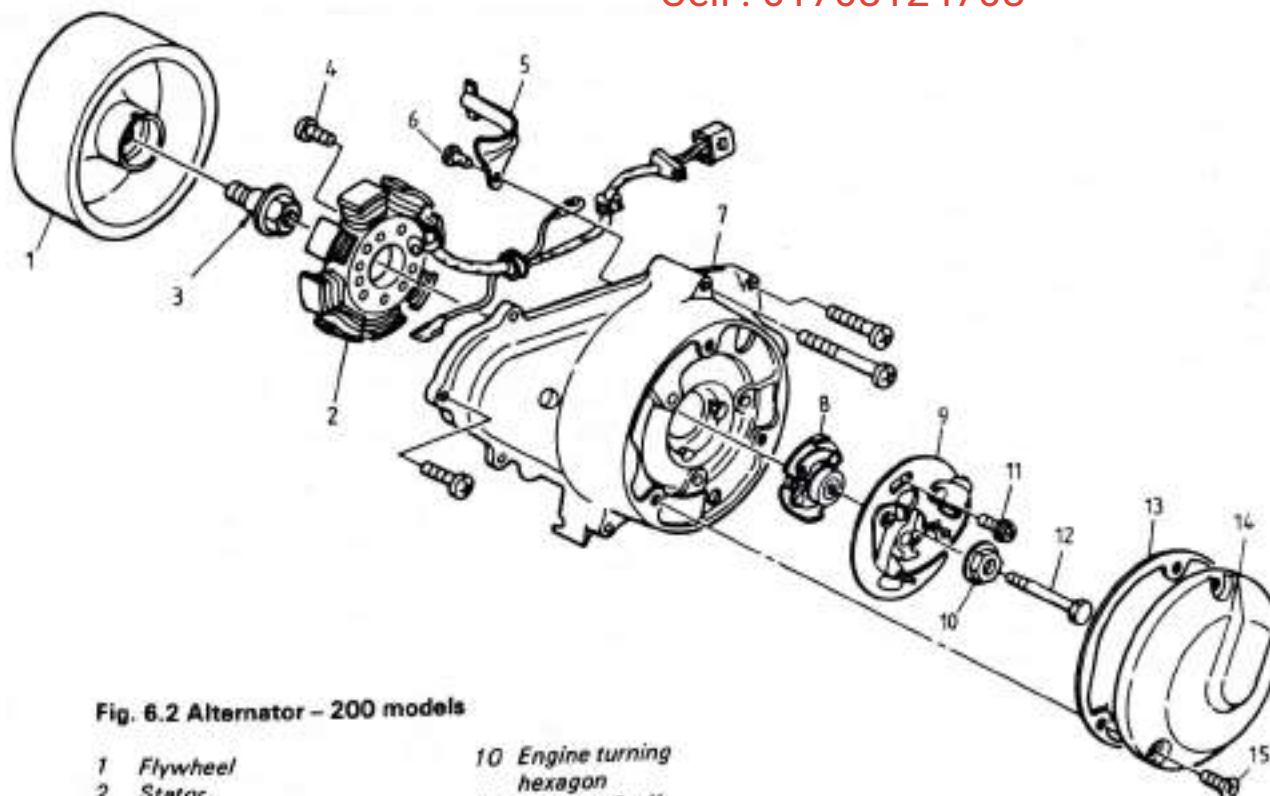


Fig. 6.2 Alternator – 200 models

- | | |
|-------------------------|---------------------------|
| 1 Flywheel | 10 Engine turning hexagon |
| 2 Stator | 11 Screw – 3 off |
| 3 Bolt | 12 Bolt |
| 4 Screw – 3 off | 13 Cover gasket |
| 5 Cable clamp | 14 Contact breaker cover |
| 6 Screw | 15 Screw – 3 off |
| 7 Left-hand casing | |
| 8 Automatic timing unit | |
| 9 Contact breaker plate | |

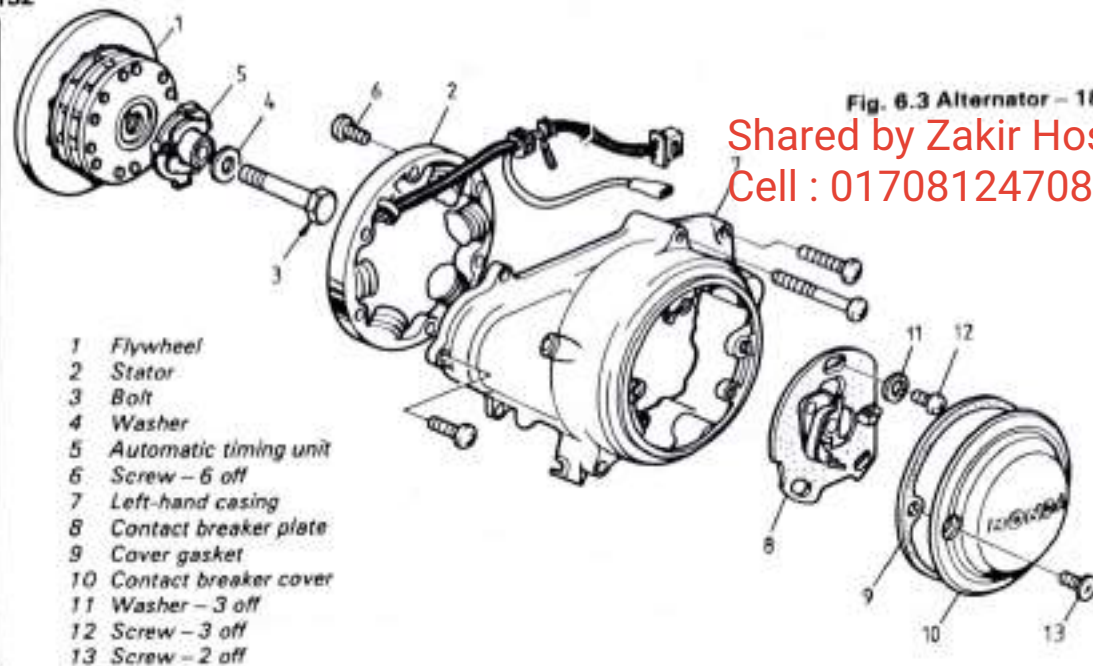


Fig. 6.3 Alternator - 185 models

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6 Battery: charging procedure

1 Whilst the machine is used on the road it is unlikely that the battery will require any attention other than routine maintenance. The alternator will keep the battery fully charged. If, however, the machine is used for a succession of short journeys only, and on each occasion that the machine is started, the electric starter motor is used, it is possible that the output from the alternator may fail to keep pace with the heavy electrical demand. Such circumstances are especially likely to result in a flat battery on the US market models, which have their lights switched on permanently, or on UK market models during the hours of darkness, when the lights are in full use, or if the machine is parked with the lights on. Should any or all of these circumstances combine to produce a flat battery, it should be removed to be charged independently.

2 The battery is located behind the right-hand side panel, fitted in a carrier and secured by a rubber strap. It is preferable to remove the battery from the machine during the charging operation. This is simply accomplished by disconnecting the leads and releasing the rubber retaining strap; the battery is now free to be lifted up and out of the carrier. Remember to remove the three vent caps from the top of the cells before commencing charging.

3 The normal charging rate for any battery is one-tenth of its rated capacity. Hence the charging rate for the 12 Ah battery is 1.2 amps. A slightly higher rate of charge may be used in an emergency. The higher charge rate should, if possible, be avoided since it will shorten the working life of the battery.

4 Make sure that the battery charger connections are correct, red to positive and black to negative. When the battery is reconnected to the machine, the black lead must be connected to the negative terminal and the red lead to positive. This is most important, as the machine has a negative earth system. If the terminals are inadvertently reversed, the electrical system will be damaged permanently.

5 Whilst the battery is removed from the machine is the best time to clean the battery top. If the terminals are corroded scrape them clean and cover them with Vaseline (not grease) to protect them from further attack.

7 Fuse: location and replacement

1 A fuse is incorporated in the electrical system to give protection from a sudden overload, as may occur during a short

circuit or current surge. The fuse is incorporated in the red lead wire from the battery positive terminal, and is enclosed in a nylon case located adjacent to the battery. On the inside of the right-hand side panel is a small rubber holder containing one spare fuse, for use in an emergency. The fuse and its spare are both rated at 20 amps.

2 If the complete electrical system is inoperative, a blown fuse should be suspected, but before the fuse is replaced the electrical system should be inspected to trace the reason for the failure of the fuse. If this precaution is not observed the replacement will almost certainly blow as well.

3 At least one spare fuse should be carried at all times. In an extreme emergency and only when the cause of the failure has been rectified and no spare is available, a 'get-you-home' repair can be made by wrapping silver paper around the blown fuse and re-inserting it in the fuse holder. It must be stressed that this is only an emergency measure and the fuse should be replaced at the earliest possible opportunity, as it affords no protection at all to the electrical system when bridged in this fashion.



6.2 Remove the battery to be charged independently



7.1a Fuse is located behind r.h. side panel in a nylon case



7.1b Spare fuse is located on inside of r.h. side panel

8 Starter motor: removal, examination and refitting

1 An electric starter motor, operated from a small push-button on the right-hand side of the handlebars, provides an alternative and more convenient method of starting the engine, than having to use the kickstart. The starter motor is mounted at the front of the crankcase, on the left-hand side, at the base of the cylinder barrels. Current is supplied from the battery via a heavy duty solenoid switch and a cable capable of carrying the very high current demanded by the starter motor on the initial start-up.

2 The starter motor drives via a free-running or one-way clutch mounted on the end of the crankshaft, behind the alternator. The clutch, which operates on the ramp and roller principle, ensures the starter motor drive is disconnected from the crankshaft immediately the engine starts.

3 The starter motor may be removed while the engine is in the frame. Disconnect the battery leads to isolate the electrical system and then detach the heavy cable from the terminal on the starter motor body. The protecting rubber boot will have to be prised off the terminal to gain access to the nut.

4 The starter motor is secured to the crankcase by four screws, two at each end of the motor casing. When these screws are withdrawn, generally with the aid of an impact driver, the motor can be prised out of position and lifted away. If necessary, a thin wooden wedge placed between the motor casing and crankcase may be used to push the starter motor backwards so that the motor boss leaves the hole in the casing.

5 Although the starter motor can be removed in the simple manner described above, refitting the motor is somewhat more time-consuming and involved to accomplish. The splined boss on the outer end of the starter motor locates with the internal splines of the drive sprocket fitted behind the left-hand outer crankcase cover. It is not possible to refit the splined boss into the drive sprocket unless the latter can be retained in position. This is best done by hand, necessitating removal of the left-hand crankcase cover to gain access to the drive sprocket. Removal of the left-hand crankcase cover requires the removal of the contact breaker baseplate (all models) and the engine timing hexagon and the ATU and its retaining bolt (200 models). Should the drive sprocket and drive chain require attention, then this too will necessitate the dismantling described above. If the driven sprocket or the one-way clutch require attention, the ATU and alternator rotor (185 models) or the rotor (200 models) must be removed. Full information on the necessary removal procedures will be found by referring to Section 10 of Chapter 1. The examination and renovation of the

various components will be found under their relevant headings. The reassembly procedure for the starter motor etc is fully described in Section 45 of Chapter 1.

6 The parts of the starter motor most likely to require attention are the brushes. The end cover is retained by the two long screws which pass through the lugs cast on both end pieces. If the screws are withdrawn, the end cover can be lifted away and the brush gear exposed.

7 Lift up the spring clips which bear on the end of each brush and remove the brushes from their holders. Each brush should have a length of between 12.2 - 12.4 mm (0.4803 - 0.4882 in). The minimum allowable brush length is 7.0 mm (0.2756 in). If the brush is shorter it must be renewed.

8 Before the brushes are replaced, make sure that the commutator is clean. The commutator is the ring of copper segments on which the brushes bear. Clean the commutator with a strip of glass paper. Never use emery cloth or 'wet-and-dry' as the small abrasive fragments may embed themselves in the soft copper of the commutator and cause excessive wear of the brushes. Finish off the commutator with metal polish to give a smooth surface and finally wipe the segments over with a methylated spirits soaked rag to ensure a grease free surface. Check that the mica insulators, which lie between the segments of the commutator, are undercut. The standard groove depth is 0.5 - 0.8 mm (0.02 - 0.03 in), but if the average groove depth is 0.2 mm (0.0079 in) or less, the armature should be renewed or returned to a Honda Service Agent for re-cutting.

9 Replace the brushes in their holders and check that they slide quite freely. Make sure the brushes are replaced in their original positions because they will have worn to the profile of the commutator. Check, and if necessary, renew the O-ring fitted behind the brush holder mounting plate. Refit the end cover, noting the washers which fit on the end of the armature, and reinstall and tighten the two long crosshead screws. Refit the starter motor cable and then refit the motor to the engine as described in Section 45 of Chapter 1. Refer also to paragraph 5 of this Section.

9 Starter solenoid switch: function and location

1 The starter motor switch is designed to work on the electro-magnetic principle. When the starter motor button is depressed, current from the battery passes through windings in the switch solenoid and generates an electro-magnetic force which causes a set of contact points to close. Immediately the points close, the starter motor is energised and a very heavy current is drawn from the battery.

2 This arrangement is used for at least two reasons. Firstly, the starter motor current is drawn only when the button is depressed and is cut off again when pressure on the button is released. This ensures minimum drainage on the battery. Secondly, if the battery is in a low state of charge, there will not be sufficient current to cause the solenoid contacts to close. In consequence, it is not possible to place an excessive drain on the battery which, in some circumstances, can cause the plates to overheat and shed their coatings. If the starter will not operate, first suspect a discharged battery. This can be checked by trying the horn (US models) or switching on the lights (UK models). If this check shows the battery to be in good condition, suspect the starter switch which should come into operation with a pronounced click.

3 The starter switch is located behind the right-hand side panel and can be identified by the heavy duty starter cable connected to it. Do not confuse the starter switch with the similarly shaped flashing indicator relay fitted to the rear of the starter switch on the 185 models. If the starter switch malfunctions, it is not possible to effect a satisfactory repair; it must be renewed.



8.5a Starter motor splined end must be relocated with drive sprocket



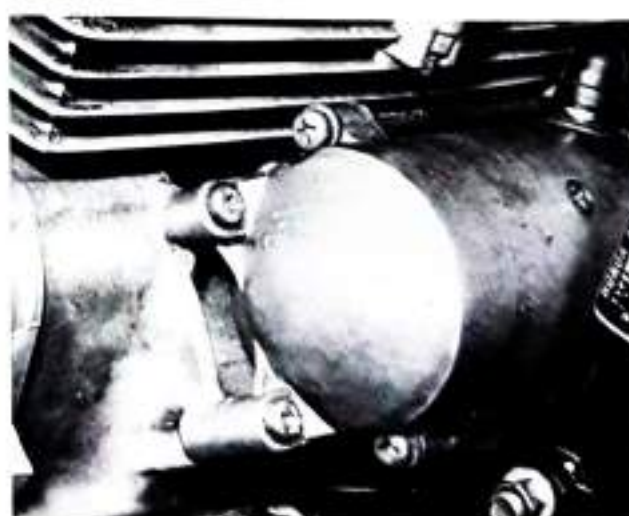
8.5b Reconnect starter motor cable before ...



8.5c ... the l.h. crankcase cover is refitted

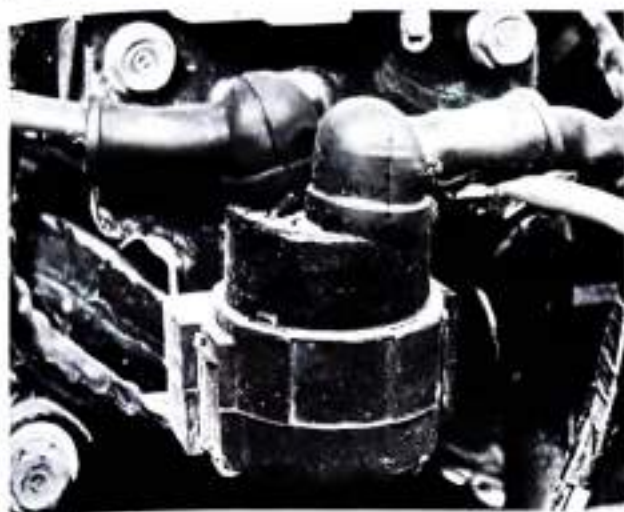


8.5d Starter motor cable is secured under guide on cover



8.5e Secure motor with two screws at each end

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9.3 Starter motor relay is rubber mounted

10 Headlamp: replacing the bulbs and adjusting beam height – UK models only

1 In order to gain access to the headlamp bulbs, it is necessary first to remove the rim, complete with the reflector and headlamp glass. The rim is retained by two crosshead screws which pass through the headlamp shell just below the two headlamp mounting bolts.

2 Pull the headlamp bulb socket from the rear of the reflector after disconnecting the spring retaining the holder to the inside of the rim. The headlamp bulb is retained by the spring loaded holder. To release the holder, depress and twist it in an anti-clockwise direction. The holder, two small springs and bulb may now be lifted from position. Twist the bulb and free it from the holder. If the bulb has failed, do not immediately throw it away, use it as a reference to get the correct voltage and wattage; there are numerous types and fittings available.

3 A reflector that accepts a pilot bulb is fitted to all models delivered to countries or states where parking lights are a statutory requirement. The pilot bulb is held in the bulb holder by a bayonet fixing.

4 Beam height adjustment is effected by tilting the headlamp shell after the mounting bolts have been loosened slightly. The horizontal alignment of the beam can be adjusted by altering

the position of the screw which passes through the headlamp rim. The screw is fitted at the 9 o'clock position when viewed from the front of the machine. Turning the screw in a clockwise direction will move the beam direction towards the right-hand side when viewed from a normal riding position.

5 To obtain the correct beam height, place the machine on level ground facing a wall 25 feet distant, with the rider seated normally. The height of the beam centre should be equal to that of the height of the centre of the headlamp from the ground when the dip switch is in the dip beam position. Furthermore, the concentrated area of light should be centrally disposed. Adjustments in either direction are made by rearranging the angle of the headlamp, as described in the preceding paragraph. Note that a different beam setting will be needed when a pillion passenger is carried. If a pillion passenger is carried regularly, the passenger should be seated in addition to the rider when the beam setting adjustment is made.

6 The above instructions for beam setting relate to the requirements of the United Kingdom's transport lighting regulations. Other settings may be required in countries other than the UK.



10.1 Headlamp rim is retained by two screws (UK models)

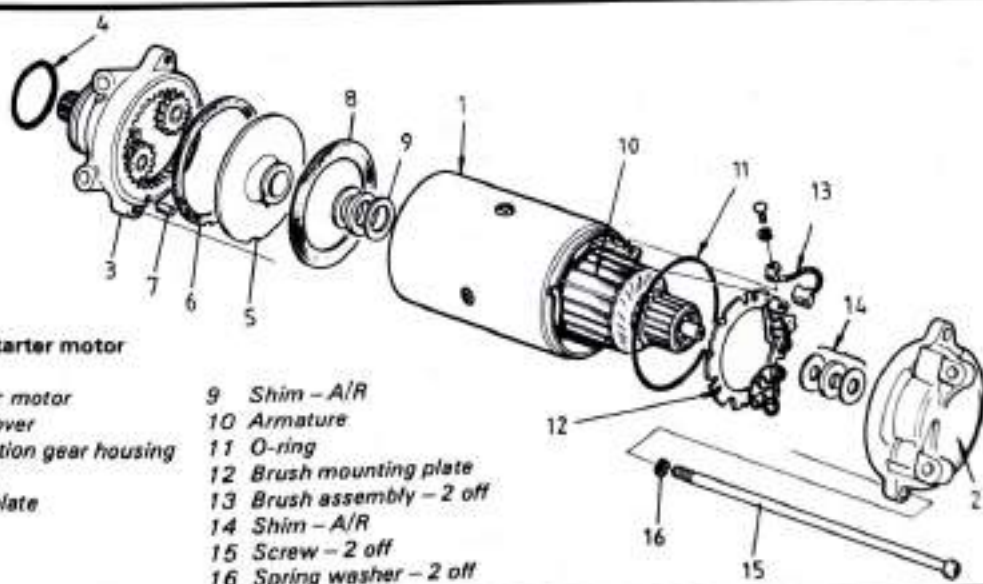


Fig. 6.4 Starter motor

- | | |
|--------------------------|---------------------------|
| 1 Starter motor | 9 Shim – A/R |
| 2 End cover | 10 Armature |
| 3 Reduction gear housing | 11 O-ring |
| 4 O-ring | 12 Brush mounting plate |
| 5 Gear plate | 13 Brush assembly – 2 off |
| 6 Seal | 14 Shim – A/R |
| 7 Pin | 15 Screw – 2 off |
| 8 Seal | 16 Spring washer – 2 off |



10.2a Disconnect bulb holder retaining spring and ...



10.2b ... pull out bulb holder from reflector



10.2c Twist and release the bulb from the holder



10.3 Pilot bulb holder will pull out. Bulb is a bayonet fitting

11 Headlamp: replacing the headlamp unit and adjusting beam height - US models only

1 The models sold on the US market (CM185T and CM200T), are equipped with a sealed beam headlamp unit. This, as its description implies, is a single unit comprising the headlamp glass, reflector and bulb filaments. Only the headlamp shell and the rim are separate items. There is no bulb as such fitted to this type of headlamp. The filaments do not have a separate glass envelope but instead use the headlamp reflector and glass unit as an envelope. Consequently, should one of the filaments of the sealed beam unit fail, the complete sealed beam unit must be replaced.

2 Remove the two crosshead screws from just below the headlamp mounting bolts on each side of the shell. This will enable the rim and headlamp unit to be pulled out of position. Remove two further screws from the 12 and 6 o'clock positions to separate the rim from the headlamp unit. Release the headlamp unit snap connectors and detach the unit. Fitting a new headlamp unit is a direct reversal of the removal procedure.

3 Adjustments of headlamp beam height and angle are made in the same way as described in paragraph 4 of the previous Section. The necessary adjustments should be made according to the relevant State or Federal regulations.

12 Tail and stop lamp: replacing bulbs

1 The combined tail and stop lamp is fitted with a double filament bulb having offset pins to prevent its unintentional reversal in the bulb holder. The lamp unit serves a two-fold purpose - to illuminate the rear of the machine and the rear number plate, and to give visual warning when the rear brake is applied.

2 To gain access to the bulb, remove the two screws (CD models) or three screws (CM models) holding the red plastic lens in position. The bulb is released by pressing inwards and with an anti-clockwise turning action, the bulb will now come out.

13 Flashing indicator lamps: replacing bulbs

1 The forward facing indicator lamps are connected to 'stalks' which are attached to the headlamp brackets. The hollow stalks carry the leads to the lens unit. The rear facing lamps are mounted on similar, shorter stalks, at a point immediately to the rear of the dualseat.

2 In each case, access to the bulb is gained by removing the plastic lens cover, which is retained by two screws. Bayonet

bulbs of the single filament type are fitted; these should be pushed in slightly and twisted for removal and replacement.

14 Flashing indicator relay: location and replacement

1 The flashing indicator relay fitted in conjunction with the flashing indicator lamps is located behind the right-hand side panel to the rear of the battery on the 185 models. The relay is fitted behind the left-hand side panel on the 200 models. The relay is rubber mounted, to isolate it from the harmful effects of vibration, and is secured by a rubber band clip. The relay is cylindrical in profile and as such should not be confused with the starter motor solenoid switch which it closely resembles.

2 When the relay malfunctions, it must be renewed; a repair is impracticable. When the unit is in working order audible clicks will be heard which coincide with the flash of the indicator lamps. If the lamps malfunction, check firstly that a bulb has not blown, or the handlebar switch is not faulty. The usual symptom of a fault is one initial flash before the unit goes dead.

3 Take great care when handling a flasher unit. It is easily damaged, if dropped.

15 Stop lamp switches: adjustment

1 All models have a stop lamp switch fitted to operate in conjunction with the rear brake pedal. The switch is located immediately to the rear of the crankcase, on the right-hand side of the machine. It has a threaded body giving a range

adjustment.

2 If the stop lamp is late in operating, turn the adjuster nut in a clockwise direction so that the switch rises from the bracket to which it is attached.

3 If the lamp operates too early, the adjuster nut should be turned anti-clockwise so that the switch body is lowered in relation to the mounting bracket.

4 As a guide, the light should operate after the brake pedal has been depressed by about 2 cm ($\frac{3}{8}$ inch).

5 A stop lamp switch operated by the front brake is also fitted. The switch is fitted in the brake lever stock and is non-adjustable. Should failure occur the switch should be renewed.

16 Instrument illumination and warning lamps: replacing bulbs

1 The illuminating bulb fitted to the speedometer head is rubber mounted and presses into the base of the instrument. To remove the bulb, simply pull out the rubber bulb holder and twist the bulb to remove it from the holder.

2 In addition, the 185 models have a column of three warning lamps incorporated in the speedometer unit. The 200 models are equipped with a small warning light console fitted to the right of the speedometer head. The three bulbs fitted to the panel/console are for warning of the flashing indicators working, the main beam in operation and the gearbox being in the neutral position. To remove these three bulbs, follow the same procedure as described for removing the speedometer illuminating bulb. The bulbs are all of a bayonet fitting type.

Fig. 6.5 Headlamp assembly

- 1 Headlamp - UK models
- 2 Headlamp - US models
- 3 Rim
- 4 Beam height adjusting screw
- 5 Spring

- 6 Nut
- 7 Reflector unit
- 8 Screw - 2 off
- 9 Nut - 2 off
- 10 Reflector retaining clip - 3 off
- 11 Bulb
- 12 Bulb holder
- 13 Pilot bulb
- 14 Grommet
- 15 Spring
- 16 Connector mounting
- 17 Connector mounting
- 18 Headlamp shell
- 19 Shouldered collar - 2 off
- 20 Spring washer - 2 off
- 21 Screw - 2 off

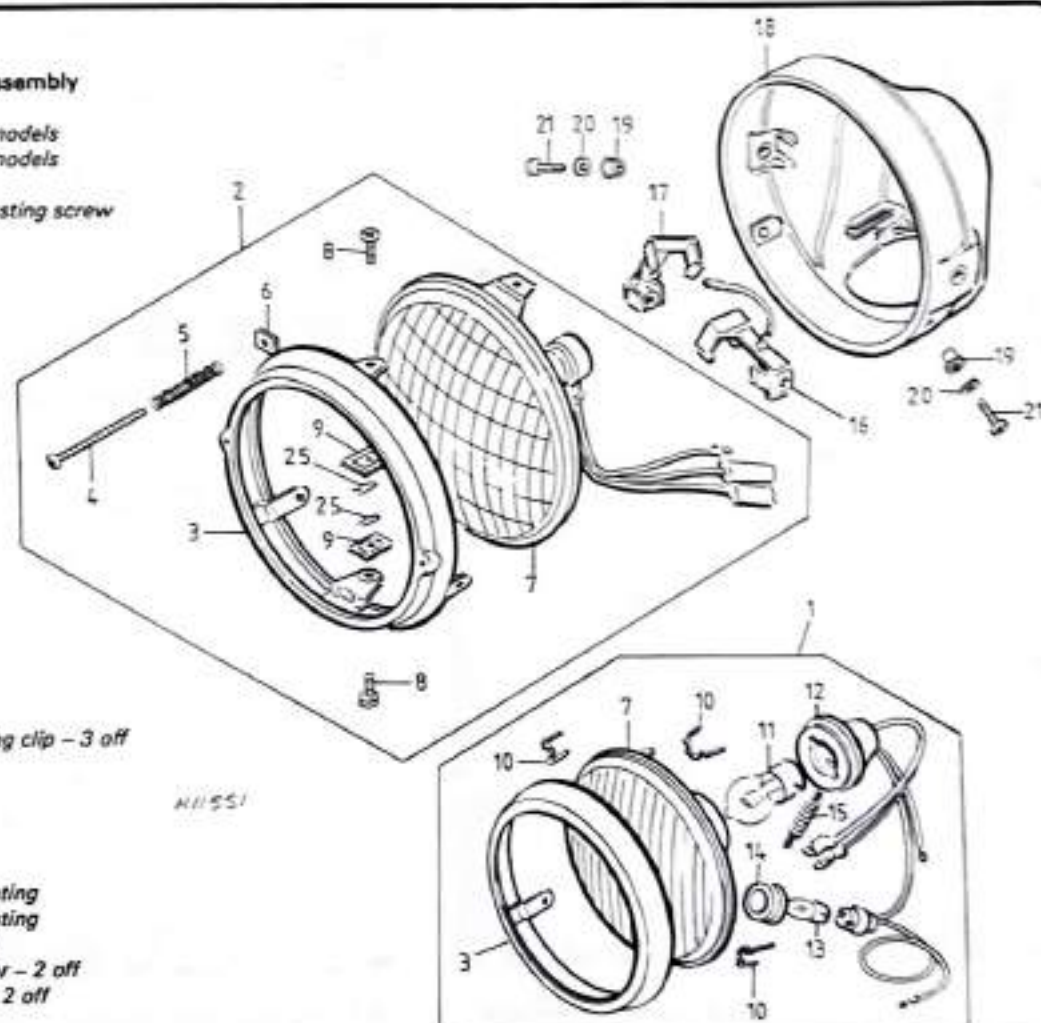
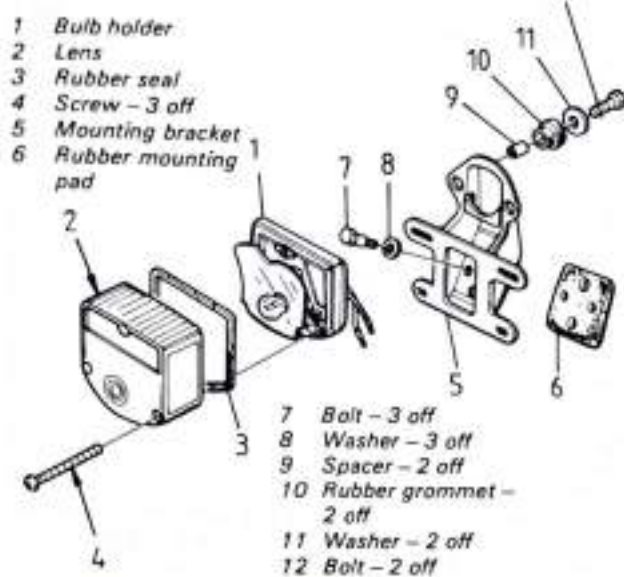


Fig. 6.6 Tail/stop lamp assembly



12.2a Remove lens to gain access to tail/stop lamp bulb



12.2b Push and twist to release bayonet fitting bulb



13.2 Indicator lenses are each held by two screws



14.1 Indicator relay is rubber mounted, behind left-hand side panel (200T models)



15.1 Stop lamp switch operation is adjusted by means of nut



16.1 Speedometer bulb holder is a push fit as are ...



16.2 ... the warning lamp bulb holders in the console (200T model shown)

17 Horn: location and adjustment

- 1 On the CD models, the horn is attached to a flexible steel strip which is secured by a single 6 mm bolt to a mounting bracket. The mounting bracket fits in front of the lower fork yoke and is secured by the two lower yoke pinch bolts. On the CM models, the flexible steel strip on the back of the horn is secured to the lower cross-member of the headlamp/front indicator lamps mounting bracket.
- 2 At the rear of the horn there is a small crosshead screw and locknut. This is the volume adjusting screw and its locknut, and the screw may need adjustment from time to time to compensate for wear inside the horn.
- 3 To adjust the horn, slacken the locknut, depress the horn button (with the ignition switched on) and screw the adjuster in or out to obtain maximum horn volume. It is suggested that this adjustment be carried out at a place of sparse population, so that unnecessary aggravation to fellow citizens may be avoided.
- 4 If the horn malfunctions, it must be replaced; it is a statutory requirement in the UK that the machine must be fitted with a horn that is in working order.

18 Ignition switch: removal and replacement

- 1 On the 185T models, the combined ignition and lighting master switch is mounted below the speedometer head, in the plastic cover which fits over the handlebar lower clamp/speedometer mounting plate. On the 200T models, the switch is mounted in an extension of the warning light console to the right-hand side of the speedometer head.
- 2 If the switch on a 185T model malfunctions, it may be removed after detaching the plastic cover, removing the two crosshead retaining screws and disconnecting the leads at the block connector. On the 200T models, remove the two crosshead screws from the underside of the warning light console, lift the console/ignition switch cover away, disconnect the leads at the block connector and remove the switch unit. Repair is rarely practicable. It is preferable to purchase a new switch unit, which will probably necessitate the use of a different key.

19 Switches and connectors: testing and repairs

- 1 In the event of any electrical fault, attention should be directed at the switch contacts and wiring connections in the circuit concerned. After long periods, these may become badly corroded, setting up a high resistance in the circuit and impairing the operation of the component controlled by them. If a multimeter is available, each switch connection and its associated wiring can be tested with the meter set on resistance.
- 2 The colour wiring diagrams at the end of this Chapter show the various switches in diagrammatic form, the lines denoting which contacts are connected in the various positions. With the machine's battery disconnected, check each of the relevant switch positions. When the two leads being checked are connected, the meter should indicate a completed circuit, with no discernible resistance, when the switch is in the 'On' position for that particular circuit.
- 3 The switch contacts can be kept clean and free of corrosion by using one of the proprietary switch cleaning fluids now available in aerosol form. These fluids can be injected into the switch body by means of the extension nozzle provided, and require that little or no dismantling is carried out. This form of preventative maintenance is especially useful during winter, when the switches are most prone to corrosion.
- 4 In the event that a switch unit fails completely, little can be done to effect a satisfactory repair. The handlebar switches can be dismantled to a certain extent, but repair is limited to cleaning the switch contacts with fine abrasive paper, where access is possible. In many instances, renewal will be the only course of action.

20 Wiring: layout and examination

- 1 The cables of the wiring harness are colour-coded and will correspond with the accompanying wiring diagram.
- 2 Visual inspection will show whether any breaks or frayed outer coverings are giving rise to short circuits which will cause the main fuse to blow. Another source of trouble is the snap connectors and spade terminals, which may make a poor connection if they are not pushed home fully, or if corrosion has occurred.
- 3 Intermittent short circuits can sometimes be traced to a chafed wire passing through, or close to, a metal component, such as a frame member. Avoid tight bends in the cables or situations where the cables can be trapped or stretched, especially in the vicinity of the handlebars or steering head.



17.2 Horn is attached by a single bolt (CD models). Note small adjuster screw and locknut



18.1 Ignition switch is mounted in warning light console (200T models)



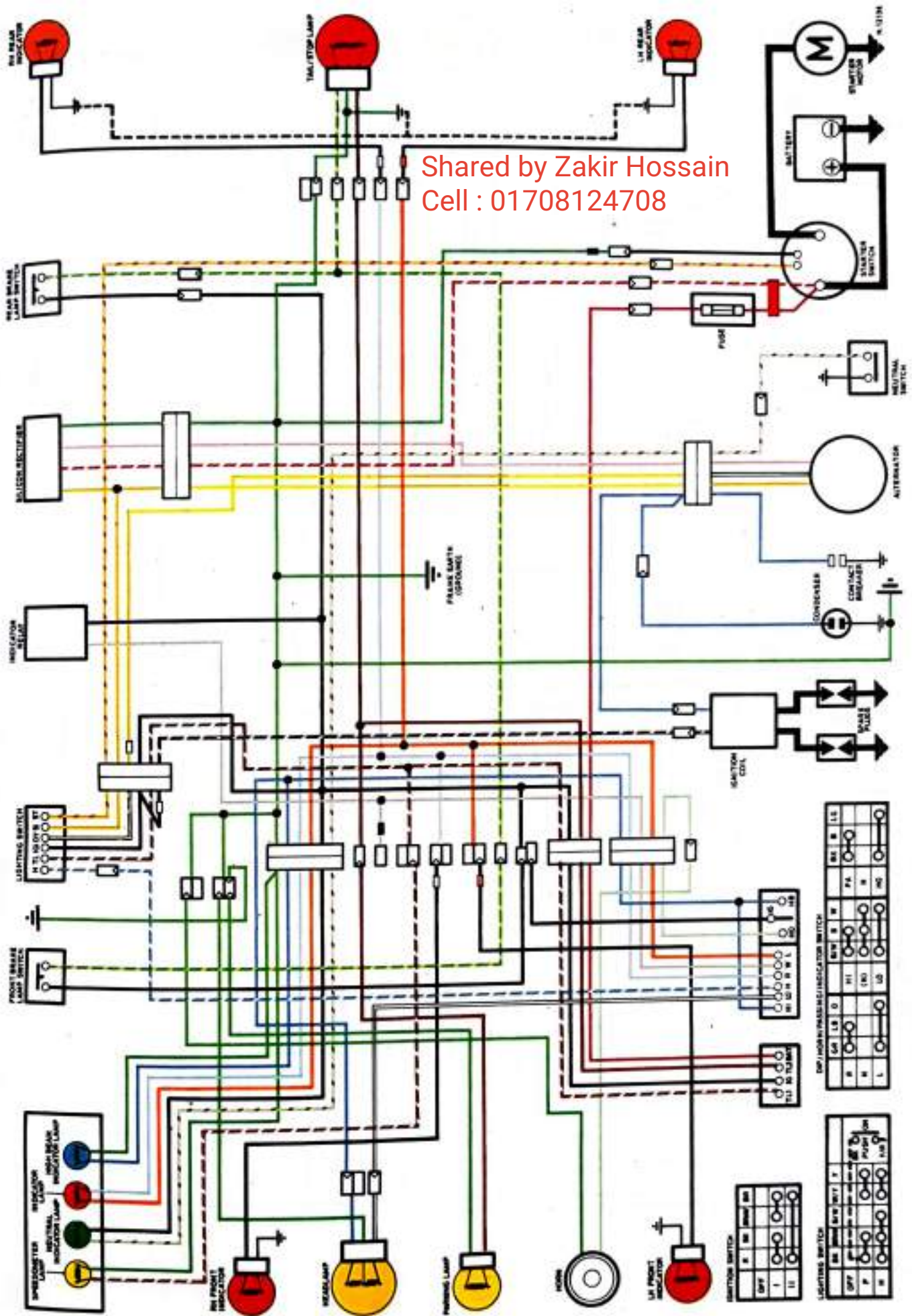
19.4a Switch clusters clamp around handlebar ends ...



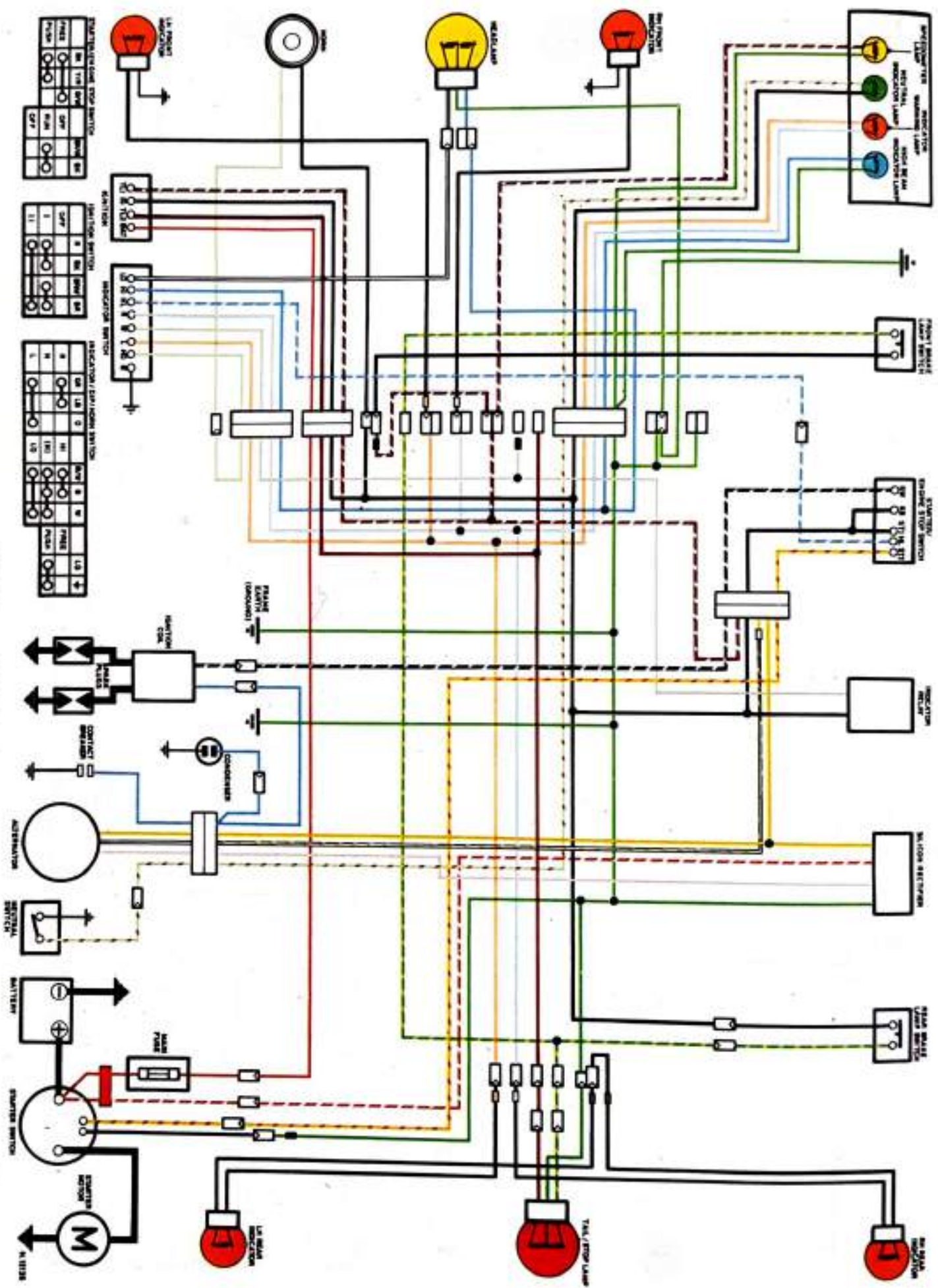
19.4b ... and may be separated for maintenance

21 Fault diagnosis: electrical system

Symptom	Cause	Remedy
Complete electrical failure	Blown fuse	Check wiring and electrical components for short circuit before fitting new 20 amp fuse. Check battery connections, also whether connections show signs of corrosion
	Isolated battery	
Dim lights, horn inoperative	Discharged battery	Recharge battery with battery charger and check whether alternator is giving correct output.
Constantly 'blowing' bulbs	Vibration, poor earth connection	Check whether bulb holders are secured correctly. Check earth return or connections to frame. Check resistor if pilot lamp bulb fails regularly.

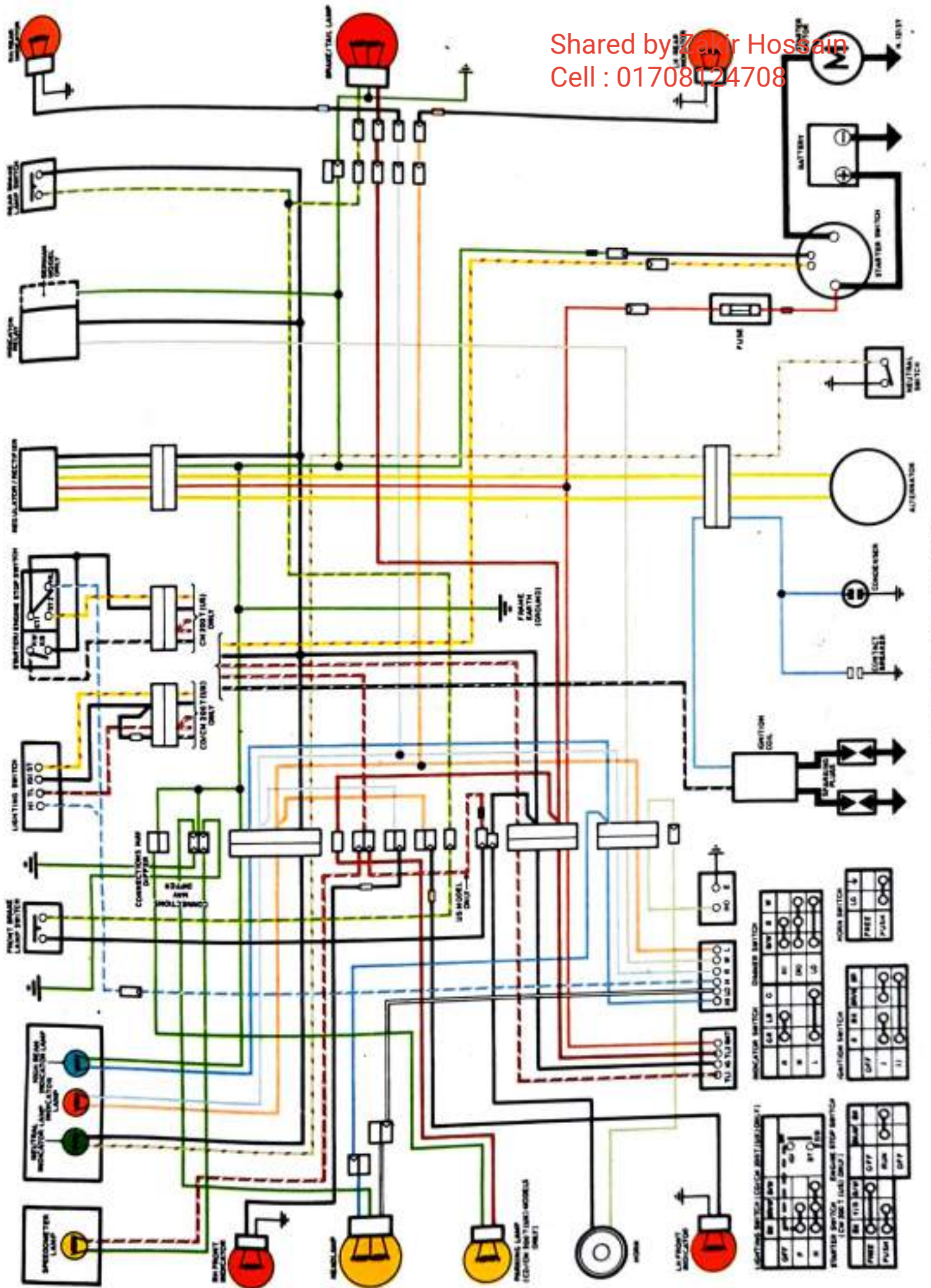


Wiring diagram - CD 185 T



Wiring diagram - CM 185 T

Shared by Zakir Hossain
Cell : 01708124708



Wiring diagram - CD and CM 200 T

Conversion factors

Length (distance)

Inches (in)	X 25.4 = Millimetres (mm)	X 0.039 = Inches (in)
Feet (ft)	X 0.305 = Metres (m)	X 3.281 = Feet (ft)
Miles	X 1.609 = Kilometres (km)	X 0.621 = Miles

Volume (capacity)

Cubic inches (cu in; in ³)	X 16.387 = Cubic centimetres (cc; cm ³)	X 0.061 = Cubic inches (cu in; in ³)
Imperial pints (Imp pt)	X 0.568 = Litres (l)	X 1.76 = Imperial pints (Imp pt)
Imperial quarts (Imp qt)	X 1.137 = Litres (l)	X 0.88 = Imperial quarts (Imp qt)
Imperial quarts (Imp qt)	X 1.201 = US quarts (US qt)	X 0.833 = Imperial quarts (Imp qt)
US quarts (US qt)	X 0.946 = Litres (l)	X 1.057 = US quarts (US qt)
Imperial gallons (Imp gal)	X 4.546 = Litres (l)	X 0.22 = Imperial gallons (Imp gal)
Imperial gallons (Imp gal)	X 1.201 = US gallons (US gal)	X 0.833 = Imperial gallons (Imp gal)
US gallons (US gal)	X 3.785 = Litres (l)	X 0.264 = US gallons (US gal)

Mass (weight)

Ounces (oz)	X 28.35 = Grams (g)	X 0.035 = Ounces (oz)
Pounds (lb)	X 0.454 = Kilograms (kg)	X 2.205 = Pounds (lb)

Force

Ounces-force (ozf; oz)	X 0.278 = Newtons (N)	X 3.6 = Ounces-force (ozf; oz)
Pounds-force (lbf; lb)	X 4.448 = Newtons (N)	X 0.225 = Pounds-force (lbf; lb)
Newton (N)	X 0.1 = Kilograms-force (kgf; kg)	X 9.81 = Newtons (N)

Pressure

Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)	X 0.070 = Kilograms-force per square centimetre (kgf/cm ² ; kg/cm ²)	X 14.223 = Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)
Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)	X 0.068 = Atmospheres (atm)	X 14.696 = Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)
Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)	X 0.069 = Bars	X 14.5 = Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)
Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)	X 6.895 = Kilopascals (kPa)	X 0.145 = Pounds-force per square inch (psi; lbf/in ² ; lb/in ²)
Kilopascals (kPa)	X 0.01 = Kilograms-force per square centimetre (kgf/cm ² ; kg/cm ²)	X 98.1 = Kilopascals (kPa)

Torque (moment of force)

Pounds-force inches (lbf in; lb in)	X 1.152 = Kilograms-force centimetre (kgf cm; kg cm)	X 0.868 = Pounds-force inches (lbf in; lb in)
Pounds-force inches (lbf in; lb in)	X 0.113 = Newton metres (Nm)	X 8.85 = Pounds-force inches (lbf in; lb in)
Pounds-force inches (lbf in; lb in)	X 0.083 = Pounds-force feet (lbf ft; lb ft)	X 12 = Pounds-force inches (lbf in; lb in)
Pounds-force feet (lbf ft; lb ft)	X 0.138 = Kilograms-force metres (kgf m; kg m)	X 7.233 = Pounds-force feet (lbf ft; lb ft)
Pounds-force feet (lbf ft; lb ft)	X 1.356 = Newton metres (Nm)	X 0.738 = Pounds-force feet (lbf ft; lb ft)
Newton metres (Nm)	X 0.102 = Kilograms-force metres (kgf m; kg m)	X 9.804 = Newton metres (Nm)

Power

Horsepower (hp)	X 745.7 = Watts (W)	X 0.0013 = Horsepower (hp)
-----------------	---------------------	----------------------------

Velocity (speed)

Miles per hour (miles/hr; mph)	X 1.609 = Kilometres per hour (km/hr; kph)	X 0.621 = Miles per hour (miles/hr; mph)
--------------------------------	--	--

Fuel consumption*

Miles per gallon, Imperial (mpg)	X 0.354 = Kilometres per litre (km/l)	X 2.825 = Miles per gallon, Imperial (mpg)
Miles per gallon, US (mpg)	X 0.425 = Kilometres per litre (km/l)	X 2.352 = Miles per gallon, US (mpg)

Temperature

Degrees Fahrenheit (°F) = (°C × $\frac{9}{5}$) + 32

Degrees Celsius = (°F - 32) × $\frac{5}{9}$

(Degrees Centigrade; °C)

*It is common practice to convert from miles per gallon (mpg) to litres/100 kilometres (l/100km), where mpg (Imperial) x 100 km = 282 and mpg (US) x 100 km = 235

Metric conversion tables

Inches	Decimals	Millimetres	Millimetres to Inches		Inches to Millimetres	
			mm	Inches	Inches	mm
1/64	0.015625	0.3969	0.01	0.00039	0.001	0.0254
1/32	0.03125	0.7937	0.02	0.00079	0.002	0.0508
3/64	0.046875	1.1906	0.03	0.00118	0.003	0.0762
1/16	0.0625	1.5875	0.04	0.00157	0.004	0.1016
5/64	0.078125	1.9844	0.05	0.00197	0.005	0.1270
3/32	0.09375	2.3812	0.06	0.00236	0.006	0.1524
7/64	0.109375	2.7781	0.07	0.00276	0.007	0.1778
1/8	0.125	3.1750	0.08	0.00315	0.008	0.2032
9/64	0.140625	3.5719	0.09	0.00354	0.009	0.2286
5/32	0.15625	3.9687	0.1	0.00394	0.01	0.254
11/64	0.171875	4.3656	0.2	0.00787	0.02	0.508
3/16	0.1875	4.7625	0.3	0.01181	0.03	0.762
13/64	0.203125	5.1594	0.4	0.01575	0.04	1.016
7/32	0.21875	5.5562	0.5	0.01969	0.05	1.270
15/64	0.234375	5.9531	0.6	0.02362	0.06	1.524
1/4	0.25	6.3500	0.7	0.02756	0.07	1.778
17/64	0.265625	6.7469	0.8	0.03150	0.08	2.032
9/32	0.28125	7.1437	0.9	0.03543	0.09	2.286
19/64	0.296875	7.5406	1	0.03937	0.1	2.54
5/16	0.3125	7.9375	2	0.07874	0.2	5.08
21/64	0.328125	8.3344	3	0.11811	0.3	7.62
11/32	0.34375	8.7312	4	0.15748	0.4	10.16
23/64	0.359375	9.1281	5	0.19685	0.5	12.70
3/8	0.375	9.5250	6	0.23622	0.6	15.24
25/64	0.390625	9.9219	7	0.27559	0.7	17.78
13/32	0.40625	10.3187	8	0.31496	0.8	20.32
27/64	0.421875	10.7156	9	0.35433	0.9	22.86
7/16	0.4375	11.1125	10	0.39370	1	25.4
29/64	0.453125	11.5094	11	0.43307	2	50.8
15/32	0.46875	11.9062	12	0.47244	3	76.2
31/64	0.48375	12.3031	13	0.51181	4	101.6
1/2	0.5	12.7000	14	0.55118	5	127.0
33/64	0.515625	13.0969	15	0.59055	6	152.4
17/32	0.53125	13.4937	16	0.62992	7	177.8
35/64	0.546875	13.8906	17	0.66929	8	203.2
9/16	0.5625	14.2875	18	0.70866	9	228.6
37/64	0.578125	14.6844	19	0.74803	10	254.0
19/32	0.59375	15.0812	20	0.78740	11	279.4
39/64	0.609375	15.4781	21	0.82677	12	304.8
5/8	0.625	15.8750	22	0.86614	13	330.2
41/64	0.640625	16.2719	23	0.90551	14	355.6
21/32	0.65625	16.6687	24	0.94488	15	381.0
43/64	0.671875	17.0656	25	0.98425	16	406.4
11/16	0.6875	17.4625	26	1.02362	17	431.8
45/64	0.703125	17.8594	27	1.06299	18	457.2
23/32	0.71875	18.2562	28	1.10236	19	482.6
47/64	0.734375	18.6531	29	1.14173	20	508.0
3/4	0.75	19.0500	30	1.18110	21	533.4
49/64	0.765625	19.4469	31	1.22047	22	558.8
25/32	0.78125	19.8437	32	1.25984	23	584.2
51/64	0.796875	20.2406	33	1.29921	24	609.6
13/16	0.8125	20.6375	34	1.33858	25	635.0
53/64	0.828125	21.0344	35	1.37795	26	660.4
27/32	0.84375	21.4312	36	1.41732	27	685.8
55/64	0.859375	21.8281	37	1.4567	28	711.2
7/8	0.875	22.2250	38	1.4961	29	736.6
57/64	0.890625	22.6219	39	1.5354	30	762.0
29/32	0.90625	23.0187	40	1.5748	31	787.4
59/64	0.921875	23.4156	41	1.6142	32	812.8
15/16	0.9375	23.8125	42	1.6535	33	838.2
61/64	0.953125	24.2094	43	1.6929	34	863.6
31/32	0.96875	24.6062	44	1.7323	35	889.0
63/64	0.984375	25.0031	45	1.7717	36	914.4

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The contents show how to remove, dismantle, inspect, repair and replace the following: engine, clutch, primary and secondary transmission; carburation and lubrication systems; ignition system; frame and suspension; wheels, brakes and tyres; and electrical system. Also included is routine maintenance, fault finding instructions, recommended lubricants and specifications.

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